Perinatal Mental Health in rural Bangladesh: A mixed methods exploration of the social determinants and perspectives of pregnant women and healthcare providers

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Abstract

Background: Pregnancy is a time of major psychological changes making pregnant women more susceptible to Perinatal Mental Health (PMH) conditions, especially antenatal depression with comorbid anxiety. Prevalence of antenatal depression in Bangladesh ranges from 18-33% compared to 7-15% in high-income countries like the UK. These disparities stem in part from poor understanding of, and stigma surrounding, PMH, which hinder pregnant women from seeking mental health care. In addition, there is limited research exploring social determinants and perspectives around PMH in rural Sylhet, Bangladesh.

Aim: This PhD research aimed to explore the social determinants and perspectives (women and healthcare providers (HCPs)) around antenatal depression in rural Sylhet, Bangladesh, to enable development of better care for underserved women in rural Bangladesh in collaboration with Maternal Aid Association (MAA).

Methods: 1) Two systematic reviews of the social determinants and perspectives surrounding PMH in South Asia; 2) Cross-sectional survey of 235 pregnant women in rural Sylhet; 3) Cross-sectional survey of 139 HCPs in Bangladesh; 4) Semi-structured interviews with 14 HCPs, and 5) mixed methods triangulation.

Results: The point-prevalence of antenatal depression was 56%. Intimate Partner Violence (adjusted odds ratio (AOR) 10.4 [95% confidence intervals (95%CI) 2.7-39.7]) and perceived husband's male gender preference (AOR 9.9 [95%CI 1.6-59.6]), were significantly associated with increased odds of antenatal depression among pregnant women. Increased family support was a significant protective factor for antenatal depression (AOR 0.94 [95%CI 0.91-0.97]). HCPs displayed non-stigmatising attitudes towards PMH, but limited knowledge regarding screening and management, and desired further training and education.

Conclusion: Antenatal depression commonly occurs in rural Sylhet, Bangladesh, highlighting the need for improved screening and management within this region. The findings recommend the integration of PMH into the MAA antenatal care services. Key activities could involve screening of antenatal depression, psychoeducation for women and families, and training and education for HCPs.

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Table of contents

Abstract i

Acknowledgements ii

List of tables viii

List of figures x

Abbreviations xi

Chapter 1. Introduction 1

1.1 Perinatal Mental Health (PMH) 1
1.1.1 Perinatal Mental Health in Bangladesh 1

1.2 Social determinants of Perinatal Mental Health 3

1.3 Healthcare providers role in Perinatal Mental Health care 5
1.3.1 Maternal Aid Association (MAA) 6

1.4 Aim and objectives 10

1.5 Outline of the thesis 10

Chapter 2. Social determinants of antenatal depression among women in Bangladesh, India and Pakistan: a systematic review & meta-analysis 13

2.1 Chapter overview 13

2.2 Introduction 13
2.2.1 Antenatal depression and anxiety in South Asia 14

2.3 Methods 16
2.3.1 Data sources and search strategy 16
2.3.2 Data extraction 22
2.3.3 Quality assessment 22
2.3.4 Data synthesis and statistical analysis 23

2.4 Results 24
2.4.1 Quality of included studies 37
2.4.2 Social determinants 38
2.4.1 Maternal Age 40
2.4.2 Maternal education/literacy 43
2.4.3 Maternal occupation 47
2.4.4 Husband's education/literacy 49
2.4.5 Husband's occupation 51
2.4.6 Household income 53
2.4.7 Intimate Partner Violence 57
2.4.8 Gender preference 59
2.4.9 Relationship with in-laws 61
2.4.10 Relationship with husband 63
2.4.11 Pregnancy intentions 65
2.4.12 Social support 67
2.4.13 Additional social determinants 69
2.4.14 Sensitivity analysis 69
2.4.15 Summary of studies from updated search between 4th January 2021 to 4th January 2023 70

2.5 Discussion 70
2.5.1 Key findings 71
2.5.2 Strengths and limitations of included studies 75
2.5.3 Strengths and limitations of this systematic review 76

2.5.4 Gaps in research ... 78
2.6 Chapter summary ... *79*

Chapter 3. Perceptions and attitudes around perinatal mental health in Bangladesh, India and Pakistan: a systematic review of qualitative data ... 80
3.1 Chapter overview ... *80*
3.2 Introduction ... *80*
3.3 Methods ... *81*
3.3.1 Search strategy ... 81
3.3.2 Quality appraisal ... 83
3.3.3 Data extraction and synthesis ... 84
3.4 Results ... *85*
3.4.1 Theme 1: Perceived causes of PMH ... 92
3.4.2 Theme 2: Perceived symptoms of PMH ... 94
3.4.3 Theme 3: Perceptions of motherhood ... 95
3.4.4 Theme 4: Accessing PMH care ... 96
3.4.5 Theme 5: Emotional sharing and coping strategies ... 100
3.4.6 Summary of studies from updated search between 4th January 2021 to 4th January 2023 ... 100
3.6 Discussion ... *101*
3.7 Conclusion ... *104*

Chapter 4. Theoretical and Conceptual framework ... 105

Chapter 5. Methodology ... 110
5.1 Chapter overview ... *110*
5.2 Research paradigm, ontology and epistemology ... *110*
5.3 Mixed methods methodology ... *112*
5.3.1 Data analysis and integration ... 116

Chapter 6. Social determinants and prevalence of antenatal depression among women in rural Bangladesh ... 119
6.1 Chapter overview ... *119*
6.2 Introduction ... *119*
6.3 Methods ... *120*
6.3.1 Study design and setting ... 120
6.3.2 Inclusion and exclusion criteria ... 120
6.3.3 Sampling ... 121
6.3.4 Survey instrument ... 122
6.3.5 Outcome variable – Antenatal depression ... 123
6.3.6 Independent variables – Social determinants ... 123
6.3.7 Data collection procedure ... 124
6.3.8 Data analysis ... 125
6.3.9 Ethical considerations ... 126
6.4 Results ... *127*
6.5 Discussion ... *138*
6.5.1 Key findings ... 138
6.5.2 Strengths and limitations of the study ... 140
6.5.3 Implications for wider PhD project ... 141
6.6 Chapter summary and contributions ... *141*

Chapter 7. Survey of healthcare providers' perinatal mental health knowledge, attitudes and learning needs in Bangladesh 143

7.1 Chapter overview 143

7.2 Introduction 143

7.3 Methods 144
7.3.1 Study design 144
7.3.2 Inclusion and exclusion criteria 144
7.3.3 Sampling and recruitment 144
7.3.4 Survey instrument and structure 145
7.3.5 Data collection 147
7.3.6 Statistical analysis 147
7.3.7 Ethical considerations 148

7.4 Results 149
7.4.1 Attitudes 151
7.4.2 Knowledge 158
7.4.3 Attitudes and knowledge 164
7.4.4 Learning needs analysis 164

7.5 Discussion 167
7.5.1 Attitudes 167
7.5.2 Knowledge 170
7.5.3 Attitudes and knowledge 172
7.5.4 Strengths and limitations 172
7.5.5 Implications for the wider PhD 174

7.6 Chapter summary and contributions 174

Chapter 8. The perceptions and experiences of healthcare providers in Bangladesh surrounding perinatal mental health: A qualitative exploration 176

8.1 Chapter overview 176

8.2 Introduction 176

8.3 Methods 177
8.3.1 Study design & setting 178
8.3.2 Participants 178
8.3.3 Topic guide 180
8.3.4 Data collection 181
8.3.5 Data analysis 183

8.4 Findings 184
8.4.1 Theme 1: Maternal healthcare sphere in Bangladesh 188
8.4.2 Theme 2: Conceptualisation of pregnancy and PMH 192
8.4.3 Theme 3: Perceived social determinants of PMH issues 197
8.4.4 Theme 4: PMH care provision 210

8.5 Discussion 221
8.5.1 Summary and interpretation of key findings 221
8.5.2 Strengths of the study 225
5.3 Limitations of the study 226
5.4 Implications for the wider PhD project 227

6. Chapter summary and contributions 228

Chapter 9. Triangulation of pregnant women's and healthcare providers' studies: A mixed methods analysis 229

9.1 Chapter overview 229

9.2 Methods 229

9.3 Findings 230

9.3.1 Agreement (Convergence) ... 237
3.2 Partial Agreement (Complimentary) ... 239
9.3.3 Silence ... 240
9.3.4 Dissonance (Discrepancy) ... 241

4. Discussion ... 242
4.1 Convergence and completeness assessment ... 242
4.2 Strengths and limitations of this mixed methods approach ... 243

5. Chapter summary and contributions ... 244

Chapter 10. Discussion & conclusion ... 246

10.1 Chapter overview ... 246

10.2 Key findings and contributions ... 246
10.2.1 Contribution 1: An investigation of rural Sylhet, Bangladesh ... 246
10.2.2 Contribution 2: High prevalence of antenatal depression in rural Sylhet ... 247
10.2.3 Contribution 3: Social determinants of antenatal depression in rural Sylhet ... 247
10.2.4 Contribution 4: Positive attitudes of HCPs towards PMH in Bangladesh ... 249
10.2.5 Contribution 5: Limited knowledge of PMH screening and management among HCPs in Bangladesh ... 250
10.2.6 Contribution 6: Implications for antenatal health care practice and policy in MAA ... 251

10.3 Strengths and limitations of the overall PhD ... 253
10.3.1 Strengths ... 254
10.3.2 Limitations ... 255

10.4 Further research & next steps ... 257
10.4.1 Qualitative studies exploring pregnant women's experiences and perceptions of PMH in Bangladesh ... 257
10.4.2 Longitudinal studies with larger sample sizes ... 257
10.4.3 Implementation of recommendations in MAA health camps ... 258

10.5 Conclusions ... 258

References ... 260

Appendices ... 283

Appendix 1 – Social determinants of antenatal depression and anxiety among women in South Asia: A systematic review and meta-analysis ... 283

Appendix 2 - Perceptions and attitudes around perinatal mental health in Bangladesh, India and Pakistan: a systematic review of qualitative data ... 308

Appendix 3 - Social Determinants and Prevalence of Antenatal Depression among Women in Rural Bangladesh: A Cross-Sectional Study ... 321

Appendix 4 – MOOSE checklist for Meta-analysis of Observational Studies ... 334

Appendix 5 - MEDLINE, Embase & PsychInfo search strategy via OVID ... 337

Appendix 6 - Data extraction form template ... 338

Appendix 7 - Newcastle-Ottawa Quality Assessment Scale adapted for cross-sectional studies ... 340

Appendix 8 - Quality assessment of all included studies ... 342

Appendix 9 – The ENTREQ checklist ... 344

Appendix 10 – Summary of CASP tool used for quality appraisal of qualitative studies ... 347

Appendix 11 – Summary of NOS tool used for quality appraisal of cross-sectional studies ... 348

Appendix 12 – STROBE checklist for pregnant women's cross-sectional survey study ... 349

Appendix 13 – Pregnant women's survey ... 352

Appendix 14 – Duke Social Support and Stress Scale scoring system ... 359

Appendix 15 – Ethics approval from Sylhet Women's Medical College Research Ethics Board *360*

Appendix 16 – STROBE checklist for HCP's cross-sectional survey study *361*

Appendix 17 – Healthcare providers' online survey *363*

Appendix 18 – COREQ checklist for qualitative study *370*

Appendix 19 – Topic guide for semi-structured interviews with healthcare providers *372*

Appendix 20 – Information sheet and consent form for healthcare providers' participating in semi-structured interviews *375*

List of tables

Table 1 Overview of thesis chapters 11
Table 2 Search term development using PICOS 18
Table 3 Inclusion and exclusion criteria for social determinants systematic review 19
Table 4 Summary characteristics of included studies 26
Table 5 Summary of social determinants identified in the review 39
Table 6 Association between antenatal depression/anxiety and maternal age 41
Table 7 Association between antenatal depression/anxiety and maternal education/literacy 44
Table 8 Association between antenatal depression/anxiety and husband's education/literacy 50
Table 9 Association between antenatal depression/anxiety and household income 54
Table 10 Inclusion and exclusion criteria for perceptions and attitudes systematic review 83
Table 11 Summary of included studies 86
Table 12 Summary of reasons for adopting a mixed-methods methodology 114
Table 13 Summary of respondent (pregnant women) characteristics 128
Table 14 Bivariate analysis of social determinants with antenatal depressive symptoms among pregnant women in MAA health camps in rural Sylhet 134
Table 15 Multivariate logistic regression model showing associated factors of antenatal depressive symptoms among pregnant women in MAA health camps in rural Sylhet 137
Table 16 Healthcare providers' attitude towards perinatal mental health 152
Table 17 Correlation between continuous demographic variables and attitude score 154
Table 18 Comparing mean attitude score of healthcare providers by gender, experience/education in mental health care, confidence in supporting pregnant women with depression/anxiety and access to perinatal mental health care 155
Table 19 ANOVA summary of mean attitude score and healthcare provider role 157
Table 20 Multiple comparisons of mean attitude score and healthcare provider role 157
Table 21 Summary of perinatal mental health knowledge questions of healthcare providers 159
Table 22 Correlation between continuous demographic variables and attitude score 160
Table 23 Independent samples t-test of mean knowledge score 161
Table 24 ANOVA summary of mean knowledge score and healthcare provider role 163
Table 25 Multiple comparisons of mean knowledge score and healthcare provider role 163
Table 26 Learning needs analysis of healthcare providers 165
Table 27 Summary of characteristics of Healthcare Providers interviewed 185
Table 28 Additional quotes supporting the theme Maternal healthcare sphere in Bangladesh 191
Table 29 Conceptualisation of pregnancy and PMH subthemes and additional quotes 196
Table 30 Perceived social determinants of PMH issues subthemes and additional quotes 207
Table 31 PMH care provision subthemes and additional quotes 217
Table 32 Convergence coding scheme for triangulation protocol 230

Table 33 Convergence coding matrix 231

List of figures

Figure 1 Map of Bangladesh with red arrow indicating the study location of Moulvibazar 9
Figure 2 PRISMA flow chart of the screening process 21
Figure 3 Forest plot of association between antenatal depression/anxiety and maternal occupation unadjusted OR 48
Figure 4 Forest plot of association between antenatal depression/anxiety and maternal occupation AOR 48
Figure 5 Forest plot of association between antenatal depression/anxiety and husband's occupation unadjusted OR 52
Figure 6 Forest plot of association between antenatal depression/anxiety and husband's occupation AOR 52
Figure 7 Forest plot of association between antenatal depression/anxiety and IPV unadjusted OR 58
Figure 8 Forest plot of association between antenatal depression/anxiety and IPV AOR 58
Figure 9 Forest plot of association between antenatal depression/anxiety and gender preference unadjusted OR 60
Figure 10 Forest plot of association between antenatal depression/anxiety and gender preference AOR 60
Figure 11 Forest plot of association between antenatal depression/anxiety and relationship with in-laws unadjusted OR 62
Figure 12 Forest plot of association between antenatal depression/anxiety and relationship with in-laws AOR 62
Figure 13 Forest plot of association between antenatal depression/anxiety and relationship with husband unadjusted OR 64
Figure 14 Forest plot of association between antenatal depression/anxiety and relationship with husband AOR 64
Figure 15 Forest plot of association between antenatal depression/anxiety and pregnancy intentions unadjusted OR 66
Figure 16 Forest plot of association between antenatal depression/anxiety and pregnancy intentions AOR 66
Figure 17 Forest plot of association between antenatal depression/anxiety and social/family support unadjusted OR 68
Figure 18 Forest plot of association between antenatal depression/anxiety and social/family support AOR 68
Figure 19 PRISMA Flow chart for qualitative systematic review 82
Figure 20 Thematic map of themes and sub-themes 91
Figure 21 Theoretical and conceptual framework 109
Figure 22 Flowchart depicting the convergent parallel mixed methods design used in my PhD 118
Figure 23 Bar graph showing the distribution of total EDPS-B scores 132
Figure 24 Demographic characteristics of the respondents; a) gender, b) healthcare provider role and c) healthcare providers primary work location 150
Figure 25 Flow diagram of HCP recruitment for interviews 179
Figure 26 Thematic map of themes and sub-themes 187

Abbreviations

ADS	Antenatal Depressive Symptoms
AKUADS	Aga Khan University Anxiety and Depression Scale
ANC	Antenatal Care
AOR	Adjusted Odds Ratio
BDI	Beck Depression Scale
CBT	Cognitive behavioural therapy
CES-D	Centre for Epidemiological Studies Depression Scale
CHW	Community Health Worker
CI	Confidence Interval
DASS	Depression Anxiety Stress Scale
DUSOCS	Duke Social Support & Stress Scale
EPDS	Edinburgh Postnatal Depression Scale
EPDS-B	Edinburgh Postnatal Depression Scale - Bangla
GDS	Goldberg's Depression Scale
HADS	Hospital Anxiety Depression Scale
HAM-D	Hamilton Depression Rating Scale
HBM	Health Brigade Members
HCP	Healthcare provider
HIC	High Income Country
IPV	Intimate Partner Violence
IQR	Interquartile Range
K-10	Kessler Psychological Distress Scale
LMIC	Low-Middle-Income Country
MAA	Maternal Aid Association
MOOSE	Meta-Analysis of Observational Studies in Epidemiology
NGO	Non-governmental organisation
OR	Odds Ratio
PHQ	Patient Health Questionnaire
PICOS	Population, Intervention, Comparison, Outcomes and Study
PMH	Perinatal Mental Health
PRTS	Pregnancy-related thoughts scale
RCT	Randomised control trials
SCAN	Schedule for Clinical Assessment in Neuropsychiatry
SD	Standard Deviation

SDG	Sustainable Development Goal
STAI	State-Trait Anxiety Inventory
WHO	World Health Organisation

Chapter 1. Introduction

1.1 Perinatal Mental Health (PMH)

Depression is a leading cause of disability worldwide (1), often presenting with comorbid anxiety (2). Globally, more than 280 and 300 million people of all ages suffer from depression and anxiety, respectively (3). Women are about twice as likely than men to develop these conditions (4), with depression and anxiety being the most common mental health conditions among women (5). There are strong links between depression and anxiety and physical health, including coronary heart disease given the neurochemical mechanisms involved in both (6), and increased risk of asthma and hypertension amongst other physical health conditions. Depression and anxiety are also associated with suicide and suicidal ideations which occur twice as much in women than in men (7). These consequences negatively affect women's quality of life, making depression and anxiety amongst women a significant public health issue (8).

Pregnancy is a time of major physiological, hormonal, and psychological changes which place women at an increased risk of mental and emotional challenges during and immediately after pregnancy (9). Perinatal Mental Health (PMH) is recognised as a worldwide public health issue, with approximately 10% of pregnant women and 13% of women who have just given birth experiencing some sort of mental health condition, worldwide (10). Depression is the most common mental health condition affecting women during the perinatal period (11), which is commonly defined as the period during pregnancy and up to a year after childbirth (12). However, most of the research related to perinatal depression and anxiety focuses on the postnatal period, after childbirth, while evidence suggests that two thirds of cases of postnatal depression actually present antenatally, before childbirth (13,14). Hence, this PhD focuses on the antenatal period as a time when women face many changes and challenges and are particularly vulnerable to depression and anxiety. Depression and anxiety during pregnancy have negative impacts on the mother and child including increased risk of postnatal depression, suicidal ideations, low birth weight and impaired fetal development (15).

1.1.1 Perinatal Mental Health in Bangladesh

In Low-Middle-Income Countries (LMICs), the prevalence of PMH conditions is even greater with approximately 15.6% of women experiencing antenatal mental

health conditions and 19.8% postnatally, giving rise to health inequalities between countries. These higher prevalence rates are a result of a number of factors including the lack of mental healthcare integration into antenatal care (ANC), poor understanding or acceptance of mental health conditions and higher socioeconomic deprivation in LMICs (16).

Bangladesh is one of the most densely populated countries and is listed as an LMIC by the World Bank (17). It has achieved enormous improvements in maternal mortality, which has declined from 322 deaths per 100,000 livebirths in 1998-2001 to 194 in 2007-2010 (18). This further improved to 123 in 2020 (19). This was largely due to increased availability and access to health facilities, enabled by policy changes and associated health system developments, creating an expansion of maternal healthcare services from within Bangladesh and internationally (18). PMH, on the other hand, has received much less attention. From a cross-sectional survey carried out in 20 sub-districts of Bangladesh between 2008-2014, it was found that the prevalence of common PMH disorders was 49% (20). A longitudinal study carried out in 2008-2009 in rural Bangladesh found the prevalence of antenatal depression was 18% (21). A more recent study published in 2022 found a prevalence rate of 53% of antenatal anxiety in Bangladesh (22). It is possible that these percentages are underestimated due the stigma attached to mental health preventing people from understanding, acknowledging and sharing the issue (23).

Despite this burden, mental health remains a neglected topic within the Bangladeshi population, in Government policy and the provision made through the health system. Only 0.5% of the total health budget in Bangladesh is allocated to mental health (24) compared to approximately 13% allocated in England. This is a small proportion considering the burden attributable to psychiatric disorders in Bangladesh is approximately 11.2% (25). Furthermore, there is only one national-level mental health institute in Bangladesh, the National Institute of Mental Health in Dhaka, and limited mental health hospitals, the majority of which are located in urban areas with the density of psychiatric beds in and around the capital Dhaka being five times higher than anywhere else (25). This raises a problem as 70% of the population live in rural areas where there is limited mobility and accessibility to mental health services as they are not located close to these regions. These areas are often served by informal providers and non-governmental organisations (NGO), such as the Maternal

Aid Association (MAA), with little governmental authority (25). These figures highlight the lack of attention mental health receives from the government resulting in limited facilities in rural areas of Bangladesh. This affects the care that pregnant women receive, where mental health is often a secondary thought, or mostly non-existent. Therefore, pregnant women in rural regions of Bangladesh do not receive adequate, holistic antenatal care which, very importantly, should include PMH care. Hence, this PhD focuses on rural Bangladesh given the limited research exploring antenatal depression in rural regions.

1.2 Social determinants of Perinatal Mental Health

Social determinants of health are complex, multidimensional factors and "conditions in which individuals are born, grow, live, work and age", as defined by the World Health Organisation (WHO) (26). The WHO further states that these factors are "shaped by the distribution of money, power and resources at global, national and international levels" (26). Over the last two decades, evidence around these social determinants has revealed them to display a significant role in shaping health, healthcare, health indicators, settings and populations (27). Social determinants of health are considered to be mostly responsible for health inequalities by creating societal stratification based on social and economic class, gender and ethnicity (26). Examples of social determinants, among many, include income, education, employment, gender and racial inequity, access to housing, food, and healthcare, social support, social norms, culture, social and family disruption, crime and violence (28). Although there is mixed evidence of the causal relationship between these social determinants and health, there is a wealth of evidence showing that they play a significant role in the underlying mechanisms within many health conditions and can influence the risk of getting sick or injured in the first place (27). Notably, there are complex interactions and feedback loops among these social determinants making it harder to analyse them and elucidate clear causal pathways (28).

Alongside tackling the underlying mechanisms and causes of many diseases, investigating social determinants plays a significant role in reducing health inequalities that are often rooted in social and economic disadvantages (29). When there are disparities in the distribution of power, money and resources, social determinants can create undesirable circumstances such as poverty, discrimination, gender and racial inequity and more (26). The Sustainable Development Goals (SDGs) encompasses

several goals related to social and economic inequalities. No poverty, zero hunger, gender equality, decent work and economic growth and reduced inequalities are examples of SDGs which can only be met if social determinants of health are addressed (30). Particularly in Bangladesh, where there are widening social and health inequalities leading to increasing poverty in rural areas compared to urban areas and gender discrimination, addressing social determinants can help better understand and address the needs of the poorest and most vulnerable groups (31). This will enable effective strategies to reduce existing gaps and accelerate progress towards equity and justice. This PhD particularly addresses SDG 3 (good health and wellbeing), 5 (gender equality) and 10 (reduced inequalities).

Mental health disorders such as depression and anxiety can be greatly shaped by the conditions in which people live, work and age and their social, economic and cultural environments (32). The same applies to antenatal depression where social factors play a huge role in the mental health of pregnant women, whether that stems from financial worries, social and family support or cultural norms (21). Further, social inequalities are associated with increased risk of many common mental health disorders (32). The WHO identifies the increasing need to raise the priority given to the prevention of mental health disorders and to promote positive mental health through actions on the social determinants of health (32). It is at the level of social determinants where policies, interventions and programmes can be better tailored and delivered to ensure prevention of depression and anxiety during pregnancy. Muñoz (33) suggests that identifying pregnant women at risk of depression and referring them to proper interventions would aid the overall health of population as depression during pregnancy can have a knock-on effect on childhood development triggering a life course effect.

Studies report some important social determinants of PMH in Bangladesh including intimate partner violence (IPV), low family support, low maternal education and male gender preference (21,34). Nevertheless, there is a paucity of current research investigating social determinants of antenatal depression in rural regions of Bangladesh. It is important to further investigate these social determinants of depression during pregnancy, especially in a country such as Bangladesh which often neglects mental health, to understand and tackle the underlying mechanisms that play a role in antenatal depression. Further, given the widespread poverty and gender

inequality in Bangladesh (ranked 142 of 188 countries in the Gender Development Index and Gender Inequality Index) (35), particularly in rural regions, it is hugely important to investigate the role of gender and wider social determinants within antenatal depression. This will enable better identification of pregnant women at risk and to prevent development of depression through tailored and effective interventions and programmes such as psychosocial counselling.

The 'Dahlgren-Whitehead model' developed in 1991 is one of the most widely used models for determining health inequalities that maps the relationship between the individual, their environment and health. This model is useful as it helps consider layers of influences on health, expanding perspectives outwards from individual factors to the possible role of wider determinants of health, and building up a complete, holistic picture. Therefore, this model will be used as a framework to guide the methods and interpretation of findings (see Chapter 4 for further details).

1.3 Healthcare providers role in Perinatal Mental Health care

Healthcare providers (HCPs) are individuals who are part of the health workforce working to deliver some form of health care to patients and communities. This includes physicians/doctors, nurses, midwives, dentists and pharmacists working to provide health care. It is important we use the term 'healthcare providers' to include Community Health Workers (CHWs) and charity workers who play a significant role in providing health care for pregnant women in Bangladesh, particularly in rural areas. These areas are often served by informal providers, such as CHWs and non-governmental organisations like MAA.

Mental health is a widely stigmatised and taboo topic in Bangladeshi communities resulting in pregnant women receiving little/inadequate healthcare for mental health conditions. Studies worldwide show that HCPs play an important role in the way mental health is perceived by communities and access to mental health care (23,36,37). Hence, it is crucial that we understand HCPs' attitudes, knowledge and learning needs around PMH and how this may impact women receiving healthcare during their pregnancy in Bangladesh.

To my best knowledge, there is only one qualitative study conducted by Williams et al (23) which investigates HCPs' attitudes towards PMH in Bangladesh. In that study,

34 medical personnel (obstetrics and gynaecology physicians, interns and nurses) were interviewed and asked about their training in mental health, protocol for diagnosis, treating depression, challenges for mental health in Bangladesh, experience with cases and general perceptions of depression in new mothers. Williams et al (23) found that 50% of the medical personnel did not take postpartum depression seriously and did not consider it to be a health concern. The study also found that medical students treat the psychiatry portion of their curriculum as a vacation. This highlights the issue of mental health not being viewed as a serious health condition and these attitudes often start from the educational level trickling into the professional life.

Williams et al (23) only interviewed doctors, nurses and medical interns. CHWs and charity workers, such as MAA health workers, play a considerable role in providing healthcare for women in rural areas where they do not have access to hospitals or health institutes. Further, midwives have not been investigated in the literature highlighting a key gap. Midwives are recently becoming more prominent within antenatal care in Bangladesh and have more direct contact with a wide range of pregnant women from all regions. Therefore, it is important to include these specific HCPs' views in these studies as they may provide different or additional views and perspectives.

1.3.1 Maternal Aid Association (MAA)

This PhD is funded through the ESRC's Northern Ireland and Northeast (NINE) Doctoral Training Programme's Collaborative Pathway. As such, my PhD is in collaboration with MAA. MAA is a volunteer-run charity with a UK and Bangladesh team, working to improve maternal health outcomes in Bangladesh. MAA has designed an original, primary maternal healthcare intervention called 'OneMaa' as a means of improving maternal health outcomes in a context of a fragmented healthcare system that currently amplifies socio-economic inequalities to the detriment of pregnant women in Bangladesh. The model consists of four components (38):

1. **Participatory Learning and Action (PLA) Women's groups (WG):** A community-based participatory intervention where local women will meet fortnightly to discuss issues surrounding maternal health and identify problems in the community which they wish to tackle.

2. **Health Brigade Members:** Made up of final year medical students who will provide regular antenatal care to pregnant women and identify high risk cases for further referral.
3. **The Clinic:** Functions as a rural health clinic where MAA-employed junior doctors (who have had further training by the Royal College of Obstetrics & Gynaecology, Bangladesh) will assess and manage high-risk pregnancies referred by HBMs.
4. **Gravida App:** The app will be used by HBMs to collect data, track pregnancies and to identify high risk pregnancies for referral to the clinic, where the Doctors will make use of the data. Thus, connecting the components of the model into a functioning system.

MAA is primarily based in Moulvibazar in Sylhet, Bangladesh. Moulvibazar (**Figure 1**) is a south-eastern district in the division of Sylhet and is composed of seven subdistricts or '*upazilas*'. Figures suggest that in Moulvibazar, approximately 46.4% of maternal deaths occur at home, and 7.1% enroute towards a health facility (39). According to the Bangladesh Demographic Health Survey 2022 (40), 2,640 women had a live birth in rural Bangladesh in the 2 years preceding the survey, compared to 970 live births in urban Bangladesh. Overall, 88% of received at least one antenatal check-up from a medically trained provider. In Sylhet 80% (n=184) of women received antenatal care from a skilled provider. This is lower than all the other divisions in Bangladesh with Khulna having the highest percentage at 91.4% (n=342). Sylhet also has the second lowest percentage of deliveries attended by a medically trained provider at 59.2% (n=142), the lowest being Mymensingh at 57.1% (n=190) and the highest being in Khulna at 86% (n=326). This highlights the differences in ANC coverage and appropriately assisted births between Sylhet and the rest of Bangladesh. Sylhet has been cited by the United Nations Population Fund (UNFPA) for having poor maternal health indicators (41) including the proportion of skilled birth attendants-attended deliveries by comparison to other divisions. Sylhet division has the highest fertility rate and lowest contraceptive use (42,43). Contraceptive prevalence rate is only 4.8% in Sylhet compared to the national average of 62.4% (44). This is a concern as high fertility rates and low contraceptive rates indicate lack of education, family planning services and gender equity within this region. Furthermore, the Sylhet region has geographical characteristics such as hilly areas which make it difficult for universal health coverage in Bangladesh (44). Despite the posting of female paramedics

intended to target emergency cases, maternal and neonatal health in Moulvibazar, Sylhet remains underwhelming in comparison to other areas in Bangladesh.

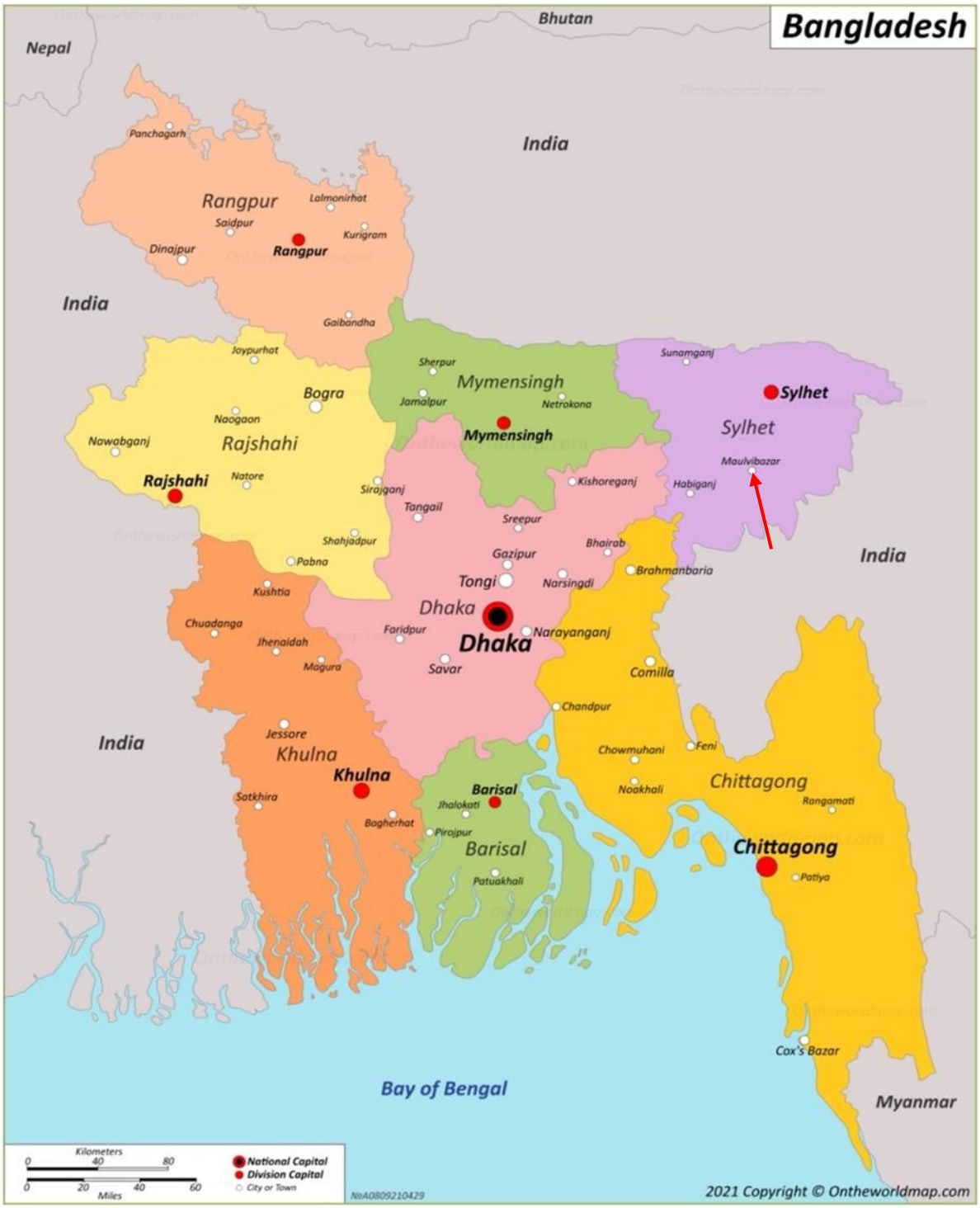


Figure 1 Map of Bangladesh with red arrow indicating the study location of Moulvibazar

Therefore, this collaboration with MAA enabled the co-production of a PhD project that focused on the region of Sylhet in Bangladesh. To the best of my knowledge, there are no known studies exploring antenatal depression in Sylhet, Bangladesh. Therefore, it is important to focus on Sylhet, one of the eight administrative regions of Bangladesh, to ensure improvement of PMH in Bangladesh as a whole and reduce health inequalities within the country. The development of health data and insights from this PhD will inform recommendations for the integration of PMH into MAA ANC services in rural Sylhet. This project effectively demonstrates global collaboration and engagement with a non-academic organisation which benefits both the research and MAA.

1.4 Aim and objectives

This PhD aims to explore the social determinants and perspectives around antenatal depression, the most prominent PMH condition, in rural Sylhet, Bangladesh to allow for better development of care for underserved women in collaboration with MAA.

Objectives:

1) To review current literature on social determinants and attitudes surrounding PMH in South Asia
2) To determine the prevalence of antenatal depression in rural Sylhet
3) To analyse social determinants associated with antenatal depression in rural Sylhet
4) To explore PMH knowledge, attitudes and learning needs of HCPs in Bangladesh
5) To form recommendations regarding integration of PMH into antenatal care services

1.5 Outline of the thesis

The purpose of this chapter has been to provide a background and justification for this research. The remaining thesis consists of nine chapters which have been summarised below in **Table *1***.

Table 1 Overview of thesis chapters

Chapter no.	Description	Target objective(s)	Contribution/novelty
Chapter 2	A systematic review and meta-analysis of the social determinants of antenatal depression and anxiety in South Asia	**Objective 1:** To review current literature on social determinants and attitudes towards perinatal depression in South Asia	The first meta-analysis investigating social determinants of antenatal depression in Bangladesh, India and Pakistan. Published in PLoS One 2022 (45) (Appendix 1).
Chapter 3	A systematic review of the perceptions and attitudes around PMH in South Asia	**Objective 1:** To review current literature on social determinants and attitudes towards perinatal depression in South Asia	The first systematic review to explore perceptions and attitudes around PMH in Bangladesh, India and Pakistan. Published in BMC Pregnancy and Childbirth 2022 (46) (Appendix 2).
Chapter 4	Theoretical and conceptual framework using the Dahlgren-Whitehead model of social determinants	N/A	Facilitated the development of a unique insight into the social determinants and theories surrounding antenatal depression in rural Bangladesh.
Chapter 5	A critical description of the mixed methods methodology	N/A	A mixed methods exploration of the social determinants and perspectives around PMH in Bangladesh.
Chapter 6	Cross-sectional survey study investigating the prevalence and social determinants of antenatal	**Objective 2:** To determine the prevalence of antenatal depression in rural Sylhet	The first study to investigate prevalence and social determinants of antenatal depression in rural Sylhet, Bangladesh. Published in the International Journal of

	depression in rural Sylhet, Bangladesh	**Objective 3:** To analyse social determinants associated with antenatal depression in rural Sylhet	Environmental Research and Public Health 2023 (47) (Appendix 3).
Chapter 7	Cross-sectional survey study investigating knowledge, attitudes and learning needs of HCPs in Bangladesh around PMH	**Objective 4:** To explore PMH knowledge, attitudes and learning needs of healthcare providers in Bangladesh	The first survey study to investigate knowledge, attitudes and learning needs of a wide range of HCPs, including midwives and CHWs, in Bangladesh around PMH.
Chapter 8	A qualitative exploration of the perspectives and experiences of HCPs surrounding PMH in Bangladesh	**Objective 4:** To explore PMH knowledge, attitudes and learning needs of healthcare providers in Bangladesh	The first qualitative study to explore perspectives and experiences of a wide range of HCPs, including midwives and CHWs, in Bangladesh around PMH.
Chapter 9	A mixed methods triangulation of findings from chapter 6, 7 & 8	**Objectives 2, 3 & 4**	The first mixed methods exploration of the social determinants and perspectives around PMH in Bangladesh.
Chapter 10	Discussion and conclusion	**Objective 5:** To form recommendations regarding integration of perinatal mental health into antenatal care services.	Informing actionable recommendations for the effective integration of PMH into MAA antenatal care services in rural Sylhet.

Chapter 2. Social determinants of antenatal depression among women in Bangladesh, India and Pakistan: a systematic review & meta-analysis

2.1 Chapter overview

This chapter aims to provide a comprehensive review and meta-analysis of the social determinants of antenatal depression and anxiety among women in Bangladesh, India and Pakistan to address objective 1 of this PhD. This will allow for a better understanding of the literature and identify the key social determinants to be further explored in subsequent chapters. Within this chapter, the rationale for conducting a systematic review and meta-analysis is explained followed by a detailed outline of the methods employed. A summary of the results is provided along with a critical discussion of the findings, strengths and limitations of this review. This systematic review was published in PLoS One in February 2022 (45).

2.2 Introduction

A systematic review is a summary and critical analysis of all the literature surrounding a particular research question. It uses explicit and reproducible methods to systematically search, appraise and synthesise results around that question (48). This makes it different from other forms of reviews which do not use such systematic or comprehensive methods. Rapid reviews are time limited and will omit some of the systematic steps to produce information quickly. Literature reviews summarise the main studies within the literature not based on inclusion or exclusion criteria to provide a foundation of knowledge on a topic. Scoping reviews tend to answer more broader topic focused questions with no need for quality assessments or critical appraisal. Hence, I chose to undertake a systematic review as it provides the best evidence for a focused social determinants research question using systematic and reproducible methods.

Meta-analysis is a statistical approach that involves combining data from several similar studies that address the same question to produce a single estimate for the effect size of treatment, or in the case of this study, social determinants. Meta-analysis uses statistical analysis methods to pool together statistical results (48). This was ideal for my study because I could pool odds ratios (OR) from the included studies

to obtain a single summary effect size for each social determinant studied. This provided an understanding of the direction and magnitude of association between the determinants and antenatal depression/anxiety. Consequently, this provided recommendations for my data collection phase where I used the knowledge of which social determinants displayed the greatest statistical impact to be the focus of the surveys and interview topic guide at a later stage in this PhD. This review also provided a better understanding of the gaps and limitations in the research, which informed my data collection in terms of study population, participant recruitment and further elements of this PhD research.

Systematic reviews and meta-analyses enable reducing of the quantity of data by summarising data from multiple sources and assist in the planning of further research or guidelines. The methods used in systematic reviews and meta-analyses limit bias and allow reliable and accurate conclusions to be drawn which enable quick and effective decision making. However, certain limitations such as quality of included studies, heterogeneity, selection of studies and publication bias can affect the quality of systematic reviews, therefore, the methods and results considered these limitations as described throughout this chapter.

2.2.1 Antenatal depression and anxiety in South Asia

As highlighted in Chapter 1, prevalence of antenatal depression and anxiety is reported to be higher in LMICs. Bangladesh, India and Pakistan are LMICs which form a major part of the Indian subcontinent of South Asia. They were once all one country until 1947 when India and Pakistan gained independence, after which in 1971, Bangladesh, formerly known as East Pakistan, gained independence (49). Although there is heterogeneity within and between these countries, they share cultural and social similarities which are reflected in a degree of commonality in the challenges faced by women living in them, hence are often studied together to understand many health phenomena within South Asia (50). Further, given the small number of studies identified in Bangladesh alone through my scoping work, it was deemed appropriate to broaden the search to include Pakistan and India.

A longitudinal study carried out in 2008-2009 within rural areas of Bangladesh found the prevalence of antenatal depression was 18% and antenatal anxiety was 29.4% (21), whereas, a higher point prevalence of antenatal depression of 33% was

established in another study in the southwest part of Bangladesh (34). Studies in India between 2013-2016 have reported a wide-ranging prevalence of antenatal depression between 9.8% and 36.7% (51–53) and a prevalence of 55.7% for antenatal anxiety (54). In Pakistan, between 2018-2019, the prevalence of antenatal depression ranged from 18% (55) to 39.5% (56) and the prevalence of antenatal anxiety was around 34.5% (57). The differing prevalence rates found between and within the countries may be a result of methodological differences in sample size, sample characteristics and region of study. It may also highlight the heterogeneity between the countries. It is likely that these reported percentages are underestimated due to the stigma attached to mental health preventing people from understanding, acknowledging and sharing these issues within these populations (23). This is particularly prominent among women who are often victims of gender-based violence, lack of family planning and low family support leading to depression and anxiety during pregnancy (16). These are common reported social determinants of antenatal depression and anxiety (16,21).

Although there are existing reviews investigating social determinants of antenatal depression/anxiety (58–60), they have not carried out meta-analyses to determine pooled summary estimates of association. Only two reviews include some studies in Bangladesh, India or Pakistan (59,60), however, they are limited as they did not capture the range of studies published in these countries as the search was only up until 2015, missing out many recent studies. To my knowledge, there are no known systematic reviews and meta-analyses investigating the social determinants of antenatal depression and anxiety in Bangladesh, Indian and Pakistan, countries in which women face increased vulnerability to mental health issues and gender inequalities. Understanding social determinants of antenatal depression and anxiety will enable holistic care for pregnant women in these countries where mental health is given less attention (24). This understanding will assist in creating and implementing tailored interventions and programmes for pregnant women to reduce symptoms of depression and anxiety during pregnancy and consequently, positively impacting mental health after childbirth.

This chapter aims to identify, synthesise and appraise the available international evidence on social determinants of antenatal depression and anxiety in women living in Bangladesh, India and Pakistan, and to identify which determinants have the most

impact on antenatal depression and anxiety and areas where more research is required.

2.3 Methods

Methods and reporting were conducted in accordance with the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) guidelines (Appendix 4). MOOSE guidelines are the formal reporting checklist of meta-analyses of observational studies. MOOSE guidelines, instead of the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) checklist used for Randomised Controlled Trials (RCTs), is preferred for meta-analyses of observational studies. Observational studies have different biases than RCTs, such as confounding, publication bias and selection bias, which need to be considered in reviews and the MOOSE guidelines acknowledge this (61). Hence, this was deemed to be the most appropriate guideline to use for this review. The review protocol was registered on the NIHR PROSPERO database of systematic reviews (reference CRD42020167903).

2.3.1 Data sources and search strategy

Observational studies (case-control, cross-sectional and cohort studies) investigating social determinants of antenatal depression and/or anxiety in any region of Bangladesh, India and Pakistan published in English or containing English translations since 1st January 2000 to 4th January 2021 were included in the systematic review. I undertook further searches to cover the time until 4th January 2023 prior to thesis submission, however, results from these additional studies were not included in the final meta-analyses which have been published, but a summary of findings (see section 2.4.15) are included in this chapter.

Causal inference is best established through RCTs, however, randomly allocating participants to certain social determinants would raise ethical concerns (62). Therefore, in risk factor research, evidence from observational studies is mostly relied upon. The date restriction was imposed because an initial scoping search of Medline showed no appropriate studies before 2000, suggesting that the investigation of risk factors of PMH in South Asia is a relatively recent area of research interest. Over time, clinical approaches and thinking towards PMH and mental health as a whole, particularly in LMICs, have developed and changed as highlighted by Rodrigues et al (63) in a 2003 paper investigating cultural validity of maternal depression, and

publications of pioneering global mental health papers in The Lancet in 2007 (64). Therefore, a date restriction would ensure that the literature in this review would still be clinically and socially relevant. Adopting this date restriction also narrowed the number of irrelevant search results returned, aiding in the screening of titles in a short time-frame whilst still capturing the most up to date literature.

Keywords and MeSH headings were developed with advice from an information specialist at Newcastle University in accordance with the PICOS framework (**Table 2**). PICOS is the most widely used framework and is endorsed by the Cochrane Collaboration (65). Although originally intended for intervention study reviews, it is being used more widely in observational study reviews and qualitative research, demonstrating its cross-design applicability (66). The PICOS framework refers to the patient, population or disease being addressed; the interventions or exposure; the comparator group; the outcome or endpoint; and the study design to be included (66). PICOS was used to give structure to the development of search terms and to ensure no aspect of the research question was left out. This framework covered all the important aspects of the search which other frameworks, such as SPIDER, would not achieve. SPIDER does not include a comparison group which is an important aspect of this review as papers must compare pregnant women with depression/anxiety to pregnant women without depression/anxiety. For this review, the intervention/exposure was taken to be the social determinants and the outcome was taken to be depression/anxiety during pregnancy. Final search strategies are provided in Appendix 5.

Table 2 Search term development using PICOS

	Patient, population or disease being addressed		**Intervention or exposure**	**Comparator group**	**Outcome/ endpoint**	**Study design**
	AND					
OR	**Country terms:** Bangladesh* India* Pakistan* "South Asia*"	**Pregnancy terms:** Maternal Perinatal Antenatal Pregnan* Postpartum Prenatal postnatal	Determinants Factors "risk factors" Dimensions Correlates *perspective** *views.mp.* *opinions.mp.* *attitude** *perception**	Pregnant women with depression/anxiety must be compared to pregnant women without depression/anxiety	Depress* Anxi* Mood Stress "mental health"	Observational studies only *Qualitative data*

*These are search terms with truncation to allow for alternative endings to the terms
Terms in italics are relevant to the second systematic review in Chapter 3 of this PhD

The review's inclusion and exclusion criteria are given in **Table *3*** below:

Table 3 Inclusion and exclusion criteria for social determinants systematic review

Inclusion criteria	**Exclusion criteria**
Full published studies (i.e. not editorials, abstracts, position pieces, research letters or posters).	Papers not published or available in full.
Studies involving observational quantitative research methods (cross-sectional, case-control and cohort studies) that investigated the social determinants of depression and/or anxiety during pregnancy.	Randomised controlled trials or intervention studies. Studies using a sub-group of pregnant women e.g. pregnant women of certain BMI, with particular conditions etc
Papers published between 1st January 2000 and 4th January 2021.	Papers published before 1st January 2000.
Papers published in the English language or contains English translations if not originally in English.	Papers not available in the English language.
Studies that use validated tools for measuring depression/anxiety.	Studies that did not use validated tools for measuring depression/anxiety.
Studies that report on the statistical association between the social determinants and depression/anxiety (i.e. odds ratios (OR) and 95% Confidence Interval (95% CI)) or report the raw numbers/proportions of pregnant women with depression/anxiety with and without each risk factor to enable calculations of ORs and 95% CI.	Studies that did not report statistical associations or provide raw data for calculations.
Studies conducted in Bangladesh, India or Pakistan as the host country.	Studies not conducted in Bangladesh, India or Pakistan as the host country.

A systematic literature search was conducted using the electronic databases Medline, Embase, PsycInfo, Scopus, Web of science and Prospero using keywords and MeSH headings (see Appendix 5 for full Medline search strategy). Search terms and subject headings were converted into the relevant format for these electronic databases. Grey literature was also searched using the Open Grey database and searching charities, non-governmental organisations and government websites, such as Building Resources Across Communities (BRAC), UNICEF, National Health Mission India and International Centre for Diarrhoeal Disease Research Bangladesh, in Bangladesh, India and Pakistan. Grey literature is important to include as there may be published results from organisations and governments which are not found in traditional scientific journals but still provide useful information. In particular, charities and non-governmental organisations in Bangladesh, India and Pakistan play a considerable role in conducting research. Supplementary searches included hand searching reference lists of studies included from databases to find any relevant studies which have been cited. Citation searches using Google Scholar, limited to the first ten pages, were carried out to identify any published studies that had cited the included studies. These steps were repeated with new studies identified until no further studies were found. Authors were contacted when full texts of studies were not found, or additional data were required for inclusion in the meta-analysis. Database searches were completed on 4th January 2023.

Search results from the electronic databases were exported into the reference manager, Zotero, and screened independently by two reviewers (including myself). Zotero provides a platform to export studies from databases, automatically remove duplicates and to conduct screening of titles. Firstly, the initial screening of titles and abstracts were carried out against the inclusion and exclusion criteria to identify studies that could potentially be included. Exclusion of studies occurred when both reviewers decided to exclude independently, otherwise it was included into the full text screening phase. Following decisions on inclusion of titles and abstracts, the full texts of these studies were screened independently by the two reviewers. Where there were disagreements between the two reviewers regarding eligibility, discussions took place to resolve it or a third reviewer from the supervisory team was included (this was not required). A PRISMA flow diagram was used to record the screening process (**Figure 2**).

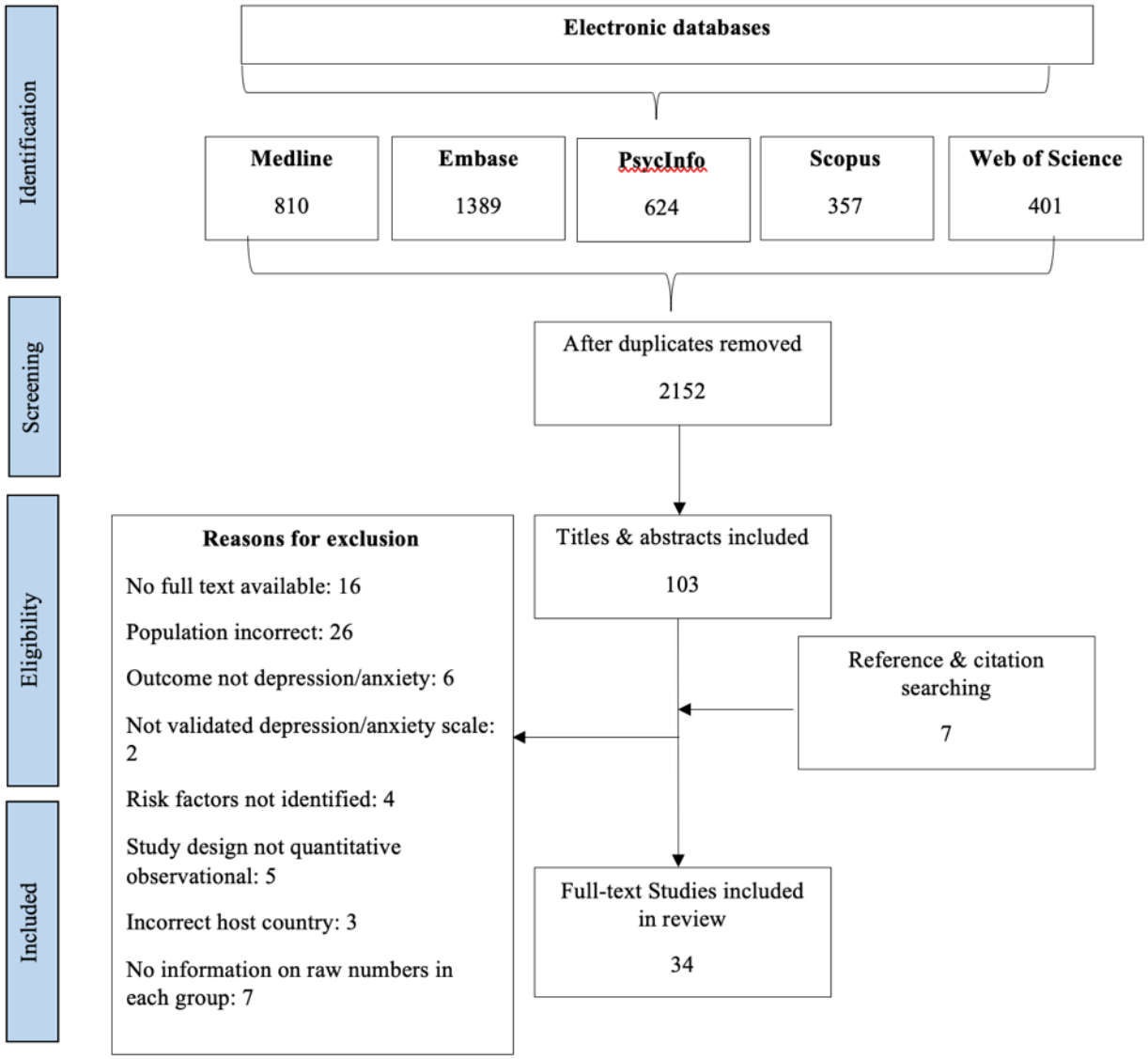


Figure 2 PRISMA flow chart of the screening process

2.3.2 Data extraction

Data extraction was carried out independently by two researchers. All independent analyses were combined, and disagreements were resolved through discussions or if it had been needed, discussion with a third reviewer (this was not required for this review). Social determinants of antenatal depression, based on the Dahlgren-Whitehead model (67), were extracted from the included studies.

The Cochrane "Data collection form for intervention review- RCT and non-RCTs" template was adapted to develop the data extraction template for this review. This is a widely used data extraction form which meets the Methodological Expectations of Cochrane Intervention Reviews (MECIR) standards for collecting and reporting information about studies for reviews and analysing their results. Although this is not an intervention review, the Cochrane data extraction process ensures that all relevant information is extracted from the included studies to write a comprehensive review. Some new sections were added, such as study setting (hospital/community-based) and country, and irrelevant sections removed from the original form such as publication type, duration of participation, race/ethnicity. The final data extraction template is provided in Appendix 6. The following information was extracted from each included study using the standardised data extraction template:

- Author, publication year
- Country
- Setting (hospital/community-based)
- Methodology
- Inclusion criteria and exclusion criteria
- Sample size of pregnant women
- Information on the outcome (depression/anxiety)
- Information on the social determinants
- Study results: unadjusted and adjusted effect sizes, whether statistically significant and factors that have been adjusted for in the analysis

2.3.3 Quality assessment

There are a few validated assessment tools applicable for observational studies. For this review, the Appraisal tool for Cross-Sectional Studies (AXIS), Joanna Briggs Institute critical appraisal tool and the Newcastle-Ottawa Scale (NOS) were

considered. A decision was made to use the NOS scale as it is the most widely used tool for observational studies and has been used for similar topic-related systematic reviews investigating risk factors for depression in patients with lung cancer (68). The NOS was adapted for cross-sectional studies, which assesses information bias, selection bias and confounding bias. The AXIS tool and Joanna Briggs Institute tool were used to inform this adaptation, along with previously adapted NOS from another systematic review (69). These steps in adapting the NOS ensured that all important aspects of bias in cross-sectional studies were covered when assessing quality of the included studies. The final quality assessment form is provided in Appendix 7. The maximum quality score a paper can receive is 10; 0-3 reflected low quality papers, a score of 4-7 was considered to be of medium quality and a score of 8-10 indicated high quality. No studies were excluded based on the quality score.

2.3.4 Data synthesis and statistical analysis

All included studies were synthesised narratively. This synthesis was categorised by social determinants. Only those social determinants that were investigated by three or more studies were included in the narrative synthesis as less than three was considered as insufficient evidence for narrative synthesis, as recommended by previous systematic reviews (58,62). A narrative summary of the included studies which investigated the social determinant is provided and I have outlined the associations between antenatal depression and the social determinant. The standardised metric used for the associations were ORs, and where this was not provided, the raw data, number of women with and without depression/anxiety in each social determinant group (e.g., unplanned pregnancy and planned pregnancy), was used to calculate the ORs, where appropriate.

For social determinants where pooling of the ORs was appropriate, meta-analyses were performed when there was a minimum of two studies as recommended by Cochrane (65). Social determinants of antenatal depression were examined using the random-effects model, which assumes that the underlying effect varies among the included studies, to pool effect sizes of the individual determinants (70). Heterogeneity between studies was measured using the I^2 statistic (0-100%) (71), with a threshold of >75% representing considerable heterogeneity as stated in the Cochrane Handbook, subject to the p-value (65). The outcome measure used for meta-analyses were the ORs, as this was the most common form of reporting associations between the

determinants and depression/anxiety. Unadjusted and adjusted ORs were pooled separately to account for the confounding factors, and both were presented to investigate the differences. Further, sensitivity analysis was performed by removing studies that were assessed to be of low quality to investigate whether this influenced the pooled results obtained. Studies that met the inclusion criteria but did not present data suitable for inclusion in the meta-analysis, for example continuous variables, no 95% CIs or standard error or no raw data on the total numbers with/without depression/anxiety in each risk factor category, were summarised narratively and compared to the meta-analysis results. The results were statistically significant when two-sided p-values were less than 0.05. Publication bias was tested by using the Egger's test ($p<0.05$ indicates the existence of publication bias) (65). The statistical analyses were conducted using RevMan 5.3 and STATA V.16.1 software.

2.4 Results

The searches identified 3,581 studies; following deduplication, 2152 studies remained for screening. Following screening and supplementary searches, a total of 34 studies met the study's inclusion criteria and are reported in this review (**Figure 2**). The sample sizes ranged from 100 (72) to 14,629 (73) pregnant women; all the studies were cross-sectional and the majority (n=30) were published since 2010. Studies were predominately from Pakistan (n=20) (14,55–57,72,74–88), followed by India (n=11) (11,51–54,89–94) and Bangladesh (n=3) (21,34,73). The majority of studies included samples of pregnant women from hospitals (n=24) (11,52–54,57,72,74–81,83–88,90,91,93,94), with the remaining being community-based samples (n=10) (14,21,34,51,55,56,73,82,89,92). Most of the studies investigated depression only (n=24) (11,14,34,51–53,56,72–74,78–82,85,86,88–94), followed by depression and anxiety together (n=9) (21,55,57,75–77,83,87) and anxiety alone (n=1) (54). The Edinburgh Postnatal Depression Scale (EPDS) was the most used measurement tool for depression/anxiety (n=11) (11,21,34,52,56,78,79,90,92–94), followed by the Hospital Anxiety Depression Scale (HADS) (n=5) (57,76,83,84,87), the Beck Depression Scale (BDI) (n=4) (53,72,81,88), Aga Khan University Anxiety and Depression Scale (AKUADS) (n=3) (55,74,77), Patient Health Questionnaire (PHQ) (73,82) and Centre for Epidemiological Studies Depression Scale (CES-D) (73,86) (n=2 each) and finally The Kessler Psychological Distress Scale (K-10) (89), Depression Anxiety Stress Scale (DASS) (75), DSM-5 (91), Hamilton Depression Rating Scale (HAM-D) (80), Pregnancy-related thoughts scale (PRTS) (54),

Goldberg's Depression Scale (GDS) (85), Schedule for Clinical Assessment in Neuropsychiatry (SCAN) (14) and State-Trait Anxiety Inventory (STAI) (21) (n=1 each) (**Table 4**).

Table 4 Summary characteristics of included studies

Author, publication year (reference number)	Country	Sample size	Hospital/ community -based	Outcome: depression/ anxiety	Ascertainment of outcome (Cut-off score)	Social determinants investigated	Quality score*
Ajinkya et al, 2013 (53)	India	185	Hospital	Depression	BDI (≥ 17)	- Age - Pregnancy intentions	4
Ali et al, 2012 (57)	Pakistan	165	Hospital	Anxiety Depression	HADS (≥ 8)	- Age - Education - Occupation - Number of household members - Pregnancy intentions - Family support - Household decision maker - IPV	6
Ayaz et al, 2019 (72)	Pakistan	100	Hospital	Depression	BDI (≥ 16)	- Age - Education - Husband's occupation	3

						- Husband's education	
Ayyub et al, 2018 (56)	Pakistan	367	Community	Depression	Urdu EPDS (≥ 12)	- Age - Education - Household income - Food insecurity - Recent job loss - Pregnancy intentions - IPV	10
Babu et al, 2018 (89)	India	823	Community	Depression	Kannada Kessler Psychological Distress Scale (K-10) (≥ 10)	- Age - Education - Occupation - Husband's education - Husband's occupation - Household income	8
Bavle et al, 2016 (90)	India	318	Hospital	Depression	Hindi EPDS (≥ 10)	- Age - Education - Occupation - Household income - Pregnancy intentions	4

Dahiya et al, 2020 (94)	India	200	Hospital	Depression	EPDS	- Age - Occupation - Education - Household income - Pregnancy intentions - Gender preference - Family support	6
Din et al, 2016 (75)	Pakistan	230	Hospital	Anxiety Depression	DASS-42	- Age - Household income - Relationship with in-laws, - IPV	5
Gausia et al, 2009 (34)	Bangladesh	346	Community	Depression	Bangla EPDS (≥ 10)	- Age - Education - Delivery location - Pregnancy intentions - Gender preference - Relationship with in-laws - Relationship with husband - Family support - IPV	8

George et al, 2016 (51)	India	202	Community	Depression	CIS-R	- Financial difficulties - Occupation - Pregnancy intentions - Gender preference - Relationship with husband - Relationship with in-laws - IPV	10
Ghaffar et al, 2017 (76)	Pakistan	750	Hospital	Anxiety Depression	Urdu HADS	- Age - Education - Occupation - Household income	8
Goyal et al, 2020 (91)	India	281	Hospital	Depression	DSM-5	- Age - Household income - Education - Relationship with husband	6
Gul et al, 2017 (77)	Pakistan	500	Hospital	Anxiety Depression	AKUADS (> 19)	- Age - IPV - Family support	7
Hegde et al, 2013 (52)	India	253	Hospital	Depression	EPDS (≥ 13)	- Occupation - Pregnancy intentions - Gender preference	2

						- Family support	
Humayun et al, 2013 (78)	Pakistan	506	Hospital	Depression	EPDS (≥ 10)	- Age - Pregnancy intentions - Family support - IPV - Fear of childbirth - Separation from husband	5
Imran et al, 2010 (79)	Pakistan	213	Hospital	Depression	Urdu EPDS (> 12)	- Household income - Pregnancy intentions - Gender preference - Relationship with in-laws - IPV - Relationship with husband	6
Jafri et al, 2017 (80)	Pakistan	300	Hospital	Depression	Hamilton Rating scale for Depression (> 7)	- Age	2
Jamal et al, 2018 (81)	Pakistan	197	Hospital	Depression	BDI (≥ 17)	- Age - Education - Household income	5

						- Pregnancy intentions - Separation from husband	
Karmaliani et al, 2009 (55)	Pakistan	1368	Community	Anxiety Depression	AKUADS (≥ 13)	- Age, - Education - Occupation - Husband occupation - Pregnancy intentions - IPV	6
Maselko et al, 2018 (82)	Pakistan	115	Community	Depression	PHQ-9	- Age - Education - Husband's education - Food insecurity	7
Mir et al, 2012 (74)	Pakistan	340	Hospital	Depression	AKUADS	- Age - Education - Occupation - Household income - Pregnancy intentions - Relationship with husband - IPV	9

Nasreen et al, 2011 (21)	Bangladesh	720	Community	Anxiety Depression	STAI (≥ 45) EPDS (≥ 10)	- Age - Education - Household expenditure - Involved in income generating activity - Pregnancy intentions - Received antenatal care - Relationship with husband - Relationship with in-laws - Family support - IPV	10
Nath et al, 2019 (54)	India	380	Hospital	Anxiety	Pregnancy-related thoughts (PRT) scale	- Age - Education - Occupation - Husband's education - Husband's occupation - Social support - Relationship with husband - IPV	8
Niaz et al, 2004 (83)	Pakistan	200	Hospital	Anxiety Depression	HADS (>7)	- Age - Education	2

Rabia et al, 2017 (84)	Pakistan	520	Hospital	Anxiety Depression	HADS (>8)	- Age - Education - Occupation - Pregnancy intentions - Relationship with husband - Relationship with in-laws - IPV	5
Rahman et al, 2003 (14)	Pakistan	632	Community	Depression	SCAN	- Age - Household income - Education - Husband's education - Family type - Husband's occupation - Relationship with husband - Relationship with in-laws - Social support	7
Sabir et al, 2019 (85)	Pakistan	450	Hospital	Depression	Goldberg's depression scale (> 21)	- Age - Household income	4
Safi et al, 2013 (86)	Pakistan	300	Hospital	Depression	CES-D (> 15)	- Age - Education	3

						- Occupation - Household income - Pregnancy intentions - Relationship with husband - IPV - Fear of childbirth	
Sheeba et al, 2019 (11)	India	280	Hospital	Depression	EPDS (>13)	- Age - Education - Husband's education - Occupation - Husband's occupation - Pregnancy intentions - Social support - Relationship with husband - IPV	8
Shehroz et al, 2019 (87)	Pakistan	200	Hospital	Anxiety Depression	HADS	- Age - Education	1
Shidhaye et al, 2017 (92)	India	302	Community	Depression	EPDS (>12)	- Age - Education - Occupation	10

						- Household income - Pregnancy intentions - Gender preference - IPV - Relationship with husband - Relationship with in-laws	
Srinivasan et al, 2015 (93)	India	100	Hospital	Depression	EPDS (≥ 10)	- Age - Education - Family type - Household income - Occupation - Pregnancy intentions - Relationship with husband - Social support - Household decision maker	6
Surkan et al, 2018 (73)	Bangladesh	14,629	Community	Depression	PHQ & CES-D	- Age - Education - Occupation - Pregnancy intentions - Gender preference - Pregnancy intentions	9

Zia et al, 2018 (88)	Pakistan	907	Hospital	Depression	BDI	- Age - Education - Occupation - Husband's education - Husband's occupation	2

*Quality assessment scores from the adapted Newcastle-Ottawa Scale reported in Appendix 8
BDI=Beck Depression Inventory, HADS=Hospital Anxiety Depression Scale, EPDS=Edinburgh Postnatal Depression Scale, K-10= Kessler Psychological Distress Scale, DASS=Depression Anxiety and Stress Scale, CIS-R=Clinical Interview Schedule Revised, DSM-5=Diagnostic and Statistical Manual of Mental Disorders, AKUADS= Aga Khan University Anxiety Depression Scale, PHQ= Patient Health Questionnaire, STAI= The State-Trait Anxiety Inventory, SCAN=Schedules for Clinical Assessment in Neuropsychiatry, CES-D=Centre for Epidemiological Studies Depression Scale
IPV= Intimate Partner Violence

2.4.1 Quality of included studies

The quality of included studies varied (for full details of quality scores for individual studies see Appendix 8 and **Table *4***). Quality scores ranged from one to ten out of a possible score of ten. Overall, 47% (n=16) of the studies scored medium; 32% (n=11) of the studies scored high; 21% (n=7) of the studies scored low. Reasons for assignment of low study quality varied. Most studies scored highly on ascertainment of outcome (antenatal depression/anxiety) with all studies using validated tools, however, 14 studies (14,52,72,74,75,78,80,81,83–85,87–89) dropped one score for not providing adequate justification for the cut-offs used for antenatal depression/anxiety. Thirteen studies (11,14,21,51,54,56,74,77,82,89,90,92,93) scored highly for ascertainment of exposure (social determinants) by using validated questionnaires; 16 (34,52,55,57,72,73,75,76,78,79,81,84,86,88,91,94) studies dropped one point out of two as they used non-validated or tested questionnaires but did describe the information collected from the questionnaire; five studies gave no description of the measurement tool (53,80,83,85,87). Representativeness of the sample scored high overall; 25 studies (11,14,21,34,51,53,54,56,73,74,76–85,89,91–94) either had a representative sample by recruiting all subjects/ random sampling or somewhat representative sample by using non-random sampling; nine studies (52,55,57,72,75,86–88,90) gave no description of their sampling strategy. Sample size justification was split with 17 studies (21,34,51,53,56,74,76–79,81,84,85,89,91–93) giving justified and satisfactory sample sizes, whereas, 17 studies (11,14,52,54,55,57,72,75,80,82,83,86–88,90,93,94) gave no justification. When considering non-respondents, the studies scored low overall with 11 studies (21,51,56,73,74,77,79,84,85,91,92) providing information on recruitment rate and meeting pre-specified target; 22 studies (11,14,34,52–55,57,72,75,78,80–83,86–90,93,94) gave no information at all and one study (76) did not achieve the pre-specified target for recruitment. When considering comparability of the groups on the basis of statistical adjustment; 15 studies (11,14,21,34,51,54–57,73–76,89,92) carried out adjusted analysis justifying which factors they adjusted for; three studies (81,82,94) adjusted for factors but did not specify which factors; 16 studies (52,53,72,77–80,83–88,90,91,93) did not carry out an adjusted analysis. Finally, quality scores varied for statistical analysis; 20 (11,14,21,34,51,54–57,72–76,78,82,89,92–94) studies carried out appropriate statistical analyses with p-values

and 95% CI; 14 studies (52,53,77,79–81,83–88,90,91) had incomplete, not described or inappropriate statistical analyses.

2.4.2 Social determinants

The review identified 12 social determinants which were investigated by three or more of the included studies: maternal age, maternal education, maternal occupation, husband's education, husband's occupation, household income, IPV, gender preference, relationship with in-laws, relationship with husband, pregnancy intentions and social/family support. **Table *5*** displays the overall trend of association for the potential social determinants.

Table 5 Summary of social determinants identified in the review

Social determinant	Total number of studies	Number of studies included in univariate meta-analysis	Pooled univariate association OR (95% CI)	Number of studies included in multivariate meta-analysis	Pooled multivariate association AOR (95% CI)
Maternal age	22	-	-	-	-
Maternal education/literacy	21	-	-	-	-
Maternal occupation	16	15	1.03 (0.76-1.41)	2	1.33 (0.55-3.21)
Husband's education	7	-	-	-	-
Husband's occupation	7	7	2.27 (0.96-5.41)	2	0.85 (0.35-2.08)
Household income	14	-	-	-	-
IPV	15	15	3.58 (2.57-5.00)*	9	2.48 (1.41-4.33)*
Male gender preference	6	6	1.84 (1.44-2.36)*	3	3.06 (1.40-6.72)*
Relationship with in-laws	7	7	4.23 (3.11-5.74)*	3	2.69 (1.25-5.80)*
Relationship with husband	13	11	3.04 (1.63-5.68)*	5	2.33 (0.90-6.02)
Unplanned pregnancy	20	19	1.88 (1.44-2.45)*	5	1.53 (1.28-1.83)*
Social/family support	10	9	2.14 (1.60-2.85)*	4	1.42 (0.78-2.62)

*Statistically significant at 0.05 significance level
OR=Odds ratio
AOR=Adjusted Odds Ratio
IPV=Intimate Partner Violence

2.4.1 Maternal Age

There were 22 studies which presented results on the association between antenatal depression/anxiety and maternal age (**Table *6***). Of these studies, 18 presented maternal age as a categorical variable with varying age group comparisons: <30 and ≥30 (14,57,94); 15-20 and ≥32 (56); <25 and ≥25 (55,81,89,90); <35 and ≥35 (34,76); <20 and ≥35 (21); <20 and ≥20 (11,54,73,78,86,92,93). Four studies presented maternal age as mean age (74,75,83,87) from the sample. The large variation in the reporting of maternal age deemed meta-analysis unsuitable for this determinant.

Six studies found that women in the higher maternal age groups had significantly higher odds of antenatal depression (56,73,86) and co-morbid antenatal depression and anxiety (18,26,47) compared to women in the lower maternal age categories. Similarly, one study (74) found that women with antenatal depression/anxiety had significantly higher mean maternal age compared to women without antenatal depression/anxiety. Conversely, three studies (75,83,87) found higher maternal age was significantly associated with lower comorbid antenatal depression & anxiety. Twelve studies (11,14,34,54,55,78,81,89,90,92–94) found no significant association between maternal age and antenatal depression/anxiety.

Table 6 Association between antenatal depression/anxiety and maternal age

Author & study year	Maternal age comparison groups	OR (95% CI)[a]	AOR (95% CI)[a]
Ali et al 2012 (57)	<30 & ≥30	3.10 (1.29-7.42)*	3.54 (1.12-10.12)*
Ayyub et al 2018 (56)	15-20 & ≥32	3.03 (1.20-7.15)*	2.77 (1.13-6.82)*
Babu et al 2018 (89)	<25 & ≥25	1.34 (0.78-2.30)	0.73 (0.42-1.27)
Bavle et al 2016 (90)	<25 & ≥25	0.93 (0.41-2.08)	-
Dahiya et al 2020 (94)	<30 & ≥30	0.22 (0.03-1.68)	-
Gausia et al 2009 (34)	<35 & ≥35	1.19 (0.62-2.29)	-
Ghaffar et al 2017 (76)	<35 & ≥35	1.23 (1.13-1.62)*	-
Humayun et al 2013 (78)	<20 & ≥20	0.37 (0.05-3.00)	-
Jamal et al 2018 (81)	<25 & ≥25	1.43 (0.46-4.40)	-
Karmaliani et al 2009 (55)	<25 & ≥25	1.09 (0.83-1.44)	-
Nasreen et al 2011 (21)	<20 & ≥35	-	3.00 (1.12-8.01)*

Nath et al 2019 (54)	<20 & ≥20	1.25 (0.75-1.07)	-
Rahman et al 2003 (14)	<30 & ≥30	1.41 (0.94-2.11)	-
Safi et al 2013 (86)	<20 & ≥ 20	2.00 (1.10-3.62)*	-
Sheeba et al 2019 (11)	<20 & ≥20	1.16 (0.67-1.99)	-
Shidhaye et al 2017 (92)	<20 & ≥20	0.73 (0.39-1.35)	-
Srinivasan et al 2015 (93)	<20 & ≥20	1.39 (0.61-3.16)	-
Surkan et al 2018 (73)	<20 & ≥20	1.15 (1.03-1.30)*	1.38 (1.12-1.70)*
Author & Study year	**Mean maternal age (years) in depressed group**	**Mean maternal age (years) in non-depressed group**	**P-value**
Din et al 2016 (75)	23.30	24.44	0.038*
Mir et al 2012 (74)	26.30	24.30	0.003*
Niaz et al 2004 (83)	25.15	27.18	<0.05*
Shehroz et al 2019 (87)	25.15	27.18	<0.05*

[a]Lower age group is the reference group;
OR=Odds Ratio; AOR=Adjusted Odds Ratio; *Statistically significant at 0.05 significance level

2.4.2 Maternal education/literacy

There were 21 studies which presented results on the association between antenatal depression/anxiety and maternal education and literacy (**Table *7***). Education level is usually defined by the highest level of formal education attained (none, primary or secondary and higher) or number of years in education. Literacy is the ability to read and write which is most often related to education (21).

Maternal education/literacy was presented as a categorical variable by 20 studies, however, with varying categories: up to intermediate & graduate (57,86); no education, primary, secondary & college/university (72,76,81,82,88,92); college & no college (11,54,89,90); 0-5 years & 5 years + (34); no education, 1-9 years & 10 or more years (55,73); illiterate & literate (14,21,74,94); <10th grade, 11-12th grade & graduate or above (93). One study, Shehroz et al (87), reported education as mean years in education. Due to this variability in the reporting of education, meta-analysis was not appropriate.

Overall, nine studies found that women within the lower education levels had significantly higher odds of antenatal depression (21,72–74,82,86,88,92,94) compared to women with higher education levels. Conversely, two studies found that women with higher education levels had higher odds of antenatal depression (90) and co-morbid antenatal depression and anxiety (55) compared to women with lower education levels. Ten studies found no significant association (11,14,34,54,57,76,81,87,89,93).

Table 7 Association between antenatal depression/anxiety and maternal education/literacy

Author & study year	Maternal education/literacy level	OR (95% CI)	AOR (95% CI)
Ali et al 2012 (57)	Graduate and above Up to intermediate	REF 0.87 (0.38-2.00)	-
Ayaz et al 2019 (72)	College or university Uneducated	REF 2.12 (2.02-3.68)*	-
Babu et al 2018 (89)	Attended college Did not attend college	REF 0.70 (0.42-1.16)	REF 0.81 (0.46-1.42)
Bavle et al 2016 (90)	Attended college Did not attend college	REF 0.46 (0.24-0.89)*	-
Dahiya et al 2020 (94)	Illiterate Literate	REF 0.25 (0.11-0.58)*	-
Gausia et al 2009 (34)	>5 years of education 0-5 years of education	REF 1.57 (1.01-2.44)*	REF 0.99 (0.58-1.67)
Ghaffar et al 2017 (76)	No education Primary Secondary Tertiary	REF 1.40 (0.70-2.12) 1.03 (0.55-1.14) 1.19 (0.62-1.03)	-
Jamal et al 2018 (81)	Intermediate/graduate and above Primary and below	REF 1.05 (0.47-2.37)	-

Karmaliani et al 2009 (55)	None 1-9 years 10 years or more	REF 1.52 (1.03-2.25)* 2.01 (1.30-3.10)*	-
Maselko et al 2018 (82)	None Primary Secondary Tertiary	REF 0.94 (0.62-1.42) 0.44 (0.30-0.64)* 0.27 (0.17-0.44)*	-
Mir et al 2012 (74)	Literate Illiterate	REF 2.50 (1.56-4.03)*	REF 1.83 (1.08-3.08)*
Nasreen et al 2011 (21)	Illiterate Literate	-	REF 0.59 (0.37-0.95)*
Nath et al 2019 (54)	Attended college Did not attend college	REF 0.86 (0.52-1.42)	-
Rahman et al 2003 (14)	Literate Illiterate	REF 0.80 (0.60-1.00)	-
Safi et al 2013 (86)	No education Intermediate and above	REF 0.18 (0.10-0.34)*	-
Sheeba et al 2019 (11)	Attended college Did not attend college	REF 0.71 (0.40-1.25)	-

Shidhaye et al 2017 (92)	Illiterate/completed primary Junior college or above	REF 0.30 (0.10-0.80)*	-
Srinivasan et al 2015 (93)	≤10th class 11th -12th class Graduate and above	REF 0.78 (0.28-2.22) 1.32 (0.51-3.46)	-
Surkan et al 2018 (73)	≤10 years in education 1-9 years None	REF 1.82 (1.37-2.40)* 2.87 (2.17-3.81)*	REF 1.45 (1.07-1.98)* 1.50 (1.05-2.15)*
Zia et al 2018 (88)	College or university Uneducated	REF 2.12 (2.0.2-3.63)*	-
Author & study year	**Mean years in education in women with depression**	**Mean years in education in women without depression**	**p-value**
Shehroz et al 2019 (87)	6.9	6.5	>0.05

REF=reference group
OR=Odds Ratio
AOR=Adjusted odds ratio
*Statistically significant at 0.05 significance level

2.4.3 Maternal occupation

There were 16 studies which presented results on the association between antenatal depression/anxiety and maternal occupation. All the studies reported maternal occupation as a binary variable: employed vs unemployed/housewife. Three studies (73,84,86) found that women who were unemployed had significantly higher odds of antenatal depression compared to women who were employed. Conversely, one study (52) found that women who were unemployed had significantly lower odds of antenatal depression compared to those employed. Twelve studies found no significant association between antenatal depression/anxiety and maternal occupation (11,21,51,54,55,57,73,74,76,89,90,92,94).

Fifteen studies met random-effects meta-analysis criteria. The pooled OR (**Figure 3**) for antenatal depression/anxiety in women who were unemployed compared to employed was found to be 1.03, (95% CI 0.76-1.41; I^2=82%; n=15 studies). The pooled AOR (**Figure 4**) was 1.33, (95% CI 0.55-3.21; I^2=73%, n=2 studies). Eggers test found no significant evidence of publication bias (p= 0.845). Both the pooled unadjusted and AORs suggest that there is no significant association between antenatal depression/anxiety and maternal occupation. The one study, George et al (51), which was not eligible for inclusion in the meta-analysis as it did not report the OR and raw data could not obtained, also found no significant association between antenatal depression and maternal occupation.

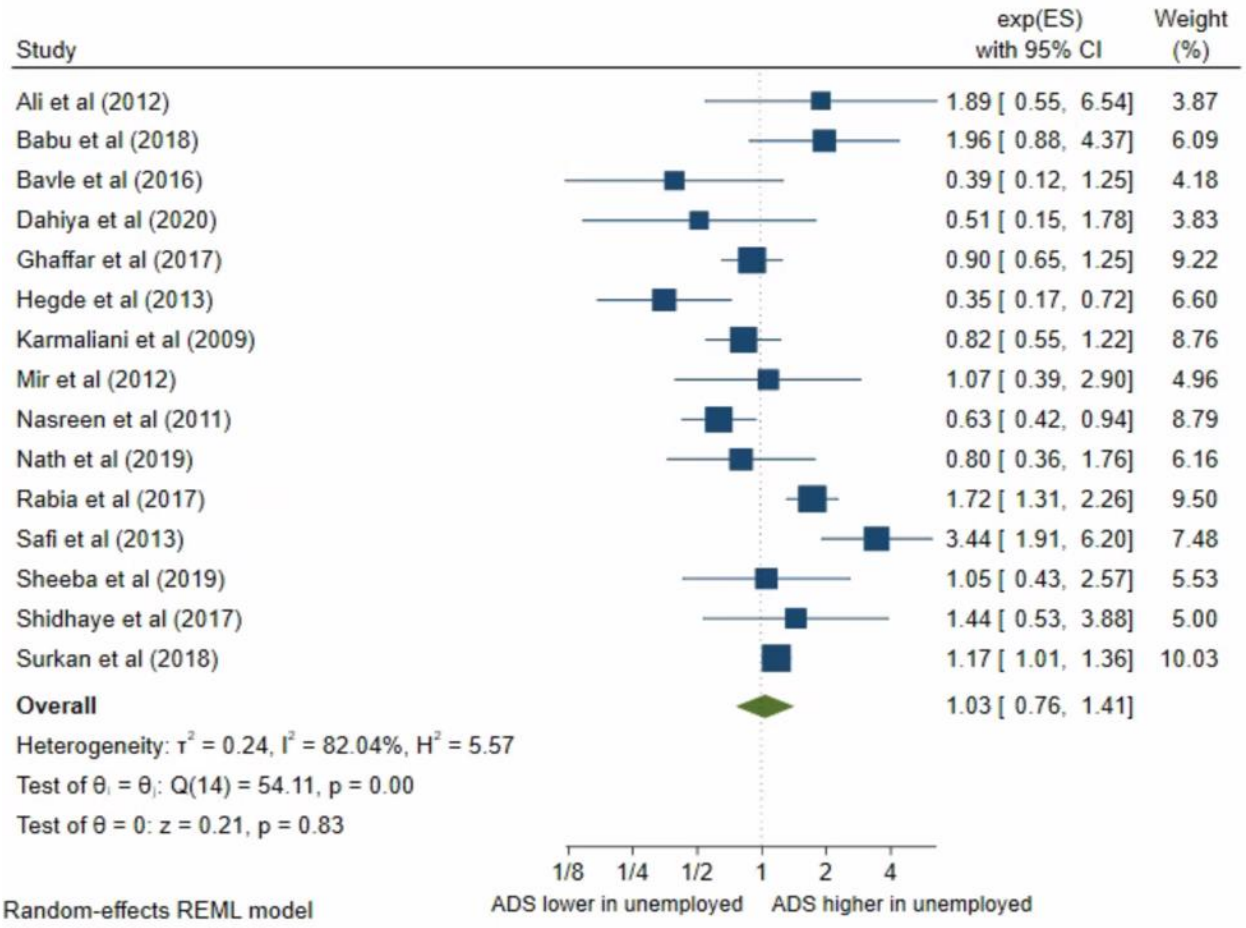


Figure 3 Forest plot of association between antenatal depression/anxiety and maternal occupation unadjusted OR

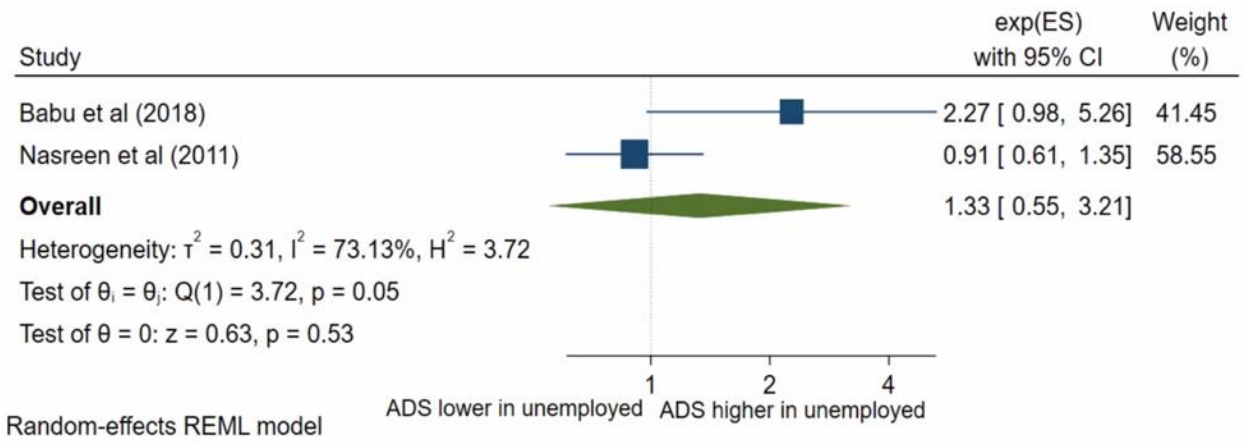


Figure 4 Forest plot of association between antenatal depression/anxiety and maternal occupation AOR

2.4.4 Husband's education/literacy

There were seven studies which presented results on the association between antenatal depression/anxiety and husband's education/literacy (**Table *8***). Husband's education/literacy was analysed as a categorical variable with varying categories in each study: no education, primary, secondary & college/university (72,82,88); illiterate, middle, high & college (89); <high school & >high school (11,54); illiterate & not illiterate (14). Due to the variability in the categories for education, meta-analysis was deemed inappropriate.

Two studies (14,82) found that women whose husbands had no literacy had higher odds of antenatal depression compared to women whose husbands were literate. Five studies found no significant association (11,54,72,88,89).

Table 8 Association between antenatal depression/anxiety and husband's education/literacy

Author & study year	Husband's education/literacy level	OR (95% CI)	AOR (95% CI)
Ayaz et al 2019 (72)	College or university Uneducated	REF 1.24 (0.30-5.24)	-
Babu et al 2018 (89)	Attended college Did not attend college	REF 1.22 (0.75-1.98)	-
Maselko et al 2018 (82)	None Primary Secondary Tertiary	REF 0.89 (0.50-1.59) 0.50 (0.31-0.78)* 0.25 (0.13-0.49)*	-
Nath et al 2019 (54)	Attended college Did not attend college	REF 1.22 (0.71-2.11)	-
Rahman et al 2003 (14)	Literate Illiterate	REF 1.60 (1.20-2.20)*	-
Sheeba et al 2019 (11)	Attended college Did not attend college	REF 1.26 (0.77-2.05)	-
Zia et al 2018 (88)	College or university Uneducated	REF 1.24 (0.30-1.34)	-

REF=reference group
OR=Odds Ratio
AOR=Adjusted odds ratio
*Statistically significant at 0.05 significance level

2.4.5 Husband's occupation

There were seven studies which analysed the association between antenatal depression/anxiety and husband's occupation. This was also a binary variable like maternal occupation: unemployed vs employed. Four studies found that women whose husbands were unemployed had significantly higher odds of antenatal depression (14,55,72,88) compared to women whose husbands were employed. Conversely, one study found that women whose husbands were unemployed had significantly lower odds of antenatal anxiety (54) compared to women whose husbands were employed. Two studies found no significant association (11,89).

All the studies were eligible for inclusion into the meta-analysis. The pooled OR (**Figure 5**) for antenatal depression/anxiety in women whose husbands were unemployed compared to employed was 2.27, (95% CI 0.96-5.41; I^2=90%; n=7 studies). The pooled AOR (**Figure 6**) was 0.85, (95% CI 0.35-2.08; I^2=83%; n=2 studies). There was evidence of publication bias from the Eggers test (p=0.0001). Both the pooled unadjusted and AOR suggest no significant association between antenatal depression/anxiety and husband's occupation.

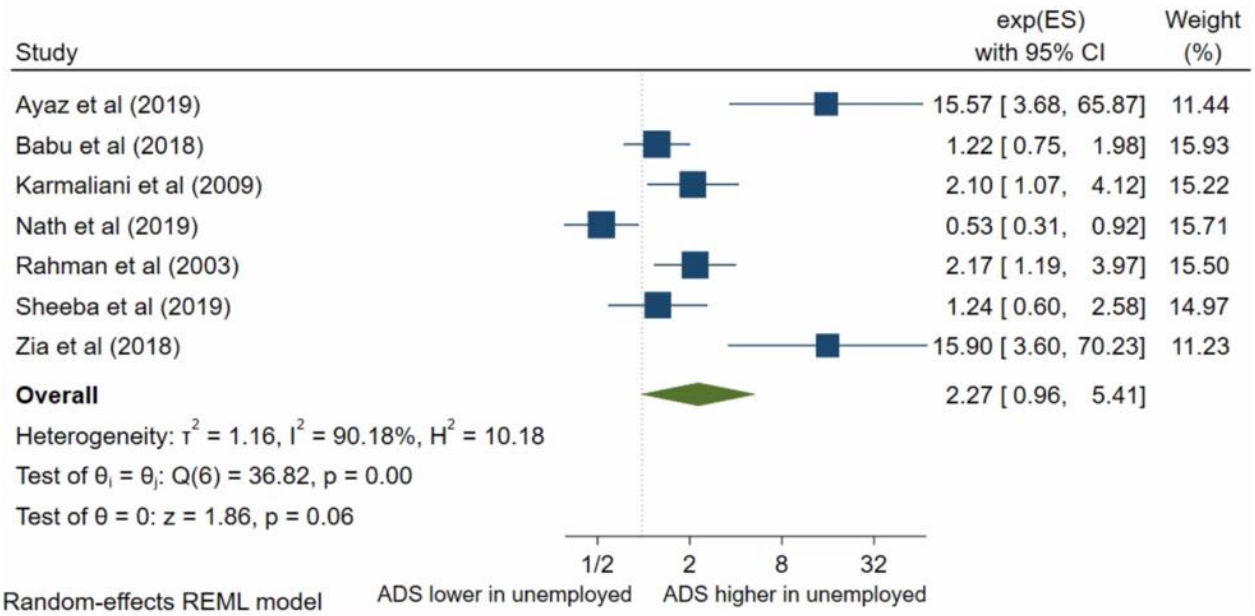


Figure 5 Forest plot of association between antenatal depression/anxiety and husband's occupation unadjusted OR

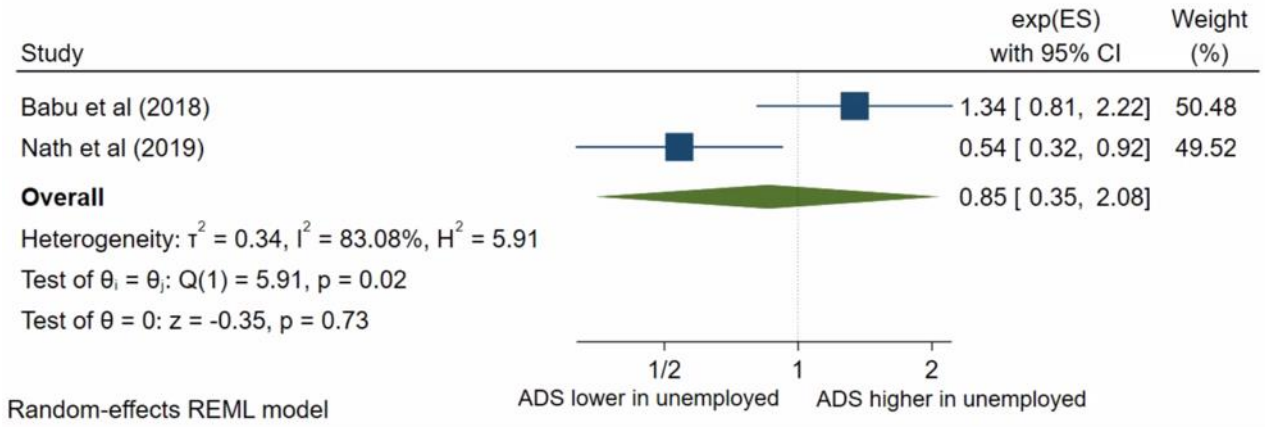


Figure 6 Forest plot of association between antenatal depression/anxiety and husband's occupation AOR

2.4.6 Household income

There were 14 studies which presented results on the association between antenatal depression/anxiety and monthly household income (**Table *9***). Household income, in Rupees or USD, was analysed as a categorical variable in 13 of the studies with varying categories: <13K & ≥13K (56); <6K, 6-10K, 10-15K & ≥15K (89); 0-2500, 2501-5K, 5-7.5K, 7.5-10K & ≥10K (90); <5K, 5-15K & ≥15K (76); <5K & ≥5K (79); <25K & ≥25K (81,85); <1K, 1-5K, 5-10K & ≥10K (74); <2.5K & ≥2.5K (14); low, middle & high (86,92); <50K & ≥50K (93); ≤13.51, 13.52-26.75, 26.76-45.03, 45.04-90.07 & ≥90.08 USD (94). One study (75) analysed household income as a continuous variable with the mean monthly household income in Rupees in women with and without antenatal depression. Due to this variability in the income categories, meta-analysis was not appropriate.

Three studies (76,81,86) found that women who live in households with lower levels of income had higher odds of antenatal depression compared to women living in households with a higher level of income. Similarly, the study (75) which analysed the mean monthly household income found that mean income was lower in those women who had co-morbid antenatal depression and anxiety compared to those women without. Ten studies found no significant association between antenatal depression/anxiety and household income (14,56,74,79,85,89,90,92–94).

Table 9 Association between antenatal depression/anxiety and household income

Author & study year	Household income groups (rupees)	OR (95% CI)	AOR (95% CI)
Ayyub et al 2018 (56)	>13,000 <13,000	-	-
Babu et al 2018 (89)	>15,000 10,001-15,000 6001-10,000 ≤6K	REF 0.91 (0.41-2.03) 1.18 (0.56-2.50) 0.52 (0.26-1.04)	REF 0.88 (0.39-2.00) 1.02 (0.47-2.20) 0.49 (0.23-1.04)
Bavle et al 2016 (90)	>10,000 7501-10K 5001-7500 2501-5000 0-2500	REF 0.40 (0.17-0.94)* 0.21 (0.05-0.75)* 0.42 (0.13-1.36) 0.62 (0.07-5.42)	-
Dahiya et al 2020 (94)	≤13.51 USD 13.52-26.75 USD 26.76-45.03 USD 45.04-90.07 USD ≥90.08 USD	REF 39.00 (0.53-2883.6) 13.00 (0.45-377.47) 4.97 (0.21-109.19) 3.56 (0.20-64.79)	-
Ghaffar et al 2017 (76)	No income <5000	REF 0.33 (0.20-0.94)*	-

	5000-15,0000 >15,000	0.56 (0.34-0.88)* 0.64 (0.17-0.99)*	
Imran et al 2010 (79)	>5000 <5000	REF 1.99 (0.74-5.34)	-
Jamal et al 2018 (81)	>25,000 ≤25,000	REF 6.24 (2.15-18.14)*	-
Mir et al 2012 (74)	>10,000 5000-10,000 1,000-5,000 <1,000	REF 1.29 (0.68-1.45) 1.30 (0.72-2.36) 1.57 (0.61-4.04)	-
Rahman et al 2003 (14)	>2,500 <2,500	REF 0.90 (0.60-1.30)	-
Sabir et al 2019 (85)	≥50,000 25,000-50,000 <25,000	REF 1.44 (0.85-2.45) 1.18 (0.69-2.02)	-
Safi et al 2013 (86)	High Middle Low	REF 0.57 (0.26-1.25) 2.83 (1.25-6.41)*	-
Shidhaye et al 2017 (92)	Low Middle High	REF 0.40 (0.10-1.00) 0.60 (0.30-1.20)	-

Srinivasan et al 2015 (93)	>1 Lakh 50,000-1 lakh ≥50,000	REF 0.49 (0.19-1.22) 0.70 (0.15-3.25)	-
Author & study year	**Mean monthly household income (thousand rupees) in depressed group**	**Mean monthly household income (rupees) in non-depressed group**	**P-value**
Din et al 2016 (75)	24.91	27.41	0.041*

REF=reference group
OR=Odds Ratio
AOR=Adjusted odds ratio
*Statistically significant at 0.05 significance level

2.4.7 Intimate Partner Violence

IPV is defined as any behaviour within an intimate relationship that results in physical, mental, or sexual harm to those involved in the relationship. This encompasses physical, verbal and sexual violence/abuse or controlling behaviour of any kind (95).

There were 15 studies which analysed the association between antenatal depression/anxiety and IPV during pregnancy. All the studies reported IPV as a binary variable: yes vs no. Eleven studies found that experiencing IPV was significantly associated with higher odds of antenatal depression (11,21,34,51,56,74,79,84) and comorbid antenatal depression and anxiety (55,57,75) compared with women who have not experienced IPV. Conversely, one study found that IPV was significantly associated with lower odds of antenatal anxiety (54) compared with no IPV. Three studies found no significant association between IPV and antenatal depression (78,86,92).

All studies met the criteria for inclusion in the meta-analysis. The pooled OR (**Figure 7**) for antenatal depression in women who had suffered from IPV during pregnancy compared to those who had not suffered from IPV was 3.58, (95% CI 2.57-5.00; I^2=61%; n=15 studies). The pooled AOR (**Figure 8**) was 2.48, (95% CI 1.41-4.33; I^2=71%; n=9 studies). Eggers test found no significant evidence of publication bias (p=0.841). Both the unadjusted OR and AOR suggest that women who experienced IPV during pregnancy had significantly higher odds of antenatal depression/anxiety compared to women who had not experienced IPV during their pregnancy.

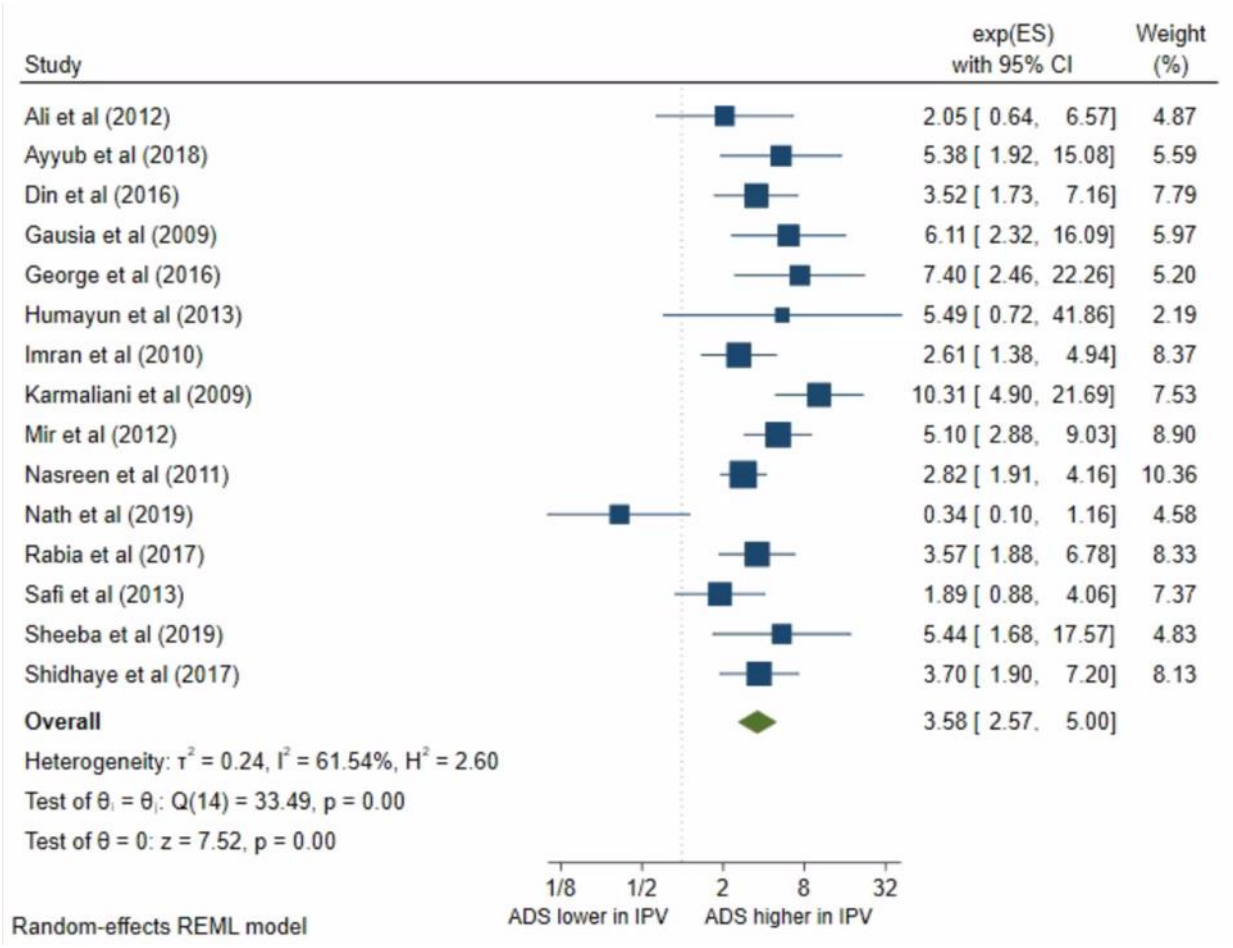


Figure 7 Forest plot of association between antenatal depression/anxiety and IPV unadjusted OR

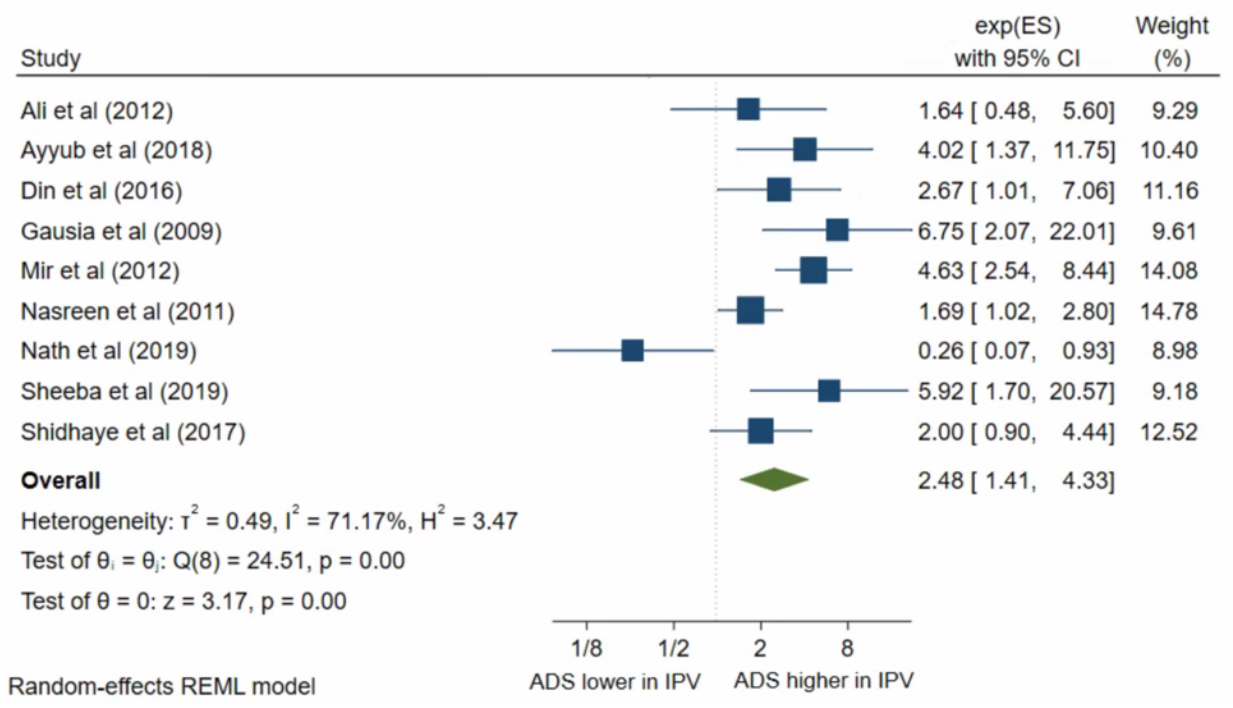


Figure 8 Forest plot of association between antenatal depression/anxiety and IPV AOR

2.4.8 Gender preference

There were six studies which analysed the association between gender preference for the baby and antenatal depression/anxiety. Gender preference was reported as a binary variable in all the included studies: preferring a male vs female child. Four studies found that preferring a male child was significantly associated with higher odds of antenatal depression (34,51,52,92) compared with wanting a female child. Two studies found no significant association (79,94).

All the studies met the inclusion criteria for meta-analysis. The pooled OR (**Figure 9**) of antenatal depression in women who wanted a male child compared to wanting a female child was 1.84, (95% CI 1.44-2.36; I^2=0%, n=6 studies). The pooled AOR (**Figure 10**) was 3.06, (95% CI 1.40-6.72; I^2=61%, n=3 studies). Eggers test found significant evidence of publication bias (p=0.027). Both the pooled unadjusted OR and AOR suggest that a male gender preference was significantly associated with increased odds of antenatal depression compared to a female gender preference.

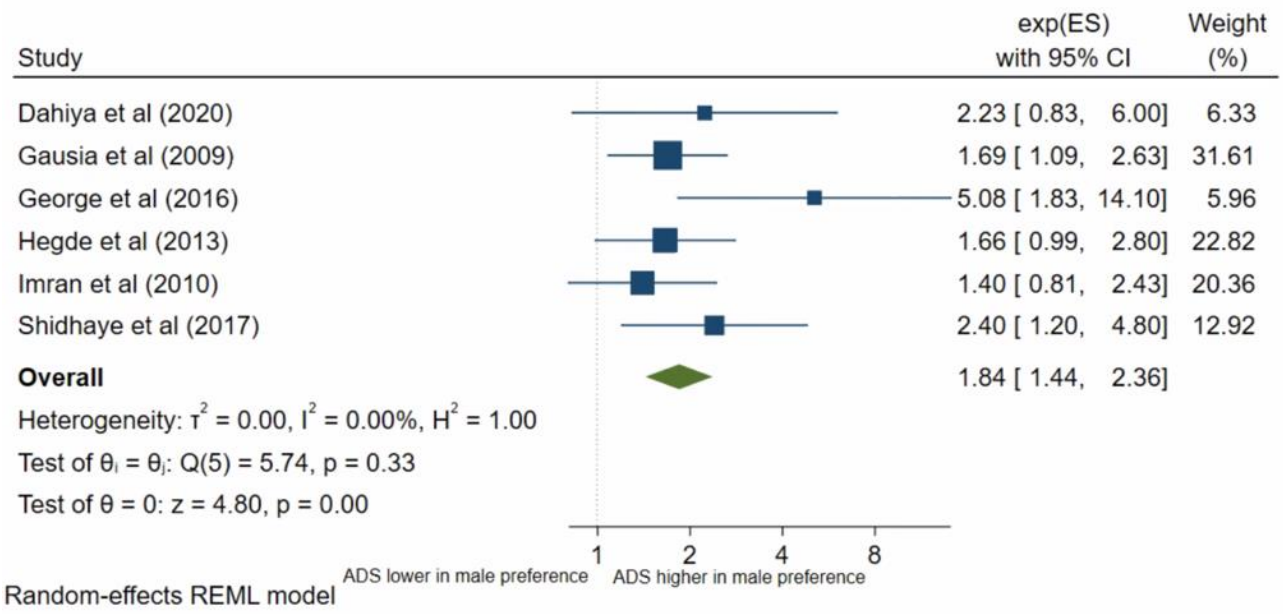


Figure 9 Forest plot of association between antenatal depression/anxiety and gender preference unadjusted OR

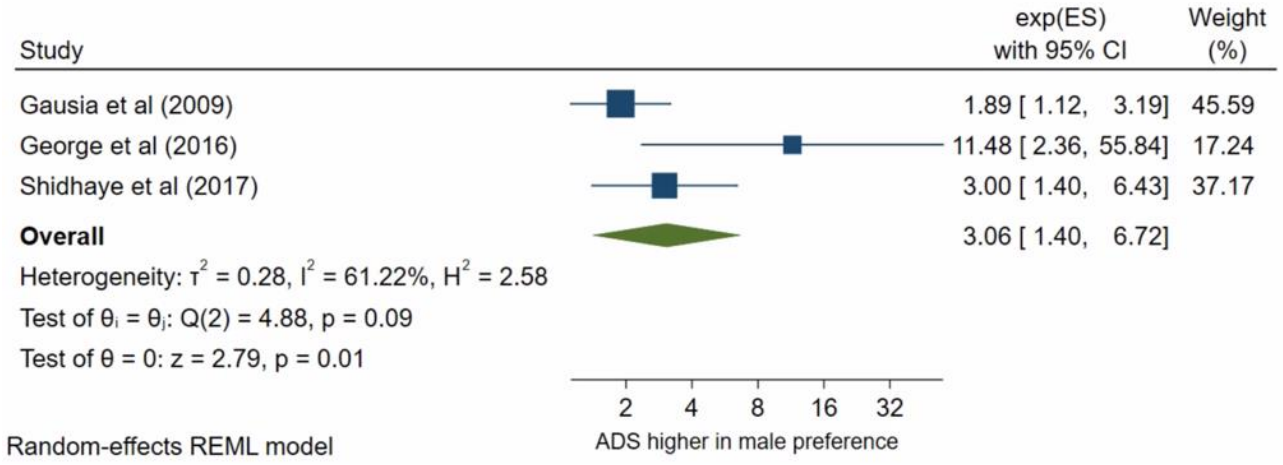


Figure 10 Forest plot of association between antenatal depression/anxiety and gender preference AOR

2.4.9 Relationship with in-laws

There were seven studies which presented results on the association between antenatal depression/anxiety and relationship with in-laws. Relationship with in-laws was reported as a binary variable in all the studies: poor vs good relationship. Four studies found that a poor relationship with in-laws was significantly associated with higher odds of antenatal depression (20,50,63) and co-morbid antenatal depression and anxiety (84) compared to having a good relationship with in-laws. Three studies found no significant association (14,34,75).

All the studies were eligible for inclusion in the meta-analysis. The pooled OR (**Figure 11**) for antenatal depression in women with a poor relationship with in-laws compared with those with a good relationship was 4.23, (95% CI 3.11-5.74; I^2=49%; n=7 studies). The pooled AOR (**Figure 12**) was 2.69, (95% CI 1.25-5.80; I^2=54%; n=3 studies). Eggers test found no significant evidence of publication bias (p=0.2174). Both the unadjusted and AOR suggest that a poor relationship with in-laws during pregnancy is significantly associated with higher odds of antenatal depression compared to experiencing a good relationship with in-laws.

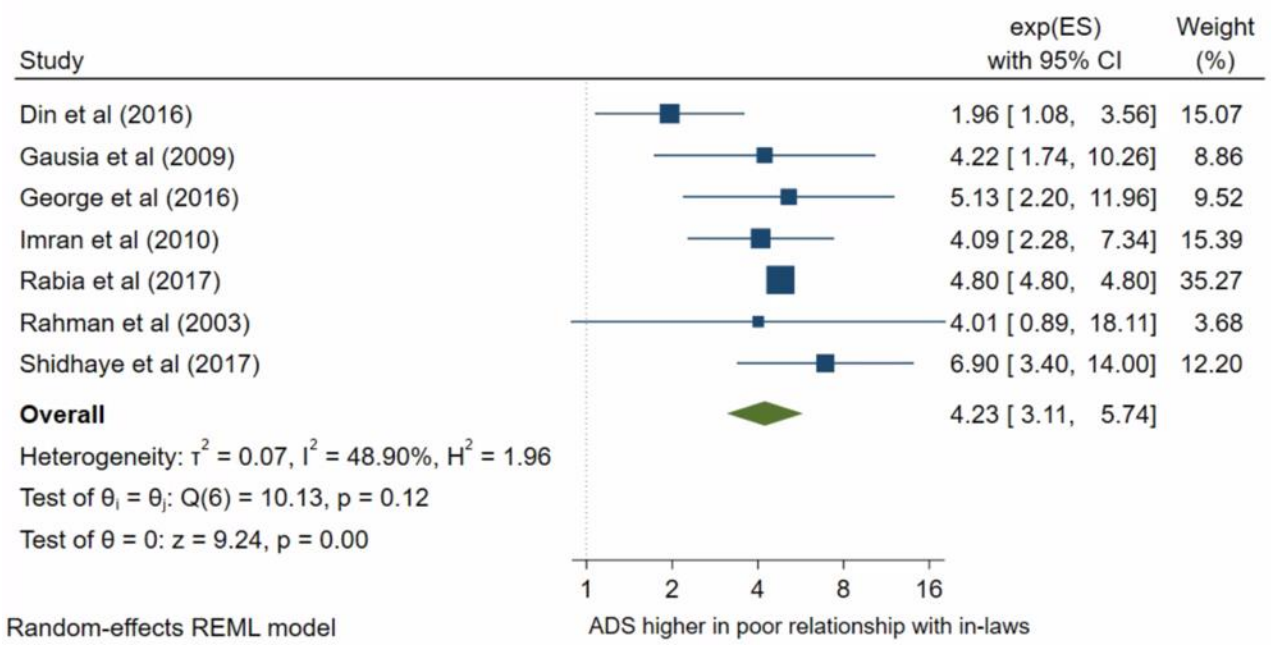


Figure 11 Forest plot of association between antenatal depression/anxiety and relationship with in-laws unadjusted OR

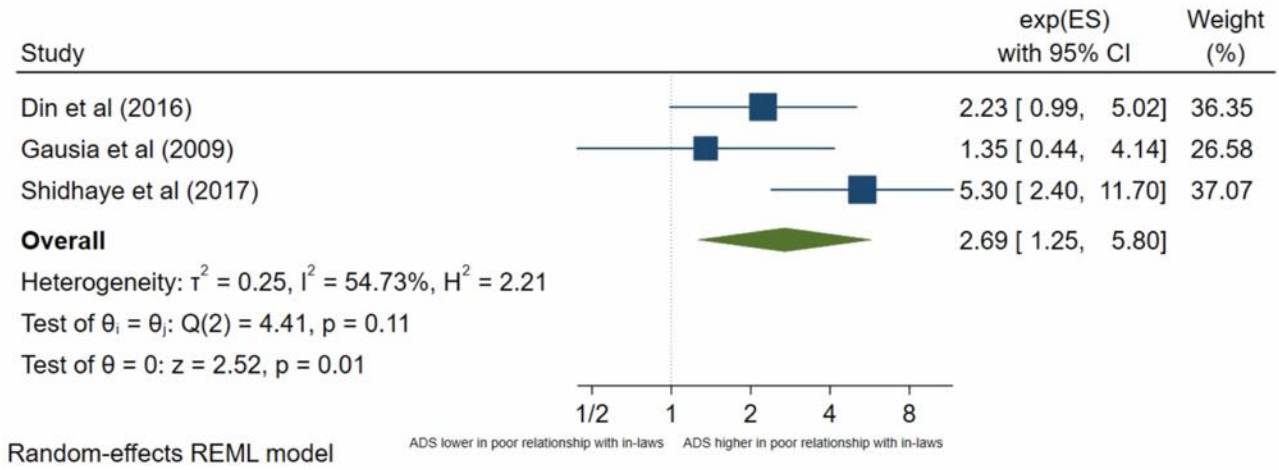


Figure 12 Forest plot of association between antenatal depression/anxiety and relationship with in-laws AOR

2.4.10 Relationship with husband

There were 13 studies which analysed the association between antenatal depression/anxiety and relationship with husband. Relationship with husband was mostly reported as a binary variable: poor vs good relationship, but two studies, Mir et al (74) and Srinivasan et al (93), reported it as a continuous variable using a marital satisfaction scale. Both these studies found that the mean marital satisfaction score was significantly lower in women with antenatal depression compared to women without antenatal depression. Eight studies found that women who considered their relationship with their husband to be poor had higher odds of antenatal depression (14,34,51,79,91,92) and co-morbid antenatal depression and anxiety (21,84) compared to those who had a good relationship with their husband. Three studies found no significant association (11,54,86).

Of the 13 studies, 11 were eligible for inclusion in the meta-analysis as Mir et al (74) and Srinivasan et al (93) reported means, therefore, could not be pooled with the majority of studies which reported ORs. The pooled OR (**Figure 13**) for antenatal depression/anxiety in women who reported a poor relationship with their husband compared to women who reported a good relationship was 3.04, (95% CI 1.63-5.68; I^2=95%, n=11 studies). The pooled AOR (**Figure 14**) was 2.33, (95% CI 0.90-6.02; I^2=89%, n=5 studies). Publication bias was evident from the Eggers test (p=0.0075). The pooled unadjusted OR suggests women who report a poor relationship have significantly higher odds of antenatal depression/anxiety compared to women who report a good relationship, however, this was no longer significant in the adjusted analysis.

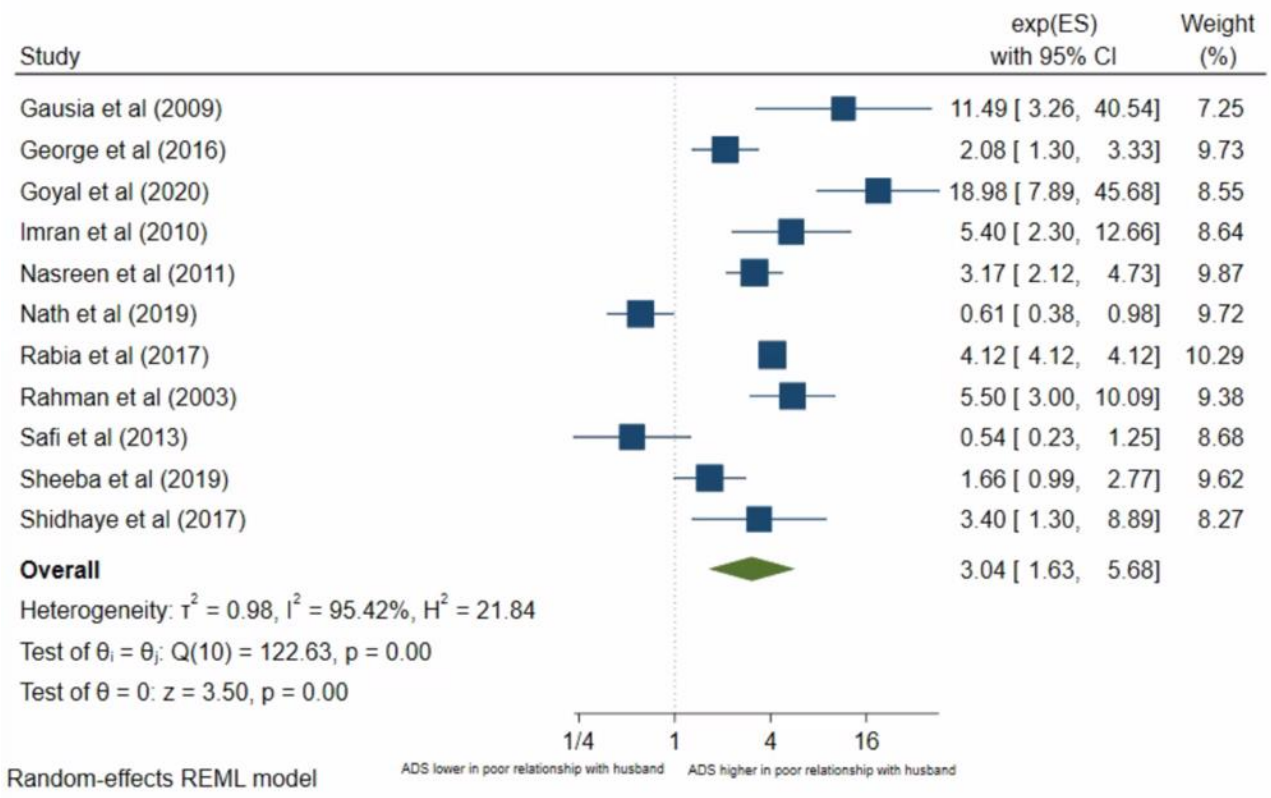


Figure 13 Forest plot of association between antenatal depression/anxiety and relationship with husband unadjusted OR

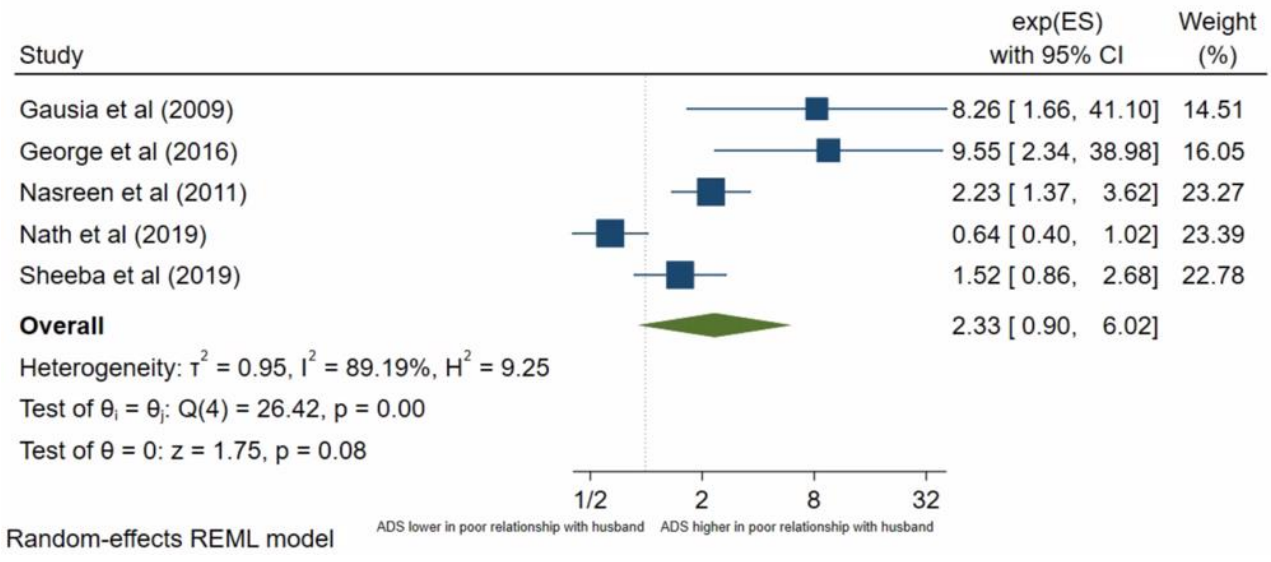


Figure 14 Forest plot of association between antenatal depression/anxiety and relationship with husband AOR

2.4.11 Pregnancy intentions

There were 20 studies which analysed the association between antenatal depression/anxiety and pregnancy intentions. Pregnancy intention was a binary variable: unplanned vs planned pregnancy. Nine studies found that women whose pregnancy was unplanned had higher odds of antenatal depression (52,53,56,73,74,81,90) and co-morbid antenatal depression and anxiety (55,84) compared to women who had a planned pregnancy. Eleven studies found no significant association (11,21,34,51,54,57,78,79,86,92,94).

Of the 20 studies, one study, George et al (51), was not eligible for inclusion into the meta-analysis as OR or raw data could not be obtained; this study found no significant association as specified in the study. Nineteen studies were included in the meta-analysis. The pooled OR (**Figure 15**) for antenatal depression/anxiety in women who had an unplanned pregnancy compared to women with a planned pregnancy was 1.88, (95% CI 1.44-2.45; I^2=91%; n=19 studies). The pooled AOR (**Figure 16***)* was 1.53, (95% CI 1.28-1.83; I^2=19%; n=5 studies). There was no significant evidence of publication bias (p=0.96). Both the pooled unadjusted and AOR suggest that women whose pregnancies were unplanned had higher odds of antenatal depression/anxiety compared to women who had planned pregnancies.

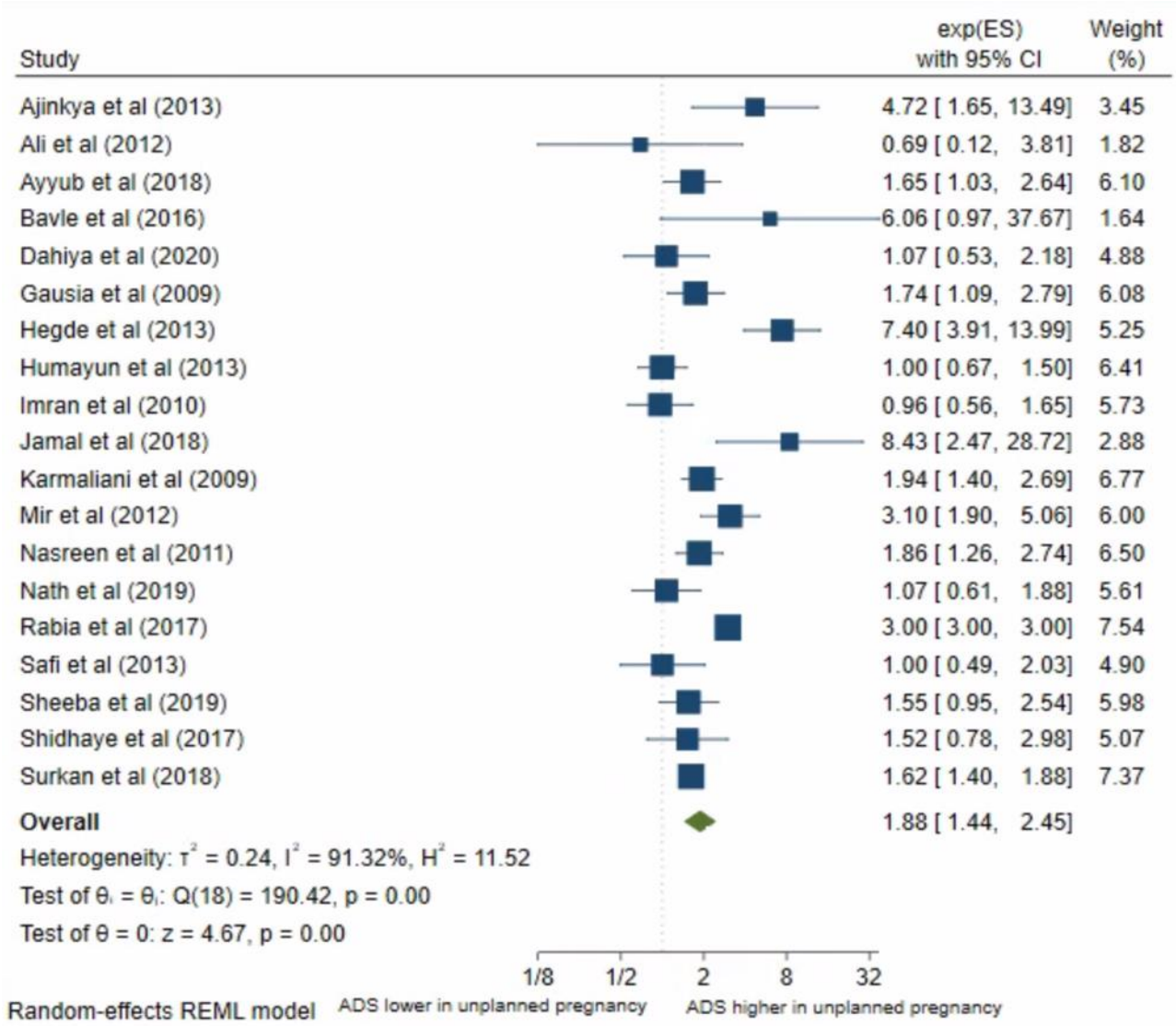


Figure 15 Forest plot of association between antenatal depression/anxiety and pregnancy intentions unadjusted OR

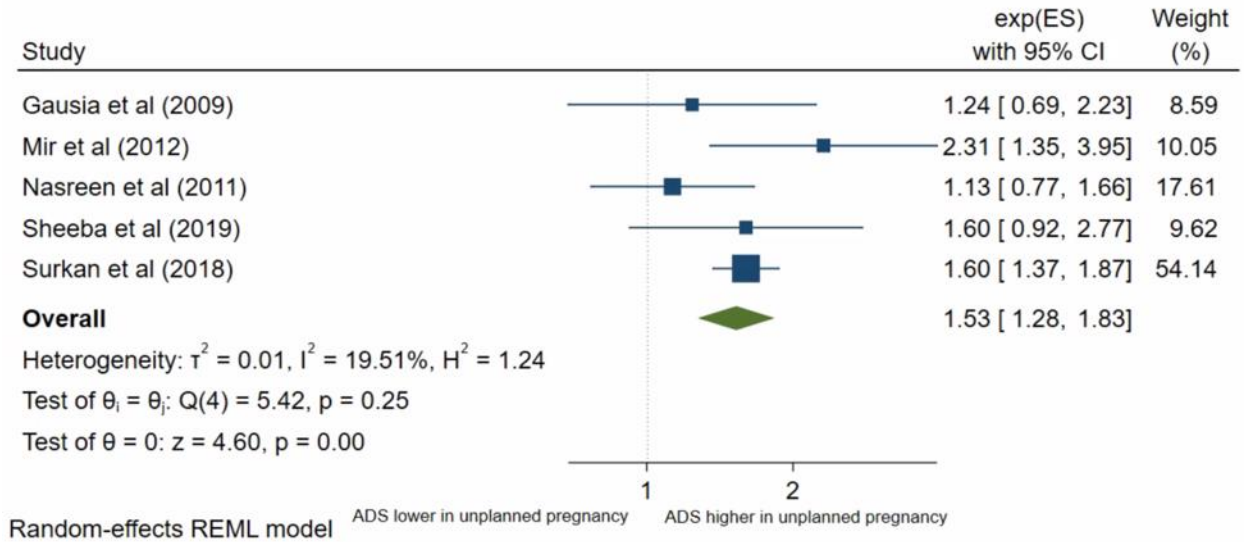


Figure 16 Forest plot of association between antenatal depression/anxiety and pregnancy intentions AOR

2.4.12 Social support

Social support is a multidimensional concept which encompasses informational support, instrumental support such as practical help and emotional support (59). This can include support from family, relatives, friends or society.

There were ten studies which analysed the association between antenatal depression/anxiety and social/ support. The majority of the studies reported support as a binary variable: low vs high support, but one study, Srinivasan et al (93), reported it as a mean on the social support scale. Four studies found that women with low social support had higher odds of antenatal depression (14,34,52) and co-morbid antenatal depression and anxiety (77) compared with women who with high social support. Six studies found no significant association (11,21,54,78,93,94).

Nine studies were eligible for inclusion in the meta-analysis. The pooled OR (**Figure 17**) for antenatal depression/anxiety in women with low social support compared to women with high social/family support was 2.14, (95% CI 1.60-2.85; I^2=60%; n=9 studies). The pooled AOR (**Figure 18**) was 1.42, (95% CI 0.78-2.62; I^2=75%; n=4). There was no significant evidence of publication bias (p=0.0516). The pooled unadjusted OR suggests that women with low social and family support had higher odds of antenatal depression/anxiety compared to women with high support, however, this was no longer significant in the adjusted analysis.

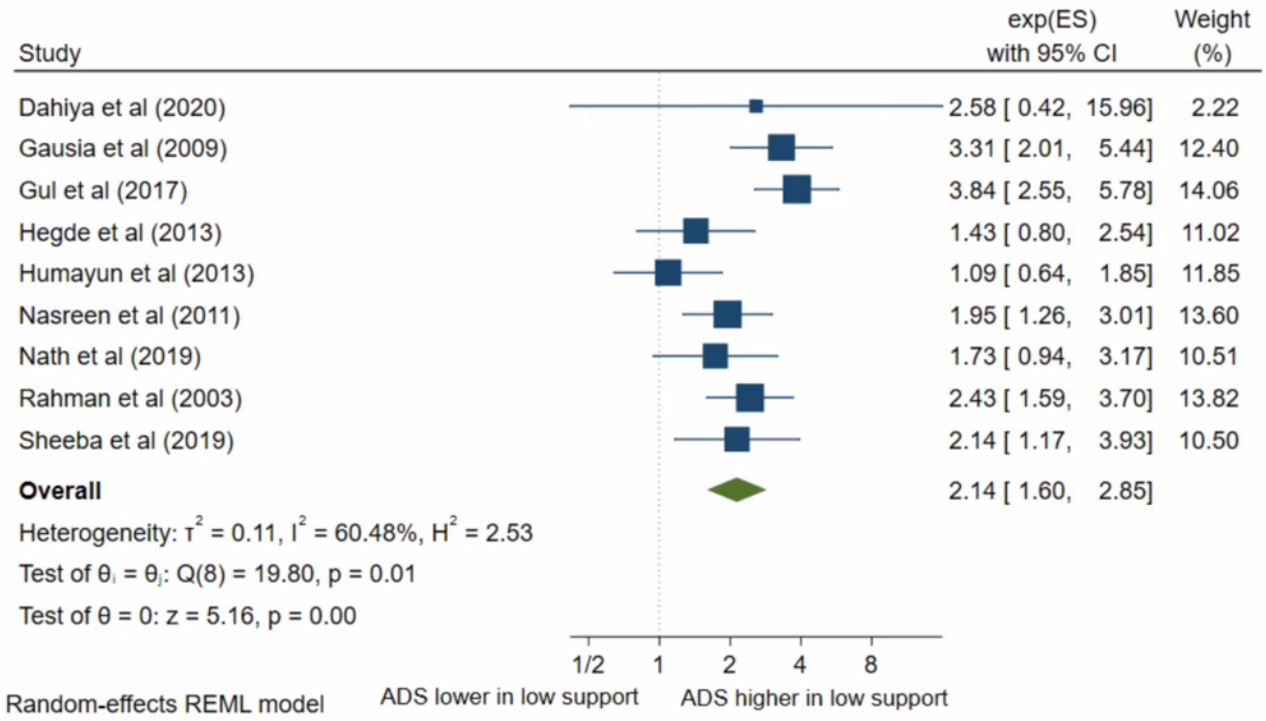


Figure 17 Forest plot of association between antenatal depression/anxiety and social/family support unadjusted OR

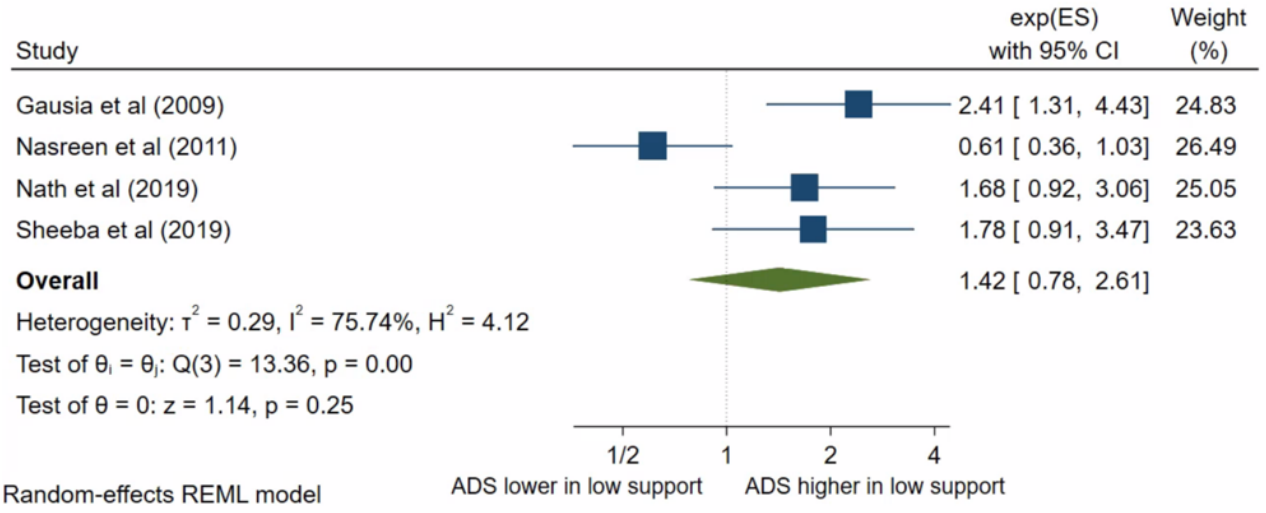


Figure 18 Forest plot of association between antenatal depression/anxiety and social/family support AOR

2.4.13 Additional social determinants

There were additional social determinants which were only investigated by one or two studies, hence, were not eligible to be included in the main narrative synthesis and analysis. The determinants that were analysed by two studies were household decision making (57,93), separation from husband (78,81) and fear of childbirth (78,86). Household decision making reflects on who holds power within the household which is often the in-laws in South Asian households. Both Ali et al (57) and Srinivasan et al (93) found no significant association between household decision making and antenatal depression. Separation from husband may highlight lack of support during pregnancy if the father of the child is not involved throughout the pregnancy. Humayun et al (78) and Jamal et al (81) found separation from husband to be significantly associated with antenatal depression. Fear of childbirth can reflect on a number of different underlying issues such fear of complications affecting the child's health and also the associated costs if there are complications and extensive medical assistance is required. Women may worry about the impact these costs will have on the family if they face financial difficulties already. Further, gender is usually revealed during childbirth in rural areas of Bangladesh, India and Pakistan. Given the pressure to have a son, women may be worried about the repercussions if they give birth to a son. As highlighted previously in this chapter, male gender preference is significantly associated with antenatal depression. Similarly, Humayun et al (78) and Safi et al (86) found that fear of childbirth is significantly associated with antenatal depression. Further research should investigate separation from husband and fear of childbirth to understand the underlying mechanisms within the associations found.

The determinants which were only investigated by one study were where women had received antenatal care (21), number of household members (57), recent job loss (56) and food insecurity (56). It may be important to further investigate these determinants to elucidate their importance in antenatal depression and anxiety once there are further studies to include.

2.4.14 Sensitivity analysis

Analyses were repeated by removing the studies with a low quality score (0-3) to investigate the effects studies with low quality had on the results. Our results did not significantly change for any of the potential social determinants, except for husband's occupation. When removing the low-quality study, Zia et al (88), from the unadjusted

analysis, the results were no longer significant (OR 1.65, 95% CI 0.88-3.10). Nevertheless, Zia et al (88) did not carry out adjusted analysis, therefore, the adjusted analysis in the meta-analysis for husband's occupation remains the same, suggesting there was still no significant association between husband's occupation and antenatal depression.

2.4.15 Summary of studies from updated search between 4th January 2021 to 4th January 2023

Five additional studies published between 4th January 2021 and 4th January 2023 were identified that matched the inclusion criteria. Due to the limited time available before thesis submission and complexity of the meta-analyses, these studies were not included in the main synthesis and analysis which have already been published in PLoS One (Appendix 1). Two studies were conducted in Pakistan (96,97), one study was conducted in Bangladesh (22), one study used population data from both Pakistan and Bangladesh (98) and one study was conducted in India (99). Three studies measured antenatal depression (97–99), one study measured antenatal anxiety (22) and one study measured both antenatal depression and anxiety (96).

IPV was investigated by two studies, and both found it to be significantly associated with antenatal depression (97,99). Relationship with in-laws was explored in the study conducted in India and found that women with poor relationships with their in-laws had higher odds of antenatal depression (99). Two studies investigated pregnancy intentions and found no significant association between unplanned pregnancies and antenatal depression/anxiety (22,99). One study investigated gender preference for child and found no significant association (99).

2.5 Discussion

This section summarises the key findings from the systematic review and meta-analyses, compares these findings with existing literature in different contexts and previous reviews, and considers the strengths and limitations of the included evidence in this review and of the review itself. This systematic review and meta-analysis investigated the social determinants of antenatal depression and anxiety in Bangladesh, India and Pakistan. This review included 34 studies and data from 26,673 pregnant women to investigate social determinants of antenatal depression and

anxiety in Bangladesh, India and Pakistan. A total of 12 social determinants were explored in this review.

2.5.1 Key findings

This review found that IPV, preference for a male child, poor relationship with in-laws and unplanned pregnancy were factors which were significantly associated with antenatal depression and/or anxiety. Sociodemographic factors such as maternal and husband's occupation showed no significant association between antenatal depression and anxiety in the meta-analysis. Two factors, relationship with husband and social support, showed a significant association with antenatal depression and anxiety in the unadjusted analysis, however, these were no longer significant in the adjusted analysis. Maternal age, maternal and husband's education and household income were not subject to meta-analysis due to large heterogeneity in the reporting of these factors. Nevertheless, narrative synthesis found inconsistent evidence, therefore, caution should be taken when drawing conclusions based on the current evidence.

The significant association between IPV and antenatal depression and/or anxiety is also reflected in previous reviews investigating risk factors for antenatal depression conducted by Lancaster et al (58), Biaggi et al (59) and Gelaye et al (60). These three reviews found IPV to be an important risk factor for antenatal depression and anxiety. These reviews included studies from HICs (58,59) and LMICs (59,60) which suggests that IPV is an important risk factor for depression and anxiety in pregnant women globally. This association may be linked to the social, emotional and physical isolation and pressure pregnant women undergo when suffering from IPV (60). Particularly in LMICs, where IPV rates are higher compared to HICs (60), women are subject to considerable amounts of social and cultural norms and stigma which prevent them from sharing or enclosing such information (100). In summary, available previous evidence and results from this review suggest that IPV is a significant social determinant for antenatal depression and anxiety in Bangladesh, India and Pakistan which should be considered in ANC services and policies.

The previous reviews (58–60) did not investigate gender preference for the baby as a social determinant of antenatal depression and anxiety. However, the significant association between preferring a male child and antenatal

depression/anxiety found in this review is reflected in observational studies conducted in China (101) and Iran (102). Conversely, Cankorur et al (103) found no significant association between gender preference and antenatal depression in a study conducted in Turkey. Although these countries tend to have similar societal preference for sons, the difference in results may be due to the sample of pregnant women in the Turkish study who largely did not have any gender preferences (103). Overall, it is surprising that gender preference appears to be a relatively under-researched determinant, as highlighted by the small number of studies included in this review which addressed it, despite it being a common societal issue within these South Asian countries (104). Male children are considered a blessing as they carry the family name and can provide for their parents in old age, whereas girls are often viewed as a burden for the family (105). Hence, indirect and direct pressure from the family and society to have a male child may produce negative effects on the mental health of pregnant women. Although this review found a significant association which is also reflected in some previous studies, the small number of studies highlights the need for further research with multivariate regressions to understand the strength and mechanisms of this association.

Similar to gender preference, relationship with in-laws was not included as a factor in the previous reviews on this topic. Most studies only discuss relationship with the husband, while neglecting the crucial interactions between in-laws and the expecting mother. Particularly in South Asian countries like Bangladesh, India and Pakistan, where women are often expected to live with their in-laws after marriage and throughout (105), these vital family dynamics can shape how women experience pregnancy making it important to investigate. There were only seven studies out of 34 which investigated relationship with in-laws in this review, further highlighting the limited research carried out on this factor. Although the pooled OR and AOR showed a significant association between poor relationship with in-laws and antenatal depression/anxiety, only three studies (34,75,92) went onto carry out multivariate analysis of which two (34,75) found an association which did not reach statistical significance. The study with the largest weight, Shidhaye et al (92), which largely contributed to the significant association was assessed to be a high-quality study. However, differences in the factors adjusted for in the multivariate analysis may contribute to the differences in the results between the three studies.

Unplanned pregnancy was found to be a significant social determinant of antenatal depression and anxiety in this review. This is reflected in previous reviews by Lancaster et al (58), which found an overall medium correlation between unintended pregnancy and antenatal depression, and Biaggi et al (59), in which 22 out of 24 studies included in the review found unwanted/unplanned pregnancy to be a significant predictor of antenatal depression. Unplanned pregnancies can reflect a lack of access to adequate family planning services and power relations between the husband and wife leading to forced sex without the use of contraception. These issues can cause significant distress to pregnant women who are not physically or mentally prepared for the pregnancy or the life after. Further, Lee et al (106) and Marchesi et al (107), suggest that unwanted pregnancy was a significant predictor of antenatal depression but mostly in the first trimester, with less importance over time. This may be due to the initial shock of an unwanted pregnancy reducing throughout the pregnancy as the woman comes to terms with the idea of having a baby. This is an interesting finding which has not been investigated in any of the included studies in this review suggesting a need for further research on this within these South Asian communities.

Maternal and husband's occupation showed no significant association with antenatal depression. Similar results are reflected in the review by Lancaster et al (58) which found that maternal unemployment was not significantly associated with depressive symptoms in bivariate studies and research was inconsistent in multivariate studies. Biaggi et al (59) concluded that there is inconsistent evidence with regards to maternal occupation and antenatal depression. However, they found that women whose partners were unemployed seem to be more likely to experience antenatal depression. In LMICs, men are the main breadwinners and usually provide the primary source of income within families, therefore, husband's unemployment, and hence lack of income for the growing family, may have a more profound negative impact on the mental health of pregnant women than their own employment status (108). Nevertheless, in this review only two studies met eligibility for the multivariate analysis for both maternal and husband's occupation highlighting the need for further research on these factors.

Relationship with husband was found to have no significant association with antenatal depression in the meta-analysis of AORs undertaken in this review. Similarly, the review by Lancaster et al (58) found that improved relationship quality was

associated with lower likelihood of depression in bivariate analysis but not in the adjusted analysis. It is possible that a poor relationship with the husband does not have such a pronounced effect on pregnant women if this is the norm throughout their marriage. Furthermore, many women in South Asian communities do not consider their relationship with their husband to be poor even if the woman faces challenges such as IPV (21). Conversely, Biaggi et al (59) found that all seven studies in their review, which investigated poor relationship with husband, found it to be a significant predictor of antenatal depression. The authors also found two studies which found marital satisfaction to be a protective factor for antenatal depression. However, the majority of these studies were bivariate which suggests the need for more studies with multivariate analysis as there may be other factors playing a role within this association.

Social support was found to have no significant association with antenatal depression in this review. Conversely, Lancaster et al (58) and Biaggi et al (59) both concluded social support to be an important risk factor for antenatal depression from their reviews. These differences in the results may be due to the differences in the populations. South Asian countries tend to encompass a larger proportion of extended families with elderly family members continuing to live with their children compared to families in western countries. Therefore, South Asian pregnant women may have wider sources of support available to them in general, hence showing no significant association. Although Gelaye et al (60) did not directly investigate social support in their review, it was stated as an additional risk factor as one study found it to be a risk factor for depression during pregnancy in Ethiopia (109) which has similar family structures to South Asian countries. Nevertheless, this study was community-based using the EPDS scale with a cut-off score of >13. In this review, the study by Nasreen et al (21), which found a non-significant result was also a community-based study, however, used a different EPDS cut-off score of >10. Sheeba et al (11), which also found a non-significant result, used the same EPDS cut-off score as the Ethiopian study, however, it was a hospital-based study. Using different cut-off scores has an effect on the ascertainment of depression, therefore, it is difficult to compare studies using different cut-offs. Pregnant women in community-based studies may have different characteristics to women from hospital-based studies as women who are able to access hospitals may be of higher socio-economic status or live in areas closer to hospitals. These factors could affect their perceived social support producing different results. Furthermore, social support is a very multidimensional factor which can be

difficult to evaluate objectively. Different studies may evaluate social support using different constructs. For example, some studies ask about practical help received, whereas others ask a more subjective question of perceived social support which can resonate differently in different populations and among different women. Hence, further studies should evaluate social support more subjectively in an attempt to understand the underlying mechanisms and factors which may influence the association.

Maternal age, maternal & husband's education and household income showed inconsistent results within this review. Similarly, Lancaster et al (58) and Biaggi et al (59) also found inconsistent results for maternal age, education and household income, with most studies suggesting no significant association. Husband's education was found to be not associated with antenatal depression in the majority of the studies in this review (five out of seven). However, given the small number of studies investigating husband's education in this review and the lack of investigation by Lancaster et al (58) and Biaggi et al (59), it is not possible to elucidate a clear conclusion from the available evidence.

2.5.2 Strengths and limitations of included studies

The included studies were mostly of high or medium quality suggesting lower levels of bias within the studies. Only seven out of the 34 studies were assessed as low quality and after these were removed in the sensitivity analysis to determine their influence, no significant changes were identified within the results.

The depression and anxiety measurement tools used in the studies have limitations as they rely on self-reported information with the questions often subjective in nature. For example, the EPDS includes the following question: 'In the past 7 days have you felt that things have been getting on top of you?' This can be subjective for the participants who may have different capacity levels. Further, all the studies used translated versions of the measurement tools. Common translated versions used were Bangla, Urdu and Hindi representing the most prevalent spoken languages in Bangladesh, Pakistan and India. Cross-cultural translations have limitations as cultural and social meanings may be lost in translation. Certain phrases and words may have different connotations and inferences in different cultural and linguistic contexts which can be complex to capture in translated items (110). However, all the included studies in this review used measurement tools which have been validated in the country and

population of study and have been shown to provide accurate screening for antenatal depression and anxiety in the study populations.

Self-reporting bias in ascertainment of the social determinants is a further limitation of the included studies. Issues such as IPV and relationship with family and husband may be sensitive topics for many women. Despite confidentiality arrangements, women may still hesitate to disclose certain information due to fear of stigma or repercussions from society and family were they ever to find out (111). This is particularly prominent in some areas of South Asian countries where women are often conditioned to be submissive to their husband, and women who are seen to stand up for her rights within the family may be stigmatised within the community. Therefore, there may be respondent bias in the form of non-reporting as most cases of spousal violence go under-reported due to social perceptions around it (111).

Another limitation in the included studies is the limited use of multivariate analysis adjusting for confounding factors. It is best practice to perform multivariate analysis by including all the factors which displayed a significant univariate or bivariate association in regression analyses. However, the majority of the studies presented only bivariate results, with only a few presenting further multivariate results. To account for this, the unadjusted and adjusted results were pooled separately which is a strength of this review as it limits heterogeneity between the data being pooled. This also allows investigation of differences between the unadjusted and adjusted results and the role of different confounding factors. Nevertheless, when pooling multivariate results, the small number of studies may limit the accuracy. This is a limitation of this review as some meta-analyses were conducted with only a few studies for example for gender preference and social support.

2.5.3 Strengths and limitations of this systematic review

Along with pooling unadjusted and adjusted results separately, this review has additional strengths. This review included a large number of studies providing evidence on many social determinants. This a considerable strength of this review as a wide range of databases were used and supplementary searches such as hand searching reference lists and citation searching were carried out to provide a comprehensive review on a large number of social determinants. Despite, this large sample size, only a few studies were available for some determinants such as gender preference,

relationship with in-laws, husband's occupation and education which only had five to seven studies. Nevertheless, this reflects on the limited research conducted on certain factors which is an important finding for future research.

Although there are differences between the different depression and anxiety scales, this review aimed to limit bias by only including studies which used validated measurement scales to determine depression and anxiety, and questionnaires which were standardised. This allowed for more appropriate comparisons to be made between the included studies. However, different cut-off scores used in different studies for the same scale may produce misclassification bias which may limit the results.

Heterogeneity was found to be significantly high, >75%, when a large number of studies were pooled for determinants like maternal occupation, relationship with husband and pregnancy intentions. This could suggest true heterogeneity between the studies due to methodological differences such as sampling of participants, sample size, adjusting for different confounding factors, use of different measurement tools and difference in characteristics between Bangladesh, India and Pakistan. Although these three countries share similar social and cultural norms and characteristics which deems it appropriate to pool them together, there is heterogeneity between them. There are differences in the predominant religions found in the three countries. In India, approximately 80% of the population are Hindu, followed by Muslims who make up 13% of the population (112). Conversely, Bangladesh is made up of 81% Muslims and 10% Hindus. There are certain differences between these religions which can influence some of the social determinants investigated within this review. For example, the system of dowry (money or assets given to the groom's family by the bride's family) is more common in Hinduism, whereas, in Islam this system does not exist. Dowry is a prominent reason why many families prefer having male children. Therefore, male gender preference may be more prevalent in India than in Bangladesh adding to the heterogeneity.

This review used random-effects model to account for the heterogeneity. However, in some cases it may be that pooling of results with such high heterogeneity was not appropriate which could limit the findings. Meta-analysis of a smaller number of studies for factors such as gender preference and relationship with in-laws found no

significant evidence of heterogeneity. However, it is possible that when there are a lot of studies, some evidence of heterogeneity will be found simply due to the high power of the test (113). Similarly, with a smaller number of studies the test has low power, hence, no significant evidence of heterogeneity may not be accurate and may just be due to the small sample size. Therefore, interpretation of heterogeneity must be carried out with caution in this review.

Finally, this review restricted the included studies to those published in or contained English translations since producing translations were not feasible which is a limitation. This may limit the number of studies included as there may be studies published in languages native to Bangladesh, India and Pakistan which were missed out of this review; this could produce publication bias. Nevertheless, our search did find one study which was originally published in the Urdu language but provided an English translated version which we included (78). Further, journals from Bangladesh, India and Pakistan tend to increasingly publish in the English language or contain translations making them accessible to a wider audience. Hence, it is unlikely that many studies were missed which were solely published in a different language other than English reducing the degree of language bias in this review.

2.5.4 Gaps in research

This review highlighted some gaps in the literature which should be incorporated in further research. Firstly, out of the 34 included studies, only three studies were conducted in Bangladesh. An additional study was identified in Bangladesh published in 2021, however, it only investigated antenatal anxiety, not depression. From these studies, prevalence of antenatal depression has been found to be high, approximately 33% (34). These studies were conducted almost 10 years ago, which highlights the need for more recent research around antenatal depression to be conducted in Bangladesh, particularly in the Sylhet division where there are poorer maternal health outcomes as discussed in Chapter 1. Studies conducted in Bangladesh suggest that mental health is still a very neglected issue, with women not receiving adequate healthcare for mental health conditions during pregnancy, particularly in rural areas (23). Hence, further research on this topic will enable recommendations for providing better care for pregnant women in rural Bangladesh.

Furthermore, the majority of the studies included in this review recruited participants from hospitals rather than community-based settings. This is important as many women in South Asian countries do not have access to hospitals during the antenatal period or childbirth (114). Women with access to hospitals may have different characteristics and face different social determinants compared to women who do not visit hospitals during their pregnancies. Lack of access to hospitals may reflect on factors such as socioeconomic status, distance and family decisions and dynamics (114). Therefore, to find results which are representative of the whole population, community-based studies should be conducted, recruiting participants from places other than hospitals.

Additional social determinants such as fear of childbirth were only investigated in a few studies. Safi et al (86) found that women with fear of childbirth were more likely to have antenatal depression than women who did not fear childbirth. This has important social and gender-based facets where women fear childbirth because of socioeconomic factors or worry about the gender of the baby. Hence, there is evidence to show this factor may be important and should be investigated in further research.

2.6 Chapter summary

This systematic review and meta-analysis contributed to the knowledge base around social determinants of antenatal depression and found that IPV, male gender preference, poor relationship with in-laws and unplanned pregnancies were the biggest determinants of antenatal depression in Bangladesh, India and Pakistan. This review included a large number of studies producing a large sample size, however, only a few studies were available for certain determinants like gender preference. Only studies using validated antenatal depression and anxiety scales were included in an attempt to overcome the limitations of screening tools. Both unadjusted and adjusted ORs were pooled separately in the meta-analysis to minimise heterogeneity, however, more studies using multivariate analyses are required. Future research should address the gap in studies conducted in rural Bangladesh and incorporate community-based data, with the addition of further factors such as fear of childbirth.

Chapter 3. Perceptions and attitudes around perinatal mental health in Bangladesh, India and Pakistan: a systematic review of qualitative data

3.1 Chapter overview

While the previous chapter provided a review of the social determinants of antenatal depression and anxiety in South Asia, this chapter aims to address the second element of this PhD project and provide a comprehensive review of the perceptions and attitudes around PMH in Bangladesh, India and Pakistan. This will allow for a better understanding of the literature and provide an exploration of the way that PMH is perceived by pregnant women, their families and HCPs. This helped inform the topics explored within the interviews with HCPs (see chapter 8). Within this chapter, a description of the methods employed is followed by a summary of the results. Finally, a critical discussion of the findings is provided followed by the strengths and limitations of this systematic review. This systematic review was published in BMC Pregnancy and Childbirth in April 2022 (46) (Appendix 2).

3.2 Introduction

Perceptions and attitudes, which are socially constructed, shape how individual's understand mental illness and the way they respond to situations, activities and relationships (23). Studies suggest that South Asian cultures, like Bangladeshi, Indian and Pakistani, are very collectivist, in that social controls such as family, culture, religion and community often dominate decision-making and the attitudes that individuals hold (115). Hence, the way such collectivist cultures perceive social roles and conceptualise identities will influence individuals' perceptions and attitudes towards certain phenomena such as PMH. A systematic review by Watson et al (116) investigating ethnic minority women's, largely South Asian and Black African, experience of PMH conditions and services in the UK found that lack of awareness, ongoing stigma and cultural expectations all impacted women's receipt of adequate PMH support in the UK (116). However, there is limited research investigating these experiences in Bangladesh, India and Pakistan where the social and structural settings are very different.

To my best knowledge, there are no existing systematic reviews which have investigated perceptions and attitudes of perinatal (pregnant and postpartum) women, their families and HCPs towards PMH in Bangladesh, India and Pakistan. These are key actors in perinatal women receiving adequate mental health care. Therefore, the aim of this systematic review was to identify, synthesise and appraise the available literature on perceptions and attitudes surrounding PMH in Bangladesh, India and Pakistan. The information provided by this review can be used to inform the development of PMH services, interventions and programmes within these countries to improve mental health of perinatal women.

3.3 Methods

This systematic review followed the Enhancing Transparency in Reporting the synthesis of qualitative research (ENTREQ) guidelines (Appendix 9).

3.3.1 Search strategy

Qualitative and quantitative studies, with elements of qualitative data, reporting perceptions and attitudes surrounding PMH in Bangladesh, India and Pakistan as the host country were eligible for inclusion, if published in English between 4th January 2000 and 4th January 2021 (see chapter 2 for justification of search dates). Searches were further updated to 4th January 2023 prior to thesis submission, however, results from these additional studies were not included in the final synthesis which has been published, but a summary of findings are provided in section 3.4.6. A full description of the search strategy is provided in Section 2.3.1 as searches for both systematic reviews in this PhD were done via one search strategy (Appendix 5). A PRISMA flow diagram was used to record the screening process (**Figure 19**).

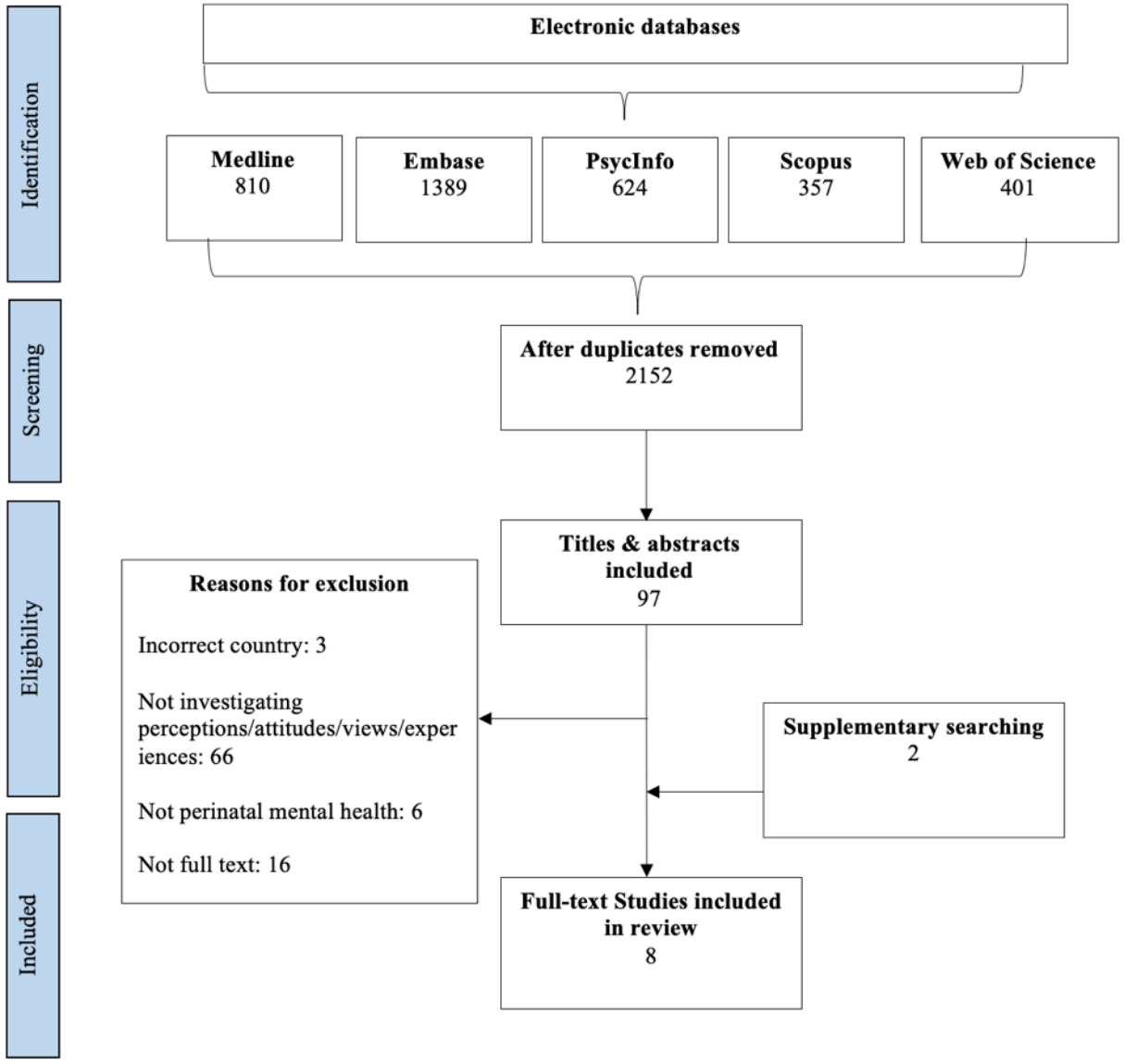


Figure 19 PRISMA Flow chart for qualitative systematic review

The inclusion and exclusion criteria for this review are given in **Table *10*** below:

Table 10 Inclusion and exclusion criteria for perceptions and attitudes systematic review

Inclusion criteria	**Exclusion criteria**
Full published studies (i.e. not editorials, abstracts, position pieces, research letters or posters).	Papers not in full (i.e. abstracts, position pieces, research letters or posters)
Qualitative studies, or quantitative studies consisting of some qualitative data or data that can be transformed into qualitative data, that explored perceptions and attitudes of pregnant women, their families or HCPs around PMH.	Studies not exploring perceptions and attitudes of either pregnant women, their families or HCPs and statistical data from quantitative studies which cannot be transformed into qualitative data
Papers published between 1st January 2000 and 4th January 2021.	Papers published before 1st January 2000
Papers in the English language or contained English translations if not originally in English.	Papers not available in the English language
Studies conducted in Bangladesh, India or Pakistan as the host country.	Studies not conducted in Bangladesh, India or Pakistan as the host country

3.3.2 Quality appraisal

Study quality was assessed using the 2018 Critical Appraisal Skills Programme (CASP) checklist for qualitative studies. The CASP is the most commonly used tool for quality appraisal in qualitative evidence synthesis and is endorsed by the Cochrane Qualitative and Implementation Methods Group (117). The CASP consists of 10 questions to guide the assessment of qualitative research (Appendix 10). A scoring of 1 mark for each question that was answered with “yes” was allocated to provide an indicator of quality and enable comparison between reviewers. Cut-off scores of ≥9 indicated ‘strong’ quality, 6-8 indicated ‘adequate’ quality and ≤5 indicated ‘weak’ quality for the CASP tool. This scoring system also used in a previous systematic review (118). There is no ‘gold standard’ tool for scoring or critical appraisal, as such

this pragmatic decision was made to enable reviewers to indicate a level of quality. No studies were excluded based on the quality score.

For the primarily quantitative studies included in this review, there are no CASP checklists for cross-sectional studies, therefore, the NOS adapted for cross-sectional studies which assesses information, selection and confounding bias was used to assess quality (Appendix 11). See chapter 2 for a critical discussion of the use of the NOS. The NOS for cross-sectional studies used a similar three level scoring system reflecting previous reviews (69); 0-3 reflected low quality papers, 4-7 was medium and 8-10 indicated high quality.

3.3.3 Data extraction and synthesis

Data including bibliographic information, country, sampling size and method, participant characteristics, setting (hospital/community based), data collection method, data analysis method, quality assessment and findings were recorded on a standardised data extraction form built using Microsoft Word. The integrated mixed methods synthesis approach developed by Sandelowski et al (119) was adopted for data synthesis. This involves transforming the data which are then integrated and synthesised together. In this review, the quantitative data were transformed into qualitative, i.e. qualitised, because the quantitative data were collected through semi-structured questionnaires or interviews and consisted of phrases/sentences that summarised certain attitudes and perceptions to form themes/sub-themes. Therefore, the nature of the quantitative data made it more appropriate for transformation as compared to the qualitative data. This method is appropriate when both quantitative and qualitative studies address the same broad research question, as in the case of this review both types of studies address perceptions and attitudes towards PMH.

Integration and synthesis of qualitative findings were carried out using thematic synthesis adapted from Thomas and Harden (120) in two iterative stages: 1) coding of extracted data: data from included studies were examined line by line for meaning and content and codes were given to capture this in each sentence. As each new study was examined, similar codes were highlighted, and new codes were added to the bank; 2) Grouping of codes into themes: list of initial codes were analysed and examined for meaning and organised into categories of similar meanings. Each category was examined, and descriptive themes were generated to reveal relationships and

theoretically link similar sub-themes together to generate analytical themes which provided deeper synthesis beyond the primary studies. This systematic analysis and synthesis were primarily conducted by me with review from supervisors. The review protocol was registered on the PROSPERO database of systematic reviews (reference CRD42020167903).

3.4 Results

Eight studies published between 2000 and 2021 met the inclusion criteria (**Table 11**). Five studies consisted of qualitative data (23,63,91,121,122) and three studies used quantitative methods (123–125). Six studies included data from populations in India (63,91,122–125), two from Bangladesh (23,121) and one from Pakistan (122). Sample sizes ranged from 21 to 332 using mostly purposive or convenience sampling. Six studies were conducted in hospital settings (63,91,122–125) with only two conducted in community-based settings (23,121). The included studies investigated perceptions and attitudes from four population groups: pregnant women (91,122,123,125), postpartum women (23,63,91,121,122,125), family members of perinatal women (63,124) and healthcare providers (23,125).

Table 11 Summary of included studies

Author, study year (reference number)	Country	Aims	Sampling size & method	Participant characteristics	Setting (hospital/community based)	Data collection method	Data analysis method	Quality
Edhborg et al, 2015 (121)	Bangladesh	"To explore and describe the experiences and concerns during the first 3-9 months following childbirth of those mothers who showed depressive symptoms 2-3 months postpartum, in a rural area in Bangladesh"	21 Purposive sampling	Mothers with depressive symptoms 2-3 months postpartum	Community	Narrative interviews	Inductive content analysis	Adequate
Goyal et al, 2020 (91)	India	"To study psychiatric morbidity, its prevalence and cultural factors influencing illness understanding, help-seeking behaviour and	27 Purposive sampling	Perinatal women with depression or anxiety screened using EPDS or PASS	Hospital	Cultural Formulation Interview	Thematic interpretations	Medium

		barriers to care in perinatal women."						
Manjrekar et al, 2018 (123)	India	"To find out the awareness and perception of mental health problems in pregnant women residing in rural areas of India"	300 Convenience sampling	Antenatal women	Hospital	Semi structured questionnaire interview	Descriptive statistics	Low
McCauley et al, 2020 (122)	India & Pakistan	"To explore what women attending for routine antenatal care (ANC) or postnatal care (PNC) at healthcare facilities in India and Pakistan consider health and ill health to be in general, and, what they consider health and ill health to be, during and after pregnancy"	130 Purposive sampling	Antenatal and postnatal women	Hospital	Focus group discussions	Thematic framework analysis	Adequate

Poreddi et al, 2020 (124)	India	"To explore the knowledge and attitudes of family members towards postpartum depression."	202 Random sampling	Family members of postpartum women	Hospital	Face to face interview using semi-structured questionnaire	Descriptive statistics	Medium
Ransing et al, 2020 (125)	India	"To assess the knowledge gap, perceptions, and misconceptions about perinatal depression at three different levels i.e. high-level service providers (e.g. specialists, general practitioners), mid-level health care providers (nurses, midwives) and service utilizers (perinatal women)."	332 Convenience sampling	270 Perinatal women 42 Nursing providers 20 Medical practitioners	Hospital	Questionnaire and online survey forms	Descriptive statistics	Medium

Rodrigues et al, 2003 (63)	India	"To describe attitudes and perceptions of mothers and husbands towards childbirth in Goa in order to explore the processes through which the relationship between social adversity and PND is mediated."	39 Purposive sampling	Postpartum mothers (19 with postpartum depression) and their husbands	Hospital	In-depth interviews	Thematic analysis	Adequate
Williams et al, 2018 (23)	Bangladesh	"To understand the cultural attitudes, from both new mothers and maternal health professionals, towards mental health post-childbirth."	70 Purposive sampling	36 postpartum mothers 34 medical personnel	Community	In-depth interviews	Thematic analysis	Adequate

Five analytical themes concerning perceptions and attitudes surrounding PMH in Bangladesh, India and Pakistan were synthesised (**Figure 20**): 1) perceived causes of PMH, 2) perceived symptoms of PMH, 3) perceptions of motherhood, 4) accessing PHM care and 5) emotional sharing and coping strategies. Themes comprised sub-themes. These themes and their sub-categories are reported in relation to the studies where they were found to be present in.

Direct quotes cited from interviews or focus groups from the relevant included studies were used to illustrate the themes and their sub-themes These quotes were identified from the studies where the sub-themes were present to best reflect the findings.

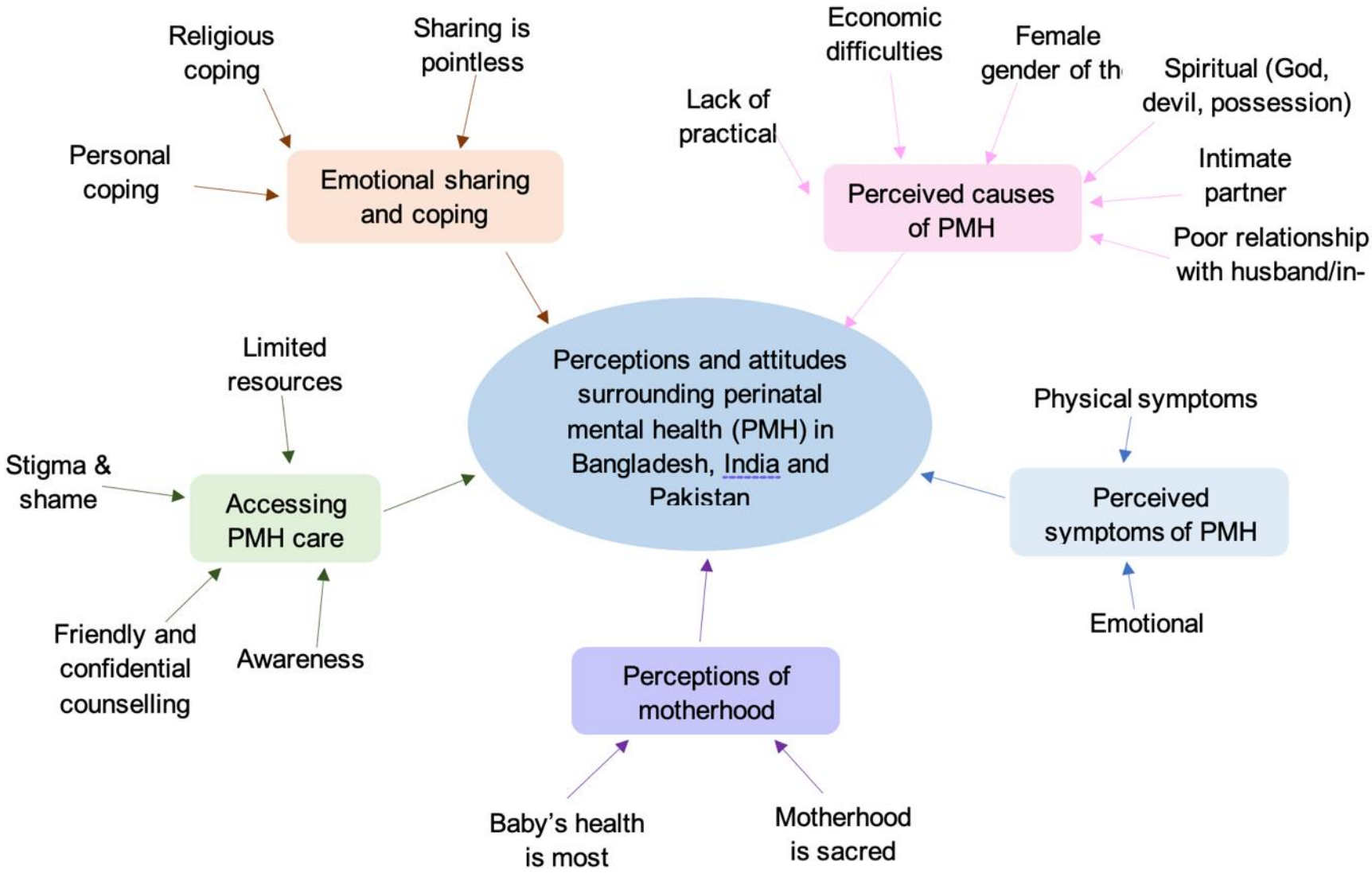


Figure 20 Thematic map of themes and sub-themes

3.4.1 Theme 1: Perceived causes of PMH

This theme explores the perceived causes of PMH conditions such as depression and anxiety. The sub-categories include a range of different psychosocial, cultural and socioeconomic issues which are perceived to be the variety of challenges that women face during and after pregnancy in Bangladesh, India and Pakistan.

Lack of practical support: Four studies reported findings suggesting lack of practical support from in-laws and husband in helping out with household chores to be a cause of PMH conditions (23,63,124,125). This view was found among postpartum women (23,63), nurse practitioners (125) and family members of postpartum women (124). This is illustrated by the quote below cited from an interview with a first-time mother in India in a study investigating motherhood and depression:

> *"I have to do everything by myself. I need someone to help me. I haven't gone to stay with mother because then my husband will have trouble managing things; he will have to cook by himself. So I'll go later and stay a month with her."* (postpartum mother in India) (63)

Poor relationship with husband/in-laws: Four studies found that a poor relationship between husband or in-laws was perceived to be a cause of PMH conditions (63,121,122,125). In particular the relationship with their husband and/or mother-in-law was suggested to be the most important (63). Both perinatal women and HCPs shared this perception. A quote cited from an interview with a postpartum mother in a study conducted in India highlights the importance women give to their relationship with their husband:

> *"Once you are married you are under the care of your husband. If your husband is not good then there is tension and you wonder why you ever got married."* (postpartum mother in India) (63)

IPV: Five studies reported IPV to be perceived as a cause of PMH conditions by pregnant women (122), postpartum women (23,63,121,122) and family members of postpartum mothers (124). IPV was found mainly to take the form of marital emotional and physical violence both during and after pregnancy, as highlighted in this quote from a postpartum mother in India.

"He used to hit me when I pregnant. We had an argument and I said I was going to leave him. He said we should both die and save our lives. I said I would give back all the money he had spent on me. Then, he hit me badly; I fell on my stomach." (postpartum mother in India) (63)

One study conducted in Bangladesh also highlighted that along with marital violence, women also feared violence from in-laws creating a very abusive family environment for the postpartum mother (121).

"I am scared of them (the parents in-law), they look at me threateningly, and they want to pick issues. When they come home, I stay in my room and don't get out." (postpartum mother in Bangladesh) (121)

However, two studies reported that, despite experience of violence, many women considered this to be "no big deal" and would not report this as marital difficulties (23,121). Concerns around husbands using drugs and taking second wives, which would often lead to IPV, were considered to be more important by the women (121).

Gender of the baby: Gender of the baby is a major concern for pregnant and postpartum women in South Asian countries and having a female child has been highlighted as a perceived cause of PMH conditions in five studies (63,121,122,124,125). This stress around gender is highlighted in this quote cited from a focus group discussion with postnatal women in a study conducted in India investigating ill health during and after pregnancy:

"This time also I have given birth to a girl child [and] because of this the family atmosphere is not good. My family members are abusing me and because of this I am going through a lot of stress." (Focus group with postnatal women) (122)

HCPs also stated that pressure from the husband and in-laws for a male child were causes of perinatal depression. This perception was more prominent in nursing providers than medical practitioners (125) as they tend to have more discussion time with perinatal women during antenatal and postnatal care visits.

Economic difficulties: Poverty was reported to be an important cause of PMH conditions by pregnant and postpartum women and their family members (23,63,121,124). In rural areas of Bangladesh, India and Pakistan, husbands often work as farmers on others' lands which does not lead to steady income due to periods of unemployment (63). For perinatal women, this results in worries around meeting medical costs during and after her pregnancy and being able to feed her baby properly so her child can develop.

> *"The main tension is because of the unemployment of my husband. Since it's summer now, he can go anywhere (and do odd jobs), but once it starts raining, where can he go…my elder son has jaundice. If one of them is better, then the other one falls ill. I'm worried because we have to buy the medicine prescribed by the doctor."* (postpartum mother in India) (63)

Spiritual: Three studies reported perceptions of spiritual causes of PMH conditions. These views were found in perinatal women (91,123) and family members (124). Many thought that mental illness during pregnancy was caused by being possessed by God or some devil. Family members stated that they thought that postpartum depression was caused by ghost possession, committing sin, black magic or witchcraft.

It is evident that communities in Bangladesh, India and Pakistan focus more on the social and cultural aspects of their lives which cause stress. Husbands and in-laws play a large role in perinatal women's lives, with many of the causes of PMH conditions stemming from these crucial relationships. Biological causes were only highlighted by medical practitioners (125).

3.4.2 Theme 2: Perceived symptoms of PMH

This theme comprises findings relating to the views of perinatal women, family members and HCPs on the symptoms of PMH conditions. The theme contained two sub-themes: physical symptoms and emotional symptoms.

Physical symptoms: Four studies highlighted that many perinatal women give importance to physical signs such as weakness, backache, headache and "brain

stops" when describing the symptoms and effects of PMH conditions (23,63,91,121). This is illustrated by this quote cited from an interview with a postpartum mother who was asked about symptoms of depression in a study conducted in India:

> *"The headaches, dizziness, cold, tiredness all are the effects of this tension."* (postpartum mother in India) (63)

It was also suggested that women tend to focus more on these physical symptoms when trying to describe PMH, especially to their families or healthcare providers (91), as they perceive these to be more acceptable and treatable symptoms rather than symptoms such as worry, fear, loneliness and stress. This also aligns with the finding that HCPs only ask pregnant women about their physical health during ANC visits, ignoring questions around mental or emotional wellbeing (122). Physical symptoms prevent women from completing household chores or looking after her baby which was also highlighted to be a concern by perinatal women (121).

Some pregnant women did not consider certain physical symptoms such change in appetite, sleep problems and feeling tired as symptoms of depression (125). Such symptoms are considered to be a normal part of pregnancy or of the postpartum period (122).

Emotional symptoms: However, some studies found that perinatal women and HCPs do consider emotional symptoms of PMH to be of importance (63,91,121,122,125). Some women described symptoms of PMH as feeling sad, low mood, being depressed, crying, having tension or being mentally weak. These symptoms were acknowledged by women, but they would not always communicate these to others (91).

> *"Things are running in my head. I can't stop worrying about everything."* (postpartum mother in Bangladesh) (121)

3.4.3 Theme 3: Perceptions of motherhood

In this theme, perceptions and views that communities hold towards the construct of motherhood and pregnancy are highlighted. The sub-themes represent

the views from perinatal women, their family members and HCPs and how these are intrinsically linked to PMH.

Motherhood is sacred: Motherhood is viewed as something very sacred and an integral part of being a woman in many South Asian communities as highlighted by three studies (23,124,125). When a woman becomes a mother, her status is elevated in her family and society, and it is acknowledged as an accomplishment as stated by a postpartum mother in a study conducted in Bangladesh:

> *"Now everything is changed, people say that now you are a mother, they give me more importance, and they are so happy with me."* (postpartum mother in Bangladesh) (23)

However, this view places many pressures on the mother or pregnant woman to live up to these expectations and status. These societal expectations of mothers can contribute to criticism of women who may appear to struggle during pregnancy or postpartum, particularly to the point where she needs to access care (23). Women seen to be struggling with PMH conditions are viewed as being weak (125). In addition, these perceptions can be subject to certain conditions, such as having female children is often perceived as a status reducing characteristic for mothers (121).

Baby's health is most important: Prominence is attached to a baby's health as an important aspect of maternal wellbeing as highlighted in five studies (23,63,121,122,124). Women during and after pregnancy assess their mothering abilities, health and success on the basis of their baby's health.

> *"The most important thing is this that baby is healthy."* (postpartum mother in Pakistan) (122)

> *"I am more worried now about the baby…if my baby stays healthy, I will be in peace."* (Postpartum mother in Bangladesh) (121)

3.4.4 Theme 4: Accessing PMH care

This theme highlights some of the key barriers and enablers of access to PMH care as perceived by perinatal women, their family members and HCPs. The theme

comprises sub-themes representing a range of cultural, social and practical factors which hinder or facilitate women to access PMH care.

Awareness: One key enabler in improving access to PMH care is awareness of mental health and the understanding that women during and after pregnancy are at risk of mental health conditions. It is important that perinatal women, HCPs and family members all have this awareness as accessing PMH care is a two-way stream which requires women themselves to be aware, and also their family members and HCPs who play a big role in women receiving PMH care.

Five studies highlighted poor awareness among communities which hindered access to PMH care (23,63,122,123,125). Many HCPs had poor awareness and understanding of PMH and perceived that such conditions were not a serious health concern (23,63,125). This lack of awareness among HCPs results in limited or no enquiry regarding PMH during ANC visits leaving any PMH conditions unidentified in pregnant women (122). This is highlighted in the quote below cited from an interview with a gynaecological intern in a study conducted in Bangladesh:

> *"There is no mental health in maternal health – we don't take mental health seriously, it's just not there."* (Medical internal in Bangladesh) (23)

Awareness among perinatal women varies with many finding it difficult to understand the links between pregnancy or motherhood and depression.

> *"Why would she be sad? I don't understand." "I don't know, this does not happen."* (Medical internal in Bangladesh) (23)

However, some women do acknowledge PMH conditions such as depression and anxiety, and would appreciate HCPs to ask them about their mental and social health. Nevertheless, they do understand that these discussions might be unacceptable because of lack of education and understanding among some women.

> *"They should ask such questions, but they do not. They should ask about family conflicts and talk about solutions."* (Postpartum mother in Pakistan) (122)

One study highlighted that the husbands of postpartum women have very limited awareness of PMH (63). Husbands are less aware of the emotional needs of women during and after pregnancy and they often feel their only duty is to provide financial support (63). Many husbands of women with postpartum depression were also ignorant to the fact that their wives were unwell in any way (63).

Stigma and shame: Stigma, and the related shame, associated with PMH in Bangladesh, India and Pakistan is a key barrier for perinatal women accessing PMH care as highlighted by seven studies (23,63,91,122–125). Women are liable to be criticised if they are seen to be mentally struggling, especially if they require access to healthcare (23). There is the attitude that emotional health problems should stay within the family and women would not tell their friends if they do attend mental health services because of the taboo attached to them (23).

Family members of perinatal women also reflect these stigmatisations regarding women with postpartum depression as not being good mothers and thinking that they should stay at home (124). Nevertheless, many are ready to help reduce depression through family interventions. Nursing providers stated that postpartum depression is a sign of weakness and "women choose to get postpartum depression." (125)

These stigmatising attitudes towards PMH from family members and HCPs prevent women from accessing adequate care. Stigma results in women being ashamed or fearful to access care due to repercussions from family or judgment from HCPs and communities (122). This fear is highlighted in the quote below cited from a focus group discussion with pregnant women in a study conducted in Pakistan:

> *'Some women will not talk about their problems as they are afraid. They think that their husband will know about it if they talk to doctor and will be angry with them."* (Focus group of pregnant women in Pakistan) (122)

Friendly and confidential counselling: This sub-theme comprises findings that many perinatal women would find counselling acceptable if it was friendly and confidential and given the stigma attached to it was removed (122). Counselling is a more appropriate form of treatment compared to drugs and medications (124,125). This attitude reflects perceptions around shame and fear and mental health not being

a serious health concern to the level that requires drugs or medications. Drugs and medications are considered to be for women who are weak (125).

Counselling and social support have been acknowledged as helpful by many women but require much more cultural attention to tackle the shame and stigma associated with it (91). Western protocols and counselling methods may not work as well in these South Asian settings as they need to consider religious, cultural and societal beliefs which shape a lot of the beliefs around PMH (23,63).

Limited resources: A key barrier in accessing adequate PMH care is the limited resources within these South Asian countries as reported by five studies. One aspect of limited resources is the lack of time that HCPs have which prevents them from focusing on PMH (23,122).

> *"A doctor has many patients…obviously, they don't have time for these details!"* (Focus group of pregnant women in Pakistan) (122)

Another element is poverty and lack of money in the healthcare system to treat PMH conditions and within families to access such treatments (63,91,123). The limited health expenditure on the healthcare system for mental health care means HCPs have to prioritise their care and, naturally, physical health takes precedence (23). The lack of mental health services within primary care also means that if women want to access PMH services they need to travel far which requires more money, hence, there is poor accessibility to PMH facilities (63,91).

> *"How can we focus on maternal depression when unattended birth and postpartum haemorrhage is such a problem, and we are a poor country?"* (Obstetrician in Bangladesh) (23)

There is also limited education on PMH with counselling not widely taught and no protocols on how to educate or identify women with postpartum depression (23). This lack of education impacts the care available to perinatal women.

3.4.5 Theme 5: Emotional sharing and coping strategies

Emotional sharing and coping strategies highlight the different ways in which perinatal women cope with their PMH conditions, and their perceptions around sharing their mental and emotions with others in the community.

Sharing is pointless: Many women reported that sharing their mental health issues is pointless or difficult because of resulting quarrels and judgment from others. (23,63,122) Family members tend to be disinterested in such matters and women feel that HCPs will not be able to do much as these are family related problems (63). Hence, emotional sharing tends to be very limited in these South Asian countries.

> *"It's not necessary to share, I feel better to just keep things to myself. I am afraid of sharing."* (Postpartum mother in Bangladesh) (23)

Personal coping strategies: With limited emotional sharing with others, perinatal women tend to resort to personal coping strategies (23,63,91). A common form is through distraction from the issues that worry women by listening to neighbours or playing with their baby (23). This is highlighted in the quote below cited from an interview with a Bangladeshi postpartum mother when asked how she copes with stress:

> *"Sometimes I go outside the gate and stand there, and then I come back."* (Postpartum mother in Bangladesh) (23)

Religious coping strategies: Aside from coping through personal strategies, perinatal women would also turn to religious coping methods through prayers, faith-healing and crying for God's help (23,63,91,123). Religion is often a source of comfort and strength for many women in these South Asian countries.

3.4.6 Summary of studies from updated search between 4th January 2021 to 4th January 2023

Two studies were identified that were published between 4th January 2021 and 4th January 2023 that fit the inclusion criteria for this review. Jawed et al (126) assessed the knowledge and practices of healthcare professionals, in a tertiary care setting in Pakistan, who deal with depressive symptoms amongst mothers. Lodha et al (127)

administered structured interviews to understand the awareness levels and attitudes of perinatal women and their families towards perinatal depression in India.

In relation to theme 1, Jawed et al (126) found that HCPs identified socioeconomic factors such as poverty, male gender preferences, lack of social support and poor relationship with husband to be factors leading to maternal depression. Causes of PMH conditions were not explored by Lodha et al (127). Neither study explored symptoms of PMH. When exploring perceptions around motherhood, Lodha et al (127) found that many perinatal women and their relatives believed that if the mother develops perinatal depression, it means that she does not love her baby. This reflects the sub-theme of motherhood being sacred and PMH conditions deem women unsuitable to be mothers. There was also a lot of emphasis put on the baby's health, particularly when discussing post-delivery priorities. Mother's mental health was given no attention. In relation to theme 4 regarding accessing PMH care, Jawed et al (126) found stigma, lack of awareness, limited resources, lack of education and training for HCPs to be barriers for delivering and accessing PMH care. Lodha et al (127) found that perinatal women and their families were unfamiliar with the concept of PMH and did not believe that pregnant women can experience mental health conditions. Neither study explored emotional or coping strategies.

3.6 Discussion

This review synthesised qualitative data from primary studies investigating perceptions and attitudes around PMH in Bangladesh, India and Pakistan, an area largely overlooked in the literature to date. This review identified five themes: 1) perceived causes of PMH, 2) perceived symptoms of PMH, 3) perceptions of motherhood, 4) accessing PMH care and 5) emotional sharing and coping strategies.

The perceptions and attitudes identified in this review are largely related to sociocultural elements reflecting cultural collectivism and social construction of gender roles. This is evident from the perceptions around causes of PMH conditions which included IPV, poor relationship with husband/in-laws, low practical support and gender of the baby. These findings relate to evidence from cross-sectional studies which found significant associations between IPV (34,58,59,128), poor relationship with husband/in-laws (16,59), low practical support (58,59) and female gender of the baby

(34,51,92) and antenatal depression among women in South Asia. Social construction of gender roles and power relations within South Asian communities often place women at a disadvantage. Women are expected to carry out all the household chores, even during pregnancy, with little practical support, and an inability to do this can result in repercussions from the family through violence. Even before a woman is born, society deems her to be lower than men as reflected by the preference for male children. These gender inequalities are rooted within many South Asian communities and highly influence individuals' perceptions and attitudes.

Spiritual causes were also discussed in the included studies mainly by perinatal women and their family members, not HCPs. These beliefs stem from the large influence of religion in these regions, combined with the stigma associated with mental health leading to the idea of punishment and possession which are used to explain women behaving in what they perceive to be an abnormal way. Such beliefs are also reflected in a study conducted in Ghana (129) where women perceived maternal mental health to have 'spiritual origins' and hence seek help from religious leaders, much like the coping methods highlighted in this review.

Emotional symptoms of PMH are acknowledged by perinatal women, however, discussing physical symptoms is much more acceptable due to the perception around women with mental health conditions being 'weak' or 'mad'. Since motherhood is perceived as a sacred position within which is it not socially acceptable to be seen as weak, women are forced to hide their emotional symptoms from family members and HCPs with the fear that their child may be removed from their care. Even HCPs' attitudes are dominated by collectivist cultural controls where they do not tend to ask about emotional or mental health symptoms as they perceive that many women may find this inappropriate. These findings are consistent with the systematic review by Watson et al (116) on minority ethnic women's, mostly South Asian and Black African, experiences of PMH in the UK (116).

Barriers and enablers to accessing PMH care operate at a number of levels ranging across sociocultural to structural factors. This review found that awareness around PMH facilitated access to PMH care for perinatal women. Perinatal women who acknowledged mental health conditions were more open to discussing these issues with HCPs and wanted them to ask questions around their mental and social health.

Family members, particularly the husband, and HCPs who had adequate awareness around the importance of PMH also enabled perinatal women to access care. This is consistent with other studies which found that lack of awareness among women, their families and HCPs led to poor recognition of symptoms and delayed management (116,118). Therefore, increased training and education, especially for HCPs as they play a key role in antenatal care, is required to increase awareness around PMH.

In addition, lack of awareness often stems from the high levels of stigma associated with mental health, particularly PMH given the perceptions around motherhood, which is a significant barrier for accessing PMH care. This again highlights the theory of collectivist cultures (130) where societal stigma found within these South Asian countries dominate the attitudes people have towards PMH resulting in women who access PMH care being criticised. Family members and HCPs who reflect this stigma prevent women from accessing adequate PMH care. Fear and shame are often associated with this stigma where women fear judgment and repercussions from family and society if they reveal their mental health conditions, or are seen to require medical assistance. These findings are highlighted in previous literature where stigma is a key barrier to accessing PMH care (116,129). Women do not want to be labelled with mental health conditions as society will deem them to be weak and unsuitable mothers. This relates to the emotional sharing and coping strategies adopted by perinatal women. Sharing is perceived as pointless as they won't be taken seriously by a society which belittles PMH, therefore, women resort to personal and religious coping strategies. Distraction and praying are two methods; however, these may not be sustainable. This calls for culturally appropriate coping interventions for long-term management.

Along with the socio-cultural factors involved in accessing PMH care, structural factors such as limited resources also create a barrier for perinatal women and HCPs. Lack of time and limited health workforce means HCPs are forced to focus only on physical health. Postpartum haemorrhage is a leading cause of maternal mortality in South Asian countries (131). Therefore, the immediate attention must be given to such physical health concerns. There are not enough HCPs to provide holistic care for perinatal women. Poverty is a big driver for the limited resources with countries like Bangladesh who only spend 0.5% of the health budget on mental health (24). There is limited money to spend on creating mental health services and facilities which are

accessible for perinatal women from all regions, and families cannot afford to travel or spend money on the limited mental health services available. These findings are consistent with other LMICs (129).

To my best knowledge, this is the first systematic review focusing on perceptions and attitudes around PMH in Bangladesh, India and Pakistan. A wide range of databases were searched using a comprehensive search strategy and supplementary searches such as hand searching reference lists and citation searching were carried out. However, the review also had some limitations. Due to resource constraints, one limitation is the restriction of journal articles to those published only in the English language which may produce language bias. Nevertheless, journals from Bangladesh, India and Pakistan tend to increasingly publish in the English language or contain translations making them accessible for a wider audience. Second, although most studies were deemed to be of 'adequate' or 'medium' quality, there were gaps in reporting, particularly with regards to possible researcher bias during data collection and analysis and what implications this may have on the findings and conclusions in the included studies.

3.7 Conclusion

There is a complex range of perceptions and attitudes around PMH which influence women's experiences and access to PMH care. Sociocultural expectations are key drivers for many of the perceptions and attitudes identified in this review including the importance of familial and societal causes of PMH, emphasis on physical symptoms, sacredness of motherhood, lack of awareness, stigma, shame, limited resources allocated for mental health and lack of emotional sharing. These interlinking perceptions highlight social construction of gender and the collectivist nature of South Asian cultures preventing proper identification and adequate care for women with PMH conditions. These findings can be used to inform policy and practice through targeted interventions to tackle stigmatising attitudes and increasing education and training for HCPs. The findings from this study also suggests the need for further research to explore perceptions and attitudes of perinatal women along with key family members, such as husband and mother-in-law, and HCPs in Bangladesh, where there is limited qualitative research, especially during the antenatal period.

Chapter 4. Theoretical and Conceptual framework

This PhD research uses the widely cited Dahlgren-Whitehead social determinants of health rainbow model (132) to explore the social determinants of antenatal depression in rural Bangladesh. Therefore, this model will be used to develop the theoretical and conceptual framework for the empirical research element of this PhD project (**Figure 21**). The Dahlgren-Whitehead model will be used to identify social determinants and to determine, to what extent and via what mechanisms, they are associated with antenatal depression amongst women in rural Bangladesh. These social determinants will act as the 'concepts' within this framework which will be situated and incorporated within key theories discussed in the systematic reviews (chapter 2 and 3). These key theories will aim to explicate the mechanisms between the social determinants and antenatal depression.

Starting from the innermost layer of the model, maternal age is a personal factor that has been explored in the literature as a potential social determinant of antenatal depression in Bangladesh (11,54,73,78,86,92,93). Although my systematic review (chapter 2) did not find an overall significant association between maternal age and antenatal depression, it should still be considered within the theoretical and conceptual framework due to the contradictory findings in the literature. Bangladesh has the highest prevalence of child marriage in South Asia and ranks amongst the top ten in the world (133). This leads to a young maternal age which is especially common in rural Bangladesh which may result in women being unprepared for pregnancies or create power differences. This may influence lifestyle factors, such as limited contraception use, resulting in unplanned pregnancies (134). My social determinants systematic review found unplanned pregnancies to be significantly associated with antenatal depression in South Asia (45).

Lifestyle factors are intrinsically linked to the social and community networks surrounding pregnant women in Bangladesh. Hierarchies are present within many Bangladeshi families, whereby the husband and mother-in-law often hold disproportionately more power and authority within the family (34). These power dynamics can be displayed in the form of certain restrictions or challenges for pregnant women creating a toxic and abusive environment. This may lead to low family/social

support, poor relationship with husband and in-laws and IPV; these factors have been associated with antenatal depression (45).

These social and community networks are often related to one's living and working conditions. Women in rural Bangladesh often lack education and many are unemployed which can exacerbate the unbalanced power dynamics. There is also limited access to antenatal care in rural areas of Bangladesh which can make pregnant women feel more disconnected from adequate care during their pregnancy (135). Social and community networks can be strengthened through more educational, employment and healthcare opportunities.

Taking a wider perspective into the general socioeconomic, cultural and environmental conditions for pregnant women in Bangladesh, we begin to establish underlying mechanisms and theories within the interrelated social determinants. Social construction of gender theory discusses the socially constructed roles for men and women which often place women at a disadvantage within society. Payne et al (136) states that women and men's identity, behaviour and expectations assigned to them reflect socially constructed notions about femininity and masculinity. West and Zimmerman state that gender is constantly created by human and social production (137). In South Asian communities, such as in Bangladesh, the traditional role of women as homemakers mean that women are socially and economically dependent on men which prescribes the lower status of women and unbalanced power dynamics (138,139). This creates intrinsic gender inequalities within society where many women are often victims of IPV leading to forced sex and unplanned pregnancies. Likewise, gender inequalities play a role in male child preference which was found to be a significant social determinant of antenatal depression (45). A son is more likely to be able to financially support the family as prescribed by socially constructed gender roles, whereas a daughter may be a burden for many families (105,140). These gender roles impact the educational and employment opportunities for women and creates the family hierarchies described above. Therefore, social construction of gender roles weaves through all the levels within the social determinant model.

Furthermore, women are unable to speak up or share their mental health concerns due to sociocultural norms and stigma, which are maintained by a collectivist culture (130), resulting in low social and family support for women in these stressful

situations. According to the Canadian sociologist Erving Goffman, the term ‘stigma’ describes the ‘situation of the individual who is disqualified from full social acceptance’ (141). Stigma surrounding mental health, which can be heightened for pregnant women as discussed in the systematic review in chapter 3, prevents effective antenatal care. Stigma existing within the maternal health sphere can prevent HCPs from screening for mental health conditions and delivering effective management. On the other side, stigma existing within families and communities prevent women from accessing and seeking care due to fear of not being accepted into society or being ostracised (116,129). This stigma and collectivist culture is embedded in South Asian societies which trickles down the levels within the social determinant model. Pregnant women have limited access to PMH care and to antenatal care in general in rural areas. Pregnant women may be seen as ‘weak’ or unfit to be mothers if they require a lot of access to care resulting in stigma against women requiring additional support. In-laws may restrict pregnant women from accessing PMH care due to this stigma and fear of what their community will say. Stigma can also contribute to the low rates of contraceptive use in rural areas as contraception is often a very taboo topic which can result in unplanned pregnancies (142). Hence, we can see here the complex and dynamic relationship between stigma and social determinants of antenatal depression at each level of the model.

Poverty is widespread in rural areas of Bangladesh and a challenging factor to explore due to its varying forms. It can be measured through household income, household expenditure, employment and education rates, crime and housing conditions among many others. My quantitative systematic review (45) (chapter 2) found that the majority of included studies found no significant association between household income and antenatal depression, however, results are inconsistent and no firm conclusion can be drawn. Nevertheless, my qualitative systematic review (46) (chapter 3) exploring perspectives of South Asian communities suggested that many perceive poverty to play an important role within antenatal depression. Poverty is interlinked with education and employment opportunities whereby low education can result in poorer employment options with lower income. Access to healthcare is also limited due to poverty through lack of resources to buy medicines or visit a HCP or travel to services. Poverty also impacts social and community networks, lifestyle and personal factors. The intersectionality between poverty and gender also illustrates the importance of poverty. Lack of education and employment opportunities for women in

rural areas of Bangladesh results in them being economically dependent on men. This can be disempowering for women and add to the gender inequalities and social hierarchies. Therefore, poverty is an important factor to consider in this framework given its complex relationship with other social determinants and theories.

Overall, the Dahlgren-Whitehead social determinants of health model, with the addition of the embedded theories of social construction of gender, stigma and collectivist cultures provides an effective framework for this PhD project. It illustrates the complex and interlinked relationships between the social determinants and explores the underlying key theories which can be used to guide the methodology, interpretation of subsequent findings and development of interventions.

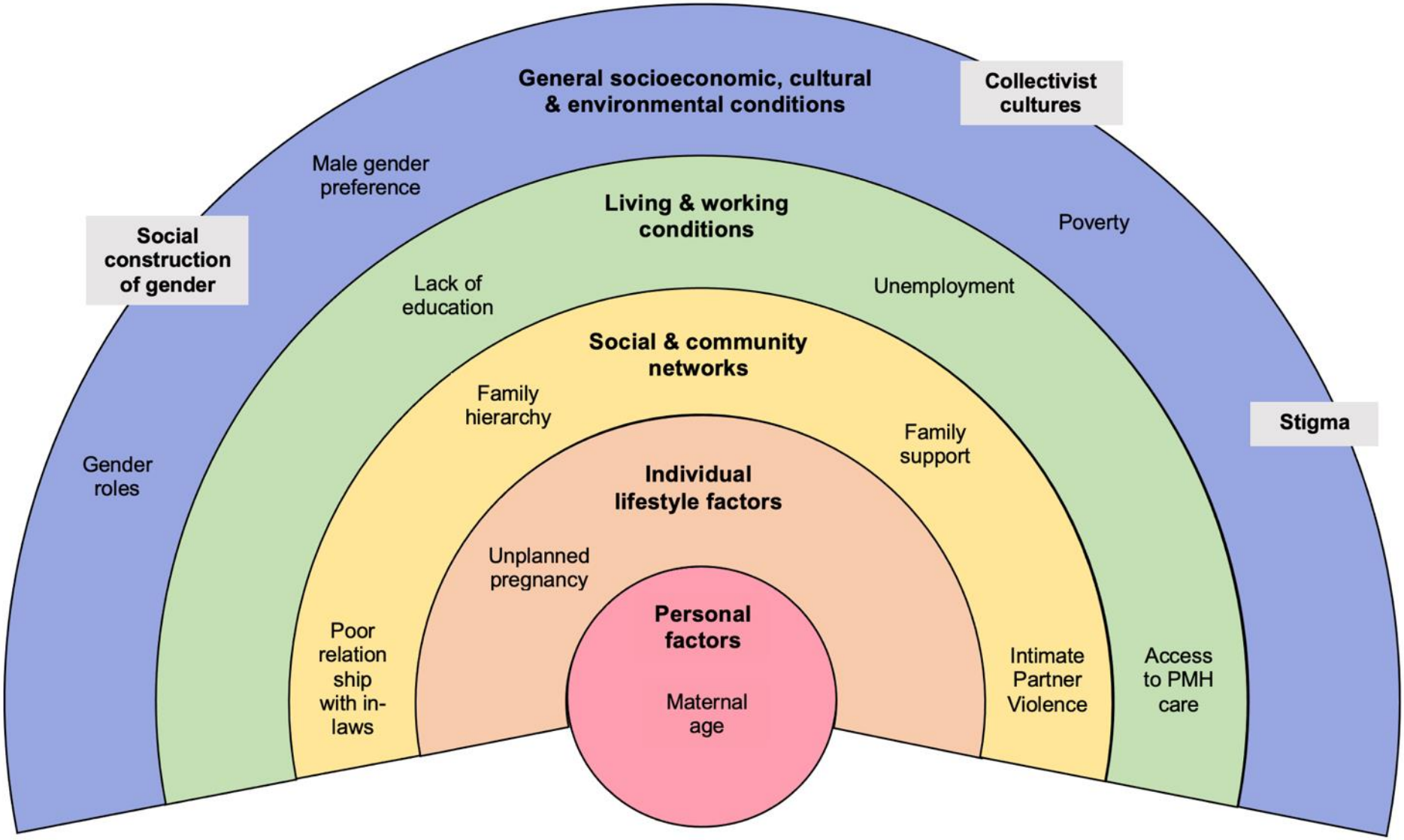


Figure 21 Theoretical and conceptual framework

Chapter 5. Methodology

5.1 Chapter overview

This chapter sets out the methodological approach taken to carry out this next stage of my PhD, starting with the overarching research paradigm within which my research is situated and the related ontological and epistemological standpoints. Subsequently, I describe and critically appraise the use of a mixed-methods methodological approach and highlight the specific design this project follows.

5.2 Research paradigm, ontology and epistemology

A research paradigm, introduced by the American philosopher Thomas Kuhn, is described as a *'set of common beliefs, shared generalisations and values between scientists regarding the nature of reality and knowledge'* (143). It is often termed as one's 'worldview' and how one thinks and makes sense of the world around them (144). Paradigms consist of two core facets, ontology and epistemology, which together contribute to one's worldview (145). These concepts feed into the way research questions are formed and the ways in which research is carried out. Ontology is the study of 'being' or 'what is' (146). It is concerned with what constitutes reality and what is possible to know about the world (146). Bryman introduced the concept of 'social ontology' which is particularly relevant for my PhD project which uses aspects of social science methodologies. Social ontology is defined as what constitutes social entities (147). There are two main opposing ontological positions: objectivism and constructionism. Objectivism assumes that *'social phenomena and their meanings have existence that is independent of social actors'*; there is a single truth or reality. In contrast, constructionism asserts that *'social phenomena are continually being accomplished by social actors'*, i.e. social phenomena are socially constructed and are constantly being revised, there is no single reality or truth that can capture this complexity (146,148).

Epistemology is concerned with theory of knowledge or '*how what is assumed to exist can be known'* (146,148). The two contrasting epistemological positions are positivism and interpretivism. The former advocates more natural sciences methods which are used to reliably and objectively measure social phenomena using valid tools. Conversely, the latter promotes more social sciences methods which aim to

subjectively interpret social meanings and phenomena to construct relevant versions of reality rather than assume one single truth (146,148). Therefore, an objectivist ontological position follows a positivist epistemology creating what is more recently known as the post-positivism paradigm (149). In contrast, a constructionist ontological position relates to an interpretivist epistemology creating the paradigm known as constructivism (149). However, given the nature of philosophy and the way we think, paradigms are not static concepts which can be so simply divided into two opposing standpoints. This is where pragmatism arises which aims to bridge the gap between post-positivism and constructivism by claiming researchers should use the methodological approach that works best for the research question (149,150). Pragmatists assert an ontological position where there can be single or multiple realities that are open to empirical inquiry. There is an objective reality which can be empirically observed, however, this reality is *'grounded in the social environment and can only be encountered through human experiences within this social environment'* (149). The epistemological position within pragmatism is that knowledge and reality are socially constructed through beliefs, habits and experiences, nevertheless, much of this knowledge can be socially shared due to *'socially shared experiences'* (149,151). Here we can see elements of both post-positivism and constructivism, where pragmatists believe that the research design and methodology selected should be based on what is most appropriate for the research question. There is a huge emphasis placed on the research question which leads the choices taken through the research process (149).

My PhD research is situated within the paradigm of pragmatism, putting the research question and aims at the forefront of deciding which methodology to use. One of the aims of this research is to identify social determinants of antenatal depression in rural Bangladesh which may be addressed through a more positivist epistemological approach which would allow quantification of the association between different social determinants, measured through empirical observation, and quantitative scores of antenatal depression. This approach will be beneficial as it allows for observations to be made through standardised and validated methods of data collection with minimal confounders to be able to generate generalisable findings (152). However, I understand that social phenomena such as PMH and the associated social determinants are socially constructed, and the reality of PMH and PMH care is intertwined with pregnant women's and HCPs' interaction with the social and cultural

environment. There are many underlying mechanisms at play within quantitative associations and multiple realities held by pregnant women and HCPs which are subjective. Hence, a constructionist and interpretivist approach are also necessary to fully explore PMH and its associated perspectives and social determinants. Therefore, this PhD adopts a pragmatist approach to enable a holistic understanding of PMH, within the context of rural Bangladesh, which advocates the use of a mixed methods methodology.

5.3 Mixed methods methodology

'Mixed methods research means adopting a research strategy employing more than one type of research method. The methods may be a mix of qualitative and quantitative methods, a mix of quantitative methods or a mix of qualitative methods' (147)

There are a variety of reasons for adopting a mixed methods methodological approach which is being increasingly recognised and encouraged in social sciences research. This PhD used a combination of quantitative and qualitative methods to enhance the richness of the findings in many ways such as corroboration, triangulation and credibility. However, there are many arguments against the use of mixed methods research which are mainly related to mixing epistemology, paradigms and research methods that sit within specific paradigms. However, as discussed above, we can see that epistemological positions and paradigms are spectrums where individuals may not have to sit at either of the two extremes. Having a pragmatistic worldview accommodates the use of mixed methods where quantitative and qualitative methods still have their own separate characteristics and stem from certain epistemological and ontological positions, but they can be integrated to produce the best answer to a research question (149).

Mixed methods research also has the challenge of potentially being more time consuming and requires more resources and training around sample selection, data collection and analysis. Nevertheless, this Doctoral Training Programme provided me with opportunities to advance my skills in both quantitative and qualitative research, with additional training on effective integration of both. Throughout this PhD, I have demonstrated the ability to provide new insights into PMH in rural Bangladesh, through

the use of mixed methods, which has not been highlighted before in the literature. Hence, although much more time consuming and resource intensive, using mixed methods has enriched my PhD project and my professional development. The reasons for using a mixed-methods methodological approach and examples within my PhD where these reasons were particularly important are shown in **Table 12**.

Table 12 Summary of reasons for adopting a mixed-methods methodology

Reasons for mixed methods research - Bryman (2016)	Importance within my PhD
Triangulation or greater validity – quantitative and qualitative methods combined to triangulate findings so they can be mutually corroborated	Corroborate findings from quantitative surveys with qualitative interviews for pregnant women and healthcare providers.
Offset – Quantitative and qualitative methods both have strengths and weaknesses so combining them can offset their weaknesses	Quantitative methods do not demonstrate perspectives of healthcare providers which qualitative interviews allowed.
Completeness – comprehensive account of the area of enquiry	Identifying significant social determinants along with the underlying mechanisms and perspectives of pregnant women and healthcare providers to get a more holistic understanding of antenatal depression.
Different research questions – quantitative and qualitative help answer different research questions	Identifying social determinants, knowledge and learning needs using standardised and validated quantitative methods. Perspectives and attitudes are best investigated using qualitative interviews.
Explanation – one of the two research methods is used to explain findings generated by the other	Statistical associations explained by underlying mechanisms identified through the qualitative interviews.

Credibility – employing both approaches enhances the integrity of the findings	Both research methods provide a more holistic overview of antenatal depression in rural Bangladesh, making it easier to develop effective interventions.
Illustration – using qualitative data to illustrate quantitative findings	Statistical associations fail to explain reasons or underlying mechanisms at play
Diversity of views – uncovering relationships between variables through quantitative research while revealing meanings among research participants through qualitative	Quantitative methods found associations between antenatal depression and social determinants while qualitative methods show how this relates to the participants and their experiences.

This PhD included both quantitative and qualitative methods. Observational quantitative methodologies were adopted to measure concepts such as antenatal depression and social determinants (IPV, social support, gender preference etc) using reliable and valid indicators which are described in detail in the subsequent individual study chapters. This enabled the investigation of associations between antenatal depression and the different social determinants in an attempt to obtain initial data on which factors are more important within antenatal depression. It also allowed me to gain a foundational understanding of HCPs' knowledge, attitudes and learning needs. This displayed an objective conception of social reality which was empirically observed and formed a foundation where initial theories and hypotheses were tested.

A qualitative descriptive approach was adopted alongside the observational quantitative methodologies for the exploration of perceptions and experiences of HCPs around PMH in Bangladesh. This approach aims to offer a comprehensive and rich summary, in everyday terms, of an experience or phenomena from the perspective of individuals or groups of individuals. In the case of this PhD, this approach allowed for a rich summary of the perceptions and experiences of providing PMH care from the perspective of a range of HCPs in Bangladesh. A qualitative descriptive approach attempts to interpret the findings without moving too far from the literal description. This was felt suitable for the purpose of this project to provide an account of the experiences and perceptions that most participants would agree are accurate. This would enable the use of such findings to influence interventions, which are of special relevance to practitioners and policy makers, which is a key element within qualitative descriptive studies. The use of a qualitative descriptive approach has been considered as particularly relevant when part of a mixed methods approach, as in this PhD, and when time and resources are limited.

5.3.1 Data analysis and integration

Descriptive and inferential statistics were used to analyse the observational quantitative data to quantify means, frequencies, prevalence rates, correlations and associations related to PMH. Adjustments were made to account for any missing data and tests were conducted to ensure there were no differences between the included and excluded data. The quantitative analysis method is outlined in detail within the relevant chapters for the quantitative studies within this PhD (Chapter 6 and 7).

For qualitative data analysis, Braun and Clarke's thematic analysis (153) was used. Thematic analysis involves latent theme development depicted by patterns of shared meanings with a central concept. Themes do not exist separate from the researcher and are mediated by their experiences, skills and research values, therefore, there was an element of reflexivity within the thematic analysis process. Thematic analysis is suitable when attempting to identify themes across a dataset to locate personal experiences within the wider socio-cultural context, as in the case of my research. The thematic analysis process is provided in greater detail in chapter 8.

The mixed methods design adopted for this study was a convergent synthesis design whereby quantitative and qualitative data were given equal priority and were collected simultaneously (154). Specifically, a parallel-results convergent design was chosen whereby qualitative and quantitative data were analysed and presented separately and integration occurred in the interpretation of results (155). The analysis from each component was then compared and merged to form an integrated whole which provided a more holistic representation of PMH in rural Bangladesh. This type of mixed methods design allowed for triangulation where two sets of findings were combined so that they can be corroborated and to gain a more complete picture. It also allowed for offsetting the weaknesses of the individual research methods by capitalising on the strengths of both. A flowchart depicting the convergent parallel mixed methods design adopted in this study is shown in **Figure 22**.

Knowledge around formal protocols of conducting mixed methods research is limited, although literature is rapidly expanding. O'Cathain et al (156) sets out three techniques to integrate data for mixed methods studies: 1) triangulation protocol, 2) following a thread, and 3) mixed methods matrix. The technique used in this PhD was the triangulation protocol as this occurs at the interpretation of results phase which aligns with the convergent parallel design. It involved listing out the findings of both quantitative and qualitative studies on the same page and considering where findings of each method agree (convergence), are complimentary (complementarity) or contradict each other (discrepancy or dissonance) (156). The six step triangulation protocol was followed: 1) sorting, 2) convergence coding, 3) convergence assessment, 4) completeness assessment, 5) researcher comparison and 6) feedback (157). These steps are discussed in detail in the mixed methods analysis in chapter 9.

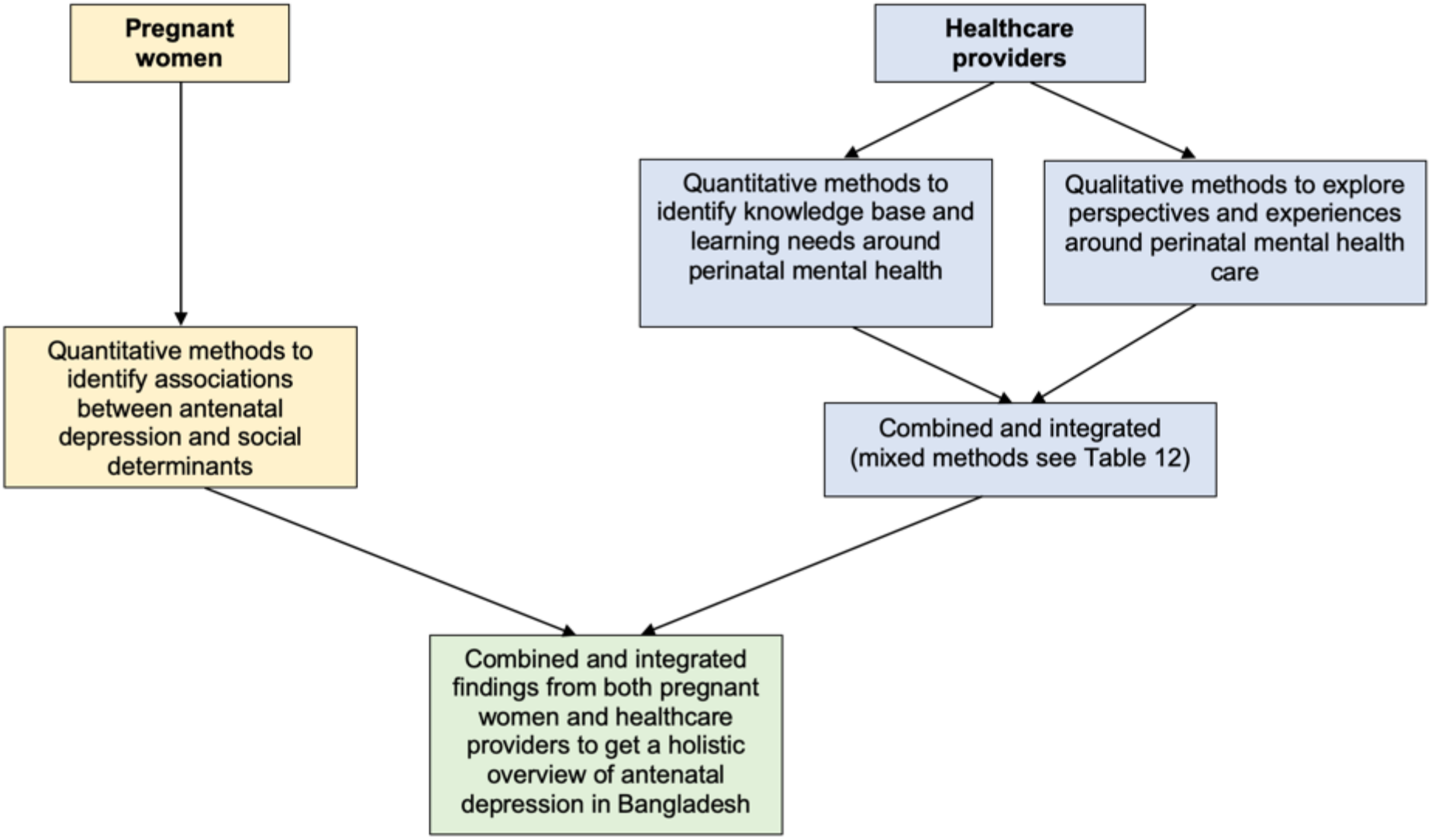


Figure 22 Flowchart depicting the convergent parallel mixed methods design used in my PhD

Chapter 6. Social determinants and prevalence of antenatal depression among women in rural Bangladesh

6.1 Chapter overview

This chapter covers the first study within this mixed methods research and aims to address objectives 2 and 3 of this PhD:

- To determine the prevalence of antenatal depression in rural Sylhet, Bangladesh
- To investigate the social determinants associated with antenatal depression in rural Sylhet, Bangladesh

There is a gap in the current literature, as highlighted by the systematic review in chapter 2, regarding social determinants of antenatal depression in rural regions of Sylhet, Bangladesh, recruiting women from community-based settings rather than hospitals. Hence, this study aims to address these gaps by conducting a cross-sectional survey in rural regions of Sylhet, Bangladesh, recruiting pregnant women visiting MAA health camps and clinics. A brief background to the study will be followed by a description and explanation of the quantitative cross-sectional methods used, results from the survey and a critical discussion of the key findings. This study has been published in the International Journal of Environmental research and Public Health in January 2023 (47) (Appendix 3).

6.2 Introduction

As highlighted in Chapter 1, prevalence of antenatal depression in rural Bangladesh ranges from 18-33%. IPV, male gender preference, poor partner relationship and low family support have been identified as significant social determinants of antenatal depression in rural Bangladesh (21,34). There is limited literature investigating the prevalence and social determinants of antenatal depression in rural regions of Bangladesh, as identified from the systematic review in Chapter 2. Nasreen et al (21) and Gausia et al (34), which do focus on rural Bangladesh, mainly focus on rural regions in and around the capital city of Dhaka. As discussed in chapter 1, data suggests that the region of Sylhet has poorer maternal health indicators (41), highest fertility rate and lowest contraception rate (42,43) in comparison to other

divisions. This makes Sylhet an important region to focus on given the increasing health inequalities within Bangladesh. A more recent study in 2021 (22) investigated the risk factors of antenatal anxiety in rural Sylhet. It did not, however, explore antenatal depression which is the most common mental health condition during pregnancy. Therefore, this study aims to investigate the prevalence and social determinants of antenatal depression in rural Sylhet, Bangladesh, to inform the development of interventions, programmes, and policies to reduce the burden of antenatal depression and improve antenatal care for pregnant women within that region.

6.3 Methods

This study followed the STrengthening the Reporting of OBservational studies in Epidemiology (STROBE) guidelines (Appendix 12).

6.3.1 Study design and setting

This study followed a cross-sectional survey design. This was deemed appropriate for this phase as it allowed for a point-prevalence to be estimated and associations to be analysed between the social determinants and antenatal depression, without the addition of interventions or manipulations. Further, given the limited time due to COVID-19 (see COVID-19 statement), a cross-sectional study was the most feasible as they are relatively faster and inexpensive when compared to longitudinal cohort studies (158). This was an important consideration as data collection took place in Bangladesh, therefore, certain activities were not practical given the resources, time and restrictions for travel.

The study setting was MAA health camps and clinics in Moulvibazar (see chapter 1 for full details of MAA and their health clinics). Moulvibazar is a town in the Moulvibazar district within the Sylhet division located in North-Eastern Bangladesh (**Figure *1***) with a population of approximately 41,358. Pregnant women attending and registering for the MAA health camps and clinics were the study population and sampling frame used for this study.

6.3.2 Inclusion and exclusion criteria

Pregnant women aged 18 or older in any trimester of their pregnancy attending the MAA health camps/clinics were eligible to take part in the survey following informed

consent. Women with a pre-existing diagnosis of any mental health conditions were excluded as I was only interested in antenatal depression which is depression brought on during the pregnancy. Women below the age of 18 were excluded due to ethical concerns around informed consent and safeguarding issues. Children may not have full ability to understand what is being asked of them from the research and to provide informed consent. There are also additional safeguarding issues around potential child abuse which this PhD do not have the resources to consider. Therefore, for both Newcastle University and Sylhet Women's Medical College ethics approval, this age restriction was imposed.

6.3.3 Sampling

Due to the study taking place within the MAA health camps/clinics, convenience sampling was used. Although it is a non-probability sampling technique, it is a very widely used and applicable sampling method in research (159). Convenience sampling involves recruiting participants that are easily available and accessible, making it a quick and inexpensive method (159). The MAA health camps/clinics are a convenient setting for this study as pregnant women are readily available and accessible for recruitment into the study. Therefore, all pregnant women at the MAA health camps/clinics that met the eligibility criteria were included into the study, given appropriate informed consent was received. Although this sampling method is subject to elements of selection bias and reduced generalisability as it missed pregnant women in rural areas who did not attend the health camps or clinics, it was the most suitable sampling technique for this study. Since this PhD project is in collaboration with MAA, the purpose was to investigate antenatal depression in the populations they serve so that interventions and programmes could be incorporated into their antenatal services to create maximum research impact. Therefore, focusing data collection within the MAA services would provide results that would be applicable for MAA. Using the sample size for prevalence studies calculation (160) (**Equation *1***) where Z is 1.96 which is the statistic corresponding to a 95% CI, P is the expected prevalence of antenatal depression in rural Bangladesh of 18% as reported by Nasreen et al (21) and d is 0.05 which is the precision of the effect size, a sample size of 227 was calculated to be adequate to estimate the prevalence of antenatal depression.

$$n = \frac{Z^2 P(1-P)}{d^2}$$

Equation 1 Sample size calculation

6.3.4 Survey instrument

The survey (Appendix 13) combined scales and sections developed after the completion of comprehensive systematic reviews (chapters 2 and 3) to understand which factors would be important to investigate. A systematic process was adopted for the development of the survey. The survey was first written in English to develop the constructs and items. The English version was sent to experts in global health, maternal health and sociology who have experience and knowledge on data collection, particularly in LMICs, as well as survey design. The experts provided feedback on the design, clarity and content of the survey providing initial face and content validity. Following feedback from the experts, some changes were made to the survey. These included changing the structure of the social support section by incorporating the Duke Social Support & Stress Scale (DUSOCS) which was deemed to be a more effective and valid scale for measuring social support (description of scale provided below). The wording of some of the questions were simplified to allow for better understanding, for example, instead of asking women if they are employed or unemployed, the question was changed to ask if they do any paid work. This better addressed the purpose of the question in understanding the autonomy of women and their participation in household income.

A systematic cultural adaptation process based on Khan & Avan (161) was used to develop, translate and validate the survey. This involved six steps and aimed to ensure that each item was assessing the construct it was attempting to understand. Each item was first written in English, and the process of adaptation was: 1) translation into Bengali independently by me and a trained researcher from the MAA team; 2) comparing these translations & assessing technical equivalence of these, then producing final translations by consensus for testing; 3) field research with five MAA staff and four mothers of young children to test understanding of the translations and to improve based on a feedback meeting I had with the MAA team. Feedback indicated that the translations matched the original English version, therefore, no changes were made to the translations at this stage; 4) finalisation and formatting of the survey for pretesting, 5) pretesting in the community to assess usability, and 6) assessor training

which included training MAA volunteers to deliver the survey in a consistent, ethical and reliable manner.

6.3.5 Outcome variable – Antenatal depression

Antenatal depression was measured using the Edinburgh Postnatal Depression Scale – Bangla (EPDS-B). The EPDS is a self-report questionnaire consisting of ten items, each with four possible responses related to the subject's mood or feelings in the last seven days (162). Although originally meant for screening of postnatal depression, studies have shown it to be just as appropriate for use in the antenatal period (163). Each item is scored between 0-3 with a total EPDS score ranging between 0-30. The Bangla translated version of the original EPDS has been validated in a Bangladeshi setting (162). The validation study on EPDS-B against standard diagnosis of depression using the Structured Clinical Interview (SCID) for DSM-IV among Bangladeshi women showed a sensitivity of 89% and a specificity of 87% at cut-off scores of 10 (162). Therefore, pregnant women scoring 10 or above in the EPDS-B were categorised as having antenatal depressive symptoms (ADS) and pregnant women scoring less than 10 were categorised as non-ADS. The EPDS-B has a Cronbach's alpha of 0.84 suggesting good internal consistency (162).

6.3.6 Independent variables – Social determinants

Along with the EPDS-B, the survey for pregnant women also included collecting information on social determinants. General background information such as trimester of pregnancy, age, marital status, highest education level, income-earning activity, and husband's employment were collected in section A of the survey.

Section B consisted of the social support section of the DUSOCS which captures an individual's perceptions of how supportive their relationships are (164). It also allows for the identification of one's most supportive relationship. The DUSOCS creates six scores: family support, family stress, non-family support, non-family stress, total social support and total social stress. For this study, I was interested in family support. A three-point scale was used ("none=0", "some=1" and "a lot=2") in which the pregnant women rate their family members as people who give personal support (12 items). The DUSOCS was slightly adapted, as recommended by literature, for this study in order to capture all the important relationships for pregnant women in rural

areas of Bangladesh (165). All scores range from 0 to 100, with higher scores indicating higher levels of support. The full DUSOCS scoring system is shown in Appendix 14. The family support has a test-retest reliability of 0.76 and Cronbach's alpha of 0.7. This suggests acceptable reliability (164).

Section C entailed questions around certain pregnancy choices based on findings from the systematic review (chapter 2) and previous literature: gender preference of both the mother and father, unwanted pregnancies and fear of childbirth. This was followed by the EPDS-B in section D of the survey.

The final section consisted of questions related to IPV. These questions covered the three main areas of violence as defined by the WHO: emotional, physical and sexual violence (166). To ensure the survey was as concise as possible due to the practical constraints of time in the MAA health camps/clinics, three questions were used to reflect the three elements of violence. A 4-point scale was used ("no=0", "yes, once=1", "yes, a few times=2" and "yes, many times=3") in which pregnant women rated whether (and how often) they had experienced each type of IPV both before their pregnancy and during their current pregnancy. The responses were coded to "yes" or "no" for all forms of IPV before pregnancy and during pregnancy to create two separate variables (IPV before and IPV during pregnancy) due to the small sample sizes for some of the 4-point responses for individual types of violence.

6.3.7 Data collection procedure

Data were collected by three MAA researchers using paper copies of the survey, due to technology and internet issues, within the health camps and the new clinic in Moulvibazar, Sylhet, from 10th March to 2nd November 2021. The MAA team are experienced in collecting health information from pregnant women visiting the health camps and receive regular training from specialist doctors and health workers. Nevertheless, I provided them with additional training on the survey for this project. The training was conducted via a single 3-hour online workshop where four MAA researchers were provided with the opportunity to go through the survey to ensure familiarity with the questions and what they mean as well as guidance on how to deliver the survey consistently across the team. This included highlighting the importance of reading out the information sheet, any risks and benefits and asking for informed consent from every pregnant woman. MAA researchers were given time to practice

delivering the survey on each other to gain confidence and receive feedback from peers. A pilot data collection from 36 pregnant women was conducted on 10th March 2021, followed by a meeting with the MAA researchers to gain feedback on clarity, design and content of the survey within the target population. No changes to the survey were deemed necessary by the MAA team, therefore, data from the pilot phase were included in the final sample and data collection was resumed.

Information regarding the study were verbally delivered to the participants in Bengali and both written and verbal informed consent were received. Survey interviews were conducted in a private area of the health camps/clinic and care was taken to ensure participants were comfortable whilst answering the questions. The MAA team have built a good rapport with the women and their families in these areas and are often a source of support and guidance during their pregnancies. This increased participants' confidence in providing honest answers given the sensitive nature of some of the questions. The interviews lasted on average 15 minutes.

Hard copies of the completed surveys were stored and locked in the MAA office in Bangladesh until final data collection. Following the completion of data collection, hard copies were both scanned and uploaded to Google drive and delivered to me in Newcastle University. The Google drive folder was password-protected and only accessible by me and Dr Reduanul Islam from MAA who sent the data. This ensured that only I had the hard copies with personal information maintaining confidentiality and copies were scanned as back-up in case hard copies were lost during delivery. Scanned copies were only accessible by me and the MAA lead researcher who transferred the data. These were deleted once the hard copies were received.

6.3.8 Data analysis

Anonymised data from the surveys were entered manually into Excel which was stored on a password-protected computer accessible only by me and backed up on Newcastle University secure One Drive. Although it is recommended to have double entry of data independently by two researchers to reduce human error, data entry was only carried out by me to maintain confidentiality and anonymity of pregnant women. However, to maintain robustness in data entry, I double checked 10% of the entries to check for any errors. No errors were identified during the double-checking process.

Data from Excel were exported into SPSS for coding and analysis. Categories were created from the total EPDS-B scores to indicate those with EPDS-B scores of 10 or more to have ADS and those with lower than 10 to have non-ADS. Continuous variables were checked for normal distribution by viewing a histogram and checking skewness (measure of symmetry of the distribution) and kurtosis (whether data are heavy-tailed or light-tailed compared to a normal distribution) values on SPSS. Means and standard deviations were used to describe normally distributed data, and median and interquartile ranges (IQR) were used for non-normally distributed data. Frequencies and percentages were used to describe categorical variables.

An independent samples t-test was used to compare means between groups. Bivariate analyses (chi-squared and Fishers exact) were conducted between each independent variable and the outcome variable of ADS and non-ADS independently to identify possible determinants that were associated with ADS. The minimum sample size for binary logistic regression was calculated to be 227 using the *pmsampsize* package on STATA as recommended by Riley et al 2020 (167). Therefore, the use of a logistic regression was justified for this study. The independent variables that were significantly associated with ADS in the bivariate analyses were considered as possible social determinants and were entered into a multivariate logistic regression model and AORs and 95% CIs were produced for each social determinant. The minimum Variance Inflation Factors (VIFs), ranging between 1.0-2.2 for the independent variables, from the regression suggested no severe multicollinearity between the independent variables. This means that none of the independent variables were correlated with each other to the point where it would produce unstable and biased standard errors leading to unstable p-values for the significance of determinants (168). VIF's of 5.0 or more indicated multicollinearity and can result in unrealistic interpretations and masking of the significance of certain variables.

6.3.9 Ethical considerations

This study contained some important ethical considerations including the safety of the pregnant women. Discussing sensitive topics such as IPV and pregnancy choices can incite certain negative emotions which may adversely impact the pregnant women participating in the study. It is also likely that we identify women with depressive symptoms, therefore, it is important to have healthcare professionals at hand to refer women to appropriate mental health care services. The doctors in MAA were on hand

to ensure that pregnant women are given adequate care. Women were informed about the voluntary nature of the study and that they could withdraw at any point.

This study was approved by the Faculty of Medical Sciences Research Ethics Committee, part of Newcastle University's Research Ethics Committee, ref 2049/4634/2020. This committee contains members who are internal to the Faculty. This study was reviewed by members of the committee, who must provide impartial advice and avoid significant conflicts of interests. The study was also approved by Sylhet Women's Medical College Research Ethics Board in Bangladesh (Appendix 15).

6.4 Results

Overall, there were 235 respondents in this study with ages ranging from 18 to 38 years and a median age of 25. The pregnant women were in varying stages of their pregnancy, with 100 women (42.6%) in their second trimester, 71 (30.3%) in their third trimester and 57 (24.2%) in their first trimester. Seven women did not know which trimester they were in. All the respondents were married, with the majority (75.8%) having no education or having ended their education before completing class 10, i.e., having not gained their SSC qualifications (equivalent to GCSE in the UK). The remaining had some level of education SSC and above. The majority of respondents (97.9%) did not undertake any paid work with the husband being the sole breadwinner for the family through work such as fishing, farming, construction or business (96.6% of respondents' husbands undertook paid work) - see **Table 13** for the full summary of respondents' characteristics.

Table 13 Summary of respondent (pregnant women) characteristics

Characteristic/social determinant	Study sample (n=235)
Age	
Age in years, median (IQR)	25 (6)
Trimester, n (%)	
First	57 (24.2)
Second	100 (42.6)
Third	71 (30.3)
Unknown	7 (3.0)
Marital status, n (%)	
Married	235 (100)
Highest level of education, n (%)	
No schooling/Pre-SSC	178 (75.8)
SSC or above	56 (23.8)
Don’t know	1 (0.4)
Paid work, n (%)	
Yes	5 (2.1)
No	230 (97.9)
Husband does paid work, n (%)	
Yes	227 (96.6)

No	8 (3.4)
Gender of baby, n (%)	
Girl	7 (3.0)
Boy	9 (3.8)
Don’t know	207 (88.1)
Prefer not to say	12 (5.1)
Gender preference, n (%)	
Girl	31 (13.2)
Boy	51 (21.7)
No preference	133 (56.6)
Prefer not to say	20 (8.5)
Perceived husband’s gender preference, n (%)	
Girl	30 (12.8)
Boy	65 (27.7)
No preference	119 (50.6)
Prefer not to say	21 (8.9)
Want pregnancy, n (%)	
Yes	142 (60.4)
No	20 (8.5)
Prefer not to say	73 (31.1)
Perceived husband wants pregnancy, n (%)	

Yes	152 (64.7)
No	17 (7.2)
Prefer not to say	66 (28.1)
Worried about childbirth/labour, n (%)	
Yes	200 (85.1)
No	17 (7.2)
Prefer not to say	18 (7.7)
Family support	
Family support score, mean (SD)	50.0 (17.3)
Intimate Partner Violence (IPV), n (%)	
Before pregnancy	
Yes	64 (27.2)
No	86 (36.6)
Prefer not to say	85 (36.2)
During pregnancy	
Yes	32 (13.6)
No	110 (46.8)
Prefer not to say	93 (39.6)
Antenatal Depressive Symptoms (ADS), n (%)	
Yes	133 (56.6)
No	102 (43.4)

SD = Standard Deviation; IQR = Interquartile range

Majority of the respondents, 207 (88.1%), did not know the gender of the baby they were having. Nine women (3.8%) stated they were having a boy and seven (3.0%) were having a girl. The rest of the women, 12 (5.1%), preferred not to say what gender they were having. Over half of the women, 133 (56.6%), stated they had no preference for the gender of the baby and half, 119 (50.6%), perceived their husband to also have no preference. Almost a quarter of the women, 51 (21.7%), stated they desired to have a boy compared to 31 (13.2%) respondents who wanted a girl, and more than a quarter, 65 (27.7%) perceived their husband's preference to be a son compared to 30 (12.8%) who perceived their husband to want a girl.

Most pregnant women, 142 (60.4%), stated that their pregnancy was wanted at this time compared to 20 (8.5%) who stated they did not want the pregnancy. Similarly, 152 (64.7%) pregnant women perceived their husband to also want the pregnancy compared to 17 (7.2%) who felt their husband did not want the baby at this time. The remaining 73 (31.1%) pregnant women preferred not to say whether the pregnancy was wanted or not and 66 (28.1%) preferred not to say whether they felt their husband wanted the baby or not. Two hundred (85.1%) pregnant women were worried about childbirth/labour compared to 17 (7.2%) who stated they were not worried. The remaining 18 (7.7%) preferred not to say.

From the DUSOCS results, the family support score was used for the analysis which ranged from 7.1 to 100.0. The kurtosis and skewness statistics were -0.3 and 0.3, respectively, indicating a close to normal distribution for this variable. The mean family support score was 50.0 (SD 17.3). The relationship with husband and in-laws were identified to be the most important from the systematic review in chapter 2. In this study, only 12.3% and 12.8% of pregnant women stated they received no support from their husband and in-laws, respectively. The remaining received some (45.5% and 66.8%) or a lot (42.1% and 20.4%) of support from their husband and in-laws, respectively.

From the IPV questions, 64 (27.2%) pregnant women stated they had experienced IPV before their current pregnancy compared to 86 (36.6%) who stated they had not experienced IPV before their current pregnancy. Furthermore, 110

(46.8%) stated they had not experienced IPV during their current pregnancy compared to 32 (13.6%) who stated they had experienced IPV during their current pregnancy.

Of the 235 respondents, 133 (56.6%) had total EPDS-B scores of 10 or above indicating a point-prevalence of ADS of 56.6% (95.5% CI 50.0-63.0%). **Figure 23** shows the distribution of total EPDS-B scores among respondents which ranged from 0 to 23. The kurtosis and skewness statistics were -0.7 and 0.2, respectively, indicating a close to normal distribution. The mean EPDS-B score among all respondents was 10.5 (SD 5.2). The mean EPDS-B score for respondents with non-ADS and those with ADS was 5.6 (SD 2.2) and 14.3 (SD 3.2), respectively. The difference of -8.8 (95% CI -9.5 – -8.0) in the mean scores was statistically significant (t=-23.7, df=233, p<0.001), indicating those with ADS had significantly higher EPDS-B scores than non-ADS.

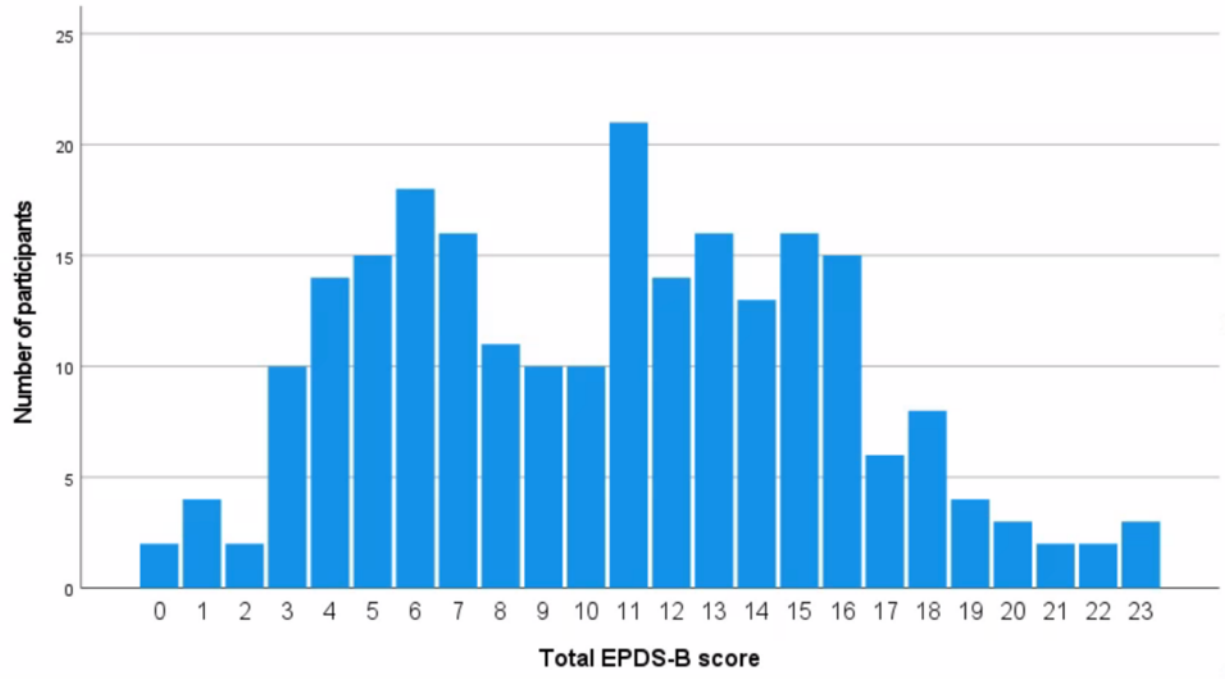


Figure 23 Bar graph showing the distribution of total EDPS-B scores

A number of independent potential risk factors for ADS were examined in this study based on the systematic review results presented in chapter 2. From the bivariate analyses, IPV before and during pregnancy, pregnant woman's gender preference, perceived husband's gender preference for the baby and family support were significantly associated with ADS (**Table 14**).

Table 14 Bivariate analysis of social determinants with antenatal depressive symptoms among pregnant women in MAA health camps in rural Sylhet

Social determinants	ADS n=133 (%)	Non-ADS n=102 (%)	p-value[c]
Trimester			
First	33 (24.8)	24 (23.5)	0.79
Second	56 (42.1)	44 (43.1)	
Third	37 (27.8)	34 (33.3)	
Education			
No schooling/Pre-SSC	106 (79.7)	72 (70.6)	0.084
SSC or above	26 (19.5)	30 (29.4)	
IPV before pregnancy			
Yes	53 (39.8)	11 (10.8)	<0.001*
No	37 (27.8)	49 (48.0)	
IPV during pregnancy			
Yes	26 (19.5)	6 (5.88)	0.002*
No	56 (42.1)	54 (52.9)	
Gender preference			
Boy	38 (28.6)	13 (12.7)	0.013*
Girl	15 (11.3)	16 (15.7)	
No preference	69 (51.9)	64 (62.7)	

Perceived husband's gender preference	51 (38.3)	14 (13.7)	<0.001*
Boy	15 (11.3)	15 (14.7)	
Girl	57 (42.9)	62 (60.8)	
No preference			
Want baby			
Yes	89 (66.9)	53 (52.0)	0.524
No	14 (10.5)	6 (5.9)	
Perceived husband wants pregnancy			
Yes	99 (74.4)	53 (52.0)	0.606
No	10 (7.5)	7 (6.9)	
Worried about childbirth/labour			
Yes	114 (85.7)	86 (84.3)	0.746
No	9 (6.8)	8 (7.8)	
Social determinants	**Depressed mean (SD)**	**Non-depressed mean (SD)**	**P-value[t]**
Age	24.33 (5.43)	24.23 (4.20)	0.871
Family support	46.4 (18.7)	52.3 (14.7)	0.009*

[C] = Chi-squared test for association or fisher's exact test
[t] = independent samples t-test
* = Statistical significance at 0.05 level
SD = Standard Deviation
Percentages do not add to 100 due as an answer of 'prefer not to say' was not included in the analyses
SSC= Secondary School Certification; ADS = Antenatal Depressive Symptoms

After performing the multivariate logistic regression (**Table 15**), IPV before pregnancy, family support and perceived husband's gender preference were found to be significant determinants. Pregnant women who had experienced IPV before their pregnancy had 10 times higher odds of ADS compared to pregnant women who had not experienced IPV before pregnancy (AOR 10.4 [95% CI 2.7-39.7]). ADS was significantly associated with low family support whereby as family support increased the odds of ADS decreased (AOR 0.94 [95% CI 0.91-0.97]). Pregnant women who perceived their husbands to want a son had almost 10 times higher odds of ADS compared to pregnant women who perceived their husband to have no gender preference (AOR 9.9 [95% CI 1.6-59.6]).

Table 15 Multivariate logistic regression model showing associated factors of antenatal depressive symptoms among pregnant women in MAA health camps in rural Sylhet

Social determinants	AOR (95% CI)	P-value
IPV before pregnancy		
Yes	10.4 (2.7-39.7)	<0.001*
No	1	
IPV during pregnancy		
Yes	1.4 (0.3-7.1)	0.656
No	1	
Gender preference		
Boy	0.21 (0.04-1.2)	0.085
Girl	0.34 (0.04-3.2)	0.348
No preference	1	
Perceived husband's gender preference		
Boy	9.9 (1.6-59.6)	0.012*
Girl	2.4 (0.29-19.1)	0.418
No preference	1	
Social determinants	**AOR (95% CI)**	**P-value**
Family support	0.94 (0.91-0.97)	<0.001*

AOR = Adjusted Odds Ratio
* = Statistical significance at 0.05 level
SD = Standard Deviation
CI = Confidence interval

6.5 Discussion

6.5.1 Key findings

This study aimed to determine the prevalence and identify social determinants of antenatal depression in rural Sylhet, Bangladesh. This study found a point-prevalence of ADS of 56.6%. Low family support was found to be a significant determinant of ADS, while perceived husband's male gender preference and IPV before pregnancy were found to significantly increase the odds of ADS.

The prevalence of ADS determined in this study is higher than previous studies conducted by Gausia et al (34) and Nasreen et al (21) which found the prevalence of ADS in rural Bangladesh to be 33% and 18.3%, respectively. A higher prevalence found in this current study may be due to methodological differences such as study location and population. Previous studies were conducted in Dhaka, the capital of Bangladesh, whereas my study was conducted in the division and district of Sylhet. As highlighted in chapter 1, Sylhet division has poorer maternal health indicators as compared to other regions of Bangladesh. Therefore, these differences may play a role in the higher prevalence of ADS in this study compared to previous studies. Nevertheless, differences may also be due to smaller sample size in this study which could affect the statistical power of the study.

The association between ADS and low family support is reflected in many previous studies (11,14,21,34,77). Family support is of special importance in Bangladeshi communities where women often live with their in-laws after marriage (21). Therefore, family dynamics play a key role in a pregnant woman's emotional wellbeing, and support from her newly formed family, especially her husband and mother-in-law, is essential in making her feel accepted and protected during her pregnancy. Furthermore, women are expected to carry out the household chores which can become more difficult during their pregnancy, therefore, practical support during this time can be a source of ease preventing ADS. Practical support can be displayed through helping with household chores like cooking, washing and cleaning. This type of support may also be seen as a sign of approval and acceptance from the in-laws which can have positive impacts on the mental health of pregnant women (21).

IPV has been well established in the literature as an important social determinant of antenatal depression, as highlighted in the systematic review in chapter 2 (45). IPV experienced before pregnancy was significantly associated with ADS and was identified to be the most important social determinant of ADS in this study. However, IPV experienced during pregnancy, although showed increased odds of ADS, was not statistically significant. Nasreen et al (21) and Gausia et al (34) both found that IPV during pregnancy was significantly associated with ADS in Bangladesh. Therefore, it would not be appropriate to conclude that IPV during pregnancy is not associated with ADS in this study as it may be due to the small sample size of the IPV during pregnancy sub-groups which may be underpowered (169). Pregnant women undergo social, emotional and physical isolation and pressure when experiencing IPV (60) which can be heightened during pregnancy due to worries about the unborn baby's health. Further studies with larger sample sizes are required to improve the power related to sub-group analyses.

Husband's gender preference for the baby as perceived by the pregnant woman has not been widely investigated in Bangladesh. A study of pregnant women in Iran (102) found that husband's preference for a male child is significantly associated with antenatal depression in women. Similarly, a study of pregnant women in China (170) found that no gender preference of spouse was associated with a significant reduction of persistent depression. This finding was also reflected in our study, whereby pregnant women who perceived their husband to want a son compared to no gender preference had higher odds of ADS. However, there was no significant association between the woman's own gender preference and ADS which is contradictory to many previous studies (34,51,92,101,102). Male gender preference is a common phenomenon in Bangladeshi communities and has been associated with antenatal depression which has been discussed in the systematic review (45) in chapter 2. This may reflect the importance of the husband in decision making and power dynamics within Bangladeshi families whereby the husband's preference is taken to be more important due to the power differences. A husband's desire for a son may create stress for many pregnant women if they are unable to give him a son, therefore, when women perceive their husband to have no gender preference it may ease them. This finding demonstrates the need for more whole-family interventions as a pregnant woman's family, in particular her husband, play a key role in her mental well-being.

6.5.2 Strengths and limitations of the study

This study has utilised a widely used and validated scale to measure ADS in pregnant women in Bangladesh (162). This increases the accuracy of measuring ADS and allows for effective comparisons to be made with previous studies. Another strength of this study is that the survey was reviewed by experts and piloted within the population of interest which allowed us to test the suitability and appropriateness of the survey in that setting. The MAA volunteers were given training in delivering the survey which reduced the chance of bias.

Furthermore, considering the likely low levels of literacy in rural Bangladesh, the survey was delivered by trained MAA volunteers in Bengali and not self-administered. This ensured that pregnant women clearly understood the survey as there was someone there to explain the instructions and questions to them. Nevertheless, this may introduce some response bias as pregnant women may provide responses they feel are socially acceptable or not give honest answers to sensitive questions due to the limited anonymity. Women may feel scared or judged whilst answering questions regarding IPV, pregnancy choices and EPDS-B items, especially if their family members were present around them. This was highlighted by the MAA volunteers after the pilot surveys who found that women were less likely or were hesitant to answer questions related to IPV if they came to the health camps with their family. Response bias may have led to an under-reporting of certain social determinants and ADS. To minimise this bias, MAA volunteers reassured pregnant women that their answers will remain anonymous and confidential, and they built a good rapport with the women beforehand.

As the study was cross-sectional, temporality and causality cannot be established as both the outcome and social determinants were measured at the same time. However, this study design was still deemed appropriate as highlighted in section 6.3.1 and addresses the objective of identifying social determinants which have an association with antenatal depression in Bangladesh which can be used to recommend the development of tailored interventions. Nevertheless, given the small sample size, results cannot be considered generalisable to the whole population. However, the results do provide an initial insight into the social determinants of antenatal depression in rural Sylhet, Bangladesh, upon which further research can be conducted.

Finally, due to ethical reasons of including children in this research, pregnant women below the age of 18 years were excluded. This does exclude findings from a very important group of pregnant women who may be more vulnerable to ADS and social determinants such as IPV. Therefore, the results may not be generalisable to that demographic of women as findings do not consider the prevalence and risk factors of ADS among pregnant women below the age of 18 years, which can be common in rural Sylhet. Nevertheless, this calls for future research with more safeguarding procedures to consider exploring ADS in this younger cohort.

6.5.3 Implications for wider PhD project

The findings from this study carry some important implications for the wider PhD project and the next phases. This study is the first to have determined a point-prevalence and identified some key social determinants of ADS in rural Sylhet, Bangladesh. The next phases of this PhD will focus on HCPs in Bangladesh by investigating their knowledge and learning needs regarding PMH and their perspectives and experiences around PMH care. Findings from these next phases will be integrated and triangulated with findings from this current study to gain a holistic exploration of PMH in Bangladesh as it is vital to understand the social determinants and perspectives from both pregnant women and HCPs. This will allow consideration of the most appropriate areas to target with interventions and policies whilst understanding the key role that HCPs play in improving PMH in Bangladesh and how we can best enhance their role.

This study also highlights several implications for policy, practice and research. For example, the need to conduct qualitative studies with pregnant women in rural Bangladesh to gain an in-depth understanding of the mechanisms behind the social determinants, and developing tailored interventions around family support and gender inequalities. These will be further discussed in detail and integrated together with implications from all the individual studies in this PhD project in chapter 10.

6.6 Chapter summary and contributions

This study explored the prevalence and social determinants of antenatal depression in rural regions of Sylhet which is one the eight administrative divisions of Bangladesh. The findings from this study contribute to the understanding of social

determinants of antenatal depression in rural Sylhet, Bangladesh, which addressed objectives 2 and 3 of my PhD. Data from the surveys in the MAA health camps/clinics in rural Sylhet revealed a point-prevalence of ADS of 56.6%. This suggests a higher prevalence of ADS in rural Sylhet compared to previous literature which is an important academic and practical contribution of this PhD highlighting the need for screening, interventions and policies to tackle such high rates. Low family support, intimate partner physical violence and male gender preference were found to be associated with ADS which reflects previous literature. This highlights the need for tailored interventions for women with low family support and/or experiencing IPV ensuring they are receiving the support they need, educating families on pregnancy support and to tackle gender inequalities within these communities. Further research should include qualitative in-depth interviews with pregnant women and their families to understand the mechanisms within these social determinants. The findings and implications of this study along with integration of the subsequent studies' findings in this PhD are discussed further in chapter 10.

Chapter 7. Survey of healthcare providers' perinatal mental health knowledge, attitudes and learning needs in Bangladesh

7.1 Chapter overview

This chapter will address objective 4: To explore PMH knowledge, attitudes and learning needs of HCPs in Bangladesh. A brief background to the study will be followed by a description and explanation of the cross-sectional methods used, the results from the online survey and a critical discussion of the findings.

7.2 Introduction

As highlighted in Chapter 1, HCPs play an important role in the way mental health is perceived and access to mental health care as they hold positions of authority and trust (171). Stigmatising attitudes among HCPs influence access to care because pregnant women may be reluctant to seek help if they feel they will be judged or ostracised by HCPs (122). It also impacts delivery of care and readiness to provide adequate and holistic interventions for pregnant women with mental health issues (172). HCPs with limited knowledge fail to adequately detect and prevent PMH problems in pregnant women. Early detection and management of PMH conditions is fundamental to promote optimal maternal health as well as fetal and childhood development (173).

To my best knowledge, there are no known cross-sectional survey studies conducted in Bangladesh investigating knowledge, attitudes and learning needs of HCPs around PMH. The majority of studies are conducted in HICs such as the UK and Australia, and focus primarily on midwives (172,174,175). Existing evidence suggests that midwives display mostly positive, non-stigmatising attitudes towards PMH and acknowledge their role to assess the mental health status of women, but negative stereotypes do still exist around disclosing mental health issues and the stigma around it. Furthermore, they expressed a strong desire for more professional education and training to be equipped with adequate knowledge and skills to manage mental health problems in pregnant women (172,174,175). There is a need for surveys to be carried out in Bangladesh which has vastly different health, cultural and social settings compared to HICs to inform what should be included within practice and policy in the Bangladeshi context. Surveys allow for the measurement of key variables in a selective

sample at one point in time which provide foundational evidence for future investigations (176). Therefore, the aim of this study was to explore the attitudes, knowledge and learning needs of HCPs around PMH in Bangladesh.

7.3 Methods

This study followed the STrengthening the Reporting of OBservational studies in Epidemiology (STROBE) guidelines (Appendix 16).

7.3.1 Study design

Given the aims of this study and its position within the overall questions being posed in my PhD, the logical choice was to undertake a cross-sectional survey. An online survey via a questionnaire was appropriate and more practical given that travel between the UK and Bangladesh was not possible due to COVID-19. A cross-sectional survey also allowed me to employ consistent measures to get a 'snapshot' of attitudes and knowledge of HCPs at one point in time, rather than to use repeated measures which would not have been feasible given the time constraints of this project. Therefore, this cross-sectional design was deemed appropriate to explore the current attitudes and knowledge base of HCPs towards PMH in Bangladesh, and provide a descriptive and explorative analysis of the relationships between the different variables in this study.

7.3.2 Inclusion and exclusion criteria

The study population comprised HCPs who were involved in the delivery or planning of antenatal healthcare in Bangladesh. This included: physicians/doctors, nurses, midwives, CHWs, charity workers and medical students who have experience in maternal healthcare in Bangladesh (see chapter 1 for full description and discussion on HCPs). HCPs were excluded if they had never provided direct healthcare to pregnant women or had never been involved in the planning of healthcare specifically for pregnant women. Individuals below the age of 18 years were excluded due to ethical concerns.

7.3.3 Sampling and recruitment

Due to the nature of the data collection through online surveys via a self-completion questionnaire, convenience sampling techniques were adopted. MAA

contacts and social media platforms, such as Instagram and Facebook, were used to recruit HCPs to complete the online survey. MAA have a wide network of HCPs which was utilised to undertake snowball recruitment of further potential participants. Organisations in Bangladesh such as the Bangladesh Midwifery Society (BMS), Obstetrics and Gynaecological Society of Bangladesh (OGSB), UNICEF Bangladesh and International Centre for Diarrhoeal Disease Research, Bangladesh (Icddr,b) were also contacted to increase the reach or recruitment and also to distribute the survey to potential participants. These organisations have access to a wide range of HCPs. Additional recruitment activity involved contacting Bangladeshi HCPs through personal contacts and social media platforms.

7.3.4 Survey instrument and structure

The survey (Appendix 17) combined an attitude scale and sections around knowledge and learning needs developed after a comprehensive review of the literature was completed (see chapter 3). Feedback on design, clarity and content was obtained from my supervisors and Dr Sunil Bhopal, paediatrician with experience of global maternal and child health, as well as HCPs within the MAA team which provided a test of face and content validity. Following this feedback, some changes were made to the survey. This included introducing a 'strongly disagree' to 'strongly agree' Likert response scale for the attitude scale as it would allow for degrees of opinion rather than a simple yes/no response. Certain items were removed or amended following feedback concerning their clarity. For example, splitting a double-barrelled item discussing women speaking to religious authorities or HCPs received some comments around which part of the item was being answered. After an iterative feedback process, the face and content validity were confirmed when no further comments were made by the experts or the MAA team. The final survey comprised four main sections: 1) sociodemographic information about the respondent; 2) attitudes and views towards PMH; 3) knowledge of PMH; and 4) perceived learning needs around PMH.

Attitudes of HCPs towards PMH were measured using a scale developed specifically for this study. Existing scales measuring HCPs' attitudes towards mental health, such as the Community Attitudes Mentally Ill inventory (CAMI-I), Mental Illness: Clinicians' Attitudes (MICA-4) scale and Opening Minds Stigma Scale for Health care Providers (OMS-HC), are limited for the purpose of this study as they do not capture attitudes towards mental health issues among pregnant women. It is important to

highlight pregnant women in this scale as they have unique healthcare needs, both physical and mental. There may be additional or different stigmas attached to mental health in pregnant women due to the presumptions around motherhood in Bangladesh (see chapter 3). In many communities, women are given an elevated status when they become a mother and are often criticised if they are seen to be struggling to do so (23). Furthermore, existing scales do not target the wide range of HCPs included in this study such as doctors, midwives, medical students. The attitude scale included nine items to be rated on a 5-point Likert scale from 0 (strongly disagree) to 4 (strongly agree). Three items out of the nine items were reverse-coded, i.e. the response scale was coded 4 (strongly disagree) to 0 (strongly disagree), to reflect the direction of stigmatisation for those items. Total attitude score could range from 0 to 36. A higher score indicated higher levels of stigma towards PMH.

Knowledge on PMH was measured using 7-multiple choice items that assessed HCPs' knowledge on the prevalence, outcomes, risk factors, symptoms, screening and management of antenatal depression. Respondents were asked to choose one of four possible answers provided for each item. A score of 1 was allocated for each correct answer given by the respondents giving a total score for knowledge ranging from 0 to 7. Items were drawn from a previous study investigating knowledge of PMH among midwives in Australia (177) and literature from the systematic review conducted (chapter 2) to make it more specific to the context of antenatal depression in Bangladesh.

PMH learning needs were ascertained from six items developed by Hauck et al (174) and adapted for use in this study. This assessment was also used in a previous study investigating midwives' knowledge and learning needs around PMH (172). A single item assessed whether HCPs felt confident in supporting pregnant women with depression or anxiety. This item had four possible responses (yes, fully confident; yes, mostly confident; no, not that confident; no, not confident at all) to capture a range of different confidence levels. Another single item (yes/no) was used to assess if HCPs felt they had sufficient access to information on antenatal depression and anxiety. HCPs were then asked to clarify their learning needs by selecting particular mental health disorders, topics and skills on which they felt that they required further education or training and to choose their preferred method of learning.

Finally, participants were given the voluntary option to provide their email address at the end of the survey if they wanted to be contacted regarding participation in the next phase of the project: semi-structured interviews with HCPs (see chapter 8). They were made aware that their email address would be used only for this purpose and would not appear in any data materials.

7.3.5 Data collection

Data collection was conducted fully online using a secure Online Surveys Newcastle University account. The online survey started with a screening section consisting of questions derived from the inclusion criteria to ensure that participants were fully eligible for the study. An information sheet describing the study, voluntary participation, data handling to ensure anonymity and confidentiality, funding and ethics statement and a consent form were included. Data were collected between 8th February and 14th April 2021.

7.3.6 Statistical analysis

Data retrieved from the online surveys were exported into a Microsoft 365 Excel spreadsheet and transferred into SPSS version 24 for statistical analysis. The raw data were cleaned by removing irrelevant data such as the mandatory consent response and personal identifiable information like email addresses of participants. Cleaning also involved converting data types to ensure numeric and string values were labelled as such and handling of missing data. Only two variables contained missing data, age and years worked in maternal healthcare, containing 11 (7.9%) and 10 (7.2%) missing values respectively. Furthermore, only four (2.9%) respondents contained a missing value for both age and years worked in maternal healthcare highlighting the limited missing data within this dataset. To determine whether the missing data were missing completely at random (MCAR) or not, Little's MCAR test was conducted which resulted in a non-significant p-value of 0.350 which accepts the null hypothesis that the data is MCAR. This means that the missingness is highly unlikely to be related to the individual being studied (178). Considering this and the small proportion of missing data, it was acceptable to use complete case analysis, i.e., listwise deletion, where SPSS would leave a respondent out of a specific analysis if there was a missing variable value. By so doing, this should not result in bias as the data are MCAR (178).

Continuous variables were checked for normal distribution by viewing a histogram and examining skewness (measure of symmetry of the distribution) and kurtosis (whether data are heavy-tailed or light-tailed compared to a normal distribution) values. Means and standard deviations were used to describe normally distributed data, and median and IQR were used for non-normally distributed data. Frequencies and percentages were used to describe categorical variables.

Pearson's correlation and Spearman's rank correlation coefficients were used to measure the relationship between parametric and non-parametric continuous variables, respectively. Independent samples t-test was used to compare two means whereas the Analysis of Variance (ANOVA) test was used to compare three means or more along with Tukey's test for multiple comparisons. Moreover, Cronbach's alpha was recorded to check the internal consistency of the attitude scale. A p-value of less than 0.05 was considered statistically significant.

7.3.7 Ethical considerations

This study contained some ethical concerns around anonymity and confidentiality of the participants; however, steps were taken to minimise these. The Online Surveys platform is a secure, password protected platform and data from the survey were exported into password protected Excel and SPSS spreadsheets. No personal identifiable information was taken from the participants apart from an email address which participants had the voluntary option to provide if they wanted to be contacted about the next phase of this PhD project. Participants were fully informed that providing an email address is completely voluntary and, if given, it would not be used for any other purposes and it would not be stored with the raw data files to maintain anonymity and confidentiality. As mentioned above, email addresses were removed at the data cleaning stage.

This study was approved by the Faculty of Medical Sciences Research Ethics Committee, part of Newcastle University's Research Ethics Committee, ref 2049/4634/2020. This committee contains members who are internal to the Faculty. This study was reviewed by members of the committee, who must provide impartial advice and avoid significant conflicts of interests. The study was also approved by Sylhet Women's Medical College Research Ethics Board in Bangladesh (Appendix 15).

7.4 Results

Overall, there were 139 respondents with ages ranging from 18 to 48 years. The median age was 24 years (IQR 3.0) and median number of years worked in maternal healthcare was 3 years (IQR 3.0). Most of the respondents were female (n= 122, 88%) and the distribution of HCPs across professions was as follows: 68 (49%) midwives, 39 (28%) doctors, 31 (22%) medical students and one (1%) nurse. The majority of the respondents carried out their primary maternal healthcare work in *Upazila* Health Complexes (n=41, 29.5%) which are the first points of primary healthcare for people living in rural areas on Bangladesh, Government medical college hospitals (n=33, 23.7%) or Non-governmental Organisation (NGO)/charity work in rural villages (n=24, 17.3%). See **Figure 24** for a summary of respondents' characteristics.

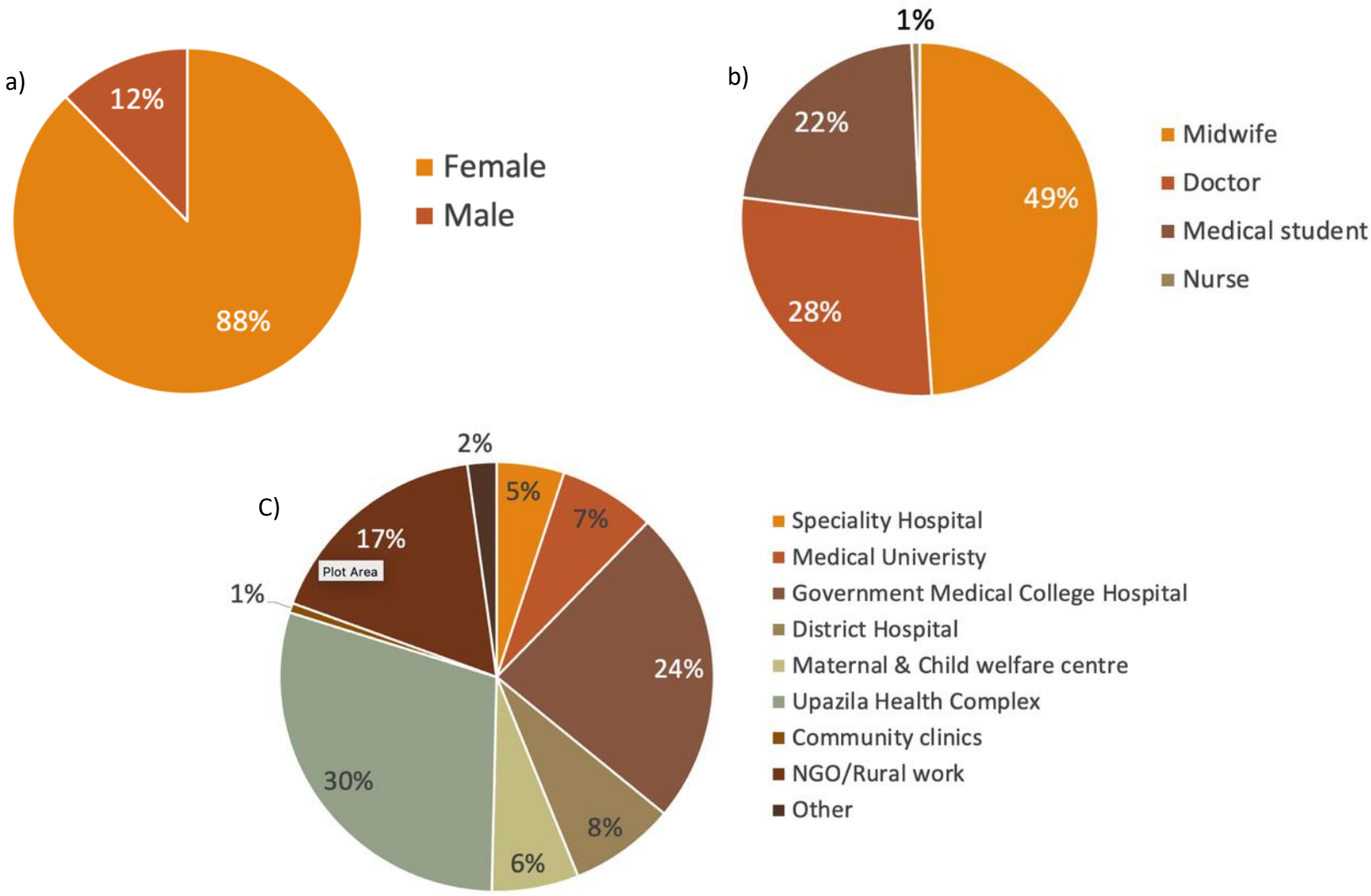


Figure 24 Demographic characteristics of the respondents; a) gender, b) healthcare provider role and c) healthcare providers primary work location

7.4.1 Attitudes

Nine items were used to investigate HCPs' attitudes towards PMH with a total possible score ranging from 0 to 36. Higher scores indicate higher levels of reported stigma. The Cronbach's alpha for this attitude scale was 0.7, indicating acceptable internal consistency. Overall, the response to the attitude construct was non-stigmatising producing a mean (SD) score of 14.68 (5.36) (lower than the mid-point of 18.0 of the total scale). As shown in **Table 16**, the mean scores of three of the nine responses from HCPs were reporting stigmatising attitudes i.e. having scores above the mid-point of 2.0 (total possible score for each item ranging from 0 to 4). Stigma was highest for the statements: "One of the main causes of mental illness is a lack of self-discipline and will power" and "A pregnant woman who shows signs of mental health issues should be immediately hospitalised", both scoring mean responses of 2.55 and 2.50 respectively. The majority of HCPs within this sample shared the view that mental illness is caused by a lack of self-discipline and will power (67.6%) and that pregnant women with signs of mental health illnesses should be immediately hospitalised (61.2%).

Table 16 Healthcare providers' attitude towards perinatal mental health

Item	Strongly disagree, n (%)	Disagree, n (%)	Neutral, n (%)	Agree, n (%)	Strongly agree, n (%)	Mean	SD
One of the main causes of mental illness is a lack of self-discipline and will power (strongly agree/agree)	16 (11.5)	15 (10.8)	14 (10.1)	64 (46.0)	30 (21.6)	2.55	1.26
I would feel comfortable talking to someone who has a mental health condition (strongly disagree/disagree)	10 (7.2)	10 (7.2)	20 (14.4)	68 (48.9)	31 (22.3)	1.28	1.38
The physical health of pregnant women is more important than their mental health (strongly agree/agree)	19 (13.7)	38 (27.3)	25 (18.0)	26 (18.7)	31 (22.3)	2.09	1.46
Mental illness is an illness like any other which requires medical attention (strongly disagree/disagree)	8 (5.8)	11 (7.9)	5 (3.6)	52 (37.4)	63 (45.3)	0.91	1.28
Pregnant women with mental health conditions are a burden to their families and society (strongly agree/agree)	31 (22.3)	39 (28.1)	9 (6.5)	25 (18.0)	35 (25.2)	1.88	1.30

A pregnant woman who shows signs of mental health issues should be immediately hospitalised (strongly agree/agree)	10 (7.2)	30 (21.6)	14 (10.1)	35 (25.2)	50 (36.0)	2.50	1.34
Mental health conditions in pregnant women are not a serious health concern that healthcare providers need to address (strongly agree/agree)	57 (41.0)	43 (30.9)	7 (5)	9 (6.5)	23 (16.5)	1.17	1.11
Pregnant women with depression or anxiety will never recover enough to have a good quality of life (strongly agree/agree)	37 (26.6)	45 (32.4)	9 (6.5)	11 (7.9)	37 (26.6)	1.57	1.15
Mental health issues should be discussed more openly within Bangladeshi families and communities (strongly disagree/disagree)	6 (4.3)	10 (7.2)	6 (4.3)	35 (25.5)	82 (59.0)	0.73	1.12

Responses in parenthesis signify the response on the Likert scale which displays stigmatising attitudes towards pregnant women with mental health conditions
SD=Standard deviation
Mean > 2.0 indicated stigmatising response to that item

Correlation analysis revealed a very weak negative correlation (r=-0.174, p=0.05) between age and attitude score. This suggests that older HCPs displayed slightly lower stigmatising attitude towards PMH. In contrast, there was a weak positive correlation (r=0.213, p=0.016) between years worked in maternal healthcare and attitude score whereby HCPs who have worked longer in maternal healthcare displayed more reported stigma towards PMH (**Table 17**).

Table 17 Correlation between continuous demographic variables and attitude score

	Correlation factor	
Variables	**Attitude score**	**p-value**
Age	-0.174	0.05*
Years worked in maternal healthcare	0.213	0.016*

*Statistical significance at 0.05 level

Independent samples t-test revealed no significant difference in the mean attitude score between females and males (14.98 vs 12.59, p=0.085). There was no significant difference between HCPs who stated they are confident in caring for pregnant women with mental health issues and those who are not (14.76 vs 13.92, p=0.593). However, there was a significant difference in the mean attitude score between HCPs with experience/education in mental health and those without (15.54 vs 11.12, p=<0.0001), whereby HCPs with experience/education have higher attitude scores suggesting more stigmatising attitudes. Further, there was a significant difference in the mean attitude score between those who stated they have sufficient access to information on PMH and those who stated they do not (16.85 vs 11.66, p=<0.0001); HCPs who stated they have access to information have higher attitude scores suggesting they display more stigmatising attitudes (**Table 18**).

Table 18 Comparing mean attitude score of healthcare providers by gender, experience/education in mental health care, confidence in supporting pregnant women with depression/anxiety and access to perinatal mental health care

Variable		**Mean attitude score (SD)**	**Mean difference (95% CI)**	**p-value[t]**
Gender	Female	14.98 (5.53)	2.39 (-0.33 – 5.11)	0.085
	Male	12.59 (3.28)		
Experience/education in mental health care	Yes	15.54 (5.26)	4.39 (2.23 – 6.54)	<0.0001**
	No	11.12 (4.24)		
Confidence in supporting pregnant women with depression/anxiety	Yes	14.76 (5.51)	0.84 (-2.25 – 3.93)	0.593
	No	13.92 (3.62)		
Access to perinatal mental health care information	Yes	16.85 (5.21)	5.20 (3.59 – 6.80)	<0.0001**
	No	11.66 (3.91)		

**Statistical significance at 0.01 level
CI=Confidence Interval
[t]=Independent samples t-test

An ANOVA test revealed a significant difference in the mean attitude score between the different HCP roles (**Table 19**). After excluding the single nurse in the sample to be able to carry out a post-hoc test, the Tukey's test for multiple comparisons revealed a significant difference in the mean attitude score between two sets: doctors and midwives (12.05 vs 17.82, p=<0.0001) and medical students and midwives (11.19 vs 17.82, p=<0.0001), whereby midwives displayed more reported stigma towards PMH (**Table 20**).

Table 19 ANOVA summary of mean attitude score and healthcare provider role

Healthcare provider role	Mean attitude score (SD)	F (bdf, wdf)	p-value[a]
Doctor	12.05 (3.66)	33.798 (2,135)	<0.0001
Medical student	11.19 (3.89)		
Midwife	17.82 (4.99)		

SD=Standard Deviation
Bdf=Between group degrees of freedom
Wdf= Within groups degrees of freedom
[a]=ANOVA

Table 20 Multiple comparisons of mean attitude score and healthcare provider role

Healthcare provider role	Comparison healthcare provider role	Difference in mean attitude score (95% CI)	P-value[a]
Doctor/physician	Medical student	0.86 (-1.66 – 3.38)	0.699
	Midwife	-5.77 (-7.87 – -3.67)	<0.0001
Medical student	Doctor/physician	-0.86 (-3.38 – 1.66)	0.699
	Midwife	-6.63 (-8.90 – -4.36)	<0.0001
Midwife	Doctor/physician	5.77 (3.67 – 7.87)	<0.0001
	Medical student	6.63 (4.36– 8.90)	<0.0001

CI=Confidence Interval
[a]=ANOVA post-hoc Tukey's test for multiple comparison test

7.4.2 Knowledge

Seven items were used to explore HCPs' knowledge around PMH in Bangladesh with a total possible score ranging from 0 to 7 (one score for each correct answer). The mean (SD) knowledge score across all respondents was 3.87 (1.51). To make general determinations about HCPs' knowledge, a topic was deemed to be known sufficiently by HCPs if at least 75% of the respondents made a correct response. This method was used in a previous study investigating midwives' knowledge of antenatal and postnatal depression in Australia (177). As shown in **Table 21**, only one of the seven questions received a correct response of more than 75%. This question covers the treatment and management method of antenatal depression, suggesting that majority (83.5%) of the HCPs in this study sample were aware that psychosocial interventions and counselling is the recommended treatment method for mild to moderate antenatal depression. The remaining six questions received less than 75% correct responses. Questions on the topics of prevalence, risk factors and screening tools received the lowest correct response percentage of 33.8%, 30.2% and 38.1%, respectively. Less than half of the respondents were aware of the estimated prevalence of antenatal depression, the common risk factors and the EPDS as a screening tool for perinatal depression.

Table 21 Summary of perinatal mental health knowledge questions of healthcare providers

Item	Correct response given, n (%)
The prevalence (percentage of the population with the condition at a given time) of antenatal depression in Bangladesh is approximately (20-33%)	47 (33.8)
Which of the following can happen as a result of antenatal depression and anxiety? (All of the above)	97 (69.8)
Which of the following is not considered a common risk factor for antenatal depression and anxiety? (High number of previous pregnancies)	42 (30.2)
Which of the following is the most common symptom of depression during pregnancy? (Feelings of isolation, loneliness and sadness)	82 (59.0)
Which one of the following statements is true? (Women with antenatal depression have higher chance of developing postpartum depression)	97 (69.8)
What is the Edinburgh Postnatal Depression Scale (EPDS)? (A screening tool for antenatal and postnatal depression)	53 (38.1)
What is the recommended treatment/management method for mild to moderate antenatal depression in Bangladesh? (Psychosocial interventions and therapy)	116 (83.5)

Responses in parenthesis indicate the correct response

Correlation analysis revealed a weak positive correlation (r=0.280, p=0.001) between age and knowledge score. This suggests that older HCPs displayed higher levels of knowledge around PMH. There was no significant correlation (r=-0.021, p=0.810) between years worked in maternal healthcare and knowledge (**Table 22**).

Table 22 Correlation between continuous demographic variables and attitude score

	Correlation factor	
Variables	**Knowledge score**	**p-value**
Age	0.280	0.001**
Years worked in maternal healthcare	-0.021	0.810

*Statistical significance at 0.05 level

Independent samples t-test revealed no significant difference in the mean knowledge score between females and males, or between HCPs who stated they are confident in caring for pregnant women with mental health issues and those who are not confident. A significant difference in the mean knowledge score was identified between HCPs with experience/education in mental health and those without (3.69 vs 4.42, p=0.026), whereby, surprisingly, those with experience/education displayed lower knowledge levels. Moreover, HCPs who stated they have sufficient access to information on PMH displayed lower knowledge levels than those who do not have sufficient access to information (3.42 vs 4.40, p=<0.0001) (**Table 23**).

Table 23 Independent samples t-test of mean knowledge score

Variable		Mean Knowledge score (SD)	Mean difference (95% CI)	p-value[t]
Gender	Female	3.81 (1.54)	-0.13 (-0.91 – 0.65)	0.742
	Male	3.94 (1.30)		
Experience/education in mental health care	Yes	3.69 (1.48)	-0.72 (-1.35 – -0.09)	0.026
	No	4.41 (1.53)		
Confidence in supporting pregnant women with depression/anxiety	Yes	3.77 (1.51)	-0.61 (-1.48 – 0.25)	0.164
	No	4.28 (1.50)		
Access to perinatal mental health care information	Yes	3.42 (1.44)	-0.98 (-1.47 – -0.49)	<0.0001
	No	4.40 (1.44)		

**Statistical significance at 0.01 level
CI=Confidence Interval
[t]=Independent samples t-test

An ANOVA test revealed a significant difference in the mean knowledge score between the different healthcare provider roles (**Table 24**). After excluding the single nurse in the sample, the Tukey's test for multiple comparisons revealed a significant difference in the mean knowledge score between two sets: doctors and midwives (4.41 vs 3.23, p=<0.0001) and medical students and midwives (4.32 vs 3.23, p=0.001), whereby midwives displayed lower levels of knowledge around PMH than doctors and medical students (**Table 25**).

Table 24 ANOVA summary of mean knowledge score and healthcare provider role

Healthcare provider role	Mean knowledge score (SD)	F (bdf, wdf)	p-value[a]
Doctor	4.41 (1.52)	11.313 (2,135)	<0.0001
Medical student	4.32 (1.38)		
Midwife	3.23 (1.35)		

SD=Standard Deviation
Bdf=Between group degrees of freedom
Wdf= Within groups degrees of freedom
[a]=ANOVA

Table 25 Multiple comparisons of mean knowledge score and healthcare provider role

Role	Role	Mean difference (95% CI)	P-value[a]
Doctor/physician	Medical student	0.09 (-0.71 – 0.89)	0.964
	Midwife	1.17 (0.51 – 1.84)	<0.0001
Medical student	Doctor/physician	-0.09 (-0.89 – 0.71)	0.964
	Midwife	1.09 (0.37 – 1.81)	0.001
Midwife	Doctor/physician	-1.17 (-1.84 – -0.51)	<0.0001
	Medical student	-1.09 (-1.81 – -0.37)	0.001

CI=Confidence Interval
[a]=ANOVA post-hoc Tukey's test for multiple comparison test

7.4.3 Attitudes and knowledge

Pearson's correlation analysis revealed a significant negative correlation between attitude score and knowledge score (r=-0.302, p=<0.0001), whereby as attitude score increases, knowledge score decreases. This suggests that healthcare providers with more stigmatising attitudes towards PMH display lower levels of knowledge around PMH.

7.4.4 Learning needs analysis

The majority of the respondents stated they felt confident in supporting pregnant women with depression or anxiety, with 65.5% fully confident and 25.2% mostly confident. Some respondents were not too confident (7.9%) and only two (1.4%) were not confident at all. More than half of the respondents (58.3%) stated they have sufficient access to information on antenatal depression and anxiety in Bangladesh.

Depression (71.2%), anxiety disorders (66.9%) and personality disorders (56.1%) were the top mental health disorders on which HCPs felt they required additional education (**Table 26**). With regards to which topics on mental health HCPs required additional education on, impact of childbearing on mental illness (71.9%), role of healthcare providers (65.5%) and signs and symptoms (57.6%) were the most commonly selected (**Table 26**). More than half of the respondents felt they required additional training on assessing mental health (69.8%), assessing risk of mental health (62.7%), clinical management of mental health disorders (67.6%), communication skills (57.6%) and working with families and carers (51.1%) (**Table 26**).

A series of seminars and workshops was the most preferred method of learning for HCPs in this sample (66.9%).

Table 26 Learning needs analysis of healthcare providers

Mental health disorders (additional education)	**N (%)**
Personality disorders	78 (56.1)
Bipolar disorder	49 (35.3)
Substance related disorders	40 (28.8)
Tokophobia (fear of childbirth)	57 (41.0)
Schizophrenia	63 (45.3)
Anxiety disorders	93 (66.9)
Depression	99 (71.2)
No further education required on any of the above mental health disorders	24 (17.3)
Other	6 (4.3)
Perinatal mental health topics (additional education)	**N (%)**
Impact of childbearing on mental health illness	100 (71.9)
Service options	46 (33.1)
Signs and symptoms	80 (57.6)
Role of healthcare providers	91 (65.5)
Child safety	77 (55.4)
No further education required on any of the topics above	24 (7.3)
Other	2 (1.4)
Skills (additional training)	**N (%)**
Handling stress and aggression	66 (47.5)

Assessment of mental health	97 (69.8)
Assessment of risk of mental illness	87 (62.6)
Clinical management of mental health disorders	94 (67.6)
Discharge planning	44 (31.7)
Communication skills	80 (57.6)
Working with families and carers	71 (51.1)
Support with breastfeeding	61 (43.9)
No further training required on the above skills	19 (13.7)
Other	3 (2.2)

% = Percentage of respondents who selected each answer option (e.g. 100% would represent that all this question's respondents chose that option)

7.5 Discussion

This study aimed to explore attitudes, knowledge and learning needs of HCPs around mental health care for pregnant women in Bangladesh. The findings of this study show relatively non-stigmatising attitudes towards pregnant women with mental health issues but overall low levels of knowledge around PMH and healthcare. HCPs desired further education and training across a range of mental health disorders, topics and skills through seminars and workshops.

7.5.1 Attitudes

Overall, HCPs' attitudes towards PMH were relatively non-stigmatising with the majority of the items on the attitude scale receiving positive, non-stigmatising, responses. This is important as HCPs play a significant role in the health seeking behaviour of pregnant women with mental health issues (23). The key areas of concern surrounding the responses to the survey were related to causes of PMH and hospitalisation of pregnant women with mental health issues. The majority of HCPs agreed that one of the main causes of mental health issues is a lack of self-discipline and will-power. This finding was also reflected in a 2018 national survey conducted in India where 60% of the respondents agreed with the same statement (179). This highlights a culture of placing blame on the individual with a mental health illness and a general misconception that these individuals are weak and unable to cope with life situations. Similarly, in Bangladesh, women seen to be struggling with PMH conditions are viewed as being weak and incapable of being a mother (125), as previously discussed in the systematic review in chapter 3. These stigmatising attitudes are maintained by a collectivist culture within many communities in Bangladesh, resulting in HCPs displaying these views which are dominated by the general shared stigma within society. This type of preconceived, judgmental attitudes from HCPs acts as a barrier to effective perinatal care (172).

The majority of the HCPs who responded to this survey agreed that pregnant women who show signs of mental health issues should be immediately hospitalised. This attitude may stem from the feeling that women with mental health conditions are 'mad' and are not fit to remain within society (116). A study by Ubaka et al (180) investigating health professional's stigma towards the psychiatric ill in Nigeria also

found similar results, where 55.1% of the sample agreed with this statement. This type of attitude can impact the care pregnant women with signs of depression or anxiety receive. In most cases, counselling and psychosocial therapy is sufficient, however, if HCPs carry this attitude, women may miss out on best treatment and be hospitalised unnecessarily which can have negative psychological consequences and socioeconomic impacts through financial burden, shame and time away from household responsibilities. Social construction of gender may play a role here, whereby signs of depression and anxiety are seen as a flaw in pregnant women who are expected to be stepping into the sacred role of being a mother. Such signs of mental health issues do not fit the socially constructed profile of a mother in many communities in Bangladesh, hence, HCPs may view it as something that needs more extreme treatment. Alternatively, this view may reflect the wider healthcare system in which there is limited resources and skilled HCPs to provide such counselling and psychosocial therapy. Thus, this view from HCPs may be positively intended, whereby they genuinely believe that the best way to treat pregnant women with signs of mental health issues is to hospitalise them where they may be in contact with more skilled mental health professionals.

This study found that older HCPs displayed lower levels of stigmatising attitudes, however, the correlation was very weak. Similar findings are reflected in the study by Ubaka et al (180) where a very weak association was obtained for younger age of health professionals and higher levels of stigma. This may reflect the idea that being older might mean more tolerant attitudes. In contrast, a study by Gaisuddin et al (181) investigating mental health stigma and attitudes among Bangladeshi medical students found that older age was associated with increased stigma towards persons with mental disorders. Nevertheless, given the very weak correlation in this study and the small age distribution in this sample, it is difficult to reach clear conclusions with regards to age.

Interestingly, despite older HCPs displaying lower levels of reported stigma, our study found that HCPs who have worked longer in maternal healthcare displayed slightly more reported stigma. This could suggest that being older may mean more tolerant attitudes, but professional experience in maternal healthcare might breed cynicism due to exhaustion or burnout in the job (182). In contrast, Ubaka et al (180)

found a weak association of less experience in healthcare and higher tendency to stigmatise. Differences in results may be a result of Ubaka et al (180) investigating attitudes towards mental health in general, whereas my study investigated mental health in pregnant women more specifically. Furthermore, my sample consisted of mostly medical students and midwives, whereas Ubaka et al (180) included only doctors, nurses and pharmacists. PMH is not adequately taught in midwifery or medical school in Bangladesh. Therefore, in Bangladesh, more years of experience in maternal healthcare may relate to getting used to a certain way of providing antenatal care which often does not involve mental health care which builds ignorance towards the unfamiliar topic of PMH. Those will less experience in maternal healthcare, may be more open to introducing mental health care into antenatal care.

Furthermore, HCPs who stated they have experience/education in mental health care and those who stated they have access to sufficient information on PMH displayed more stigmatising attitudes compared to those who do not. Some studies suggest that experienced individuals become realistic and sense the danger along with mental illnesses (183,184). However, this finding is contradictory to that found by Ubaka et al (180) who showed that health professionals working directly with patients with mental health illnesses showed more positive attitudes. Increased contact with patients and education around mental health care may reduce desired social distance and perceived danger as you increase familiarity (185). Nevertheless, another study found no significant association between the familiarity of a health professional with mental illness and positive attitudes (171). Therefore, the association between experience and attitudes is not well established and may be influenced by the type of experience HCPs receive in mental health care. Experience of mental health care within maternal health may be inadequate given the lack of mental health integration within perinatal care in Bangladesh (16). Hence, HCPs who state they have experience/education may, in fact, not be receiving adequate levels of positive experience. Furthermore, HCPs who stated they have access to sufficient information on PMH may not be accessing information which promotes a non-stigmatising attitude towards pregnant women. Also, with more access to information, HCPs may possess a better ability to discriminate clinical mental health conditions and, hence, view patients with non-clinical presentations of antenatal depression or different symptoms as ‘weak’.

This study found an association between the HCP role and stigmatising attitudes whereby midwives displayed more stigmatising attitudes compared to both doctors and medical students. There are no known studies investigating midwives' attitudes towards mental health in Bangladesh, hence, this study adds original findings to the knowledge base on midwives' attitudes around PMH. A systematic review investigating the approaches to HCP education and professional development in perinatal depression by Legere et al (186) found that the majority of the literature, mostly from HICs, highlighted little education and training with regards to PMH which negatively impacts levels of awareness on these issues, especially within midwives. This may be reflected in Bangladeshi midwives where less awareness compared to doctors and medical students may result in more stigma around PMH. Nevertheless, in this study midwives attitude score was still less than the mid-point suggesting that, while they may have slightly more stigmatising attitudes as compared to doctors and medical students, they still have relatively non-stigmatising attitudes towards PMH. Midwives have a lot of face-to-face time with pregnant women and are critical in antenatal care, therefore, it is vital that midwives have positive, non-stigmatising attitudes so pregnant women receive the best antenatal care. However, further studies need to be conducted with Bangladeshi midwives to form clear conclusions around their experiences and attitudes in PMH care.

7.5.2 Knowledge

This survey found low levels of knowledge around antenatal depression among HCPs in this sample. This aligns with the learning needs analysis where most HCPs stated they require further education on antenatal depression. Only one topic area, management and treatment of mild to moderate antenatal depression, was deemed to be adequately known by the respondents as it received a correct response rate of over 75%. Counselling or psychosocial interventions were well acknowledged as the treatment of choice. This is reflected in literature from other South Asian countries such as a study in India by Ransing et al (125) where service providers, medical practitioners and nurse providers, stated counselling and psychotherapy as well-accepted treatment modalities for perinatal depression. In contrast, Williams et al (23) found that medical personnel in Bangladesh stated that mental health counselling is not widely appreciated or taught as a learned skill by health providers and that most

Bangladeshi doctors were not familiar with counselling. Differences in findings between our study and the qualitative study may result from different study designs, whereby a survey can often be subject to response bias. HCPs in Bangladesh may not be skilled at counselling but may be more aware of this method than other options in the survey and, therefore, were more inclined to choose this option. This highlights the need for in-depth qualitative studies on HCPs' knowledge and understanding of counselling as an effective form of treatment for mild to moderate PMH conditions.

Common risk factors and screening tools of antenatal depression were less well known to the HCPs in my sample. It is important to note that out of all the knowledge items, questions on risk factors and screening tools are two of the most essential topics for HCPs to be informed about as it impacts their ability to identify pregnant women with risk or symptoms of antenatal depression. Early identification of women with risk and signs of antenatal depression is crucial for prevention and effective management. Formal screening of antenatal depression or protocols to identify pregnant women with depression are not commonly incorporated into ANC services in Bangladesh (23). Screening tools, such as the EPDS, are not well known by HCPs, despite there being a Bangla validated version. Lack of knowledge around the common risk factors also limits screening protocols as HCPs are not aware of which factors to look out for during ANC appointments. This aligns with the learning needs analysis where the majority of HCPs stated they require further training on assessing mental health and risks of mental health issues during pregnancy.

Older HCPs displayed higher levels of knowledge around PMH than younger respondents in this study. This is reflected in another study which suggests that as people grow older, they are exposed to more experience and, therefore, may acquire more knowledge (187). However, our study found that HCPs who stated they have experience/education in mental health care and those with sufficient access to information on PMH displayed lower levels of knowledge. Nevertheless, this may reflect the inadequate education and experience in PMH of HCPs in Bangladesh which does not cover a lot of the vital information regarding PMH. Information that is available may have gaps or not represent up to date research. Therefore, more education/experience in mental health care or access to PMH information may not result in acquisition of the right knowledge around PMH in Bangladesh.

This study found an association between the HCP role and knowledge whereby midwives displayed lower levels of knowledge compared to both doctors and medical students. There are no known studies which investigate midwives' knowledge around PMH in Bangladesh, as such, this study fills a gap in research. This finding is reflected in previous studies conducted in high-income countries where midwives generally have poor or inadequate knowledge around PMH (174,175,186). An Australian study (177) found that many midwives reported that their educational preparation was inadequate for the effective care of those with perinatal depression; there is little or no emphasis on the assessment and management of women with antenatal depression. This reflects the learning needs of the HCPs within our sample. Nevertheless, more large-scale studies in Bangladesh are required to understand midwives' knowledge around PMH and how this compares with other HCPs.

7.5.3 Attitudes and knowledge

In this study, a negative correlation was established between knowledge and attitude; higher levels of knowledge were associated with less stigmatising attitudes. Similar results were found in the literature (172,188,189) suggesting that individuals with higher levels of knowledge may have had the chance to receive adequate education and, as a result, may be more understanding of the topic and carry more positive attitudes. Therefore, educating HCPs in Bangladesh regarding PMH and providing them with the correct information could potentially promote attitudes towards pregnant women with mental health issues. Specific seminars and workshops should be introduced into the curriculum and training of HCPs.

7.5.4 Strengths and limitations

This study included a wide range of HCPs including doctors, medical students and midwives. Midwifery is a relatively new profession in Bangladesh, with the first diploma level midwives becoming licenced in 2016 (190). Therefore, the addition of midwives in this survey fills a large research gap regarding midwives and PMH. The nature of the online survey facilitated the anonymity and confidentiality of the respondents which would reduce self-reporting bias. Self-reporting bias is a key limitation of surveys in general that can present in the form of social desirability

responses or recall bias. Furthermore, there were limited missing data and data that were missing were missing completely at random, therefore, there is little bias from missing data.

However, the findings of this study must be viewed in the context of its methodological limitations. The convenience sampling technique adopted may result in selection bias as the survey will only be available by those who have access to a computer/mobile and the internet. This may miss out HCPs who do not have this access such as CHWs who play a key role in maternal healthcare in rural areas of Bangladesh. Thus, this sampling method, along with the small sample size, reduces the power and generalisability of the results to the whole population of HCPs in Bangladesh. However, the purpose of this survey was more descriptive and explorative and not to form any causal or inferential conclusions about all HCPs or between certain groups.

Although effort was taken to validate the attitude scale developed using face and content validity methods and measuring internal consistency, a full validation was not carried out including measurements like criterion or construct validity. Performing criterion or construct validity would have given more confidence that the scale is assessing stigma around PMH and that it is also related to other similar constructs. Not conducting a full validation may impact the scores that were reported as the scale may have been further altered to better assess the outcome, I was interested in. Therefore, it may be difficult to compare findings with other studies measuring the PMH stigma if we cannot be sure this study does the same. Nevertheless, given the scope of this PhD, it was not feasible to carry out a full validation study of the attitude scale which can be time consuming. This was a descriptive and explorative study and the current scale allowed me to explore some important items around PMH stigma.

Finally, the survey was only available in the English language due to the Online Surveys platform used. This may limit the respondents' understanding of some of the questions as their mother tongue is Bengali. Nevertheless, a strength of this survey was that I designed it using simple English and it was piloted with HCPs in Bangladesh to get feedback on clarity ahead of launching it.

7.5.5 Implications for the wider PhD

The results from this study carry some important implications for the wider PhD project and the next steps. This study has identified some initial findings on HCPs' attitudes, knowledge and learning needs around PMH in Bangladesh, however, there is a need for further research to build on this. Qualitative studies are crucial to gain deeper cultural and social understanding of the perspectives and experiences of HCPs on PMH care. This study has shed some light on the views and levels of knowledge that HCPs hold and some important associations between attitudes and knowledge and experience/education, access to PMH information, HCP role and years worked in maternal healthcare. However, a qualitative study will allow for more in-depth understanding of the mechanisms underlying these associations and begin to provide explanations as to if and why, for example, midwives display higher stigmatising attitudes or why there are low levels of knowledge around certain topics.

This study also highlights several implications for practice and policy such as increasing education, resources and training for HCPs; these will be discussed in detail and brought together along with implications of the other studies within this PhD in chapter 10.

7.6 Chapter summary and contributions

The findings from this study contribute to our understanding of HCPs' attitudes, knowledge and learning needs around PMH in Bangladesh which addresses research objective 4 of this PhD. Data from the online survey revealed relatively non-stigmatising attitudes towards pregnant women with mental health issues but overall low levels of knowledge around PMH and healthcare. Results also showed HCPs with experience/education in mental health and access to PMH information displayed higher reported stigma, but low levels of knowledge. This suggests the need for further investigation, through qualitative means, into the types of experience/education and information HCPs in Bangladesh have access to. Midwives displayed higher levels of reported stigma and poorer knowledge compared to doctors and medical students. This is an important and novel finding as midwives in Bangladesh have not been investigated before with regards to attitudes and knowledge around PMH. There is a need for subsequent, qualitative research to gain deeper cultural and social

understanding of the perspectives and experiences of HCPs, including midwives and CHWs, on PMH care. The findings and implications of this study, along with integration of the other study findings in this PhD, are discussed further in chapter 9 and 10.

Chapter 8. The perceptions and experiences of healthcare providers in Bangladesh surrounding perinatal mental health: A qualitative exploration

8.1 Chapter overview

This chapter covers the final study within this mixed methods project and aims to further address objective 4 of this PhD: To explore PMH knowledge, attitudes and learning needs of HCPs in Bangladesh. The following gaps and implications highlighted by the work reported in chapter 7 are addressed in this study: 1) the need for qualitative methods to provide a richer understanding of how HCPs in Bangladesh view and experience PMH and PMH care, and 2) gaining insights from a wider range of HCPs, including doctors, midwives, medical students and CHWs. A brief background to the study will be followed by a description and explanation of the qualitative approach and the methods used, the findings from the thematic analysis of the semi-structured interviews, and a critical discussion of the findings.

8.2 Introduction

As highlighted in chapter 1, HCPs play an important role in the way mental health is perceived and in the way care is accessed and delivered as they hold positions of authority and trust (23,36,37). Stigma and negative attitudes surrounding PMH can impact women accessing and receiving adequate PMH care (23). Therefore, it is imperative to explore the conceptualisation of PMH within the maternal healthcare sphere in Bangladesh to ensure that women receive holistic care during and after their pregnancy. The findings from the HCPs' survey in chapter 7 found that there is a need for qualitative research to gain deeper understanding of the cultural and social context affecting PMH from the perspective, and, based on the experiences, of HCPs. Midwives are particularly important as a focus as they were found to have higher levels of stigma and lower levels of knowledge surrounding PMH care. Furthermore, given the limitations with online survey recruitment, the survey failed to recruit CHWs who are important actors in maternal healthcare in rural settings. Therefore, there is a need to capture the perceptions and experiences of this important group of HCPs.

Existing literature on perceptions and experiences of HCPs in Bangladesh surrounding PMH is limited, as identified in the systematic review in chapter 3, with

only one study by Williams et al (23) investigating cultural attitudes towards postpartum depression in Dhaka, Bangladesh. The study used in-depth interviews conducted in 2014 with maternal health workers including 12 obstetric-gynaecological physicians, 12 interns and 11 nurses, who were asked about their training in mental health, the protocol for diagnosing and treating depression, challenges for mental health in Bangladesh, experiences with depression cases and general perceptions of depression in new mothers. The HCPs were all recruited from private tertiary health facilities in Dhaka. This further supports the need for a study which includes HCPs recruited from different health settings such as *upazila* (sub-district) primary health complexes and rural communities. In their study, Williams et al (23) found high levels of stigma, lack of awareness and poor understanding of PMH within the maternal healthcare sphere. HCPs also highlighted that they had no protocols on how to educate women on postpartum depression, screen or diagnose women with depression or appropriate counselling methods. This prevents women from receiving holistic care, which entails both physical, mental and social health care, during and after their pregnancy.

As such, there is a need for further qualitative research including a wider range of HCPs to reflect the growing and diverse maternal healthcare sphere in Bangladesh. In particular, midwives and CHWs play an important role in maternal healthcare in rural regions of Bangladesh (191). Midwives have more frequent interactions with and greater accessibility to pregnant women, compared to doctors (175). Since licensed midwives have been recently introduced in Bangladesh, there is a scarcity of research on this group. CHWs are key in providing maternal healthcare in rural regions where there is limited access to formal healthcare. Hence, they may provide better insights into PMH and PMH care in rural settings. Therefore, the aim of this study was to explore the perceptions and experiences of HCPs, including doctors, midwives, medical students and CHWs, in Bangladesh around PMH.

8.3 Methods

This study followed the Consolidated criteria for reporting qualitative studies (COREQ) guidelines (Appendix 18).

8.3.1 Study design & setting

This study used a qualitative descriptive approach (see chapter 5 for full discussion of this methodological approach). Key informant semi-structured interviews were conducted virtually with HCPs in the maternal healthcare sphere in Bangladesh between 28th February and 5th May 2021. This method of data generation was chosen as it enabled direct interaction and communication with participants, as opposed to focus groups, because it would allow deeper and more open discussions around perceptions and experiences (192,193). A semi-structured nature was adopted as it provided participants and the researcher with some guidance on the key topics to discuss, but it also allowed for flexibility to discover or elaborate on information important to the participants that may not have been previously considered as pertinent by the researchers (192).

Interviews were conducted by me over Zoom or telephone in a virtual private room to ensure privacy of the participants. However, full privacy could not be guaranteed from the participants end as they may not have had access to a private space. Nevertheless, participants were asked at the start if they were happy and willing to proceed with the interview. Developing a good relationship and rapport with the participants was essential to the success of the interviews as it would enable them to comfortably discuss their experiences and views. In order to achieve this, Rubin and Rubin's responsive interviewing method was adopted which is based on forming a conversational partnership between the researcher and the participant (194,195). Especially since both the participants and I share the same ethnic background, Bangladeshi, and the majority of the participants were of a similar age to me, I felt it was important to reach a level of empathy and reciprocity during the interviews. This involved disclosing a level of personal information to establish some commonalities and build a connection with the participants. This positions me and the participant as equal contributors to the co-construction of knowledge which reflects a constructivist approach (196).

8.3.2 Participants

HCPs who participated in the online survey (see chapter 7) were given the voluntary option to provide their email address if they consented to being contacted with information regarding the interview phase of the project. Participants were informed that their email address would be used for the sole purpose of recruitment for

interviews. After initial interest via the online survey, snowballing and opportunistic techniques were also employed to identify and recruit participants. Of 139 participants in the online survey, 126 provided their email address consenting to be contacted and, of these, six participants went on to provide informed consent to participate in the interviews. The rest of the sample for this study were recruited via snowballing and through MAA contacts (see **Figure 25** for recruitment flow diagram). Recruitment took place between 8th February and 5th May 2021, with interviews ongoing during this time period as and when participants were recruited. Participants were chosen purposively to ensure the development of a rich understanding of attitudes and experiences of PMH care across a diverse range of HCPs. This approach meant sampling participants according to a range of different characteristics appropriate to the research question.

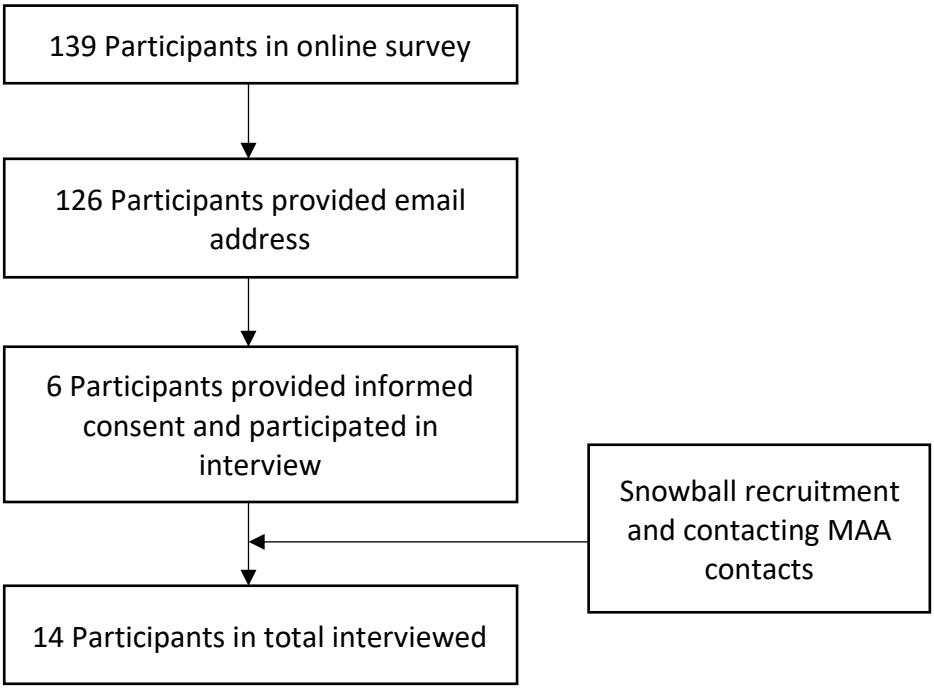


Figure 25 Flow diagram of HCP recruitment for interviews

HCPs were sampled according to: 1) type of HCP (doctors, midwives, medical students and CHWs), and 2) location of healthcare (rural vs urban). Type of HCP was a key factor for sampling as it was important that the findings reflect the growing and diverse maternal healthcare sphere in Bangladesh. The characteristics of different HCPs vary in terms of education, experience and working environment and such a diverse range has not been explored in the existing literature. It was felt that interviewing HCPs who work in rural areas, and those in urban areas, would be vital to understanding how perspectives and experiences differ with regards to locality, as rural areas have different social, cultural and economic contexts compared to urban areas. The locality of HCPs was determined through the online survey for participants that were recruited via the survey; for participants recruited via snowballing, the locality was asked prior to recruitment. Although some HCPs may not be based exclusively in rural or urban areas, it was ensured that they mostly practiced in a given region and carried significant experience.

Participants were also recruited based on their capability to take part in an online interview on Zoom or over the telephone. This was dependent on their access to technology, proficiency in using online video platforms or network/internet coverage. Participants were recruited sequentially until data saturation occurred. This was the point at which no new topics appeared to be emerging from the interviews. Data saturation is a common approach to determining when the generation of qualitative data is sufficient to provide for a rigorous analysis with the aim of attaining depth of understanding rather than a statistically representative sample associated with positivist quantitative methodologies (197).

8.3.3 Topic guide

The topic guide (Appendix 19) was a flexible tool with key topics and prompts to guide the interview. It enabled me to capture the HCPs' responses on important topics identified from my qualitative systematic review (see chapter 3). These topics included exploring HCPs perceived social determinants of PMH, their perceptions of motherhood and PMH, and their views on key barriers and facilitators for PMH care provision. The topic guide provided a flexible space for participants to expand on or

introduce topics important to them which is a key element of semi-structured interviews and reflects the responsive interviewing approach used. The topic guide also acted as a cue to probe participants further, for example, asking them to provide examples or elaborate, to develop a deeper understanding of the participants' beliefs and experiences. To assess flow and clarity of the topics and prompts, the topic guide was piloted on three doctors from the MAA team and four peers with a medical healthcare/research background. This pilot group had a good mix of individuals who understood the Bangladeshi culture, were experienced in providing maternal healthcare, had knowledge of PMH care and were skilled in qualitative research. Hence, collectively, the group provided feedback on the flow and clarity of the topic guide. Following the pilot, the topic guide was amended and refined, for example, it was clarified that we were interested in the views and experiences of the HCPs themselves when discussing how open pregnant women are in talking about their mental health. There were also some changes made to the way the questions were worded to ensure they were not leading the participants to a particular answer or thought. For example, instead of asking participants what social determinants they feel are associated with PMH, participants were asked if they feel social factors play a role in PMH and then they were probed for examples. The topic guide was originally developed in the English language for ease and later translated into Bengali for any participants who wanted the interview to be conducted in Bengali. I carried out the first translation into Bengali which was then sent to a member from the MAA team, who is also a native Bengali speaker, who translated the topic guide back into English. I reviewed the back translation to ensure that the translation did not lose any important meanings. In addition to sociodemographic questions, the topic guide included five topics: 1) knowledge base, attitudes and perceptions of mental health overall; 2) experiences and views on PMH; 3) social determinants of PMH; 4) management for pregnant women who experience mental health issues and 5) suggestions for change to improve PMH care in Bangladesh.

8.3.4 Data collection

HCPs who showed interest in the interview phase of the study, by providing their email address in the online survey, were emailed the participant information sheet and consent form (Appendix 20). Details about the study aims, procedure, potential risks, benefits and information around anonymity and confidentiality were provided to the participants. These were written in both English and Bengali to accommodate all

participants. The participants were given the opportunity to ask questions and make an informed decision. An appointment was then scheduled at a convenient time for the participants and a meeting link was emailed to the participants. It was important to consider the six/five-hour time difference between the UK and Bangladesh when organising the interviews. The majority of the interviews were conducted in the afternoon in the UK which equated to the evening in Bangladesh which seemed to be the most convenient time for participants. Most of the interviews were conducted over Zoom in Bengali (as I am a fluent Bengali speaker), with one interview conducted over the telephone, and two interviews carried out in English. Interviews lasted on average 30 minutes. Zoom audio recordings of the interviews were saved on a password protected computer, after which they were translated and transcribed into English.

Translation of the interviews conducted in Bengali were carried out by me in parallel to transcribing them in English. Translators should have a deep familiarity with the interview topic, context of the interviews and culture of the participants (198), making me the most suitable translator. Although, it is ideal to have a team of translators and cycles of translation and back translation, due to time and resource constraints this was not possible. It was decided that interview recordings were best kept within the core research team (i.e. myself, Professor Judith Rankin and Professor Simon Forrest), for confidentiality and anonymity of the participants, therefore, being the only Bengali speaker, I carried out the translations myself. Nevertheless, to ensure semantic and conceptual equivalence of the translations, I reviewed all the English transcriptions alongside the original recording.

The approach taken for transcription was based on the research question and what information I was interested in. Firstly, the required completeness of the transcript was determined (198). For this content-focused study, I opted for ‘naturalised’ transcription whereby certain verbal elements were omitted or not highlighted to a great extent such as long pauses, “ers” and “erms”, repetition of words or when an individual mis-speaks or if the participant completely goes off topic due to interruptions or technical issues. These elements were not analysed for the purpose of this research question. This form of transcription is sometimes termed ‘intelligent verbatim’ as you adapt the oral to written norms making the transcription easier to read and analyse (198,199). Secondly, the level of detail required was determined (198). It was decided that only audio data would be used as this aligned with the ethics approval obtained

for this research and visual data was deemed unnecessary for this study. Exploration of visual cues are more appropriate for studies exploring topics such as social relationships or sensitive topics that participants might find difficult to convey purely verbally. In such studies, it may be useful to identify certain non-verbal cues such as hand actions, facial expressions and body language. However, for this current study, such non-verbal features would not add to answering the research question which is focused on exploring views and experiences.

8.3.5 Data analysis

Interview data were analysed using the six phases of thematic analysis as outlined by Braun and Clarke (153) consisting of: 1) familiarising yourself with the data; 2) generating initial codes; 3) searching for themes; 4) reviewing themes; 5) defining and naming themes, and 6) producing the report. Thematic analysis is widely used in qualitative research for identifying, analysing, organising, describing and reporting themes found within a dataset (153). When carried out rigorously, thematic analysis can produce trustworthy and insightful results (153).

Transcribing the interviews manually aided in the first stage of thematic analysis as it developed my familiarity with the interviews and allowed me to become fully immersed in the data throughout the data analysis phase. Transcripts were re-read in an active manner whereby initial thoughts on patterns and meanings were shaped. Subsequently, transcripts were coded manually by systematically highlighting phrases/sentences deemed to be interesting features regarding the research objective and assigning them a keyword or phrase as an initial code. Notes were written on the text to indicate potential patterns and my initial thoughts. The codes for each transcript were collated in Microsoft Excel along with the respective quotes and initial notes to develop a comprehensive database of all the codes. Particular attention was given to the quotes extracted for each code to ensure that the context was not lost which is a common criticism of coding. Codes were reviewed for each transcript and across the dataset and sorted into potential overarching themes and all the relevant data extracts were collated. Tables and mind maps were used to help sort the codes into themes and sub-themes. Once a set of candidate themes and sub-themes were developed, these were reviewed for refinement. Some themes which shared similar meanings and ideas were combined to form larger overarching themes and some themes were dismissed as it became evident there was not enough data to support them.

Connections and relationships were analysed between the themes, yet ensuring the themes were distinct and accurately address the research objective. A thematic map was devised after which the themes were defined and named by identifying what the theme is about and the 'story' that is tells in relation to the overall research question. The scope and content of each theme was described, and an appropriate name was assigned. It should be noted that, although I have discussed these steps of thematic analysis in a linear fashion to aid the understanding of each step, the actual process was iterative whereby codes and themes were revisited, revised and refined throughout the entire analysis.

8.4 Findings

A total of 14 HCPs (**Table 27**) participated in this interview-based study (six midwives, three medical students, two doctors, two MAA CHWs and one maternal field mentor; of whom 12 were female and two male). All participants took part in a one-to-one, semi-structured key informant interview with 13 being conducted over Zoom and one as a telephone interview. Participants were aged between 21 and 28 years of age. Six participants had experience of working in rural settings and providing community healthcare within NGO projects, four midwives worked in an '*Upazila*' health complex providing routine primary level perinatal care to women in sub-units of districts, three worked in government medical college hospitals and one midwife worked in a Rohingya camp (**Table 27**). Most participants had between one to six years of experience in providing maternal healthcare. Twelve participants were interviewed in Bengali and two were interviewed in English.

Table 27 Summary of characteristics of Healthcare Providers interviewed

Participant	Healthcare provider role	Age	Type of maternal healthcare work	Location
Participant 1	Midwife	27	Upazilla Health Complex	Urban
Participant 2	Midwife	28	Upazilla Health Complex	Urban
Participant 3	Midwife	26	Upazilla Health Complex	Urban/rural
Participant 4	Maternal field mentor	24	Upazilla Health Complex	Urban
Participant 5	Midwife	28	Rohingya Camp	Rural
Participant 6	Midwife	24	Government Medical College Hospital	Urban
Participant 7	Midwife	26	NGO	Rural
Participant 8	Medical student	26	Government Medical College Hospital	Urban
Participant 9	Medical student	21	Government Medical College Hospital	Urban
Participant 10	Doctor	25	NGO	Rural
Participant 11	Medical student	23	NGO	Rural
Participant 12	Doctor	26	NGO	Rural
Participant 13	MAA Community Health Worker	24	NGO	Rural
Participant 14	MAA Community Health Worker	24	NGO	Rural

MAA = Maternal Aid Association
NGO = Non-governmental organisation

Four key themes were synthesised from the data and mapped onto a thematic map with the sub-themes (**Figure 26**): 1) maternal healthcare sphere in Bangladesh, 2) conceptualisation of pregnancy and PMH, 3) perceived social determinants of PMH issues, and 4) PMH care provision. The themes and sub-themes will be described and explored in this section along with key quotes from the data to support the findings. Additional quotes will be provided in a table for each theme for further evidence from the data.

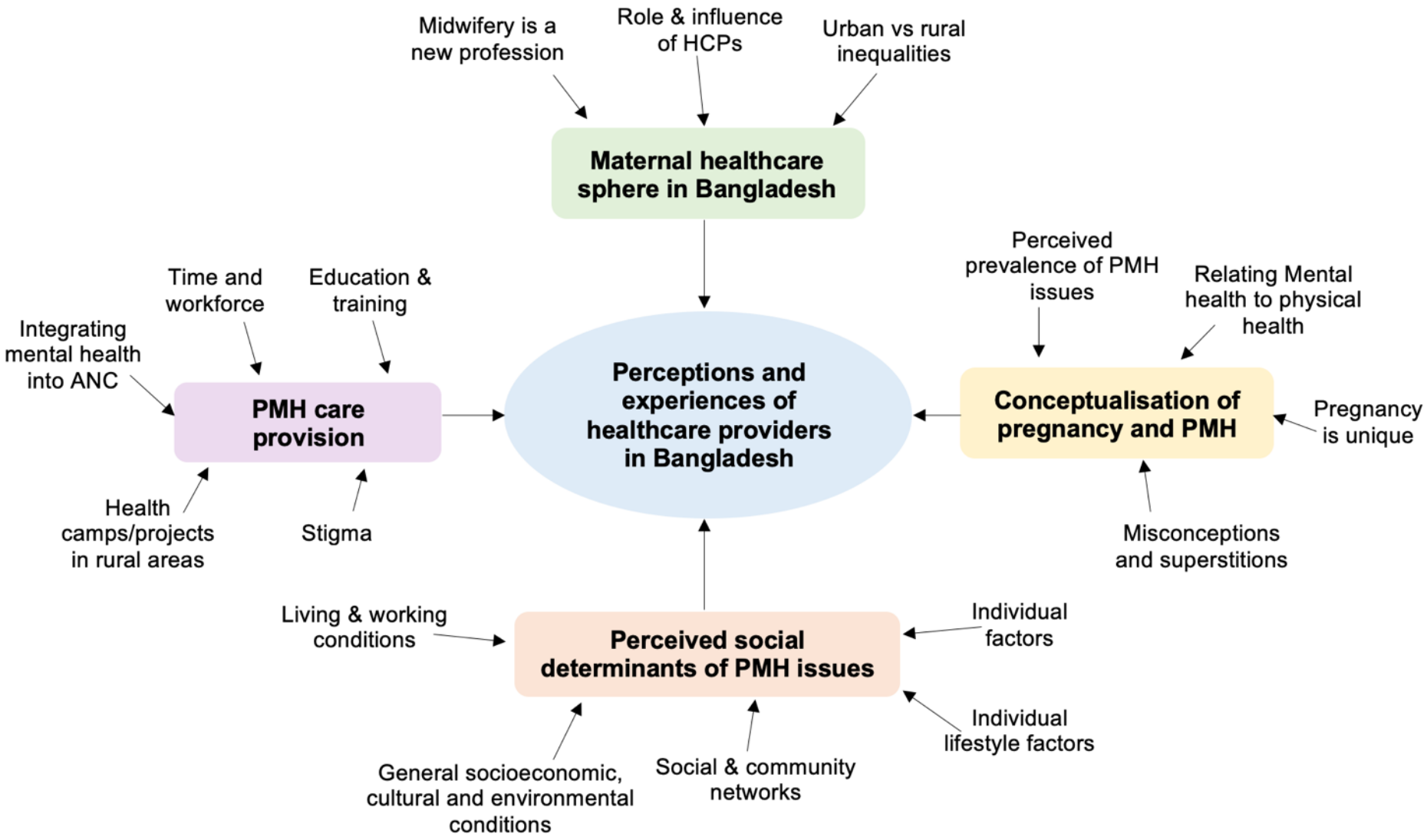


Figure 26 Thematic map of themes and sub-themes

8.4.1 Theme 1: Maternal healthcare sphere in Bangladesh

This theme highlights the views of the HCPs regarding the key actors within the maternal healthcare sphere in Bangladesh and their role in providing PMH care, along with exploring the different maternal healthcare spheres in urban versus rural regions of Bangladesh.

Midwifery is a new profession: The range of HCPs in the maternal healthcare sphere in Bangladesh is diversifying with the establishment of licensed diploma level midwives in 2016, making midwifery a relatively new profession in Bangladesh. This was reflected in the interviews with midwives where many expressed their unfamiliarity with the course and profession (**Table 28**: Q1). As participant 1 (a midwife with five years of experience) suggests, there is typically a strong sense of the novelty of midwifery as a licensed profession:

> *"When we entered midwifery we were the first batch so we actually didn't know what to expect (.) when we passed our first year then we started to understand what the situation was going to be like (.) so at one point when we came to know that it was working with mothers and children" (P1, Midwife)*

Role and influence of HCPs: HCPs in this study, particularly midwives, expressed a common view regarding their role of providing some form of counselling to women through giving support, advice and motivation on physical, social and mental health issues (**Table 28**: Q2, Q3). One midwife with six years of experience recognised her role in providing as much support as she can, despite not being a professional psychologist:

> *"I'm not a professional psychologist but I try and give as much support as I can from my side" (P5, Midwife)*

However, one midwife shared that their counselling is limited when it comes to certain social issues such as gender-based violence for which they can provide support and advice, but beyond that they cannot provide a long-term solution for the root issue:

> *"For example, a woman comes with a lot of problems with her husband she doesn't get access to a lot of things or help but to diagnose her in anyway like*

they are getting treatment and advice and then after that we can't really do anything as such actually" (P2, Midwife)

This may suggest that a range of actors are required in the maternal healthcare sphere in Bangladesh to be able to tackle the different social aspects and to transform care beyond support and advice to sustainable, long-term changes. Nevertheless, the majority of the HCPs in this study acknowledged their authority and influence on pregnant women suggesting that women listen to their advice and tend to share their problems when encouraged by HCPs (**Table 28**: Q4, Q5). This is reflected by participant 5, a midwife, who highlighted her importance within the maternal healthcare sphere as women view midwives as doctors and are often closest to the mothers:

"especially since women view us as doctors so when we say something to them it effects them a lot so I try…Midwives are closest to the mothers" (P5, Midwife)

From this extract it is not clear how the participant defined the term 'closest' which could be in terms of access to pregnant women and/or empathy and understanding. Either way, it suggests that midwives recognise their role within the maternal healthcare sphere in relation to other actors and have an advantage in being able to provide effective support and advice. Nevertheless, this quote also suggests a hierarchy amongst HCPs, where doctors come at the top and may have the most authority and influence.

Urban vs rural inequalities: Many interviewees addressed the different maternal healthcare spheres in urban versus rural regions of Bangladesh. It was widely expressed that there is limited perinatal healthcare, and therefore PMH care, and access to health professionals in rural areas (**Table 28**: Q6). For example, participant 13, a MAA CHW, stated that in rural areas the majority of home deliveries are with traditional birth attendants rather than health professionals. This may cause concern for pregnant women in rural regions as they may worry about the health of their child if any complications arise during delivery:

"it works differently there (rural regions) like you've got dais and traditional birthing attendants rather than healthcare professionals who are medically

trained and so yep I would definitely say the access to the health camps erm sorry to antenatal checks is fairly limited" (P13, MAA CHW)

A midwife working in a Government Medical College Hospital stated that she is unclear about what the Government is going to do to improve the access to midwives in rural areas of Bangladesh. Limited access to midwives in rural regions directly impacts the PMH care that can be available for pregnant women. This suggests that there is limited information around Government interventions in rural areas which may reflect the lack of Government involvement in these regions:

"Actually a lot of mothers we have live in rural villages but we can't do much for them because there are no midwives and I don't know what the government is going to do" (P6, Midwife)

Those interviewees working in rural areas highlighted the importance of NGOs in providing PMH care, training and increasing awareness, as well as providing free and specific women's health care in rural areas of Bangladesh (**Table 28**: Q7). As highlighted by participant 14, pregnant women in rural regions were incredibly grateful of the free health camps delivered by MAA which suggests the huge impact of NGO involvement in maternal healthcare in Bangladesh, particularly for outreach into rural regions. Such outreach into rural regions can be utilised to implement PMH interventions and programmes as well.

"I think probably the majority of women that were there had never seen like had probably not been seen in a hospital environment like a healthcare professional in a hospital environment…I think the overriding emotion was just gratitude like they were so so grateful that we were there, that we were providing the service for free" (P14, MAA CHW)

Table 28 Additional quotes supporting the theme Maternal healthcare sphere in Bangladesh

Subtheme	Quotes	
Midwifery is a new profession	Q1	“When I first came into this profession I didn’t realise that this was a midwifery course and I didn’t have much knowledge about this course” (P7, Midwife)
Role and influence of HCPs	Q2	“Now patients do come to the ANC they see the benefits of coming to ANC we do tests and we tell them what to eat that you have to eat nutritious foods otherwise the baby will have such and such problems” (P4, Midwife)
	Q3	“We give them motivation about what is normal and what isn’t normal we tell them these things to make them understand” (P2, Midwife)
	Q4	“we tell them to bring their husbands and when we say it they actually listen” (P3, Midwife)
	Q5	“within my friend circle and other people I know they tell me things because I am in the medical field they like to tell me their problems” (P9, Medical Intern)
Urban vs rural inequalities	Q6	“those who are from the very rural areas will do home delivery and then they get PPH (postpartum haemorrhage) and eclampsia and then they have to come to the hospital when it’s too late (.)" "in higher societies they go to ANC and there their midwives and doctors will tell them that you need to sleep and eat such things and do such and such activities” (P4, Maternal field mentor)
	Q7	“IFMSA is an international organisation which works with medical students so when I was there there was a mental health training course a workshop around brain health and burnout which was organised by a couple of students from Lebanon during this pandemic so a few days ago I did this course” (P10, Doctor)

8.4.2 Theme 2: Conceptualisation of pregnancy and PMH

This theme highlights the way that pregnancy and, in turn, PMH is viewed and conceptualised by the HCPs interviewed.

Pregnancy is unique: HCPs expressed the idea that pregnancy is a unique time that no-one else, except the woman going through the pregnancy, can understand, and acknowledged that women are more vulnerable to emotional and psychological changes during this time (**Table 29**: Q1).

> *"after a woman gets pregnant the feelings she goes through actually no one else can feel" (P3, Midwife)*

From the interviewees' experience, however, they witnessed members from older generations in families such as mothers and mothers-in-law compare their pregnancies to their daughters and daughters-in-law and expect them to have the same pregnancy experience and not complain (**Table 29**: Q2). As highlighted in this quote from participant 8, there seem to be certain expectations surrounding pregnancy and motherhood which are passed inter-generationally. This reflects the unique nature of pregnancy compared to other phases in a woman's life which have these culturally embedded expectations.

> *"a lot of the time mothers who are first time pregnant don't have any experience so they are dependent on their mothers-in-law and mothers who are from an older generation, so we see that they say that there is no need to take the pregnant woman to the hospital, 'I also had a child at home so you will also have your child at home'" (P8, Medical student)*

Nevertheless, the importance of pregnant women in society was highlighted by a medical student and that more attention should be given to them within the healthcare sphere in Bangladesh.

> *"A pregnant woman is a major factor in society and I thought that really we should do more work on this" (P8, Medical student)*

Misconceptions and superstitions: HCPs in this study expressed that many women, and their families', carry misconceptions and superstitions around pregnancy such as not feeding the pregnant woman too much as the baby might become too big and will require a caesarean, not eating duck as the baby's voice will be like a duck, eating certain fish will make the baby's personality moody and angry and giving honey instead of milk (**Table 29**: Q3).

> *"actually I read in one of the subjects we had in villages there is just so much superstition and preconception that they think that fish and meat or erm if she eats duck meat then the baby's voice will be like a ducks so they don't let the mother eat ducks meat at all and then there is certain fish as well like if she eats this fish then the baby will be very angry and moody personality so there is a lot of superstition like this" (P4, Maternal field mentor)*

The interviewees acknowledged these beliefs as 'superstitions' which may suggest that they themselves do not carry these beliefs regarding pregnancy and aim to educate pregnant women, and their families, who come to their care. These misconceptions and superstitions directly impact the care that pregnant women receive from their families, and they may feel lack of support impact her mental health. Also, not being fed enough can incite worry in pregnant women about the development of her child. Therefore, understanding these misconceptions and superstitions is important to understanding how families and communities perceive pregnancy and pregnant women which plays a role in their care and status impacting their mental health.

Perceived prevalence of PMH issues: PMH issues were perceived to be a widespread issue by the HCPs in this study who stated that it was common to find depression and anxiety among pregnant women (**Table 29**: Q4).

> *"yes this happens a lot almost maximum mothers have this, they suffer from depression" (P5, Midwife)*

It should be noted, however, that it is not clear whether these pregnant women are clinically diagnosed with depression, or whether interviewees just perceive them to have symptoms of depression. It was also suggested by interviewees that PMH issues were believed to be more prevalent in socioeconomically deprived areas, however, even women in wealthier regions were perceived to be susceptible to such issues and the prevalence is rising.

> *"I won't just say remote areas but even in affluent areas we see depression rising" (P10, Doctor)*

Relating mental health to physical Health: When discussing PMH, HCPs in the interviews often related it to physical health seemingly in order to make better sense of it. The importance of mental health was expressed by highlighting the physical, somatic symptoms such as headache and fatigue, and physical health consequences of poor mental health, such as women not eating enough, which have adverse impacts on physical health (**Table 29**: Q5).

> *"after they have become pregnant their mental health such as they get morning sickness, headache, fatigue, sleepiness a lot of such complications they come with (.) you can come to see that some suffer from a lot of depression (.) morning sickness is a big problem in them I think because vomiting and nausea happen in pregnant women a lot" (P2, Midwife)*

This reflects the well-established link between psyche and soma which describes how our psyche (i.e., mind, emotions and spirituality) interact with the world around us to induce changes in our soma (i.e., body). However, it also raises the question of how mental health is defined by the interviewees, seemingly by using physical health, and why this is defined in this way.

It can be suggested from the quote below that HCPs put more emphasis on discussing physical health to make pregnant women take mental health more seriously.

“I think if you emphasise enough that not taking care of their mental health has a detrimental effect on the rest of their health that fear will probably make them take it more seriously” (P14, MAA CHW)

This quote highlights the cultural view of mental health and physical health being connected, but at the same time conceiving them implicitly as separate. This connection may be used as a clinical tactic to engage pregnant women in discussing mental health which reflects the perceived status of mental health amongst women. Further, this quote also raises a discussion around health promotion whereby an approach that utilises amplifying concerns is used to get a patient’s attention. This aligns with the health belief model in which perceived severity (i.e., a person’s feelings on the seriousness of contracting an illness or leaving it untreated) is a key construct used for health promotion.

Table 29 Conceptualisation of pregnancy and PMH subthemes and additional quotes

Subtheme	Quotes	
Pregnancy unique	Q1	“not be angry with her because of her changing mentality because there is now a difference between her previous life and her life now, now she has a child in her belly, she doesn’t feel well physically so she won’t feel mentally either” (P5, Midwife)
	Q2	“we also had children we didn’t do this! (.) when she vomits (.) this is natural there is nothing to be scared about why are you doing this didn’t we have children!?” (P1, Midwife)
Misconceptions and superstitions	Q3	“the mother-in-law would say things about how she should eat like don’t eat like this the baby will become too big then you will have to have a caesarean the baby won’t be delivered normally a lot of things like this” (P7, Midwife)
Perceived prevalence	Q4	“we will probably find this (mental health issues) in maximum women…we feel that almost everyone has this issue” (P3, Midwife)
Relating mental health to physical health	Q5	“physical health relies a lot on mental health…if I am upset I won't eat properly (.) the weight I need to gain won't happen if I'm always upset or feeling lonely” (P3, Midwife)

8.4.3 Theme 3: Perceived social determinants of PMH issues

This theme highlights the perceptions of HCPs regarding the most important social determinants of PMH issues in Bangladesh. It is important to consider HCPs' understanding of social determinants as it will reflect on how they identify pregnant women at risk of PMH issues and it will inform what management methods they use. It will also inform what further education on risk factors HCPs need. The factors are organised and discussed here in respect to the Dahlgren and Whitehead 'rainbow model' of social determinants of health (200) (see chapter 4 for the theoretical and conceptual framework).

Individual factors: young age of marriage and maternal age was highlighted as a risk factor for PMH issues by a few of the HCPs. One MAA CHW in the quote below stated that the average age of pregnant women at their health camp was 19 years and there were some women as young as 15 years of age. These young women are perceived to very often to be uneducated about sexual and reproductive health and conceive children soon after their marriage resulting in a lack of preparation and also apprehension about what is to come (**Table 30**: Q1).

> *"The average age of a women when she is pregnant is also much younger than say in the UK for example and so that lack of preparation can lead to you know do they or do they not want to continue with the pregnancy and that can feed into and throughout their gestation and after as well…we were taking the age down and some were like 15 and one person said 14 (.) the average age was around 19 or 20" (P13, MAA CHW)*

There is also indication of societal and cultural differences between Bangladesh and Western countries, like the UK, whereby young age at marriage may be more common and seemingly more accepted in Bangladesh. This may be indicative of important health inequalities for women between countries. It is important to consider age of pregnant women when tailoring interventions for PMH as younger women may have differing needs and understanding of pregnancy.

Individual lifestyle factors: Interviewees highlighted that family planning, and the use of contraception is limited in Bangladesh, although change is occurring over time (**Table 30**: Q2). This can lead to unplanned pregnancies which was highlighted

as a key risk factor for PMH issues by majority of the HCPs in this study (**Table 30**: Q3). As expressed by a doctor in the quote below, unplanned or unwanted pregnancies can lead to women not being prepared for the pregnancy which can lead to antenatal depression.

> *"what we see is that most of them are not ready for the pregnancy like its unwanted pregnancy so she is not prepared for that pregnancy and because of that there is some depression" (P10, Doctor)*

Lack of preparation as mentioned by the doctor could involve not knowing what to expect during pregnancy, not understanding how to look after themselves and the financial considerations of a pregnancy and a child. All these factors may contribute to pregnant women suffering from anxiety or depression during their pregnancy.

HPCs perceived women to be unaware of the types of contraception available and how to use them properly (**Table 30**: Q3, Q4). This was highlighted by a MAA CHW in the quote below who states that women are not as aware of Intrauterine Devices (IUDs), coils or the pill due to the lack of access and education.

> *"what they weren't so aware of are the different types so you know things like IUDs or the coil all of that or the pill because it is not so widely accessible in Bangladesh and also the education just isn't there about that" (P13, MAA CHW)*

HCPs also perceived women to be hesitant to use certain forms of contraception due to restrictions from their husbands. For example, as stated by a maternal field mentor in the extract below, many women do not use an IUD, although promoted by midwives as an effective method of contraception, as it might be uncomfortable for their husband.

> *"if she does use IUD then maybe the husband will be uncomfortable and if he finds out then there will be problem with that so injectables no one can tell" (P4, Maternal field mentor)*

This raises a question around pregnancy intention discordance between the husband and the wife which may be linked to lack of contraception use and, hence,

unplanned pregnancies. Pregnancy intention discordance has also been associated with perinatal depressive symptoms in Bangladesh, particularly when the husband did want the pregnancy but the woman did not (73).

Social and community networks: Interviewees expressed that, after marriage, the husband and in-laws become a woman's main family playing a huge role in their wellbeing (**Table 30**: Q5). As indicated by the extract below, this is particularly prominent in communities from a lower socioeconomic class.

> *"Especially those in the lower class after they get married their husband and in-laws become the main people so they control a lot" (P4, Maternal field mentor)*

The above extract also suggests that there are power relations at play as the interviewee highlights the idea of 'control' on the pregnant woman. This is reflected by many HCPs who suggested that the mother-in-law and husband carry the most importance and authority. This could reflect the family hierarchy in Bangladeshi communities placing the daughter-in-law lower. For example, it was reported by interviewees that mothers-in-law often restrict food for the pregnant woman, makes decisions about their healthcare and force women to carry out household chores even during pregnancy as indicated in the extract below. This may be in an attempt to demonstrate their power and authority, and women are not in a position to question their mother-in-law (**Table 30**: Q6, Q7).

> *"so one day someone was saying that, 'sister, they make me do a lot of work and I can't say anything to the mother-in-law' so I said that when you come to the next visit to bring the mother-in-law and she actually started crying" (P7, Midwife)*

HPCs in this study also suggested that husbands carry a lot of power, particularly in rural areas, as women are economically and socially dependent on them (**Table 30**: Q8). Interviewees stated that many husbands choose their partners' contraception methods, can restrict access to antenatal healthcare and stand in a position to give a divorce for reasons such as gender of the baby (**Table 30**: Q9, Q7). As highlighted by a doctor in the extract below, counselling for a woman's permanent

contraception goes through the husband and requires his consent which often does not work if he does not have a son.

> *"with the husbands what I face in the hospital is that if they do not have a male child then in that case our counselling for permanent contraception does not always work (.) they do not want their wife to have permanent contraception" (P10, Doctor)*

Poor relationship with in-laws and husbands can create an abusive and toxic environment for pregnant women where many suffer from violence and insults from their in-laws as highlighted in the extract below. This medical student suggests that the in-laws are a direct cause of perinatal depression for the pregnant woman through mental and physical abuse.

> *"mothers-in-law or sisters-in-law who may be keeping her mentally depressed and a lot of the times that might turn into physical torture as well which is even more painful" (P8, Medical student)*

Along with abuse from in-laws, domestic violence from the husband has also been stated to be a key issue in Bangladeshi communities, particularly in rural areas as stated by the maternal field mentor in the extract below.

> *"so the mothers who would come from the villages mainly there would be a lot of domestic violence" (P4, Maternal field mentor)*

Within such abusive and toxic family environments, HCPs in this study felt that pregnant women faced a lack of family support which was also highlighted as a key reason for pregnant women suffering from depression and anxiety (**Table 30**: Q10). As highlighted in the quote below, midwives have experience caring for pregnant women who share that they feel a lack of care from their family.

> *"I think most feel lack of care they think they are ill and they don't get the support from their family that 'I am ill why aren't they looking after me'" (P3, Midwife)*

Living & working conditions: HCPs in this study stated that many of the pregnant women they care for, particularly in rural areas, are uneducated (**Table 30**: Q11). One interviewee, as highlighted in the quote below, stated that this lack of education often makes pregnant women unaware of how to look after themselves during pregnancy, for example how long to sleep. This can have adverse consequences on women's mental health as they do not even get the support from others.

> *"for those who are extremely poor they don't get much support because they have to work they don't get to rest or have people who can support them or because they are uneducated they don't even know that like they have to sleep for at least 2 hours in the afternoon, so no one really gives this support" (P4, Maternal field mentor)*

Lack of education is also evident from women's limited knowledge of contraception resulting in unplanned pregnancies (**Table 30**: Q3). Majority of women are also unemployed and are, therefore, economically dependent on their husbands (**Table 30**: Q8). From the suggested family hierarchy and power relations within the family, unemployment can reduce women's decision-making power in many aspects including access to antenatal care, family planning and carrying out household chores which were highlighted above.

Access to antenatal healthcare and health professionals is limited, especially in rural regions. One MAA CHW felt that antenatal care is a very niche area in Bangladesh, hence, why it is not readily available in many local areas. This raises discussions around the idea that antenatal care is a women centred service and given many of the gender inequalities in Bangladesh, there may be little attention to providing universal antenatal care.

> *"antenatal checks are something that is quite specific its quite a niche form of healthcare to receive and so I can understand why in a country like Bangladesh you wouldn't have that readily available in your local area" (P13, MAA CHW)*

Given that antenatal care is not readily available, the services that are available are not usually accessible for the majority of women. This was highlighted by the MAA

CHW who shared stories about pregnant women facing major travel issues, which require time and money, to reach antenatal care services which is a huge barrier to accessing antenatal care.

> *"one of the biggest barriers to healthcare erm is travel (.) simply getting like a rickshaw or I think I was speaking to … and he said you know one of the mothers came and she took a boat and then a rickshaw and then walked for a while with her son and came to the camp at the end of the day, so she was like travelling for eight hours so I think the biggest barrier to you know getting access to this healthcare is travel" (P13, MAA CHW)*

Nevertheless, from the extract above, it can be inferred that pregnant women want antenatal care as many are willing to travel such far distances to receive the free healthcare. It is imperative that all pregnant women can receive the care they need and want easily without such barriers. Lack of antenatal care places pregnant women under stress and fear regarding childbirth, the health of their child and child development (**Table 30**: Q12). The quote below from a medical student who worked in community health camps expressed the worry that pregnant women have regarding their child's development due to lack of healthcare throughout their pregnancy.

> *"we also see in our health camps that they cross the first trimester and this patient has not received folic acid and are worried about what their child development will be like so there are a lot of reasons for depression like this" (P11, Medical student)*

One midwife expressed that if every pregnant woman had access to a midwife "they would be mentally and physically satisfied." (**Table 30**: Q13).

General socioeconomic, cultural and environmental conditions: Poverty was highlighted by the majority of the participants as a major reason for perinatal women suffering from depression and anxiety (**Table 30**: Q14, Q15). As stated in the quote below, financial issues can result in limited access to perinatal healthcare as families cannot afford two or three antenatal check-ups which can be a reason for PMH issues. It can be suggested that pregnant women are either not able to afford the

service itself or they cannot afford to travel to access the service. Either way, poverty creates many barriers.

> *"a lot of families are not capable financially to have two or three antenatal visits at the doctors so this is another reason (for PMH issues)" (P11, Medical student)*

Therefore, women worry about the development of their unborn child and subsequent delivery arrangements if any medical complications arise requiring medical attention and, hence, money. One example given by a midwife is that medical complications requiring an emergency caesarean may prevent pregnant women to give birth at home which is the chosen cheaper delivery method for many women.

> *"they also face financial problems because if it's a normal delivery then what God has willed that's very good, a lot of mothers have malpresentations or they go into premature labour then they need to have an emergency caesarean which requires money" (P6, Midwife)*

Financial difficulties also mean that husbands must work longer hours or more days to run their family resulting in low participation during a woman's pregnancy. HCPs in this study felt that this plays a role in making women feel lonely and isolated during their pregnancy (**Table 30**: Q16). This links back to the idea of lack of family support which may sometimes be due to such practical reasons such as husband's having to work rather than hierarchy or power reasons.

> *"the biggest thing is that what they expect from their husband they don't get and that's why they suffer they feel lonely, for example a women comes by herself and she will say 'look I have come by myself my husband didn't come with me what would happen if he didn't go into work for one day if he came with me I would feel very good'" (P3, Midwife)*

Furthermore, women undergo stress during their pregnancy, sometimes to the point of suicidal ideations, because they "cannot afford to feed another mouth" and look after another child (**Table 30**: Q17).

Social construction of gender and associated beliefs about gender roles have been suggested from the interviews as an important factor in women experiencing PMH issues in Bangladesh. Women are expected to carry out household chores and an inability to do so, like during pregnancy, is taken to imply that she is not fulfilling her responsibilities. HPCs perceive that this places pregnant women under immense physical and mental stress followed by lack of morale when she is unable to participate in household responsibilities as highlighted in the extract below (**Table 30**: Q18).

> *"in a family the mothers play a big role so if their working capabilities reduce because they need more resting or they cannot participate in chores as much as they used to be able to so this brings them very morally down which is pretty much very common" (P11, Medical student)*

One midwife expressed that in South Asian communities and culture the responsibility of contraception is placed entirely on the woman. Hence, if a baby is unintentionally conceived and the husband does not want the child, the blame falls on the woman.

> *"it's women's responsibility there's nothing for men here so when a baby is automatically conceived whether that's a mistake or however else then the blame goes on the woman" (P1, Midwife)*

Notably, as discussed previously, decisions around contraception are mostly from the husband given the suggested power relations and hierarchy. Nevertheless, the responsibility to use it seems to be with women which may place women in very difficult situations when husband's do not give consent for contraception.

The majority of HCPs stated that many pregnant women tell them that they want to have a son. This male gender preference has been identified as a key social determinant of PMH issues as HCPs in this study felt that pregnant women are often concerned about the gender of their unborn child which is a big reason for depression (**Table 30**: Q19).

> *"in a lot of cases especially in rural areas a lot of people want a male child so there is a concern around whether I am going to have a male child or not*

especially if someone has daughters then they are worried about having another daughter and what will happen, so this is common in a lot of people and is a big reason for depression" (P11, Medical student)

From the interviews it can be suggested that this depression and concern often stems from the in-laws and husbands who place a huge amount of pressure on the woman to give them a son and if she fails there may be problems within the family as expressed by a midwife in the extract below (**Table 30**: Q20).

"so family pressures such as they might expect a son but actually find out that the woman is having a daughter so that creates a pressure from the family" (P6, Midwife)

HCPs stated that many families want a son as they will be able to earn and take the family responsibility when they grow up (**Table 30**: Q21). Daughters on the other hand are often seen as a burden for families. Nevertheless, HCPs in this study do not reflect this attitude they see within families. HCPs stated that the main concern is to have a healthy baby regardless of gender. This is important to note as it is crucial such gender preferences do not exist within the healthcare sphere as they have a role in reducing inequalities and can influence families.

"son or daughter we have no control over this (.) whether it's a son or a daughter the main thing we want is a healthy baby (.) so many people aren't able to have kids and you are having a baby so there is no reason to be upset so accept what God has given you" (P6, Midwife)

A medical student in this study shared an experience where the in-laws of a pregnant woman did not even come to see her after delivery as she had given birth to a girl. It can be suggested from this extract that the in-laws are so focused on having a son that they often give little attention to the pregnant woman, placing the woman's status below the baby, given that the baby is a boy.

"they had a girl child and then she was kept in the PNC ward for a few days and then we can see that no one from the husband's family came to see the mother and we can see the mother and see that she is in depression because their

family is not happy with a daughter so we see things like this" (P9, Medical intern)

Table 30 Perceived social determinants of PMH issues subthemes and additional quotes

Subtheme	Quotes	
Individual factors	Q1	“sometimes very young girls get married who don’t even know about the situation so from that side they are scared as well” (P1, Midwife)
Individual lifestyle factors	Q2	“yes yes I know the access to contraception has increased over the past decade or so and there is much better education around that… compared to the awareness say in a country like the UK erm there still is a long way to go” (P13, MAA CHW)
	Q3	“there are a lot of uneducated people who do not know about family planning and so they end up having unplanned pregnancies they don’t use any family planning methods” (P7, Midwife)
	Q4	“so they might have kids aged 3 months or 4 months or 6 months and they kept taking the pill but unfortunately they still got pregnant because they made a mistake they didn’t understand how to take the pill properly” (P5, Midwife)
Social and community networks	Q5	“in rural areas things like the family plays a big role and has a big impact on the mother” (P11, Medical student)
	Q6	“Mother-in-law say the baby will get too big so they don't let me eat (.) I will go back to my mothers house (.) over there my mother will me eat” (P3, Midwife)
	Q7	“so like if I asked any mother that why didn’t you come to ANC and they would say that no actually my husband and my in-laws wont let me come” (P4, Maternal field mentor)
	Q8	“and just structurally the way Bangladesh is it is hard for poor people to not be poor and definitely hard for women to do anything for their family they are so dependent on men” (P14, MAA CHW)

	Q9	“I have done a delivery myself where before I could even let her go she was divorced, she has three girls already and then at the labour table she said that if she doesn’t have a son then there is a lot of suffering written for her so tried to support her but after the delivery we left her in the postnatal observation and then after coming back the next day we heard that her husband had divorced her” (P5, Midwife)
	Q10	“when they conceive then it can be said that they don’t receive much support from their husband and the in-laws the mother-in-law” (P1, Midwife)
Living and working conditions	Q11	“maximum mothers here aren't actually that educated which is the main problem” (P3, Midwife)
	Q12	“There are tensions like this or when I have the delivery will I be able to give birth without any complications so there are a lot of tensions like this that mothers go through” (P7, Midwife)
	Q13	“now what is needed is that there should be a midwife with every pregnant woman its needed then they I think they will be mentally and physically satisfied” (P6, Midwife)
General socioeconomic, cultural and environmental conditions	Q14	“we do see some women depressed then and are in tension especially because of their financial situation which is a cause of their mental sickness” (P9, Medical intern)
	Q15	“I feel like the first that comes to mind is just generally low socioeconomic background so how much their general income is because you know we were seeing a range of mothers coming from real depravity (.) erm from villages where there is limited access to they just had the bare basics you know just shelter food and they had a general but low income coming in” (P13, MAA CHW)
	Q16	"husbands don’t come that often (.) for example if I have 100 patients then maybe 10 will bring their husbands…husband participation is very little low…some will take time off (work) to bring their wife but

		maximum are poor (.) so like we will tell a lot to have one egg per day but they say how can we have eggs we don't have money their husbands will bring food on the day they eat" (P3, Midwife)
	Q17	"she had asked the mother you know which pregnancy is this and the mother explained this her fourth third or fourth pregnancy erm (.) and she had come to the health camp with the soul of aim asking how she can terminate the pregnancy because she with the main reason was she just couldn't afford to feed another mouth" (P13, MAA CHW)
	Q18	"erm a lot of the times we see that the husband might not be as helpful the family members are not that helpful so the woman has to do all the work herself a lot of heavy weight jobs" (P8, Medical student)
	Q19	"I'm having another girl I wish I had a son then that would be good" (P3, Midwife)
	Q20	"when we did the diploma we would see situations where a woman has three girls and if she doesn't have a boy next then there will be lots of problems in the family so this causes a lot of depression" (P5, Midwife)
	Q21	"a son will be able to feed us they really need a son (.) when a son grows up they will start earning money and be able to take on the families responsibilities" (P7, Midwife)

8.4.4 Theme 4: PMH care provision

This final theme highlights the key enablers and barriers, ranging from structural to sociocultural factors, to PMH care provision, along with suggestions made by HCPs regarding what could be done to improve PMH and PMH care in Bangladesh.

Education and training: A key enabling factor for the provision of better PMH care that appeared from the interviews was HCPs' awareness around PMH and the understanding that it is important and deserves medical attention (**Table 31**: Q1). As highlighted by a midwife in the extract below, when PMH is acknowledged to be a medical disorder by HCPs, there is a willingness to want to provide separate treatment which enables better care.

> *"this (perinatal depression and anxiety) is actually a medical disorder so in that there should be a separate treatment or a separate counsellor" (P3, Midwife)*

Mental health counselling was widely acknowledged by many of the interviewees and was viewed as a crucial, if not the most important, element in the management of PMH issues as highlighted in the quote below by a doctor (**Table 31**: Q2, Q3).

> *"to improve the mental health of pregnant women I think counselling is the most important" (P12, Doctor)*

Nevertheless, HCPs felt that they were given inadequate formal training and education around PMH (**Table 31**: Q4, Q5, Q6). HCPs expressed a desire for more education and training specifically around screening and diagnosis of PMH issues and management methods as highlighted by a midwife and a medical student in the extracts below.

> *"we need I would say especially in order to improve as midwives I would say that we need to have a specific better course on mental health within our training and to add a subject on mental health then that will be very useful then a midwife will become more knowledgeable about perinatal mental health…more education on diagnosing depression and anxiety" (P7, Midwife)*

“it is in education but not that much and not a lot. We get taught a little bit but it would be good to learn about the different ways of managing mental health” (P9, Medical student)

The source of PMH knowledge and education for HCPs is currently limited and inadequate with one midwife saying she learned mostly from self-study and another midwife stating she would learn from watching senior colleagues conduct mental health counselling (**Table 31**: Q7, Q8). HCPs in this study stated that they want more institutional training integrated within their curriculum, as highlighted by a midwife in the extract below, and a medical student stated how regular seminars would be beneficial to gain knowledge around counselling (**Table 31**: Q9, Q10). There is also a desire, as suggested in the extract below, for formal procedures to be put in place that HCPs can follow.

“we who are healthcare providers should be trained up (.) a lot of us might know a lot already but there is always something lacking so if we get some institutional training then that will be good so we can follow procedures” (P1, Midwife)

Along with education for themselves, HCPs in this study also highlighted awareness of PMH within perinatal women, their families and communities as a key enabler for PMH care provision. A maternal field mentor suggested that ANC visits are an ideal time to deliver PMH education for women as they rarely visit HCPs before this period. This also suggests there is limited outreach to women and families in many regions of Bangladesh where education can be delivered.

“I think when they come to ANC only then we can (educate mothers) because mothers don’t really come before” (P4, Maternal field mentor)

Alternatively, a MAA CHW suggested educational interventions around PMH at a community level, using existing women’s groups, where women may be more comfortable discussing sensitive topics. This highlights the importance of outreach into communities to deliver education rather than waiting for women to come to services themselves. MAA CHWs also highlighted that women in their health camps value health education and listen to the information and advice given by HCPs (**Table 31**:

Q11, Q12). It may be that women value the time and effort taken for HCPs to deliver education at a community level making it accessible for all.

> *"main way would be to find a way to introduce mental health education into communities and the way that I think would be best in terms of maternal mental health so within women's groups so locating women's groups that already exist…identifying communities where women are comfortable talking about certain sensitive topics erm and then targeting those groups with education seminars about mental health" (P13, MAA CHW)*

Time and workforce: Lack of time due to workforce shortage was conveyed by many of the HCPs interviewed as a challenge to assess and manage women with PMH issues. With physical health issues such as postpartum haemorrhage and pre-eclampsia being such main causes of maternal death in Bangladesh, HCPs must prioritise their limited time, and physical health often takes precedence (**Table 31**: Q13, Q14).

> *"so I think the priority was the clinical side of things like we were screening for preeclampsia that was the leading causes of death right so it was like seeing whether their blood glucose was under control their blood pressure was under control (.) so there was a lot more emphasis on the clinical side of things rather than mental health I think" (P14, MAA CHW)*

Increasing workforce, particularly in rural areas, was highlighted as an important way to ensure that pregnant women have access to HCPs and will enable them to prioritise time towards PMH (**Table 31**: Q15).

> *"how we can provide the care, for that we need to increase workforce (.) there are very few of us compared to how many we need" (P3, Midwife)*

Along with increasing the numbers of HCPs in the maternal healthcare sphere, many midwives suggested task shifting by training local leaders, community family planners and midwives in counselling so they can provide formal counselling to perinatal women in all areas (**Table 31**: Q16). The extract below from a midwife suggests that midwives in Bangladesh can act as key players in PMH care as they are

being positioned into many sectors and may have a wider skillset, if provided with the training, to cover many health areas such as PMH counselling.

> *"Us midwives are going into a lot of sectors like we can become counsellors if we receive the training, we are midwives we can become counsellors as well" (P3, Midwife)*

Stigma: Many HCPs in this study acknowledged the high levels of stigma associated with PMH, and women's health topics in general, which they felt prevents women from seeking PMH care (**Table 31**: Q17, Q18).

> *"so I think we need mental health advocates who can remove the mental health stigma so that patients come to us doctors because I think it is because of the stigma that no one wants to come" (P10, Doctor)*

As highlighted in the extract above, HCPs felt that this stigma embedded within the culture and community can create a barrier for women to share their emotions and mental health concerns with them. Reasons for this can be due to shame and/or fear of judgment and repercussions if their family or the community find out as expressed by the maternal field mentor below (**Table 31**: Q19, Q20). This fear may also reflect the power relations within the family where the husband and mother-in-law can use this fear and threats of violence to restrict women from seeking PMH care.

> *"regarding mental health actually they don't speak about so openly because there is always that fear that if husband or mother-in-law gets to know then they will hit her because why did she say these things" (P4, Maternal field mentor)*

Interviewees stated some key qualities they felt they should possess to facilitate emotional sharing from pregnant women. A key quality highlighted was trust; a midwife expressed that if women feel they can trust HCPs to not judge or tell anyone, they are more open to sharing confidential or sensitive matters. This reflects the fear of repercussions many pregnant women carry, therefore, trust and confidentiality are key factors.

"so when we say that if you have any problems you can share with me and you can trust me and consider me a friend I wont tell anyone, so if you have any problems you can tell me, so we say things like this and at that point many do start to share" (P7, Midwife)

Being friendly and building a good rapport with women was also highlighted as a key quality HCPs should possess (**Table 31**: Q21). HCPs also suggested that community level mental health advocacy and publicity is required to increase awareness and reduce stigma. This can include advertisements on television, word of mouth publicity and dedicated advocates (**Table 31**: Q22, Q23).

"we can firstly organise different workshops or do advertising on television (.) seeing advertisement can increase attention" (P1, Midwife)

Health camps/projects in rural areas: From the interviews it was suggested that having specific maternal health related health camps/projects in rural areas, where distance to and costs of healthcare are key challenges, is an enabler for easier access to ANC services and HCPs. One midwife expressed that health projects in villages allow better outreach of HCPs to rural areas.

"if we can give them midwives through some sort of health projects in the upazila then all the villages in the upazila can know about these midwives…a lot of time there isn't anyone just to give this advice to pregnant women in villages" (P6, Midwife)

It was also suggested from the interviews that these health camps make pregnant women feel more comfortable and empowered that they are being recognised and given importance in society encouraging them to share more about their health. Such projects are often delivered through NGOs, such as MAA, who provide specific and free healthcare to pregnant women (**Table 31**: Q24).

"in MAA there is a follow on system which means that mothers are more interested to come…and in other health camps they are only health camps for everyone not just specifically for pregnant women like MAA so in that case MAA is very specific" (P8, Medical student)

MAA CHWs highlighted how pregnant women had a good experience in the health camps, left with a positive mindset and were grateful for the care (**Table 31**: Q25, Q26).

> *"but when the mothers leave the health camps we really feel they leave with a positive mindset and they have learnt a lot about what they need to do" (P11, Medical student)*

Integrating mental health into ANC: Suggestions were made regarding actively integrating mental health into ANC visits, as indicated by the extract below. During ANC visits pregnant women receive more face-to-face time with HCPs which provides an ideal time to discuss mental health issues to ensure that women receive holistic care during their pregnancy. This medical student also suggests using this time to provide education for families too.

> *"I think that in ANC when mothers come and they bring their husband and if their mother-in-law and others come as well, then from there we take them in groups of two or three and we can talk to them about postnatal depression" (P9, Medical student)*

Integration can be achieved by introducing mental health booths, stations or sections within the ANC visit, as highlighted by a doctor in the extract below, where HCPs can specifically target PMH concerns with pregnant women and their families where appropriate (**Table 31**: Q27). This can work particularly well in remote areas to ensure that all services are joined up in one area making it more accessible for pregnant women and their families.

> *"I see that our MAA health camps are in very remote areas so at that time at the health camps we can put in place a mental health booth in which we can talk to family members" (P10, Doctor)*

HCPs in this study stated that they do not use any scales or tools to screen for antenatal depression or anxiety. This suggests that there is a lack of formal screening/diagnosis tools for antenatal depression and anxiety in Bangladesh (**Table**

31: Q28, Q29). The quote below from a midwife also suggests that HCPs have no clear protocol on screening for antenatal depression and may often be based on just giving advice on general mental health concerns.

> *"no we don't maintain any scales for depression but whatever type of depression someone says we give advice on that" (P2, Midwife)*

Nevertheless, HCPs did express the need for formal PMH screening protocols, for example, adding mental health questions alongside the physical health check-ups would enable them to pick up on symptoms and 'red flags' for PMH issues which otherwise may be missed (**Table 31**: Q30-32). The extract below suggests that HCPs may not remember mental health during antenatal check-ups, therefore, adding it into existing systems would act as a reminder for HCPs to screen for PMH issues. This also raises a question around changing the mindset of HCPs slowly to incorporate mental health into ANC, starting with consistent deliberate prompts and progressing to more natural and autonomous integration of mental health into ANC.

> *"in our database we collect lots of information like LMP, BP, pulse etc so if there are some questions added around mental health then we will be forced to ask these questions to them (.) so, if there are protocols like this then we will begin to know more about mental health" (P5, Midwife)*

Furthermore, HCPs highlighted the importance of mental health experts, such as psychiatrists and psychologists, to be an integral part of the maternal healthcare sphere (**Table 31**: Q33). This suggests the need for more joined-up, holistic healthcare in Bangladesh where there is a continuum between different services.

> *"we do all the basic antenatal investigations and minimal treatments so along with this if we can just include one psychiatrist or not even a psychiatrist but a psychologist will do as well who understands mental health then they can give mental support and talk to them and assess their mental health conditions" (P10, Doctor)*

Table 31 PMH care provision subthemes and additional quotes

Subtheme	Quotes	
Education and training	Q1	“in my professional life I try to support mothers and understand their emotional feelings however little I can, I’m not a professional psychologist but I try and give as much support as I can from my side” (P5, Midwife)
	Q2	“make sure we do at least ten minutes of counselling because counselling is the most important” (P4, Maternal field mentor)
	Q3	“we need to give them lots of counselling actually counselling can remove someone’s depression” (P7, Midwife)
	Q4	“No I haven’t received any training or anything on mental health…we didn’t get taught anything” (P1, Midwife)
	Q5	“after joining here we didn’t receive any such training…there was no extra or separate anything we received on mental health” (P3, Midwife)
	Q6	“yes our educational curriculum includes a psychiatry module…it was not specifically around maternal mental health it was just overall psychiatry wards and lectures so that was included in the curriculum” (P11, Doctor)
	Q7	“after joining here we didn’t receive any such training…there was no extra or separate anything we received on mental health…the amount we did learn was during our study time” (P3, Midwife)
	Q8	“we would see how our seniors would do the counselling or the older sisters how they do it so from there we learnt” (P7, Midwife)
	Q9	“so if we have such training or introduce it into the curriculum itself” (P9, Medical intern)

	Q10	“I mean firstly there is nothing that beats education and knowledge (.) If I want to counsel someone them first of all I have to know myself so it would be nice to have regular seminars around maternal mental health or videos which have audio and visual things or organise some training these things will be very beneficial for us” (P11, Medical student)
	Q11	“I see that mothers care a lot more and listen to the health education very attentively” (P8, Medical student)
	Q12	“I remember sitting in one of the workshops and the women were hanging onto every word because somebody had taken the time to educate them on something so I think if we do that with mental health like it is possible” (P14, MAA CHW)
Time and workforce	Q13	“maximum time in Bangladesh midwives don’t give that much time to mothers which is the truth but if we give them the time then they don’t us what they are feeling” (P5, Midwife)
	Q14	“what we see that they (doctors) are often so busy with other things that mental health does go to the side a bit” (P9, Medical intern)
	Q15	“actually a lot of mothers we have live in rural villages but we can’t do much for them because there are no midwives” (P6, Midwife)
	Q16	“who are the leaders of the villages or local areas we can counsel them then maybe the situation will improve” (P1, Midwife)
Stigma	Q17	“if we do (mental health) counselling then the mothers don’t normally come anymore” (P4, Maternal field mentor)
	Q18	“the second will be kind of difficult but sort of shifting a mindset within a community that is so tight lipped around mental health” (P13, MAA CHW)

	Q19	“it happens a lot that they feel shame to talk to me…actually you know that Bangladesh is actually quite conservative that’s maybe why they feel shame” (P1, Midwife)
	Q20	“no still they don’t (share) still it is stigmatised and they don’t talk about these problems still” (P10, Doctor)
	Q21	“I can tell that the patient wants to tell me something privately away from the mother-in-law” (P3, Midwife)
	Q22	“I think that we need mental health advocates who can do mental health advocacy and at least remove the stigma and so people will mental health issues will approach hospitals because if they do not come to hospital then we cannot treat them” (P10, Doctor)
	Q23	“we have to advertise this one person will tell another person who will tell another” (P6, Midwife)
Health camps/projects in rural areas	Q24	“in MAA there is a follow on system which means that mothers are more interested to come…and in other health camps they are only health camps for everyone not just specifically for pregnant women like MAA so in that case MAA is very specific” (P8, Medical student)
	Q25	“it’s so like you think this many people are going to come and so many more women come and its amazing because you want to help as many people as possible” (P14, MAA CHW)
	Q26	“I think the overriding emotion was just gratitude like they were so so grateful that we were there that we were providing the service for free” (P14, MAA CHW)
Integrating mental health into ANC	Q27	“we need to add mental health station into the services which is very important” (P12, Doctor)
	Q28	“no not formally actually we don’t take this as a separate thing (.) when we see that and understand from ourselves or the patient might sometimes share their problems only then discuss depression and anxiety but not otherwise as a formal thing” (P9, Medical intern)
	Q29	“No us midwives don’t have such protocols right now, we don’t have do such things” (P5, Midwife)

	Q30	“erm like this (depression) scale that you mentioned can be introduced and the answers we get from the scales we can go a bit deeper with the mothers” (P8, Medical student)
	Q31	“we talk about red flags symptoms in MAA but it’s all physical it’s all red flag symptoms we can see whether its protein in the urine or high blood glucose but we don’t talk about red flag symptoms for mental health so maybe if there was an equivalence for that whether its er a mothers tone of voice” (P13, MAA CHW)
	Q32	“I think when the women first presents to the clinic or the healthcare professionals I think they could do a mental health screen” (P14, MAA CHW)
	Q33	"I think there needs to be mental health professionals incorporated within the team” (P14, MAA CHW)

8.5 Discussion

8.5.1 Summary and interpretation of key findings

This qualitative study aimed to explore the perceptions and experiences of HCPs, including doctors, midwives, medical students and CHWs, in Bangladesh surrounding PMH. The maternal healthcare sphere in Bangladesh is expanding with the introduction of licensed midwives and more involvement of NGOs and CHWs, such as MAA, with specific maternal health camps and projects. This creates an opportunity to incorporate these HCPs in PMH care, potentially becoming key actors as they appear to have better access to pregnant women, in particular gaining access to rural regions of Bangladesh where there is limited workforce and access to maternal healthcare. A review by Coates et al (175) investing the role and competence of midwives in supporting perinatal women with mental health issues suggested that midwives are well placed to respond to PMH issues and play a primary role in facilitating PMH reduction through effective screening and interventions, as they frequently interact with women. Ganjekar et al (201) highlighted that programmes addressing the social determinants of PMH should include the involvement of NGOs as key actors. Midwives and CHWs within NGOs are also capable of taking on the role of counsellors through task shifting as expressed in this current study. Previous studies have found evidence to suggest that task shifting interventions are a valuable tool for implementing mental health care in settings with limited professionals (202). Hence, the expanding maternal healthcare sphere in Bangladesh tackles the barrier of limited workforce and time, which was highlighted by HCPs in this study, and creates a platform for new initiatives to improve PMH care.

HCPs viewed pregnancy as a unique phase during which women are more vulnerable to emotional and psychological changes. They acknowledged that PMH is a widespread issue and that it deserves medical attention. These findings contradict those from the study in Bangladesh by Williams et al (23) in which medical personnel did not take postpartum depression seriously and did not consider it to be a health concern they should address. These differences may be due to the time difference between the studies. Williams et al's (23) study was conducted in 2014. Since then, we have seen an increase in knowledge and awareness of PMH, not only in Bangladesh but globally as well. Furthermore, the age groups of the respondents differ whereby the average age of HCPs in this study is less compared to that in Williams et

al (23). Younger age is related to more awareness of mental health which may be due to the expansion of social media which has promoted open discussions within communities to normalise mental health problems (203,204). The different attitudes compared to previous literature may also reflect the changes in society where mental health in general is more accepted and acknowledged. A review by Lien et al (205) found that HCPs' attitudes towards mental health have become more positive over time.

Despite awareness of PMH, HCPs felt they had a lack of education and training around PMH screening/diagnosis and management methods. There are also no formal protocols within ANC on identifying women with PMH issues or how to counsel them. This finding is reflected in the study by Williams et al (23) and studies in different settings. McCauley et al (129) found that HCPs in Ghana felt they were not adequately trained to assess and manage maternal mental health, and a study in Australia by Noonan et al (172) found that midwives did not feel equipped to support pregnant women with PMH issues and desired more education and training to be fully prepared. This highlights the need for improved education and training courses for all HCPs, ideally integrated within the curriculum at an early stage. Additional seminars should be introduced to allow for continuous professional development. Along with educational programmes, HCPs should have access to PMH care pathways including specific protocols for screening and managing PMH issues, in particular antenatal depression and anxiety. Screening questions should be integrated into ANC to prompt HCPs to screen for symptoms and "red flags" of PMH issues alongside physical health checks. A good example of this could be the EPDS-B which has been validated in a Bangladeshi setting (162).

In order to screen for PMH issues, HCPs need to be aware of the risk factors and social determinants to identify pregnant women who may be at risk of developing antenatal depression or anxiety. HCPs in this study expressed their thoughts on what they perceived to be the most important social determinants of PMH issues in Bangladesh. At an individual level, young age of marriage was highlighted as a risk factor as young girls are unprepared for marriage, marital relations and bearing a child. A study by Santhya et al (134) investigating early marriage and young women's marital and reproductive outcomes in India found that young women who married below the age of 18 years were less likely to have used contraceptives to delay their first

pregnancy and were more likely to have experienced IPV. Unplanned pregnancies, a consequence of low contraceptive use, and IPV are well documented social determinants of PMH issues (21,34,60,73), and have also been highlighted by HCPs in this current study.

Shifting from an individual level to social and community networks, family dynamics and hierarchy come into play. Limited access to ANC, increased household chores, restricting food and making decisions on contraception were some of the examples of how mothers-in-law and husbands display their power and authority. HCPs expressed how these factors place pregnant women in a toxic familial environment and many complain of lack of family support and abuse resulting in PMH issues. These findings are reflected in previous literature where poor relationship with in-laws and husband and lack of family support were highlighted to be significant social determinants of PMH conditions (11,14,21,34). HCPs are exposed to family dynamics when pregnant women attend antenatal check-ups, and many stated that they are well placed to provide counselling and education for families.

The impacts of social and community networks are further heightened by the fact that the majority of pregnant women, particularly in rural areas, are uneducated as stated by many of the participants in this study. Education is a tool for empowering women and studies have found that women's decision-making autonomy increases when they are empowered with education (206,207). Hence, pregnant women in Bangladesh who have received little/no education are more susceptible to the negative family dynamics and unequal power relations. HCPs in this study have stated the importance of educating women on maternal health, including PMH, and have acknowledged their role in providing this within ANC visits and at a community level targeting existing women's groups. Many HCPs have experienced the positive impact they have when they provide health education to pregnant women as it gives them autonomy and makes them more aware of their wellbeing during and after their pregnancy.

Widening out into the general socioeconomic, cultural and environmental factors, HCPs viewed poverty as playing an important role in women's experience of pregnancy. Those HCPs who had experience working in rural areas of Bangladesh expressed that poverty in these regions is a big reason for pregnant women suffering

from depression and anxiety as they worry about their child's development due to lack of access to ANC and medical complications resulting in a caesarean which will be expensive. It is important to acknowledge that financial worries and concerns about the baby are issues that many pregnant women undergo worldwide, however, in rural Bangladesh, these are heightened by poor access to antenatal care, cultural and social norms and stigma. Some midwives discussed financial incentives for pregnant women to attend ANC visits which have also been highlighted in previous literature. A review conducted by Till et al (208) found that pregnant women receiving financial incentives were more likely to attend prenatal visits on a regular basis and more likely to deliver by caesarean. Majority of women in this review were from rural, low-income, overwhelmingly Hispanic communities in Central America. Such incentives could be further developed in Bangladesh to increase uptake of ANC, especially in rural areas.

Aside from financial incentives, providing free ANC close to home through initiatives such as MAA health camps and clinics is a good way of increasing uptake of ANC in rural regions of Bangladesh. MAA CHWs stated that their health camps were very well attended by pregnant women who were so grateful for the free check-up, health education and medications, and left with a positive mindset; this has positive impacts on their mental health during their pregnancy. Bhondve et al (209) found that community health camps are a reasonable and practical approach to providing universal health coverage.

The sociocultural environment in Bangladesh creates many challenges for HCPs to provide PMH care and for pregnant women to access the care. HCPs in this study, although not expressing stigmatising attitudes themselves, acknowledged the existing stigma in Bangladeshi communities associated with PMH which prevents women from seeking care. Pregnant women who feel that their HCPs carry stigma around PMH do not openly discuss their mental health issues. Therefore, HCPs play an important role in reducing this stigma by reassuring women there is nothing to be ashamed about, as recognised by many of the participants in this study. Women also don't share confidential information in the fear that their community or families will find out resulting in serious repercussions. Hence, HCPs reiterated the importance of rapport building and being friendly to ensure women feel safe to share their emotions and situations. These findings are reflected in previous literature where stigma around PMH is consistently evident across different settings and highlighted to be a key barrier

to the provision of PMH care (23,63,91,123,129). It was also stated in this study that the younger generation of HCPs carry less stigma compared to the older generation who are not as keen on talking about mental health. This reflects the discussion above on younger age and positive attitudes towards mental health. Members from the younger generation could act as mental health advocates to create discussions within communities and reduce the stigma associated with mental health. Community level activities are important with the role of CHWs and NGOs being prominent in this initiative to tackle stigma (210).

Stigma around mental health is often heightened for PMH as gender comes into play. Social construction of gender in Bangladesh dictates the roles and responsibilities of women in society which primarily include household chores and child-rearing, particularly in rural communities (140). The fulfilment of these responsibilities can determine the status of women and, hence, an inability to do so can be detrimental. Women with PMH issues are often labelled as ‘mad’ or ‘not right in the head’ which can impact the way society views them as mothers. This can reduce the status of women within these communities who view motherhood as sacred and an important factor in determining the status of women.

8.5.2 Strengths of the study

This study explored the perceptions and experiences of HCPs in Bangladesh surrounding PMH and, to my knowledge, this is the first study to explore this topic within a wide spectrum of HCPs in Bangladesh, including midwives and CHWs who have not been represented adequately within the literature. The study participants had a wide range of experiences within maternal healthcare in Bangladesh ranging from working in rural areas, health camps, *Upazila* health complexes, refugee camps and Government hospitals. This enabled a variety of different experiences to be captured.

The interview topic guide was piloted on members of the MAA team and HCPs from the maternal healthcare field to identify any issues with the effectiveness of the topic guide. For example, checking the appropriateness and understanding of the questions, identifying additional topics of importance, and gaining more confidence in conducting the interviews. The pilot interviews ensured the topic guide flowed logically and that questions were worded appropriately.

Many of the interviews were conducted in Bengali to allow for better relationship and rapport building which supports the responsive interviewing method. This enabled participants to become immersed in the interviews and express their views as they were able to talk in the language, they are more comfortable with. This is especially important given the virtual nature of the interviews between two different countries to help bridge the gap.

To maintain robustness in the thematic analysis process, one transcript was coded independently by three researchers (myself and my two supervisors) and codes were brought together in a data analysis meeting, discussed and verified. This ensured that the coding was validated, and initial analysis involving the start of theme generation was discussed.

5.3 Limitations of the study

The translation of the interviews from Bengali to English was not translated by two independent researchers; it was only done by me due to resource constraints and to maintain confidentiality of the participants as personal information were mentioned in the interviews. Some studies suggest that two bilingual translators should be used to help validate the translation of general content and to reduce the possible errors in translation (211,212). Conversely, Twinn (213) suggested the use of one translator as consistency is obtained and reliability in the analysis of the data can be maximised. Overall, although two translators were not used given the reasons above, the interviews were double checked i.e., the recordings were matched against the English translation to check for any errors, and I ensured that the meaning of the original interviews were not altered given my fluency in the Bengali language.

The data may also be subject to potential biases such as response bias in the form of social desirability. This is the tendency of participants to provide answers or behave in a manner they feel will be viewed favourably by the interviewer (214). Topics such as mental health, domestic violence, and gender inequalities can provoke controversial opinions and participants may feel they need to respond in a certain way. This can lead to biased responses in the interviews which may not reflect the true views. To minimise this bias, participants were reassured there were no right or wrong answers and anonymity would be strictly maintained.

Since interviews were conducted online, there is a potential impact on recruitment and characteristics of the sample in this study. Only HCPs who had access to the internet and a computer were eligible for the study which may have missed out certain CHWs who may not have had this access. It could also explain the relatively young sample given that the younger generation may be more comfortable and proficient in using virtual video platforms. Also, those with access to technology may be more exposed to social media and, as discussed previously, social media has promoted open discussions within communities to normalise mental health problems and reduce stigma (203,204). This could lead to a biased sample who may display less stigma towards mental health, hence, biasing the findings. Furthermore, the young age range in the sample may be a limitation as discussed above, younger individuals may display less stigma towards mental health which can bias the findings. Nevertheless, the findings are still relevant and important as technology and social media has become very widespread in recent years, and this study reflects the views and experiences of a generation that will become the future of the maternal healthcare sphere in Bangladesh.

5.4 Implications for the wider PhD project

The findings from this qualitative study carry some important implications for the wider PhD project. It provides deeper understanding of some of the factors identified from the survey study (chapter 7) such as reasons for lower knowledge levels, types of training and education desired and certain differences between different HCPs which will be further discussed in the mixed methods study chapter (chapter 9). Given the study (chapter 6) on social determinants of antenatal depression identified social determinants from the perspective of pregnant women, the findings from this study provide an alternative perspective, from HCPs, on the social determinants of PMH in Bangladesh. This allows for an interesting discussion and comparison around the understanding of social determinants from both perinatal healthcare users and HCPs which is further discussed in chapter 9 and 10. This reflects the aim of this PhD project to explore perinatal mental health in Bangladesh, and a qualitive study allows for that deeper and more holistic exploration.

This study also highlights several implications for policy, practice and research such as the need to increase educational opportunities for HCPs, better integration of mental health into ANC and wider social factors such as tackling poverty, gender

inequalities and stigma. These will be further discussed and tied together with implications from all previous studies within this PhD in chapter 10.

6. Chapter summary and contributions

The findings from this study contribute to a deeper understanding of HCPs' views, experiences and knowledge around PMH care in Bangladesh which addresses objective 4 of this PhD. Thematic analysis of interview data from 14 participants revealed 4 key themes: 1) maternal healthcare sphere in Bangladesh, 2) conceptualisation of pregnancy and PMH, 3) perceived social determinants of PMH issues, and 4) PMH care provision. In summary, the maternal healthcare sphere in Bangladesh is expanding with the addition of licenced midwives and key involvement of CHWs providing essential healthcare in rural areas of Bangladesh, where there is limited access to perinatal healthcare. HCPs are aware of PMH and the importance of targeted healthcare which should be integrated into existing antenatal healthcare systems and processes, including addition of screening tools and protocols and mental health experts. Unplanned pregnancy, IPV, male gender preference, poverty, poor relationship with in-laws and husband and low social support were highlighted to be important social determinants of PMH issues in Bangladesh. These findings will be integrated and corroborated with findings from the HCPs' survey and pregnant women's survey to provide a holistic exploration of PMH in Bangladesh in chapter 9. Implications for policy, practice and research from this wider mixed methods analysis will be discussed in chapter 10.

Chapter 9. Triangulation of pregnant women's and healthcare providers' studies: A mixed methods analysis

9.1 Chapter overview

This chapter presents the findings from a mixed methods analysis of the pregnant women's study (chapter 6) and HCPs' studies (chapters 7 and 8). A triangulation protocol was used to combine and corroborate the findings in order to form a more holistic overview of PMH in Bangladesh. A description of the methods will be followed by a critical discussion of the results.

9.2 Methods

The convergent mixed methods methodology is outlined in chapter 5 along with the reasons behind adopting this methodology. My PhD used the triangulation protocol based on the approach by Farmer et al (157) which is a six step method: 1) sorting, 2) convergence coding, 3) convergence assessment, 4) completeness assessment, 5) researcher comparison, and 6) feedback.

Firstly, key findings and themes from the three studies were sorted and separated from the rest of the data into a separate file to allow for comparisons. These themes form the rows of the convergence coding matrix. The three studies were compared with regards to the themes identified and were coded in terms of the level of convergence (**Table 32**). Agreement was assigned when two or more of the studies found converging findings on a particular theme. Partial agreement was assigned when two or more studies found complimentary findings on a particular theme. Findings were determined to be complimentary when either one could be used to explain/discuss the other or an interpretation of the findings lead to similar conclusions. Silence meant that only one study explored a particular theme, whereas the other two did not find any information regarding that theme. When one or more studies found discrepant findings on a particular theme compared to the other studies, this was assigned as dissonance.

Table 32 Convergence coding scheme for triangulation protocol

Convergence code label	Definition
Agreement	Two or more of the studies found converging findings
Partial agreement	Two or more of the studies found complimentary findings
Silence	Finding does not appear in two studies
Dissonance	One or more studies found discrepant findings

Subsequently, a convergence assessment was completed whereby a global level of convergence between the three studies was determined. This was based on the percentage of agreement and partial agreement compared to silence and dissonance. Following this, a completeness assessment was performed to create a summary of the unified findings from all three datasets. This highlights how complete the findings were with regards to each theme in terms of the different perspectives outlined and the overall picture they develop. Steps five and six deal with validating the analysis by comparing assessment from multiple researchers and finally feedback of results to the research team or stakeholders for review and clarification. This was achieved by holding a meeting with my supervisors and reviewing the themes and convergence codes. Findings were also discussed with the MAA team for review and feedback.

9.3 Findings

Table 33 presents the convergence coding matrix which highlights the level of convergence among the three studies regarding the 18 key themes extracted from each study. The majority of the themes indicate agreement or partial agreement among the studies. Four themes only have one study that explores each theme demonstrating silenced themes. Dissonance is highlighted within two of the themes.

Table 33 Convergence coding matrix

Theme	Pregnant women's survey (chapter 6)	Healthcare providers' survey (chapter 7)	Healthcare providers' interviews (chapter 8)	Convergence code
High prevalence of PMH issues in Bangladesh	Point-prevalence of 56% of antenatal depression in rural Sylhet, Bangladesh	-	(Theme 2) HCPs expressed that majority of pregnant women suffer from depression	Agreement
Low family support associated with PMH	Higher family support reduced the odds of antenatal depression - AOR 0.94 (95% CI 0.91-0.97) (**Table 15**)	-	(Theme 3) HCPs stated that low family support is a key reason for antenatal depression	Agreement
Male gender preference associated with PMH	Women who perceive their husband to have no gender preference had reduced odds of antenatal depression compared to women who perceived their husband to want a son – AOR 9.9 (95% CI 1.6-59.6) (**Table 15**)	-	(Theme 3) HCPs stated that many families want a son, and this can cause stress for pregnant women	Agreement

Hierarchy and power relations within Bangladeshi families	During data collection, pregnant women were more hesitant to answer questions regarding sensitive topics such as intimate partner violence if their husband/mother-in-law were present at the health camp	-	(Theme 3) HCPs stated that the mother-in-law and husbands carry a lot of power and authority within the family which they felt can create restrictions for pregnant women	Partial Agreement
Unwanted/unplanned pregnancies	Only 9% of pregnant women stated their pregnancy was unwanted (**Table 13**)	-	(Theme 3) HCPs expressed that many pregnancies in Bangladesh are unwanted/unplanned	Dissonance
Intimate partner violence (IPV) is associated with PMH	There was a significant association between IPV before pregnancy and antenatal depression (AOR 10.4 95% CI 2.7-39.7) (**Table 15**)	-	(Theme 3) HCPs stated that there is a lot of domestic violence in villages which they felt result in antenatal depression	Agreement
Poverty	-	-	(Theme 3) HCPs highlighted that a lot of families are not financially	Silence

			capable and suffer from poverty	
Lack of education and employment among women	98% of pregnant women were unemployed/no-paid work and 75% had no-schooling/formal qualifications (**Table 13**)	-	(Theme 3) HCPs stated that the majority of women are uneducated in rural villages	Agreement
Young age of marriage	Median age of the pregnant women was 25 years, suggesting they got married before this age (**Table 13**)	-	(Theme 3) HCPs stated that women get pregnant at a very young age, often around 19 or 20 years	Agreement
Limited access to ANC	-	-	(Theme 3) HCPs expressed that ANC is limited in rural areas of Bangladesh and not readily available	Silence
Relatively non-stigmatising attitudes from HCPs	-	Response to the attitude construct was non-stigmatising producing a mean (SD) score of 14.68 (5.36) (lower than the mid-	(Theme 4) HCPs acknowledged that PMH is a medical disorder that deserves medical attention	Agreement

		point of the total scale of 18.0)		
PMH stigma across the range of HCPs	-	Midwives displayed more stigmatising attitudes compared to doctors and medical students (**Table 20**)	(Theme 1) Midwives expressed that they have a role in providing support and understanding their emotional wellbeing, suggesting no stigma	Dissonance
PMH knowledge levels across the range of HCPs	-	Midwives displayed lower levels of PMH knowledge compared to doctors and medical students (**Table 25**)	(Theme 1) Midwives expressed that midwifery is a very new profession and did not know what to expect. Midwives stated they have limited education and training on PMH	Partial Agreement
Learning needs for HCPs – PMH screening and management	-	70% of HCPs felt they required additional training on assessing PMH, and 67% wanted further training on clinical	(Theme 4) HCPs expressed a desire for more education and training around screening and diagnosis of PMH	Agreement

		management of PMH issues (**Table 26**)	issues and management methods	
Lower PMH knowledge was associated with higher stigmatising attitudes towards PMH	-	Significant negative correlation between attitude score and knowledge score (r=-0.302, p=<0.0001)	-	Silence
Acknowledgement of PMH being a medical disorder	-	82.7% of HCPs agreed/strongly agreed that mental illness is an illness that requires medical attention (**Table 16**)	(Theme 4) HCPs acknowledged PMH as a medical disorder which require separate treatment	Agreement
Acknowledgement of counselling as an effective PMH management option	-	83.5% of HCPs correctly identified counselling as the recommended management method for mild to moderate PMH issues (**Table 21**)	(Theme 4) mental health counselling was widely acknowledged as a crucial, if not the most important, element of PMH management	Agreement

Urban vs rural inequalities in provision of ANC	-	-	(Theme 1) HCPs expressed that there is limited provision of ANC and access to health professionals in rural areas of Bangladesh. Care is mostly provided by unskilled traditional birth attendants	Silence

- The theme was not explored in this study
PMH – Perinatal Mental Health
ANC – Antenatal Care
HCPs – Healthcare providers
Agreement – Two or more studies found converging findings on the theme
Partial Agreement – Two or more studies found complimentary findings on the theme
Silence – Only one study explores the theme, the other two do not explore the theme
Dissonance – One or more studies found discrepant findings on the theme

9.3.1 Agreement (Convergence)

High prevalence of PMH issues was reflected in both the pregnant women's survey findings and interviews with HCPs. The survey found a point-prevalence of antenatal depression of 56% in rural Sylhet, Bangladesh. HCPs expressed that PMH issues "happen a lot almost maximum mothers have this; they suffer from depression." Therefore, the widespread issue of PMH that is acknowledged by HCPs is corroborated by quantitative findings from the pregnant women's survey. This suggests that PMH is a public health concern in rural regions of Bangladesh with many pregnant women experiencing depressive symptoms during pregnancy. Thus, further work must be done to enable improvement of PMH in rural Bangladesh.

Low family support and male gender preference were found to be significantly associated with ADS in the pregnant women's study. These findings were also highlighted in the interviews with HCPs where they expressed that they felt that low family support and male gender preference were causes of pregnant women experiencing PMH issues during their pregnancy. This suggests that HCPs are aware of and acknowledge these prominent issues in Bangladeshi communities that are associated with antenatal depression. The interviews with HCPs help reveal underlying mechanisms at play within these associations. Low family support is often displayed through "lack of care", limited assistance with household chores and "feeling lonely" during a psychologically and physically challenging time. This can lead to a woman feeling depressive symptoms during her pregnancy. HCPs also highlighted that many Bangladeshi families may expect a son who can take on family responsibilities resulting in pressures from the family on pregnant women which creates a lot of stress for the mother.

IPV was found to be significantly associated with increased odds of antenatal depression in the pregnant women's study. This was also highlighted by HCPs in the interviews where many expressed that domestic violence is common in rural villages of Bangladesh which can result in pregnant women being "mentally depressed". Interviews also helped to reveal that many pregnant women may also face violence from their in-laws including the mother- and sister-in-law which can exacerbate violence from the husband and creates a very toxic environment for pregnant women. This links back to the lack of family support as it is likely that pregnant women living in an abusive environment will feel lack of support from her family.

The pregnant women's survey highlighted that many women were uneducated (no schooling or pre-SSC) (75.8%) and did not participate in any paid work/employment (97.9%). This finding is also reflected in the interviews with HCPs who stated that the majority of women are uneducated. This may relate to the young age of marriage commonly found in Bangladesh which was expressed by HCPs in the interviews and evident from the survey where the median age of the pregnant women was 25 and the majority (66.4%) were 25 or younger. Within many Bangladeshi communities, it is normal to marry off a daughter at a young age to reduce the burden on the family; once married, a woman is no longer her parents' responsibility. Thus, opportunities for education, and hence employment, is limited for many women especially in rural regions (215).

Both the HCPs survey and interviews suggested relatively non-stigmatising attitudes towards PMH issues and acknowledged that PMH issues deserve medical attention. From the survey (chapter 7), the mean attitude score was 14.68 which was lower than the mid-point (18.0) suggesting non-stigmatising attitudes. The majority of the HCPs (82.7%) agreed that mental health illness deserves medical attention like any other illness, and 83.5% correctly identified counselling to be the recommended management method for mild/moderate antenatal depression. This may have been due to response bias whereby HCPs may have just been familiar with counselling compared to other options provided in the survey, however, after analysis of the interview data, these findings were corroborated. HCPs interviewed expressed that they felt that PMH "is actually a medical disorder" and that "there should be a separate treatment or a separate counsellor". They also acknowledged the importance of mental health counselling to improve the mental health of pregnant women. This may reflect the role of HCPs in providing support and motivation as perceived by HCPs in the interviews which enables them to recognise the importance of mental health and counselling. Both my studies exploring HCPs' attitudes found contradictory results to a previous study by Williams et al (23) exploring HCPs' attitudes towards PMH in Bangladesh. Williams et al (23) found that HCPs did not take PMH seriously or consider it to be a health concern that they needed to be addressed and were not familiar with counselling. As discussed in chapter 8, this may be due to the younger age of the samples in both my studies reflecting the changing attitudes towards mental health within younger generations. Nevertheless, there are limited studies investigating

attitudes of HCPs regarding PMH in Bangladesh. Therefore, more quantitative and qualitative studies are needed to provide opportunities for adequate comparison and discussion around changes in attitudes towards PMH in Bangladesh over time and between different generations. However, it can be concluded that there is a willingness amongst HCPs in Bangladesh to promote improvements in PMH which should be used to accelerate action.

HCPs' PMH knowledge levels were found to be relatively low from the survey study which can be partly explained by the qualitative study. HCPs in the interviews expressed that they have limited education and training regarding PMH integrated within their curriculum. Their primary source of knowledge is from self-study or learning from their seniors which can be limited and not always accurate. The key learning needs for HCPs were identified to be screening and management methods for PMH. The survey highlighted that only 38% of the respondents were familiar with the EPDS-B and correctly identified its purpose as an effective screening tool for perinatal depression. Alongside this, approximately 70% of the respondents felt they required additional training around screening for PMH conditions and clinical management of PMH disorders. This suggests that knowledge around screening and management of PMH issues is limited among HCPs. The interviews also highlighted that HCPs desired "more education on diagnosing depression and anxiety" and "different ways of managing mental health" indicating that their knowledge on screening and management is currently limited or inadequate.

3.2 Partial Agreement (Complimentary)

Family hierarchy and power relations was a key theme identified within the interviews with HCPs who expressed how the husband and mother-in-law carried the most authority and power within a familial network. HCPs stated how this can often lead to restrictions and toxic environments for pregnant women which can result in PMH issues during pregnancy. Although the pregnant women's survey did not directly explore family hierarchy and power relations, feedback during the pilot phase elucidated ideas related to this theme. MAA researchers highlighted that women were more hesitant to answer questions regarding sensitive topics such as IPV if their husband or mother-in-law were with them during the ANC visit. This may be suggestive of fear of repercussions which is indicative of family hierarchies and power relations as

expressed by HCPs. It is important to conduct further in-depth interviews with pregnant women to gain an understanding of these familial dynamics from their perspective.

This PhD included data from a range of different HCPs including midwives, doctors, medical students and CHWs, therefore, it was interesting to explore differences in their knowledge, attitudes and experiences surrounding PMH through both quantitative and qualitative methods. Midwives were found to have lower levels of PMH knowledge compared to doctors and medical students from the survey study. Although the qualitative study did not find convergent findings, it provided complimentary findings whereby midwives expressed how they were very new to this profession and “didn’t know what to expect”. This may provide an explanation to midwives displaying lower PMH knowledge levels in the survey study; since midwifery is new profession, the curriculum may not have established the same level of depth regarding PMH. In the interviews, midwives stated how they felt they had very limited education and training within their curriculum and that most of their knowledge about PMH was from self-study or learning from seniors on the job. Doctors and medical students, on the other hand, have a more specific psychiatry module during their medical degree and experience on wards during which they may have gained some basic knowledge. This highlights the need for improved education and training for midwives who play a key role in ANC and are often more involved than doctors or medical students in community-based care within rural regions through outreach activities. Effective screening and management of PMH introduced during midwifery-lead ANC appointments may lead to better diagnosis and improved care for pregnant women.

9.3.3 Silence

Poverty, access to ANC, urban vs rural inequalities and correlation between PMH knowledge and attitudes were regarded as silenced themes, as for each there was only one study which explored it. Poverty and limited access to ANC were expressed by HCPs in the interviews to be common problems in rural Bangladesh which play a role in PMH. The pregnant women’s dataset was silent on this theme as poverty was deemed to be a complex factor to measure given the nature of the survey and the capacity of the health camp/clinic. Furthermore, many previous studies found no significant association between poverty and antenatal depression (85,89,93,94). Therefore, to minimise the time taken to complete the survey and the complexity of it

for pregnant women, the construct of poverty was not added into the survey. The HCPs survey was also silent on this theme as it was not the objective of this survey. Further research, including qualitative and quantitative studies, with a focus on poverty and access to ANC is needed to provide more clear conclusions on their role in PMH.

The HCPs survey found a significant negative correlation between PMH knowledge levels and stigmatising attitudes whereby the higher the knowledge levels, the lower the levels of stigmatising attitudes. The qualitative study with HCPs was not appropriate for this type of quantitative finding as it is difficult to identify correlations through qualitative data, therefore, this study was silent on this theme. Likewise, the pregnant women's survey was also silent on this theme as it was not the objective of this survey.

9.3.4 Dissonance (Discrepancy)

HCPs within the interviews expressed that many women are "not ready for the pregnancy...its unwanted" due to many people being "uneducated who do not know about family planning, so they end up having unplanned pregnancies". This was highlighted to be an important issue particularly in rural regions of Bangladesh where there is limited use of contraception. In contrast, however, the pregnant women's survey study revealed that only 8.5% of the pregnant women had an unwanted pregnancy, the vast majority stated their current pregnancy was wanted. This contradiction between the HCPs and pregnant women's perspectives may be due to potential response bias in the survey study. Many pregnant women may not want to disclose their pregnancy choices in a survey setting due to fear of judgment or repercussions from their family and the interviewer. Therefore, the percentage of unwanted pregnancies may be underestimated from the survey. This type of information may be more easily shared with HCPs during antenatal appointments as HCPs are in a position of trust for many pregnant women. Another explanation may be that majority of pregnant women surveyed were in their second or third trimester, by which time, if it was an unwanted pregnancy at first, they may have come to terms with their pregnancy and impending motherhood may dampen those feelings of an unwanted pregnancy. The systematic review of social determinants (chapter 2) found that unintended pregnancies were significantly associated with increased odds of antenatal depression (45). Therefore, further qualitative research is required to

understand the complexities of unwanted/unplanned pregnancies and their role in PMH.

While both HCPs' studies were in partial agreement with midwives displaying lower levels of PMH knowledge, there is discrepancy in the finding that midwives display more stigmatising attitudes. Analysis of the survey data revealed that midwives displayed significantly more stigmatising attitudes compared to doctors or medical students. However, this result should be interpreted with caution due to the large number of midwives in the survey study compared to doctors and medical students. Although, it is not possible to identify significant associations from qualitative findings, the interview data largely indicated that midwives displayed non-stigmatising attitudes towards PMH. Midwives were very vocal in expressing that they "try to support mothers and understand their emotional feelings" and acknowledged PMH "is actually a medical disorder" and that "there should be a separate treatment or separate counsellor". Nevertheless, due to the nature of qualitative studies, it is difficult to determine whether midwives displayed more or less stigma towards PMH than doctors or medical students. At most, it can be concluded that all the HCPs interviewed seemingly had very similar positive attitudes towards PMH, and no differences between roles were apparent from the interviews.

4. Discussion

4.1 Convergence and completeness assessment

Overall, there was a large amount of convergence between the three study findings given that there was partial or full agreement on 67% of the themes. Only 22% of the themes showed an instance of presence in one dataset but not in the other two studies and only 11% of the themes showed instances of disagreement in the comparison of the studies. It is likely that these differences relate to differences in study objectives or nature of the study designs.

Based on the convergence assessment, it is evident that there are many themes in which the three studies agree on and confirm the core themes. The survey data generally added an overview of different themes, whereas the interview data provided more in-depth illustration and explanation regarding some of the underlying mechanisms and complexities at play. For example, the pregnant women's survey

identified, IPV, low family support and male gender preference as two factors that increased antenatal depressive symptoms in pregnant women. The HCPs interview data helped to corroborate this along with providing some potential reasons and meanings behind this association. In another example, the HCPs survey study identified an overall low level of PMH knowledge. This again was corroborated by the interviews along with adding further detail regarding the lack of education and training and the areas where HCPs wanted further training. In addition, the three datasets provided an opportunity for triangulation of different perspectives. Pregnant women's and HCPs' perspectives were triangulated to enable completeness and development of a comprehensive account of PMH in Bangladesh. For example, it is evident that certain social determinants, such as IPV, low family support and male gender preference, are highlighted by both HCPs and pregnant women. Therefore, there is a potential for interventions that integrate the role of both actors in improving PMH. HCPs come from a perspective of providing care and education regarding family dynamics and gender inequalities, whereas pregnant women are on the receiving end of this care. Both need to be in harmony for a holistic care approach. Thus, each study on its own partly answers the research question, but together, they contribute a higher level of analysis and broader understanding of PMH in women living in Bangladesh.

4.2 Strengths and limitations of this mixed methods approach

The mixed methods analysis largely followed a systematic process as set out by Farmer et al (157), whilst adapting the process for the needs of my PhD. This encouraged me to think through the mixed methods analysis logically by firstly identifying meta-themes, creating a convergence matrix and conducting convergence and completeness assessments. The analysis was further reviewed by my supervisors and the wider research team adding to the credibility of the analysis.

The three PhD studies do vary in their content due to differences in their purpose and focus. This resulted in differences in the coverage of themes across the studies, therefore, some differences in the presence and absence of themes might be due to these inherently different qualities of the studies or methods themselves and not representing legitimate differences. Thus, it was challenging to determine the source of dissonance or silence. In one way this may be a strength as it allows for a broader set of perspectives and methods from which findings can be drawn, however, it can

pose a challenge to the confirmatory aims of triangulation as it can be difficult to determine the level of convergence.

Furthermore, there were situations when one study was privileged or weighted more than the others due to their ability to address a particular research question. For example, themes which were related to associations between social determinants and PMH may have been better suited to the pregnant women's survey as quantitative studies aim to determine such associations through statistical significance. Although the HCPs' interview study did highlight the perspectives of HCPs on IPV, low family support and male gender preference, which was triangulated with the survey study, one cannot determine direct significance from qualitative studies. Therefore, to answer whether IPV, low family support and male gender preference do result in higher levels of PMH issues in pregnant women, the quantitative study provides a more direct answer, albeit with limitations as discussed in chapter 6, whereas the qualitative study can provide some insight into the underlying mechanisms and perspectives surrounding it. Nevertheless, this does not reduce the importance of any study, but merely highlights the complexities of a mixed methods analysis.

5. Chapter summary and contributions

The findings from this mixed methods analysis contribute to a more holistic understanding of the social determinants and prevalence of antenatal depression in Bangladesh and the perspectives, knowledge and experiences surrounding PMH care which address all the objectives of this PhD. Overall, a large amount of convergence was found between the three PhD studies on majority of the key themes. These themes were: a high prevalence of PMH, IPV, low family support and male gender preference are associated with antenatal depression, lack of education and employment among women in rural regions of Bangladesh, young age of marriage, HCPs have relatively non-stigmatising attitudes towards PMH including the acknowledgement of PMH as a medical disorder and the importance of counselling, and screening and management of PMH being the key learning needs for HCPs in Bangladesh. Mixed methods analysis allowed for the triangulation and corroboration of findings from different perspectives enhancing the credibility of these findings. Further qualitative research focusing on poverty, access to ANC, unwanted pregnancies and family dynamics from the

perspective of pregnant women in Bangladesh are needed to fill the gap in our understanding of these themes.

Chapter 10. Discussion & conclusion

10.1 Chapter overview

This PhD research consists of six studies: 1) a meta-analysis of the social determinants of antenatal depression in South Asia, 2) a systematic review of qualitative data exploring perceptions and attitudes around PMH in South Asia, 3) a cross-sectional survey study investigating the prevalence and social determinants of antenatal depression in rural Sylhet, Bangladesh, 4) a cross-sectional survey study investigating the knowledge, attitudes and learning needs of HCPs in Bangladesh, 5) a qualitative study exploring the perceptions and experiences of HCPs in Bangladesh around PMH, and 6) a mixed methods triangulation of the pregnant women's and HCPs' studies.

This chapter aims to provide an overall discussion of the key findings, in light of the theoretical and conceptual framework presented in chapter 4, and highlight the specific contributions of my PhD to the field. Following this, the strengths and limitations of the overall PhD will be discussed. Finally, further research and next steps stemming from this work will be highlighted.

10.2 Key findings and contributions

10.2.1 Contribution 1: An investigation of rural Sylhet, Bangladesh

I was not able to find, through extensive systematic reviews, any other studies which have investigated the prevalence of antenatal depression in rural Sylhet, Bangladesh. Previous studies have focused on women who live in and around the capital of Bangladesh, Dhaka, recruiting women though existing databases or surveillance systems maintained by the non-governmental organisation BRAC (formerly Bangladesh Rural Advancement Committee) (21) and International Centre for Diarrhoeal Disease Research, Bangladesh (ICDDR,B) (34). As highlighted in chapter 1, Sylhet is one of the eight administrative divisions in Bangladesh with a different socioeconomic and demographic environment compared to the other divisions and has poorer maternal health outcomes. Therefore, this PhD addressed this gap in global health literature by exploring PMH in women living in Sylhet, Bangladesh, to determine the prevalence, social determinants and the interventions and practices needed to tackle antenatal depression. This is important in order to reduce health inequalities within Bangladesh.

10.2.2 Contribution 2: High prevalence of antenatal depression in rural Sylhet

The pregnant women's survey study in chapter 6 found a high prevalence of 56.6% of ADS in the MAA health camps/clinics in rural Sylhet. This is higher than prevalence's of 18% (21) and 33% (34) determined in previous studies in rural regions near the capital city of Dhaka. All the studies, including mine, used the EPDS-B scale to measure ADS with the same cut-off scores and the scale was administered through structured interviews with the pregnant women by trained interviewers. Therefore, given the similarities in the study methodologies, the higher prevalence may be due to the differences in the study locations. This may suggest that rural Sylhet has a higher prevalence of antenatal depression, however, this study was only conducted in the MAA health camps and clinics with a small sample size, reducing the generalisability of the findings. More studies in different villages with larger sample sizes will be required to form a clear conclusion and to understand reasons behind the higher prevalence rates. Nevertheless, my PhD research findings highlighted a major concern around antenatal depression in the MAA health camp/clinics regions in rural Sylhet, which may be applicable to other rural regions in Sylhet, and advocates the need for this to be a priority on the public health agenda.

10.2.3 Contribution 3: Social determinants of antenatal depression in rural Sylhet

One of the key objectives of this PhD research was to explore the social determinants of antenatal depression in rural Sylhet, Bangladesh. Both the systematic reviews and original studies contribute to the identification of key social determinants which is of importance for the development of interventions, practices and policies. Male gender preference, particularly the husband's preference, IPV and family support were highlighted to be of most significance. As outlined in the theoretical and conceptual framework, these fall in the more distal social determinant categories of general cultural environment and social and community networks. This suggests that for pregnant women's mental health in Bangladesh, her cultural environment and her societal and familial networks have a larger influence than her lifestyle or individual factors. This is reflective of the strong collectivist culture present within Bangladeshi communities, whereby, attitudes, behaviours and feelings of pregnant women are dominated by the culture, family and community surrounding her. This may also reflect the poor autonomy of women in Bangladesh whereby her decisions and feelings are

informed by her family, community and culture. This is especially highlighted by the fact that perceived husband's male gender preference had a greater influence on antenatal depression than the woman's own preference, further demonstrating the importance of the husband in decision making.

These social determinants are complex and interrelated, sharing similar underlying mechanisms and theories. Socially constructed gender roles favour and elevate the status of men creating gender inequalities in Bangladesh (140). These gender inequalities can be displayed through IPV where men feel they have power over women, pressure on pregnant women to give birth to a son who can provide for the family in the future and lack of family support during pregnancy. Stigma around mental health and women's health also prevents women from speaking up about violence, family issues and their emotions resulting in many pregnant women feeling a lack of support and being vulnerable to depressive symptoms during their pregnancy. Further, motherhood is perceived to be a status raising characteristic for women in many Bangladeshi communities. Therefore, if pregnant women are seen to be struggling with their mental health during their pregnancy, this can question their capability of being a mother, reduce her status and create a stigma against her. Additionally, when placed within the socioeconomic context of rural Bangladesh where there is widespread poverty, lack of education and poor access to antenatal care, these social determinants can be further heightened. For example, women may be more vulnerable to IPV when there are financial pressures within the family and lack of education can reduce women's autonomy (204,205). Poverty may also encourage male gender preference within families given the financial burden that daughters can be in Bangladeshi communities (103,138).

These findings are reflected widely in previous literature. My systematic review and meta-analysis of social determinants of antenatal depression in South Asia suggested that IPV and male gender preference were significant determinants (45). Although my meta-analysis found no significant association between social support and antenatal depression, it did find poor relationship with in-laws to be a significant predictor which may reflect family support which is one element of social support. Social support is a multidimensional factor and can be evaluated using many different parameters such as emotional support, practical support, relationship with family and community support among many others. From my meta-analysis alone, we cannot

exclude social support as no clear conclusion was formed. It may be that the familial aspect of social support plays a bigger role in antenatal depression than other wider dimensions of social support. Gausia et al (34) and Nasreen et al (21) also highlighted IPV, male gender preference and family support to be the most important social determinants of antenatal depression in rural regions of Bangladesh. Therefore, family support is a very important factor, along with IPV and male gender preference, which should be explored in-depth to understand the mechanisms and dimensions involved.

10.2.4 Contribution 4: Positive attitudes of HCPs towards PMH in Bangladesh

My PhD research with HCPs is the first to explore attitudes of such a wide range of HCPs including doctors, medical students, midwives and CHWs towards PMH in Bangladesh. In particular, this PhD fills a large gap in the literature by representing midwives and CHWs who have not been investigated adequately regarding PMH, despite the fact they are key actors in providing antenatal care to pregnant women in rural regions of Bangladesh. Furthermore, this PhD suggests contradictory findings to previous literature where only doctors and nurses were explored. Williams et al (23) was the only study identified in the systematic review (chapter 3) to have explored attitudes of doctors and nurses towards postpartum depression in Bangladesh. The study (23) found largely negative, stigmatising attitudes towards postpartum depression where medical personnel did not take postpartum seriously and mental health counselling was not appreciated. There was also a lack of concern around not having a formal PMH protocol.

Conversely, both studies (chapter 7 and 8) investigating HCPs attitudes in this PhD found largely positive attitudes towards PMH. HCPs, including midwives and CHWs, acknowledged that PMH is a medical disorder that requires medical attention and expressed that counselling is the most effective management method. This may suggest a shift in attitudes towards PMH in recent years which may be due to the prominent use of social media promoting mental health and women's health in Bangladesh. Given that the HCPs that participated in this PhD research were younger than in the previous study, they may be more exposed to social media and hence developed more positive attitudes towards PMH. This study also included midwives and CHWs, whose attitudes were not explored before, who may work more closely with pregnant women in rural regions of Sylhet and provide more emotional support

compared to doctors and nurses. This close one-to-one contact may positively influence their attitudes towards PMH.

Therefore, this PhD research has added to the limited knowledge base on HCPs attitudes towards PMH in Bangladesh and highlighted a potential shift in attitudes which may reflect cultural and societal changes in Bangladesh. This is particularly important in light of the theoretical and conceptual framework (chapter 4), where stigma plays an integral role in the social determinants of antenatal depression. Stigma around PMH can prevent HCPs from delivering the best PMH care for pregnant women and can create a barrier for pregnant women to approach HCPs. Therefore, it is important to understand HCPs attitudes as they will play a key role in developing and delivering the interventions and practices to tackle PMH issues in Bangladesh. This is an area which should be explored in greater depth to develop a better understanding.

10.2.5 Contribution 5: Limited knowledge of PMH screening and management among HCPs in Bangladesh

As well as HCPs' attitudes to PMH, this PhD research also explored HCPs' knowledge around PMH in Bangladesh. While Williams et al (23) explored cultural attitudes, there was no exploration of knowledge levels. Therefore, the findings from this PhD study will be situated in literature from other countries. Both the qualitative and quantitative exploration of HCPs knowledge around PMH in this PhD suggest limited knowledge levels. In particular, HCPs knowledge of screening and risk factors was limited, and most were not educated or experienced with any formal protocols. Nevertheless, HCPs desired more education and training, particularly around screening and managing PMH. This is promising as it indicates that HCPs take PMH seriously and want to provide effective care for which education and training is a necessity.

These findings are reflected in previous literature. McCauley et al (129) found that HCPs in Ghana felt they were not adequately trained to assess and manage maternal mental health. Australian studies by Jones et al (177), Noonan et al (172) and Hauck et al (174) reported knowledge deficits relating to assessment and treatment of PMH and found that midwives did not feel equipped to support pregnant women with PMH issues and desired more education and training to be fully prepared. Here we

see that HCPs limited knowledge around PMH issues, particularly among midwives, is reflected across the globe.

Limited knowledge of PMH may stem from the cultural stigma attached to mental health resulting in it not being effectively integrated into the curriculum for medical or midwifery courses. It is crucial for PMH to be introduced into the educational curriculum and within community-based educational programmes early on and covered in enough detail to equip HCPs with adequate knowledge to be able to assess and manage PMH problems.

10.2.6 Contribution 6: Implications for antenatal health care practice and policy in MAA

This PhD research not only contributes to academic knowledge, but also carries practical contributions in the form of PMH care development, implementation and impact. This PhD has highlighted the importance of, and the need for, integration of PMH care into existing antenatal health care structures and systems. As part of the collaboration with MAA, this learning can be utilised in MAA health camps and clinics in Bangladesh to tackle the high prevalence of antenatal depression found in those regions. The WHO guide (216) for integration of PMH in maternal and child health services outlines an evidence-informed approach to integrating mental health services for women during the perinatal period. This report will be used to guide the integration of PMH care in MAA maternal health services.

A stepped-care approach can be utilised which can result in an efficient use of HCPs and resources within MAA. This approach starts with the promotion of good mental health, which includes the least resource-intensive interventions, and increases to preventative interventions for vulnerable women, treatment for mild-to-moderate conditions and treatment for moderate-to-serious conditions. For effective PMH care, it is essential to create a safe, women-centred (217), and non-stigmatised environment in which pregnant women feel respected and listened to. This was highlighted by HCPs in the interviews conducted in this PhD. This may be achieved by creating a private space within the health camps and clinics for mental health screening and for women to speak about their mental health. By normalising conversations around mental health, pregnant women will be more inclined to speak up during their antenatal check-ups.

Promotion and prevention of PMH conditions includes supporting pregnant women in developing coping strategies, creating an environment that supports mental well-being and building awareness around the risk factors and social determinants. These interventions should target both pregnant women, men and wider family members. Firstly, PMH care screening can be integrated into the antenatal health care protocol using the EPDS-B validated scale which has already been successfully used during the data collection phase. Alongside this, questions around the important social determinants such as IPV, low family support and male gender preference, both the woman's and her husband's, should be included to identify pregnant women who may be at an increased risk of developing PMH issues. Secondly, upstream interventions and policies that target the social determinants should also be developed which include targeting men as key actors. Interventions can include educating men on gender equality and pregnant women's mental health through community-based outreach and actively encouraging men to be involved in perinatal care. Once example can be introducing cash incentives for the family when both mother and father attend antenatal check-ups which would remove one important barrier of missing work for men.

The WHO guide mentions that CHWs and midwives can be trained and supported to promote and prevent PMH conditions (216). This was also reflected in the findings from this PhD where midwives suggest task-shifting as an effective way to utilise the existing workforce. Therefore, HCPs within the MAA team can be given adequate training and education around antenatal depression screening and social determinants to be able to identify women with antenatal depression or those at risk. Education and training were desired by HCPs in this PhD and is key to the effective integration of PMH care into antenatal care.

Educating and training HCPs will also encourage and create opportunities for psychoeducation for pregnant women and their families which is highlighted in the WHO guide (216). MAA already deliver perinatal health education to pregnant women. PMH can be integrated into these education programmes and existing women's groups to empower women to recognise PMH and its symptoms, inform women of the coping strategies and treatment options and reduce stigma surrounding PMH. Given the influence and authority that HCPs carry in Bangladesh, as highlighted in the interviews within this PhD, psychoeducation delivered by MAA HCPs may be taken more seriously by pregnant women. Highlighting the impact of mental health on physical

health and the baby's health can be used as a clinical tactic to engage pregnant women in discussing mental health as indicated by HCPs in the interviews (chapter 8). Families of pregnant women, especially her husband and mother-in-law, should also be given psychoeducation including information around how to support pregnant women during the perinatal period and discuss topics such as male gender preference and IPV. In addition, local pregnant women's groups can be utilised, such as the Participatory Learning and Action Women's groups, which are community-based participatory intervention where local women will meet fortnightly to discuss issues surrounding maternal health and identify problems in the community which they wish to tackle. These groups, where women feel comfortable, can then be targeted with psychoeducation. This will also increase social support for pregnant women who may be experiencing low family support. Thus, psychoeducation is an effective low-resource intervention which can improve PMH knowledge, reduce stigma and target the key social determinants of antenatal depression.

Finally, treatment and management of PMH should be integrated into antenatal care for pregnant women who have screened positive for antenatal depression. HCPs interviewed in this PhD highlighted the importance of psychological interventions, such as counselling, and expressed their willingness to be trained in such. A systematic review of the effectiveness of psychological interventions in the treatment of perinatal depression found that Cognitive Behavioural Therapy (CBT) was the most effective intervention (218). The Thinking Healthy Programme by WHO is an evidence-based approach describing how CHWs can reduce prenatal depression through evidence-based CBT techniques recommended by the Mental Health Gap Action Programme (mhGAP). A study in rural Pakistan found that the integration of CBT based Thinking Healthy Programme intervention into the routine work of CHWs reduced the rate of depression in prenatally depressed women compared to the control group who did not receive CBT (219). This can be replicated within the MAA antenatal services to provide effective management of antenatal depression.

10.3 Strengths and limitations of the overall PhD

This section aims to highlight the key strengths and limitations of the PhD as a whole. Strengths and limitations of individual studies and methodologies are discussed in detail in their respective chapters.

10.3.1 Strengths

This PhD research is a collaborative undertaking with MAA which is a key strength of this work. Collaboration with MAA enabled improved understanding of the study setting, population and context of maternal health in rural Sylhet. This guided the development of the research objectives and data collection materials for which content was validated, translated and piloted by MAA researchers where necessary. The collaboration also adds to the impact of this PhD project as findings and recommendations can be directly used by MAA to further develop their maternal health services in rural regions of Bangladesh. These benefits of collaboration have been highlighted by ESRC (220). Although MAA were collaborators, they did not influence decisions. They contributed and provided advice at various points of the PhD; however, the final decision was taken by me with advice from my supervisors. This, along with the fact that they did not provide any financial contribution, reduces the bias they may have had on the research. As health actors in Bangladesh, MAA may have been understandably perceived as supportive and even trusted by research participants, but that will likely only have enhanced engagement. However, one cannot guarantee that the women weren't influenced by their appreciation of the health camps. Also, many of the HCPs in the survey and interviews were recruited via MAA contacts which may play a role on their experiences and views. However, this PhD was not evaluating MAA or their services, therefore, there it is unlikely for HCPs to bias the results based on their links with MAA.

The mixed methods nature of this PhD research adds credibility and integrity to the findings as they are corroborated and triangulated with each other. Not only were the quantitative findings integrated with qualitative findings, but findings from pregnant women were also triangulated with findings from HCPs which allows for a holistic understanding of PMH from two different, but very important perspectives. It also allowed for a diversity of views as the quantitative findings help identify associations and the qualitative findings helped to understand some of the underlying mechanisms and reasons. Therefore, employing mixed methods approaches enhances the novelty and importance of this PhD project.

Situating this PhD research within the Dahlgren-Whitehead social determinants of health model, with the addition of key theories, has facilitated the development of a

unique insight into antenatal depression in rural Sylhet, Bangladesh. It enables us to explore the different levels of social determinants, from a personal to a structural and cultural level and establish the complex relationships between these. This is particularly useful when trying to develop tailored holistic interventions for pregnant women in rural Bangladesh as one can identify at which level interventions will be most beneficial. In the case of this PhD, it is evident from the findings that the outer layers consisting of culture, living and working conditions and social networks seem to be of most importance.

All of the data collection materials, i.e. the surveys and topic guide, were developed after extensive literature reviews using validated scales (EPDS-B, DUSOCS and WHO IPV questionnaire) and piloted in appropriate settings. This reduced measurement bias of the surveys and ensured that flow and clarity of the interview topic guide was established in a similar population group. It also aided the comparison of findings from previous literature which used the same tools to be able to present a critical discussion of the findings.

My two systematic reviews (chapter 2 and 3) are published in well-established journals, *PLoS One* (Appendix 1) and *BMC Pregnancy and Childbirth* (Appendix 2), respectively. This adds to the credibility and rigour of the reviews, upon which the theoretical & conceptual framework and data collection materials were developed on, as they have been peer-reviewed and undergone critique. The meta-analysis also received first prize abstract at the Society for Social Science and Population Health annual conference in 2021 which further emphasises the strengths of the review. Furthermore, the cross-sectional survey study investigating social determinants of antenatal depression in rural Sylhet (chapter 6), has also been published in the *International Journal of Environmental Research and Public Health* (Appendix 3) in the maternal perinatal mental health special issue in January 2023 (47).

10.3.2 Limitations

Data collection for this PhD started from February 2021 which is after the initial waves of COVID-19 and the major lockdown restrictions. COVID-19 has had devastating consequences in Bangladesh including a decline in household income and food security in both urban and rural regions of Bangladesh during the lockdown (221). A study found that 62% of respondents in rural areas have suffered an income drop

(222). There was also an increase in unemployment with 55% of main earners in rural areas rendered economically inactive (222). In rural regions, the main source of income comes from informal sectors such as rickshaw driving, farming and fishing which were all negatively impacted by COVID-19. Due to lockdowns and restrictions, people could not ride rickshaws and people were not coming to the markets so farming produce and fish would not sell (223). Financial problems sparked an increase in IPV with 55% of women in rural areas reported having experienced such abuse; many cases of IPV would have gone unreported (221). These consequences of COVID-19 and the restrictions may have had unprecedented effects on the mental health and well-being of women in rural Sylhet prior to the data collection period. This may have skewed the prevalence of antenatal depression within the population of interest who were overcoming the effects of COVID-19. Therefore, it is important to consider the findings from this PhD in the light of COVID-19 which should be further explored in future studies. Nevertheless, this PhD still highlights the important relationships between IPV, family support, male gender preference and antenatal depression which exist despite COVID-19.

Data collection was carried out via surveys and interviews with the participants which can be subject to self-report bias. Especially, sensitive topics such as mental health, family relationships and IPV can incite strong emotions and views in participants and, therefore, they are less likely to give accurate or honest answers or provide responses they feel are socially acceptable. Nevertheless, collaborating with the MAA researchers, who have a good rapport with the pregnant women and HCPs in those communities, for data collection was a great way to reach out to participants and create a comfortable space for participants to provide their responses. The online nature of the HCPs' survey also promoted anonymity which reduced self-report bias. Interviews were conducted in Bengali as it enabled HCPs to express more during interviews as many of them felt more comfortable talking in Bengali. Therefore, efforts were taken throughout the PhD to minimise self-report bias.

This PhD research was conducted in one region of rural Sylhet, Bangladesh, using purposive or convenience sampling for the recruitment of participants. Therefore, whilst the findings may have high internal validity, caution should be taken when generalising the findings to the whole population due to poor external validity. Furthermore, sample sizes for the surveys were relatively small which can reduce the

statistical power and, thus, limits the interpretation and generalisability of the findings. Nevertheless, whilst I conducted inferential statistics and offered interpretations of associations, which provide useful contributions when situated within the literature, the importance of this PhD lies in the novel mixed methods descriptive exploration of the social determinants and HCPs' knowledge and attitudes surrounding PMH. These findings are applicable to MAA for development of their ANC services.

Due to the scope of this PhD, it was not possible to include perspectives of men, in particular the husbands of the pregnant women, within the findings. This does exclude a very important perspective given the social determinants identified, such as husbands' male gender preference, IPV and low family support, which are highly influenced by men. Understanding men's role and perspectives would enable us to tailor interventions to include them as well as targeting pregnant women.

10.4 Further research & next steps

10.4.1 Qualitative studies exploring pregnant women's experiences and perceptions of PMH in Bangladesh

Due to COVID-19, it was not possible to conduct the semi-structured interviews with pregnant women as had originally been proposed in my funding application (see COVID-19 impact statement). Therefore, future research and next steps stemming from this PhD research would be to conduct these qualitative interviews with pregnant women and their husbands and families in rural Sylhet, Bangladesh. This would provide a more in-depth understanding of antenatal depression from the perspective of pregnant women and their husbands and be able to establish some of the underlying mechanisms of the social determinants identified in the quantitative survey study. Key topics in the topic guide would include attitudes towards mental health, knowledge of PMH, family dynamics and relationships, dimensions and source of support during pregnancy, experiences of IPV and gender preference and discordance within the family.

10.4.2 Longitudinal studies with larger sample sizes

To gain a better understanding of causal relationships between social determinants such as IPV, family support and gender preference, longitudinal studies should be conducted. This can be achieved by recruiting a cohort of pregnant women

in rural Bangladesh who do not have antenatal depression and assessing the social determinants at the start of their pregnancy. The cohort can then be followed throughout their pregnancy prospectively and screened for antenatal depression using the EPDS-B at various points. It will then be possible to explore the impact of the social determinants, measured at the start of the pregnancy, on antenatal depression. Since antenatal depression developed after the measurement of the social determinants, some level of causal relationship can be explored unlike in cross-sectional studies.

It is important, however, for this type of study, to ensure the cohort of pregnant women include some that are exposed to the social determinants and some that are not. For example, some pregnant women and their husbands should have a male gender preference, and some should state they have low family support. However, with a number of different social determinants this becomes more complex as associations can become confounded or altered. It is ideal for the cohort of pregnant women be similar in all aspects expect for one social determinant. Therefore, longitudinal studies are more complex, time consuming and resource intensive to carry out given the nature of the research question.

10.4.3 Implementation of recommendations in MAA health camps

My collaboration with MAA is a long term one in which I hope to be able to continue my work with them to implement some of the recommendations and practices based on the findings from this PhD.

10.5 Conclusions

This PhD aimed to explore the social determinants and perspectives around PMH, with a focus on antenatal depression, in women living in rural Sylhet, Bangladesh, to allow for better development of care for underserved women in rural Bangladesh in collaboration with MAA. The aim and objectives were met, and some key findings were highlighted.

This research has made an important contribution to the understanding of antenatal depression in rural Sylhet being the first known study to be conducted in this region. The prevalence of antenatal depression was found to be higher than those reported in other regions of Bangladesh such as Dhaka, highlighting the need for

further investigation within rural Sylhet, wider screening programmes and tailored interventions. IPV, low family support and male gender preference were found to be significant determinants of antenatal depression which should be considered when developing tailored interventions and psychoeducation programmes. The complex interlinks between the social determinants, underlined by stigma, social construction of gender and collectivist cultures, as highlighted in the theoretical and conceptual framework, facilitated the development of a holistic insight into antenatal depression. Contradictory to findings in the literature, HCPs in this PhD displayed positive attitudes towards PMH indicating a promising way forward for the integration of PMH into ANC and reducing PMH stigma in Bangladeshi communities. Nevertheless, HCPs require further training and education on screening and management as indicated by the low levels of knowledge identified in this PhD.

This PhD not only contributes to the academic knowledge base and understanding of antenatal depression in Bangladesh, but it also has practical contributions through informing recommendations for the integration of PMH in MAA antenatal services. Integration involves planning the budget and activities, preparing HCPs and structures, providing all women with the support, and monitoring & evaluating the intended outcomes. Based on the findings from this PhD, key activities will involve screening of antenatal depression, training and education of HCPs, psychoeducation for women and families and CBT based management.

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Appendices

Appendix 1 – Social determinants of antenatal depression and anxiety among women in South Asia: A systematic review and meta-analysis

PLOS ONE

RESEARCH ARTICLE

Social determinants of antenatal depression and anxiety among women in South Asia: A systematic review & meta-analysis

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Abstract

Background

Pregnancy is a time of major psychological changes making pregnant women more susceptible to depression and anxiety. Prevalence is higher among women living in Bangladesh, India and Pakistan, compared to high-income countries, due to poor understanding and lack of mental health integration within antenatal care. Antenatal depression/anxiety is associated with adverse outcomes including postnatal depression, low birth weight and impaired fetal development. Existing systematic reviews provided only limited information on the social determinants of antenatal depression/anxiety in these South Asian countries.

Objective

This review aimed to identify, synthesise and appraise the evidence on the social determinants associated with antenatal depression and anxiety in women living in Bangladesh, India and Pakistan.

Methods

We searched five databases (MEDLINE, Embase, PsycINFO, Scopus, Web of Science) and PROSPERO. Observational studies published between 1st January 2000 and 4th January 2021 were included if they were in the English language, used validated tools for measuring depression/anxiety in pregnant women and reported statistical associations or raw numbers. Summary estimates were obtained using random-effects model. Heterogeneity and publication bias was measured using the I^2 statistic and Egger's test, respectively. This review was registered on PROSPERO (reference: CRD42020167903).

Results

We included 34 studies (with 27,379 women). Meta-analysis of Adjusted Odds Ratios (AOR) found that Intimate partner violence (AOR 2.48, 95% CI 1.41–4.33), unplanned pregnancy (AOR 1.53, 95% CI 1.28–1.83), male gender preference (AOR 3.06, 95% CI 1.40–

6.72) and poor relationship with in-laws (AOR 2.69, 95% CI 1.25–5.80) were significantly associated with antenatal depression/anxiety.

Conclusion

The review identified a complex range of social determinants of antenatal depression and anxiety in Bangladesh, India and Pakistan. Screening tools to identify pregnant women at high risk should be integrated within antenatal care to prevent adverse outcomes. Knowledge of these social determinants will inform the development of such screening tools and interventions.

Introduction

Pregnancy is a time of major physiological, hormonal, and psychological changes placing women at an increased risk of emotional changes during and immediately after pregnancy [1]. Maternal mental health is recognised as a worldwide public health issue, with approximately 10% of pregnant women and 13% of women who have just given birth experiencing some sort of mental health condition, particularly depression and comorbid anxiety [2]. Evidence suggests that two thirds of cases of postnatal depression actually present antenatally, before childbirth [3, 4], making the antenatal period a key stage to target in interventions. Depression and anxiety during pregnancy results in adverse outcomes in the mother and child including increased risk of postnatal depression, low birth weight and impaired fetal development [5].

Prevalence of antenatal depression ranges from 7 to 15% in high-income countries (HICs) and 19 to 25% in Low- and Middle-Income Countries (LMICs) [6]. These disparities result from the lack of mental healthcare integration into antenatal care, poor understanding and acceptance of mental health conditions and higher socioeconomic deprivation in LMICs [7]. It is likely that these reported percentages in LMICs are underestimates due to the stigma attached to mental health preventing people from understanding, acknowledging and sharing these issues within these populations [8]. This is particularly prominent among women who are often victims of gender-based violence, experience lack of family planning and low support leading to depression during pregnancy [7]. These are commonly reported social determinants of maternal depression [7, 9]. Social determinants are "the conditions in which people are born, grow, live, work and age". They are crucial to improving health since they are "mostly responsible for health inequalities" [10].

Bangladesh, India and Pakistan are LMICs which form a major part of the Indian subcontinent of South Asia. Although there is heterogeneity within and between these countries, they share cultural and social similarities which are reflected in a degree of commonality in the challenges and social inequalities faced by women living in these countries. For this reason, they are often studied together to enhance understanding of health phenomena within South Asia [11].

A longitudinal study carried out in 2008–2009 in Bangladesh found the prevalence of antenatal depression was 18% and antenatal anxiety was 29.4% [9]. Studies in India have reported a wide-ranging prevalence of antenatal depression between 9.8 and 36.7% [12–14] and a prevalence of 55.7% for antenatal anxiety [15]. In Pakistan, the prevalence of antenatal depression ranges from 18% [16] to 39.5% [17] and the prevalence of antenatal anxiety is approximately 34.5% [18]. The varying prevalence rates found between and within these countries may be a

result of methodological differences in sample size, sample characteristics and region of study. It may also highlight the heterogeneity between the countries.

Existing systematic reviews investigating risk factors of antenatal depression/anxiety [6, 19, 20] do not include meta-analyses to determine pooled summary estimates of association. Only two reviews included any studies in Bangladesh, India or Pakistan [6, 20], however, they did not capture the range of studies published in these countries and only included papers published to 2015. To our knowledge, there are no known systematic reviews and meta-analyses investigating the social determinants of antenatal depression in Bangladesh, Indian and Pakistan where women face increased vulnerability to mental health issues and social inequalities. Understanding social determinants of antenatal depression and anxiety will enable holistic care for pregnant women in these countries where mental health is given less attention [21]. This understanding will assist in the development and implementation of tailored interventions and programmes for pregnant women to reduce symptoms of depression and anxiety during pregnancy and, consequently, positively impacting health after childbirth. This study aimed to identify, synthesise and appraise the available evidence on social determinants of antenatal depression and anxiety in women living in Bangladesh, India and Pakistan, and to identify which determinants have the most impact on antenatal depression and anxiety.

Methods

Methods and reporting were conducted in accordance with the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) guidelines (S1 Table). Observational studies investigating social determinants of antenatal depression or anxiety in any region of Bangladesh, India and Pakistan published in, or translated into, the English language since 1st January 2000 to 4th January 2021 were included in this review. To be included, studies had to use validated tools for measuring antenatal depression/anxiety and either reported the odds ratios (OR) and 95% Confidence Intervals (95% CI) or provide sufficient raw data for ORs to be calculated. Randomised controlled trials and studies using only a specific sub-group of pregnant women were excluded.

A systematic literature search was conducted via the electronic databases MEDLINE, Embase, PsycINFO, Scopus, Web of science using keywords and MeSH headings developed with an information specialist in accordance with the PICOS framework (see S1 Fig for full search strategy). PROSPERO was also searched to identify if there were any similar systematic reviews conducted already. Search results from the electronic databases were exported into the reference manager Zotero and screened independently by two investigators. Zotero provides a platform to export studies from databases, automatically remove duplicates and to conduct screening of titles. Where there were disagreements between the two reviewers regarding eligibility, discussions would occur to resolve it, or a third reviewer was included, however, this was not required in this review. Grey literature was searched using the Open Grey database and also via charities and non-governmental organisations such as Building Resources Across Communities (BRAC), UNICEF, National Health Mission India and International Centre for Diarrhoeal Disease Research Bangladesh. Supplementary searches included hand searching study reference lists included from databases and citation searches using Google Scholar. Authors were contacted when full texts of studies were not found, or additional data were required for inclusion in the meta-analysis. Database searches were completed on 31st January 2020 and updated on 4th January 2021.

Data extraction was carried out independently by two researchers using a standardised protocol (S2 Table). The Newcastle-Ottawa Scale adapted for cross-sectional studies which assesses information bias, selection bias and confounding bias (S3 Table; S2 Fig) was used to assess study quality.

Data synthesis and statistical analysis

Data synthesis was grouped by social determinants. Only those social determinants that were investigated by three or more studies were included in the review as fewer than three was considered as insufficient evidence for narrative synthesis, as recommended by previous systematic reviews [19, 22]. A supplementary table (S4 Table) reports the results for social determinants investigated by only one study. Although it would be preferable to investigate depression and anxiety individually, they were grouped together under one outcome in the studies. As they are comorbid common mental disorders and there was only one included study with anxiety as a separate outcome, we combined the outcomes depression and anxiety. This approach has been practiced in previous studies [23, 24]. We considered this to be the most appropriate outcome for this synthesis given the number of studies.

For determinants where pooling of the ORs were appropriate, meta-analyses were performed using the random-effects model [25], regardless of the heterogeneity. This is a slight deviation from the protocol as after screening of full texts we expected heterogeneity between the studies, thus, opted for random-effects model from the start. Heterogeneity was measured using the I^2 statistic (0–100%) [26], with a threshold of >75% representing considerable heterogeneity as stated in the Cochrane Handbook [27]. Unadjusted and adjusted ORs were pooled separately to account for the confounding factors and both were presented to investigate the differences (see S5 Table for factors included in the included studies adjusted models). Sensitivity analysis was performed by removing studies that were assessed to be of low quality. The results were statistically significant when two-sided p-values were less than 0.05. Publication bias was tested for using the Egger's test ($p<0.05$ indicates the existence of publication bias) [28]. The statistical analyses were conducted using STATA V.16.1. The review protocol was registered on the PROSPERO database of systematic reviews (reference CRD42020167903).

Results

The searches identified 3,372 studies; following deduplication, 1,987 studies remained for screening. Following screening and supplementary searches, a total of 34 studies met the inclusion criteria and were included in this review (Fig 1).

The sample sizes ranged from 100 [29] to 14,629 [30] pregnant women; all the studies were cross-sectional and the majority (n = 30) have been published since 2010. Studies were predominately conducted in Pakistan (n = 20) [4, 16–18, 29, 31–45], followed by India (n = 11) [12–15, 46–52] and Bangladesh (n = 3) [9, 30, 53]. The majority of studies included samples of pregnant women recruited from hospitals (n = 24) [13–15, 18, 29, 31–38, 40–46, 48, 49, 51, 52], with the remaining being community-based samples (n = 10) [4, 9, 12, 16, 17, 30, 39, 47, 50, 53]. Most studies investigated depression only (n = 24) [4, 12–14, 17, 29–31, 35–39, 42, 43, 45–53], followed by co-morbid depression and anxiety (n = 9) [9, 16, 18, 32–34, 40, 44] and anxiety alone (n = 1) [15]. Quality scores ranged from one to ten out of a possible score of ten. Overall, 47% (n = 16) of the studies scored medium; 32% (n = 11) of the studies scored high; 21% (n = 7) of the studies scored low (Table 1, S3 Table).

The review identified 12 social determinants: maternal age, maternal education, maternal occupation, household income, husband's education, husband's occupation, relationship with husband, intimate partner violence (IPV), pregnancy intentions, relationship with in-laws, social/family support and gender preference. Table 2 displays the overall trend of association for the social determinants.

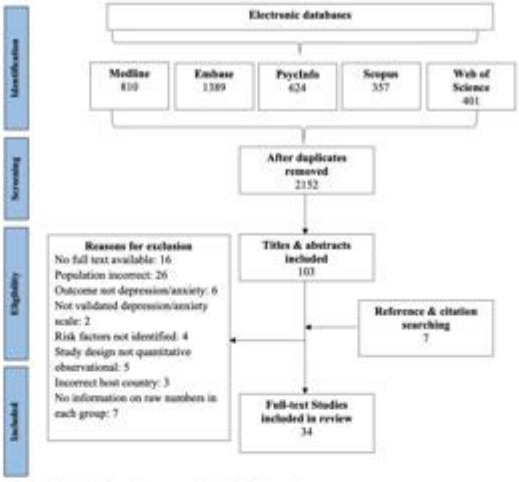


Fig 1. PRISMA flow diagram of included studies.

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Maternal age

Twenty-two studies presented results on the age of the pregnant women (Table 3). The large variation in the reporting of maternal age made it unsuitable for meta-analysis. Six studies found women in the higher maternal age groups had significantly higher odds of antenatal depression [17, 30, 43] and co-morbid antenatal depression and anxiety [18, 26, 47] compared to women in the lower maternal age categories. Similarly, one study [31] found women with antenatal depression/anxiety had significantly higher mean maternal age compared to women without antenatal depression/anxiety. Conversely, three studies [32, 40, 44] found higher maternal age was significantly associated with lower comorbid antenatal depression & anxiety. Twelve studies [4, 15, 16, 35, 38, 46–48, 50–53] found no significant association. Nevertheless, despite non-significance, seven out of these twelve studies found an effect size that indicated higher maternal age groups had higher odds of antenatal depression/anxiety [4, 15, 16, 38, 46, 51, 53]. Whereas five studies found effect sizes that suggested the opposite trend [35, 47, 48, 50, 52]. Therefore, majority of the studies (n = 14) suggest a trend in higher risk with of antenatal depression/anxiety with higher maternal age.

Maternal education/literacy

Twenty-one studies presented results on maternal education and literacy (Table 4). Due to the variability in the reporting of education, meta-analysis was not appropriate. Nine studies found that women with the higher education/literacy levels had significantly lower odds of antenatal depression [9, 29–31, 39, 43, 45, 50, 52] compared to women with lower education/literacy levels. Conversely, two studies showed women with higher education/literacy levels had higher odds of antenatal depression [48] and co-morbid antenatal depression and anxiety [16]. Ten studies found no significant association [4, 15, 18, 33, 38, 44, 46, 47, 51, 53]. Out of these ten studies, nine studies [4, 15, 18, 33, 44, 46, 47, 51, 53] found a trend that indicated women with lower education/literacy levels had lower odds of antenatal depression/anxiety.

Table 1. Descriptive summary of the 34 included studies.

Author, publication year (reference number)	Country	Sample size	Recruitment source	Outcome	Ascertainment of outcome (Cut-off score)	Quality score*
Ajinkya et al, 2013 [14]	India	185	Hospital	Depression	BDI ≥ 17	4
Ali et al, 2012 [18]	Pakistan	165	Hospital	Anxiety & Depression	HADS ≥ 8	6
Ayaz et al, 2019 [29]	Pakistan	100	Hospital	Depression	BDI ≥ 16	3
Ayyub et al, 2018 [17]	Pakistan	367	Community	Depression	Urdu EPDS ≥ 12	10
Babu et al, 2018 [47]	India	823	Community	Depression	Kannada Kessler Psychological Distress Scale (K-10) ≥ 10	8
Bavle et al, 2016 [48]	India	318	Hospital	Depression	Hindi EPDS ≥ 10	4
Dahiya et al, 2020 [52]	India	200	Hospital	Depression	EPDS	6
Din et al, 2016 [32]	Pakistan	230	Hospital	Anxiety & Depression	DASS-42	5
Gausia et al, 2009 [53]	Bangladesh	346	Community	Depression	Bangla EPDS ≥ 10	8
George et al, 2016 [12]	India	202	Community	Depression	CIS-R	10
Ghaffar et al, 2017 [33]	Pakistan	750	Hospital	Anxiety & Depression	Urdu HADS	8
Goyal et al, 2020 [49]	India	281	Hospital	Depression	DSM-5	6
Gul et al, 2017 [34]	Pakistan	500	Hospital	Anxiety	AKUADS > 19	7
				Depression		
Hegde et al, 2013 [13]	India	253	Hospital	Depression	EPDS ≥ 13	2
Humayun et al, 2013 [35]	Pakistan	506	Hospital	Depression	EPDS ≥ 10	5
Imran et al, 2010 [36]	Pakistan	213	Hospital	Depression	Urdu EPDS > 12	6
Jafri et al, 2017 [37]	Pakistan	300	Hospital	Depression	Hamilton Rating scale for Depression > 7	2
Jamal et al, 2018 [38]	Pakistan	197	Hospital	Depression	BDI ≥ 17	5
Karmaliani et al, 2009 [16]	Pakistan	1368	Community	Anxiety & Depression	AKUADS ≥ 13	6
Maselko et al, 2018 [39]	Pakistan	115	Community	Depression	PHQ-9	7
Mir et al, 2012 [31]	Pakistan	340	Hospital	Depression	AKUADS	9
Nasreen et al, 2011 [9]	Bangladesh	720	Community	Anxiety & Depression	STAI ≥ 45	10
					EPDS ≥ 10	
Nath et al, 2019 [15]	India	380	Hospital	Anxiety	Pregnancy-related thoughts (PRT) scale	8
Niaz et al, 2004 [40]	Pakistan	200	Hospital	Anxiety	HADS >7	2
				Depression		
Rabia et al, 2017 [41]	Pakistan	520	Hospital	Anxiety	HADS >8	5
				Depression		
Rahman et al, 2003 [4]	Pakistan	632	Community	Depression	SCAN	7
Sabir et al, 2019 [42]	Pakistan	450	Hospital	Depression	Goldberg's depression scale > 21	4
Safi et al, 2013 [43]	Pakistan	300	Hospital	Depression	CES-D > 15	3
Sheeba et al, 2019 [46]	India	280	Hospital	Depression	EPDS >13	8
Shehroz et al, 2019 [44]	Pakistan	200	Hospital	Anxiety	HADS	1
				Depression		
Shidhaye et al, 2017 [50]	India	302	Community	Depression	EPDS >12	10
Srinivasan et al, 2015 [51]	India	100	Hospital	Depression	EPDS ≥ 10	6
Surkan et al, 2018 [30]	Bangladesh	14,629	Community	Depression	PHQ & CES-D	9
Zia et al, 2018 [45]	Pakistan	907	Hospital	Depression	BDI	2

*Quality assessment scores from the adapted Newcastle-Ottawa Scale reported in S3 Table

BDI = Beck Depression Inventory, HADS = Hospital Anxiety Depression Scale, EPDS = Edinburgh Postnatal Depression Scale, K-10 = Kessler Psychological Distress Scale, DASS = Depression Anxiety and Stress Scale, CIS-R = Clinical Interview Schedule Revised, DSM-5 = Diagnostic and Statistical Manual of Mental Disorders, AKUADS = Aga Khan University Anxiety Depression Scale, PHQ = Patient Health Questionnaire, STAI = The State-Trait Anxiety Inventory, SCAN = Schedules for Clinical Assessment in Neuropsychiatry, CES-D = Centre for Epidemiological Studies Depression Scale

Outcome of Anxiety & Depression indicates that the measurement scale used does not discern between anxiety and depression and are used to measure both under as common mental health disorders

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Table 2. Summary of social determinants identified in the review.

Social determinant	Total number of studies	Number of studies included in univariate meta-analysis	Pooled OR (95% CI)	I^2	Number of studies included in meta-analysis of AOR	Pooled AOR (95% CI)	I^2
Maternal age	22	-	-	-	-	-	-
Maternal education/ literacy	21	-	-	-	-	-	-
Maternal occupation	16	15	1.03 (0.76–1.41)	82%	2	1.33 (0.55–3.21)	73%
Household income	13	-	-	-	-	-	-
Husband's education	7	-	-	-	-	-	-
Husband's occupation	7	7	2.27 (0.96–5.41)	90%	2	0.85 (0.35–2.08)	83%
Relationship with husband	13	11	3.04 (1.63–5.68)*	95%	5	2.33 (0.90–6.02)	89%
Intimate Partner Violence	15	15	3.58 (2.57–5.00)*	61%	9	2.48 (1.41–4.33)*	71%
Unplanned pregnancy	20	19	1.88 (1.44–2.45)*	91%	5	1.53 (1.28–1.83)*	19%
Relationship with in-laws	7	7	4.23 (3.11–5.74)*	49%	3	2.69 (1.25–5.80)*	54%
Social/family support	10	9	2.14 (1.60–2.85)*	60%	4	1.42 (0.78–2.62)	75%
Male gender preference	6	6	1.84 (1.44–2.36)*	0%	3	3.06 (1.40–6.72)*	61%

*Statistically significant at 0.05 significance level

OR = Odds ratio

AOR = Adjusted Odds Ratio

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Maternal occupation

Sixteen studies presented results on maternal occupation. Fifteen met random-effects meta-analysis criteria. The pooled OR (Fig 2) for antenatal depression/anxiety in unemployed women compared to employed was 1.03, (95% CI 0.76–1.41; I^2 = 82%; n = 15 studies) and the pooled AOR (Fig 3) was 1.33, (95% CI 0.55–3.21; I^2 = 73%, n = 2 studies). Both the pooled unadjusted and AORs indicate no significant association between antenatal depression/anxiety and maternal occupation. The one study, George et al [12], which was not eligible for inclusion in the meta-analysis also reported no significant association.

Household income

Thirteen studies analysed household income (Table 5). Due to the variability in the income categories, meta-analysis was not appropriate. Three studies [33, 38, 43] found that women who live in households with lower levels of income had higher odds of antenatal depression compared to women living in households with higher levels of income. Similarly, a study [32] which analysed the mean monthly household income found that mean income was lower in those women who had co-morbid antenatal depression and anxiety compared to those women without. Nine studies found no significant association [4, 31, 36, 42, 47, 48, 50–52]. Of these nine studies, five studies [4, 47, 48, 51, 52] suggest a trend in reduced odds of antenatal depression/anxiety with lower household income levels, while four studies [31, 36, 42, 50] suggest the opposite trend.

Table 3. Association between antenatal depression/anxiety and maternal age in the included studies.

Author & study year	Maternal age comparison groups	OR (95% CI)[a]	AOR (95% CI)[a]
Ali et al 2012 [18]	<30 & ≥30	3.10 (1.29–7.42)*	3.54 (1.12–10.12)*
Ayyub et al 2018 [17]	15–20 & ≥32	3.03 (1.20–7.15)*	2.77 (1.13–6.82)*
Babu et al 2018 [47]	<25 & ≥25	1.34 (0.78–2.30)	0.73 (0.42–1.27)
Bavle et al 2016 [48]	<25 & ≥25	0.93 (0.41–2.08)	-
Dahiya et al 2020 [52]	<30 & ≥30	0.22 (0.03–1.68)	-
Gausia et al 2009 [53]	<35 & ≥35	1.19 (0.62–2.29)	-
Ghaffar et al 2017 [33]	<35 & ≥35	1.23 (1.13–1.62)*	-
Humayun et al 2013 [35]	<20 & ≥20	0.37 (0.05–3.00)	-
Jamal et al 2018 [38]	<25 & ≥25	1.43 (0.46–4.40)	-
Karmaliani et al 2009 [16]	<25 & ≥25	1.09 (0.83–1.44)	-
Nasreen et al 2011 [9]	<20 & ≥35	-	3.00 (1.12–8.01)*
Nath et al 2019 [15]	<20 & ≥20	1.25 (0.75–1.07)	-
Rahman et al 2003 [4]	<30 & ≥30	1.41 (0.94–2.11)	-
Safi et al 2013 [43]	<20 & ≥ 20	2.00 (1.10–3.62)*	-
Sheeba et al 2019 [46]	<20 & ≥20	1.16 (0.67–1.99)	-
Shidhaye et al 2017 [50]	<20 & ≥20	0.73 (0.39–1.35)	-
Srinivasan et al 2015 [51]	<20 & ≥20	1.39 (0.61–3.16)	-
Surkan et al 2018 [30]	<20 & ≥20	1.15 (1.03–1.30)*	1.38 (1.12–1.70)*
Author & Study year	**Mean maternal age (years) in depressed group**	**Mean maternal age (years) in non-depressed group**	**P-value**
Din et al 2016 [32]	23.30	24.44	0.038*
Mir et al 2012 [31]	26.30	24.30	0.003*
Niaz et al 2004 [40]	25.15	27.18	<0.05*
Shehroz et al 2019 [44]	25.15	27.18	<0.05*

[a]Lower age group is the reference group

OR = Odds Ratio

AOR = Adjusted Odds Ratio

*Statistically significant at 0.05 significance level

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Husband's education/literacy

Seven studies presented results on husband's education/literacy (Table 6). Due to the variability in the categories for education, meta-analysis was deemed inappropriate. Two studies [4, 39] found that women whose husbands were illiterate had higher odds of antenatal depression compared to women whose husbands were literate (Table 5). Five studies found no significant association [15, 29, 45–47], however, they all suggest a trend in higher odds of antenatal depression in women with husband's who have lower levels of education.

Husband's occupation

Seven studies analysed husband's occupation; all were eligible for inclusion into the meta-analysis. The pooled OR (Fig 4) for antenatal depression/anxiety in women whose husbands were unemployed compared to women whose husbands were employed was 2.27, (95% CI 0.96–5.41; $I^2 = 90\%$; n = 7 studies). The pooled AOR (Fig 5) was 0.85, (95% CI 0.35–2.08; $I^2 = 83\%$; n = 2 studies). There was evidence of publication bias (p = 0.0001) in the adjusted analysis. Both the pooled unadjusted OR and AOR suggest no significant association between antenatal depression/anxiety and husband's occupation.

Table 4. Association between antenatal depression/anxiety and maternal education/literacy in the included studies.

Author & study year	Maternal education/literacy level	OR (95% CI)	AOR (95% CI)
Ali et al 2012 [18]	Graduate and above	REF	-
	Up to intermediate	0.87 (0.38–2.00)	
Ayaz et al 2019 [29]	College or university	REF	-
	Uneducated	2.12 (2.02–3.68)*	
Babu et al 2018 [47]	Attended college	REF	REF
	Did not attend college	0.70 (0.42–1.16)	0.81 (0.46–1.42)
Bavle et al 2016 [48]	Attended college	REF	-
	Did not attend college	0.46 (0.24–0.89)*	
Dahiya et al 2020 [52]	Illiterate	REF	-
	Literate	0.25 (0.11–0.58)*	
Gausia et al 2009 [53]	>5 years of education	REF	REF
	0–5 years of education	1.57 (1.01–2.44)*	0.99 (0.58–1.67)
Ghaffar et al 2017 [33]	No education	REF	-
	Primary	1.40 (0.70–2.12)	
	Secondary	1.03 (0.55–1.14)	
	Tertiary	1.19 (0.62–1.03)	
Jamal et al 2018 [38]	Intermediate/graduate and above	REF	-
	Primary and below	1.05 (0.47–2.37)	
Karmaliani et al 2009 [16]	None	REF	-
	1–9 years	1.52 (1.03–2.25)*	
	10 years or more	2.01 (1.30–3.10)*	
Maselko et al 2018 [39]	None	REF	-
	Primary	0.94 (0.62–1.42)	
	Secondary	0.44 (0.30–0.64)*	
	Tertiary	0.27 (0.17–0.44)*	
Mir et al 2012 [31]	Literate	REF	REF
	Illiterate	2.50 (1.56–4.03)*	1.83 (1.08–3.08)*
Nasreen et al 2011 [9]	Illiterate	-	REF
	Literate		0.59 (0.37–0.95)*
Nath et al 2019 [15]	Attended college	REF	-
	Did not attend college	0.86 (0.52–1.42)	
Rahman et al 2003 [4]	Literate	REF	-
	Illiterate	0.80 (0.60–1.00)	
Safi et al 2013 [43]	No education	REF	-
	Intermediate and above	0.18 (0.10–0.34)*	
Sheeba et al 2019 [46]	Attended college	REF	-
	Did not attend college	0.71 (0.40–1.25)	
Shidhaye et al 2017 [50]	Illiterate/completed primary	REF	-
	Junior college or above	0.30 (0.10–0.80)*	
Srinivasan et al 2015 [51]	$\leq 10^{th}$ class	REF	-
	11th -12th class	0.78 (0.28–2.22)	
	Graduate and above	1.32 (0.51–3.46)	
Surkan et al 2018 [30]	≤10 years in education	REF	REF
	1–9 years	1.82 (1.37–2.40)*	1.45 (1.07–1.98)*
	None	2.87 (2.17–3.81)*	1.50 (1.05–2.15)*
Zia et al 2018 [45]	College or university	REF	-
	Uneducated	2.12 (2.0.2–3.63)*	

(*Continued*)

Table 4. (Continued)

Author & study year	Mean years in education in women with depression	Mean years in education in women without depression	P-value
Shehroz et al 2019 [44]	6.9	6.5	>0.05

REF = reference group
OR = Odds Ratio
AOR = Adjusted odds ratio
*Statistically significant at 0.05 significance level

https://doi.org/10.1371/journal.pone.0263760.t004

Relationship with husband

Thirteen studies analysed a women's relationship with her husband. Relationship with husband was mostly reported as a binary variable: poor vs good relationship, but two studies, Mir et al [31] and Srinivasan et al [51] reported it as a continuous variable using a martial satisfaction scale. Both found that the mean marital satisfaction score was significantly lower in women with antenatal depression compared to women without antenatal depression. The other 11 studies were eligible for inclusion in the meta-analysis. The pooled OR (Fig 6) for antenatal depression/anxiety in women who report a poor relationship with their husband compared to women who report a good relationship was 3.04, (95% CI 1.63–5.68; I^2 = 95%, n = 11 studies). The pooled AOR (Fig 7) was 2.33, (95% CI 0.90–6.02; I^2 = 89%, n = 5 studies). Publication bias was evident from the Eggers test (p = 0.0075) for both analyses. The pooled unadjusted OR suggests women who report a poor relationship have significantly higher odds of antenatal depression/anxiety compared to women who report a good relationship, however, this was no longer significant in the adjusted analysis. To assess the different significance level of the pooled OR and pooled AOR, sub-group analysis was conducted to compare the pooled raw OR and pooled AOR including only the five studies that reported both a raw OR and AOR. The sub-group analysis found that when pooling only these studies, the pooled OR was 1.82 (95% CI 1.45–2.28, I^2 = 89%, n = 5), which is still significant. This suggests that the non-significant pooled AOR is not due to the different number of studies, the study characteristics or sample size, but likely due to the confounding factors.

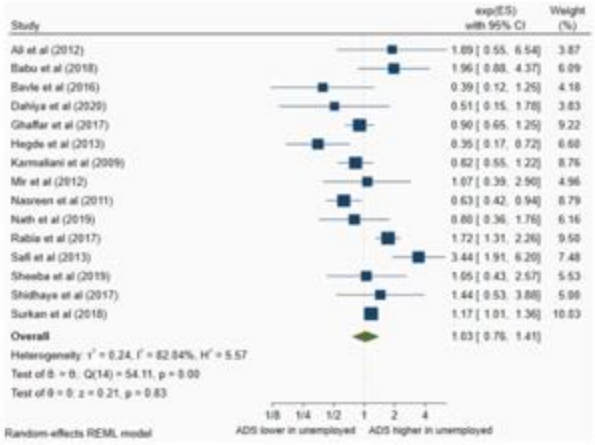


Fig 2. Forest plot of association between antenatal depression/anxiety and maternal occupation unadjusted OR.

https://doi.org/10.1371/journal.pone.0263760.g002

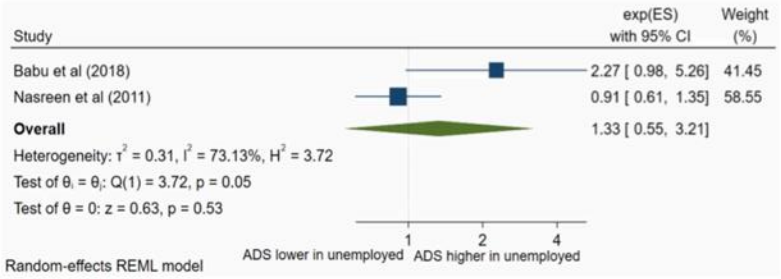


Fig 3. Forest plot of association between antenatal depression/anxiety and maternal occupation AOR.

https://doi.org/10.1371/journal.pone.0263760.g003

Intimate partner violence

Fifteen studies presented results on IPV during pregnancy. All studies met the criteria for inclusion in the meta-analysis. The pooled OR (Fig 8) for antenatal depression in women who had suffered from IPV during pregnancy compared to those who had not was 3.58, (95% CI 2.57–5.00; I^2 = 61%; n = 15 studies). The pooled AOR (Fig 9) was 2.48, (95% CI 1.41–4.33; I^2 = 71%; n = 9 studies). Both the unadjusted OR and AOR suggest that women who experienced IPV during pregnancy have significantly higher odds of antenatal depression/anxiety compared to women who have not experienced IPV during their pregnancy.

Unplanned pregnancy

Twenty studies analysed pregnancy intentions. George et al [12] was not eligible for inclusion into meta-analysis; this study found no significant association. Nineteen studies were included in the meta-analysis. The pooled OR (Fig 10) for antenatal depression/anxiety in women who had an unplanned pregnancy compared to women with planned pregnancy was 1.88, (95% CI 1.44–2.45; I^2 = 91%; n = 19 studies). The pooled AOR (Fig 11) was 1.53, (95% CI 1.28–1.83; I^2 = 19%; n = 5 studies). Both the pooled unadjusted and AOR suggest that women who had unplanned pregnancies have higher odds of antenatal depression/anxiety compared to women who had planned pregnancies.

Relationship with in-laws

Seven studies analysed a women's relationship with her in-laws. The pooled OR (Fig 12) for antenatal depression in women with a poor relationship with in-laws compared with those with a good relationship was 4.23, (95% CI 3.11–5.74; I^2 = 49%; n = 7 studies). The pooled AOR (Fig 13) was 2.69, (95% CI 1.25–5.80; I^2 = 54%; n = 3 studies). Both the unadjusted and AOR suggest that a poor relationship with in-laws during pregnancy is significantly associated with higher odds of antenatal depression compared to perceiving a good relationship with in-laws.

Social/family support

There were ten studies which analysed social support. Nine studies were eligible for inclusion in the meta-analysis. The pooled OR (Fig 14) for antenatal depression/anxiety in women with low social/family support compared to women with high social/family support was 2.14, (95% CI 1.60–2.85; I^2 = 60%; n = 9 studies). The pooled AOR (Fig 15) was 1.42, (95% CI 0.78–2.62;

Table 5. Association between antenatal depression/anxiety and household income in the included studies.

Author & study year	Household income groups (rupees)	OR (95% CI)	AOR (95% CI)
Babu et al 2018 [47]	>15,000	REF	REF
	10,001–15,000	0.91 (0.41–2.03)	0.88 (0.39–2.00)
	6001–10,000	1.18 (0.56–2.50)	1.02 (0.47–2.20)
	≤6K	0.52 (0.26–1.04)	0.49 (0.23–1.04)
Bavle et al 2016 [48]	>10,000	REF	-
	7501-10K	0.40 (0.17–0.94)*	
	5001–7500	0.21 (0.05–0.75)*	
	2501–5000	0.42 (0.13–1.36)	
	0–2500	0.62 (0.07–5.42)	
Dahiya et al 2020 [52]	≤13.51 USD	REF	-
	13.52–26.75 USD	39.00 (0.53–2883.6)	
	26.76–45.03 USD	13.00 (0.45–377.47)	
	45.04–90.07 USD	4.97 (0.21–109.19)	
	≥90.08 USD	3.56 (0.20–64.79)	
Ghaffar et al 2017 [33]	No income	REF	-
	<5000	0.33 (0.20–0.94)*	
	5000–15,0000	0.56 (0.34–0.88)*	
	>15,000	0.64 (0.17–0.99)*	
Imran et al 2010 [36]	>5000	REF	-
	<5000	1.99 (0.74–5.34)	
Jamal et al 2018 [38]	>25,000	REF	-
	≤25,000	6.24 (2.15–18.14)*	
Mir et al 2012 [31]	>10,000	REF	-
	5000–10,000	1.29 (0.68–1.45)	
	1,000–5,000	1.30 (0.72–2.36)	
	<1,000	1.57 (0.61–4.04)	
Rahman et al 2003 [4]	>2,500	REF	-
	<2,500	0.90 (0.60–1.30)	
Sabir et al 2019 [42]	≥50,000	REF	-
	25,000–50,000	1.44 (0.85–2.45)	
	<25,000	1.18 (0.69–2.02)	
Safi et al 2013 [43]	High	REF	-
	Middle	0.57 (0.26–1.25)	
	Low	2.83 (1.25–6.41)*	
Shidhaye et al 2017 [50]	Low	REF	-
	Middle	0.40 (0.10–1.00)	
	High	0.60 (0.30–1.20)	
Srinivasan et al 2015 [51]	>1 Lakh	REF	-
	50,000–1 lakh	0.49 (0.19–1.22)	
	≤50,000	0.70 (0.15–3.25)	
Author & study year	**Mean monthly household income (thousand rupees) in depressed group**	**Mean monthly household income (rupees) in non-depressed group**	**P-value**

(*Continued*)

Table 5. (Continued)

Din et al 2016 [32]	24.91	27.41	0.041*

REF = Reference group
OR = Odds Ratio
AOR = Adjusted Odds Ratio
*Statistically significant at 0.05 significance level

https://doi.org/10.1371/journal.pone.0263760.t005

I^2 = 75%; n = 4). The pooled unadjusted OR suggests that women with low social and family support had higher odds of antenatal depression/anxiety compared to women with high support, however, this was no longer significant in the adjusted analysis. To assess the different significance level of the pooled OR and pooled AOR, sub-group analysis was conducted to compare the pooled raw OR and pooled AOR including only the four studies that reported both a raw OR and AOR. The sub-group analysis found that when pooling only these studies, the pooled OR was 2.24 (95% CI 1.73–2.91, I^2 = 75%, n = 4), which is still significant. This suggests that the non-significant pooled AOR is not due to the different number of studies, the study characteristics or sample size, but likely due to the confounding factors.

Gender preference

Six studies analysed gender preference for the baby. All the studies met the inclusion criteria for meta-analysis. The pooled OR (Fig 16) of antenatal depression in women who wanted a male child compared to wanting a female child was 1.84, (95% CI 1.44–2.36; I^2 = 0%, n = 6 studies). The pooled AOR (Fig 17) was 3.06, (95% CI 1.40–6.72; I^2 = 61%, n = 3 studies). Both

Table 6. Association between antenatal depression/anxiety and husband's education/literacy in the included studies.

Author & study year	Husband's education/literacy level	OR (95% CI)	AOR (95% CI)
Ayaz et al 2019 [29]	College or university	REF	-
	Uneducated	1.24 (0.30–5.24)	
Babu et al 2018 [47]	Attended college	REF	-
	Did not attend college	1.22 (0.75–1.98)	
Maselko et al 2018 [39]	None	REF	-
	Primary	0.89 (0.50–1.59)	
	Secondary	0.50 (0.31–0.78)*	
	Tertiary	0.25 (0.13–0.49)*	
Nath et al 2019 [15]	Attended college	REF	-
	Did not attend college	1.22 (0.71–2.11)	
Rahman et al 2003 [4]	Literate	REF	-
	Illiterate	1.60 (1.20–2.20)*	
Sheeba et al 2019 [46]	Attended college	REF	-
	Did not attend college	1.26 (0.77–2.05)	
Zia et al 2018 [45]	College or university	REF	-
	Uneducated	1.24 (0.30–1.34)	

REF = reference group
OR = Odds Ratio
AOR = Adjusted odds ratio
*Statistically significant at 0.05 significance level

https://doi.org/10.1371/journal.pone.0263760.t006

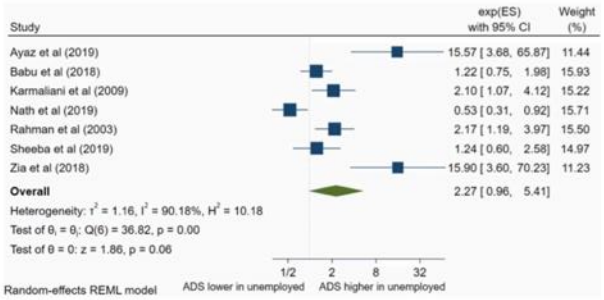


Fig 4. Forest plot of association between antenatal depression/anxiety and husband's occupation unadjusted OR.

https://doi.org/10.1371/journal.pone.0263760.g004

the pooled unadjusted OR and AOR indicate that a male gender preference was significantly associated with increased odds of antenatal depression compared to wanting a female child.

Sensitivity analysis

Analyses were repeated by removing the studies which scored low (0–3) in the quality assessment. The findings only significantly changed for husband's occupation. When removing the low-quality study, Zia et al [45], from the unadjusted analysis, the results were no longer significant (OR 1.65, 95% CI 0.88–3.10). Nevertheless, Zia et al [45] did not carry out adjusted analysis, therefore, the adjusted analysis in the meta-analysis for husband's occupation remains the same, suggesting no significant association between husband's occupation and antenatal depression.

Discussion

To our knowledge this is the first systematic review and meta-analysis to investigate the social determinants of antenatal depression and anxiety in Bangladesh, India and Pakistan. The meta-analysis found IPV, preference for a male child, poor relationship with in-laws and unplanned pregnancy to be statistically significantly associated with antenatal depression and anxiety.

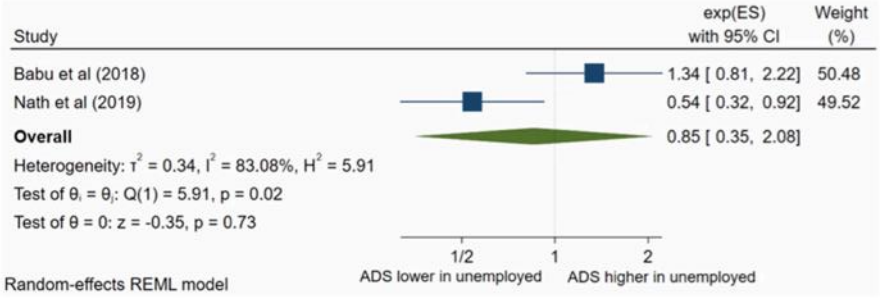


Fig 5. Forest plot of association between antenatal depression/anxiety and husband's occupation AOR.

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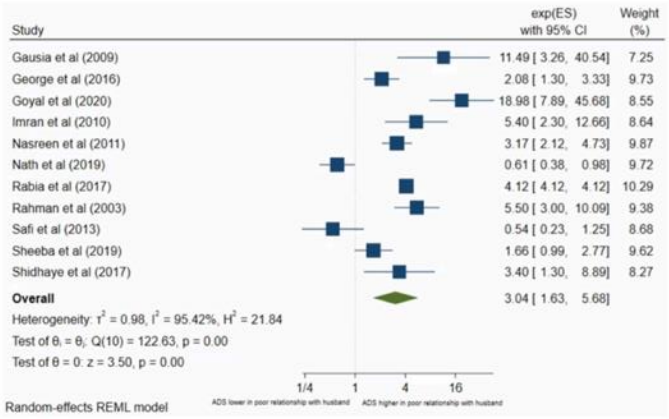


Fig 6. Forest plot of association between antenatal depression/anxiety and relationship with husband unadjusted OR.

https://doi.org/10.1371/journal.pone.0263760.g006

Previous reviews [6, 19, 20] have consistently found IPV to be an important risk factor for antenatal depression/anxiety globally, supporting the finding in this present review. This may be linked to the social, emotional and physical isolation and pressure pregnant women experience when suffering from IPV [6]. Particularly in LMICs, where IPV rates are higher compared to HICs [6], women are subject to social and cultural norms which prevent them from sharing such information [54]. Furthermore, there may be forced economic dependency, which is associated with IPV, on the perpetrator where many women in LMICs are not able to leave their partners [55]. One study found that many women in South Asian communities do

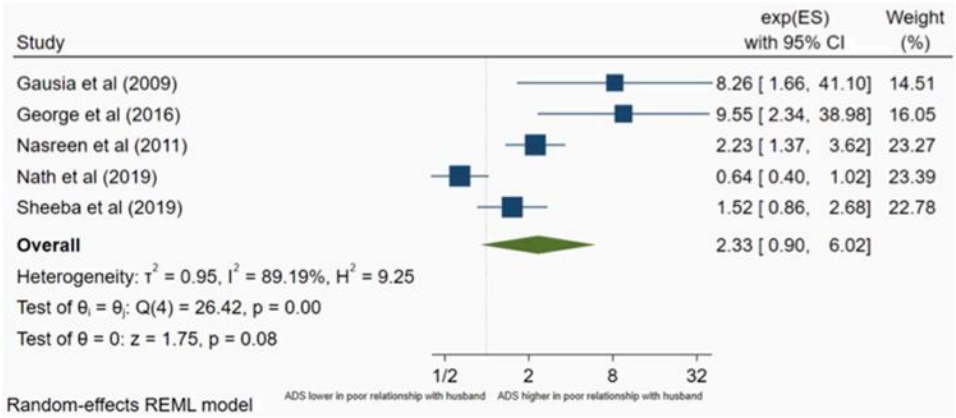


Fig 7. Forest plot of association between antenatal depression/anxiety and relationship with husband AOR.

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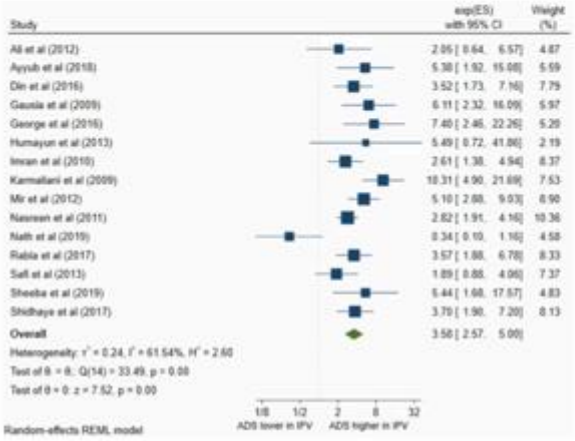


Fig 8. Forest plot of association between antenatal depression/anxiety and IPV unadjusted OR.

https://doi.org/10.1371/journal.pone.0263760.g008

not consider their relationship with their husband to be poor even if the woman faces challenges such as IPV [9]. There are many underlying mechanisms within the relationship with husband which should be further investigated through qualitative studies.

The present review found unplanned pregnancy to be a significant social determinant of antenatal depression and anxiety which is reflected in previous reviews [19, 20]. Unplanned pregnancies can reflect a lack of access to adequate family planning services [56] and power relations between the husband and wife leading to IPV through forced sex without the use of contraception [57]. These issues can cause significant distress to pregnant women. Further, Lee et al [58] and Marchesi et al [59], suggest that unwanted pregnancies, which is a pregnancy

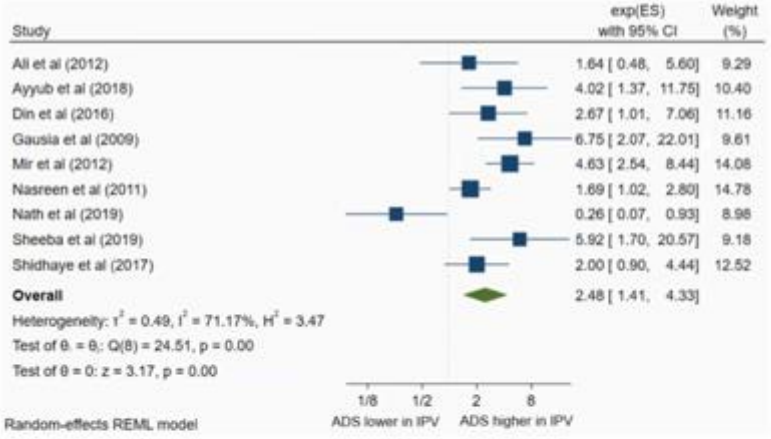


Fig 9. Forest plot of association between antenatal depression/anxiety and IPV AOR.

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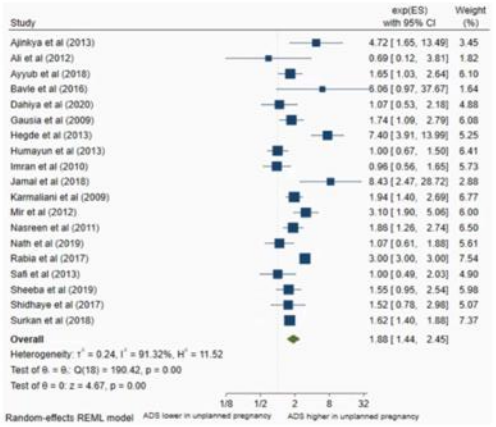


Fig 10. Forest plot of association between antenatal depression/anxiety and pregnancy intentions unadjusted OR.

https://doi.org/10.1371/journal.pone.0263760.g010

intention subset of unplanned pregnancy [60], were a significant predictor of antenatal depression, but mostly in the first trimester, with less importance over time. This may reflect the initial stress of an unwanted pregnancy reducing throughout the pregnancy as the woman comes to terms with the idea of having a baby. This is an interesting finding which has not been investigated in any of the included studies in this review suggesting a need for further research on this within these South Asian countries.

Many studies focus on the relationship with husband, while neglecting the crucial interactions between in-laws and the expecting mother. Particularly in South Asian countries, where women are often expected to live with their in-laws after marriage and throughout [61], these vital family dynamics can shape how women experience pregnancy. Although the pooled OR and AOR showed a significant association between poor relationship with in-laws and antenatal depression/anxiety, only three studies [32, 50, 53] went onto carry out multivariate analysis of which two [32, 53] found a non-significant association. The study with the largest weight, Shidhaye et al [50], which largely contributed to the significant association, was assessed to be

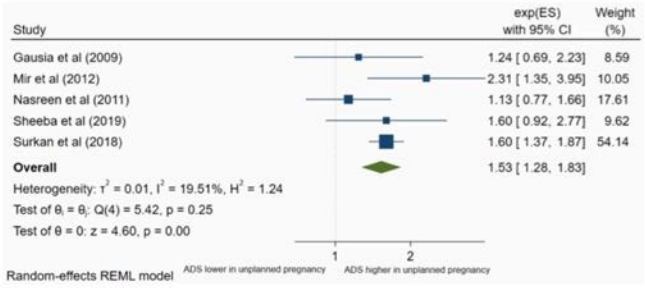


Fig 11. Forest plot of association between antenatal depression/anxiety and pregnancy intentions AOR.

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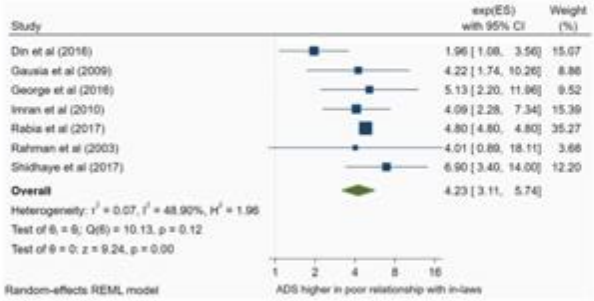


Fig 12. Forest plot of association between antenatal depression/anxiety and relationship with in-laws unadjusted OR.

https://doi.org/10.1371/journal.pone.0263760.g012

a high-quality study. However, differences in the factors adjusted for in the multivariate analysis may contribute to the differences in the results between the three studies.

Gender preference for the child appears to be a relatively under-researched determinant in relation to antenatal depression/anxiety, as highlighted by the small number of studies included in this review which addressed it, despite it being a common societal issue within these South Asian countries [62]. The significant association between preferring a male child and antenatal depression/anxiety found in this review is reflected in observational studies conducted in China [63] and Iran [64]. Male children are considered a blessing as they carry the family name and can provide for their parents in old age, whereas female children are viewed as a burden for the family [61]. Hence, indirect and direct pressure, along with internalised expectations, from the family and society to have a male child may produce adverse effects on the mental health of pregnant women.

Strengths and limitations of the study

A wide range of databases were used to systematically search for papers between January 2000 and January 2021 using the PICOS framework. The date restriction was imposed because an initial search of Medline showed no appropriate studies before 2000, suggesting that investigation of risk factors of maternal depression in South Asia is a relatively recent area of research interest. Overtime clinical approaches and thinking towards maternal depression and

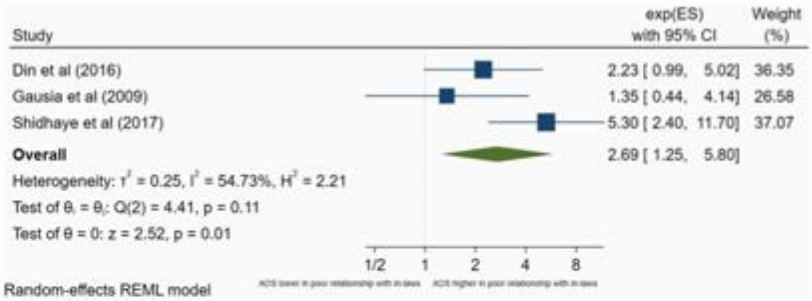


Fig 13. Forest plot of association between antenatal depression/anxiety and relationship with in-laws AOR.

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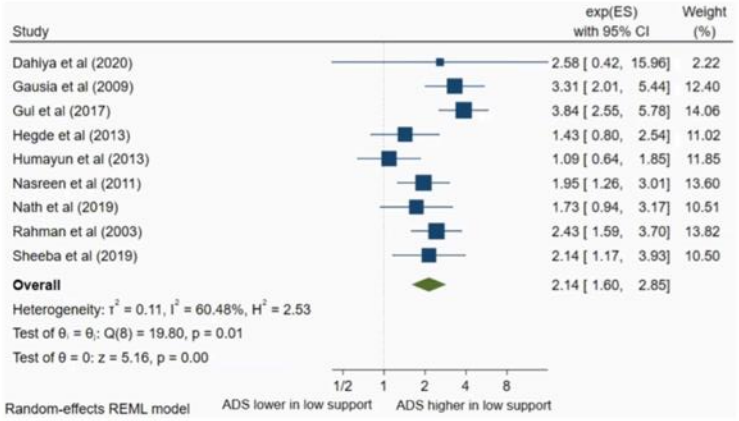


Fig 14. Forest plot of association between antenatal depression/anxiety and social/family support unadjusted OR.

https://doi.org/10.1371/journal.pone.0263760.g014

depression as a whole, particularly in LMICs, have developed and changed [65]. Therefore, a date restriction ensured that the literature in this review would still be clinically and socially relevant. Supplementary searches such as hand searching reference lists, citation searching and contacting authors, were carried out to provide a comprehensive review on a large number of social determinants. Procedures to minimise human error and subjectivity included double independent screening, data extraction and quality assessment. The included studies were mostly of high or medium quality suggesting lower levels of bias within the studies. Only seven out of the 33 studies were assessed as low quality and after these were removed in the sensitivity analysis to determine their influence, no significant changes were identified within the results. Although there are differences between the different depression and anxiety scales, this review aimed to further limit bias by only including studies which used validated measurement scales to determine depression and anxiety and questionnaires which were standardised. This allowed for more appropriate comparisons to be made between the included studies.

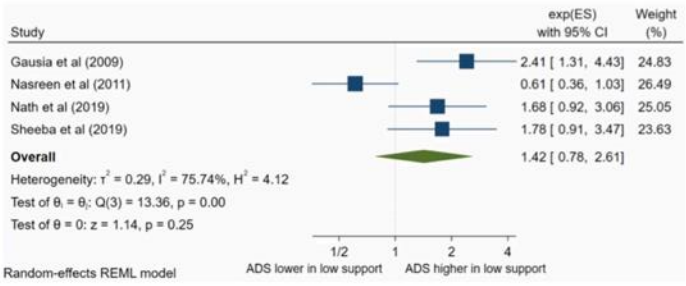


Fig 15. Forest plot of association between antenatal depression/anxiety and social/family support AOR.

https://doi.org/10.1371/journal.pone.0263760.g015

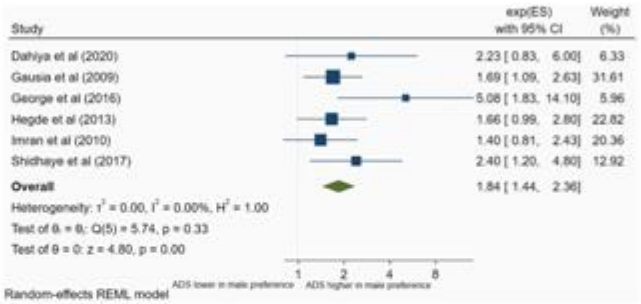


Fig 16. Forest plot of association between antenatal depression/anxiety and gender preference unadjusted OR.

https://doi.org/10.1371/journal.pone.0263760.g016

Publication bias was found in the meta-analyses for husband's occupation and relationship with husband, therefore, the results should be interpreted with caution. Nevertheless, the results from the meta-analyses for these two factors were non-significant which suggests that the results from the included studies are not biased towards a particular direction, given the non-significant pooled AORs.

Heterogeneity was found to be high when a large number of studies were pooled. This could be due to methodological differences such as sampling of participants, sample size, use of different measurement tools and difference in characteristics between Bangladesh, India and Pakistan. Although these three countries share similar social and cultural norms and characteristics which deems it appropriate to pool them together, there is heterogeneity between them. There are differences in the predominant religions found in the three countries. In India, approximately 80% of the population are Hindu, followed by Muslims who make up 13% of the population [66]. Conversely, Bangladesh is made up 81% Muslims and 10% Hindus. There are certain differences between these religions which can influence some of the social determinants investigated in this review. For example, the system of dowry (money or assets given to the groom's family by the bride's family) is more common in Hinduism and is a prominent reason why many families prefer having male children. Therefore, male gender preference may be more prevalent in India than in Bangladesh adding to the heterogeneity. This review used random-effects model to account for the heterogeneity.

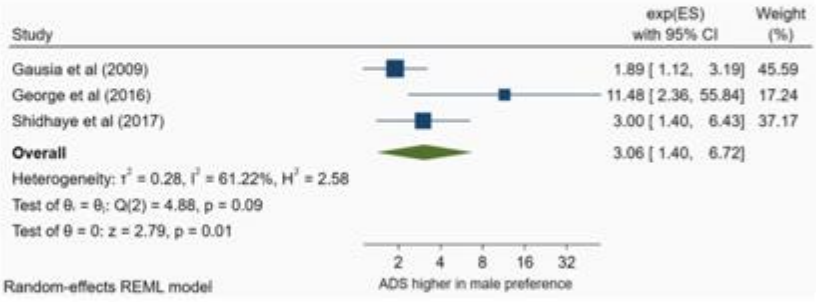


Fig 17. Forest plot of association between antenatal depression/anxiety and gender preference AOR.

https://doi.org/10.1371/journal.pone.0263760.g017

Finally, this review restricted the included studies to those published in or containing English translations since producing translations was not feasible. Therefore, studies published in languages native to Bangladesh, India and Pakistan may have been missed from this review. Nevertheless, journals from Bangladesh, India and Pakistan tend to increasingly publish in the English language or contain translations making them accessible for a wider audience.

Conclusion

This systematic review and meta-analysis found that IPV, male gender preference, poor relationship with in-laws and unplanned pregnancies were significant predictors of antenatal depression in Bangladesh, India and Pakistan. The findings are important for the development of targeted interventions for the prevention and screening of antenatal depression and anxiety in South Asia. These social determinants highlight the inequalities between gender where many women are victims of IPV, forced sex and unplanned pregnancies, and male children are often preferred over female children. Such inequalities should be tackled through policy, education and advocacy within South Asian communities. Further research involving large scale studies incorporating factors which this review found inconsistent evidence on and qualitative studies to understand the relationships are needed.

Supporting information

S1 Checklist. PRISMA 2009 checklist.
(DOC)

S1 Fig. Search terms for OVID database.
(DOCX)

S2 Fig. Adapted Newcastle-Ottawa Scale for cross-sectional studies.
(DOCX)

S1 Table. MOOSE checklist for meta-analysis of observational studies.
(DOCX)

S2 Table. Data extraction protocol.
(DOCX)

S3 Table. Quality scores for all included studies.
(DOCX)

S4 Table. Results for social determinants investigated by only one study.
(DOCX)

S5 Table. Factors included in the included studies' adjusted models.
(DOCX)

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Appendix 2 - Perceptions and attitudes around perinatal mental health in Bangladesh, India and Pakistan: a systematic review of qualitative data

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Perceptions and attitudes around perinatal mental health in Bangladesh, India and Pakistan: a systematic review of qualitative data

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Abstract

Background: Perinatal mental health (PMH) is a worldwide public health issue crossing cultural boundaries. However, the prevalence of PMH conditions vary considerably. These disparities stem in part from poor understanding and stigma surrounding PMH which hinder pregnant women from seeking mental health care and may exacerbate their conditions. Bangladesh, India and Pakistan are South Asian countries with a higher burden of PMH conditions than in the Global North-West and very different social and cultural norms around gender and mental health. The aim of this systematic review (PROSPERO Ref: CRD42020167903) was to identify, synthesise and appraise the available literature on perceptions and attitudes of perinatal (pregnant and postpartum) women, their families and healthcare providers surrounding PMH in Bangladesh, India and Pakistan.

Methods: Five electronic databases, MEDLINE, Embase, PsycINFO, Scopus and Web of science, and grey literature were searched using predefined search terms. Qualitative or quantitative articles with a qualitative component reporting perceptions and attitudes surrounding PMH in Bangladesh, India and Pakistan were eligible for inclusion, if published in English between January 2000 and January 2021. The Critical Appraisal Skills Programme Qualitative Research Checklist and Newcastle–Ottawa Scale for cross-sectional studies were used to assess study quality. Findings were synthesised using thematic synthesis, as described by Thomas and Harden 2008.

Results: Eight studies were included. Five overarching themes comprising 17 sub-categories were identified. These descriptive themes were: perceived causes of PMH, perceived symptoms of PMH, perceptions of motherhood, accessing PMH care and emotional sharing and coping strategies. Sociocultural expectations underpin many of the themes identified in this review including the importance of familial and societal causes of PMH, emphasis on physical symptoms, sacredness of motherhood, lack of awareness, stigma, shame, limited resources allocated for mental health and lack of emotional sharing.

Conclusions: There is a complex range of perceptions and attitudes around PMH which influence women's experiences and access to PMH care. These findings will inform policy and practice through targeted interventions to tackle stigmatising attitudes and increasing education and training for healthcare providers.

Keywords: Perinatal mental health, South Asia, Perceptions, Attitudes

Background

Perinatal mental health (PMH) conditions, particularly depression and anxiety being the most common PMH conditions, have been long considered a Western biomedical construct [1]. It was thought that the

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rise in modern obstetric practices in Western countries, which isolated women from guidance and social support throughout their pregnancy and after childbirth, was related to maternal depression [2]. This medicalisation of pregnancy often shifted the care of women away from sociocultural practices such as mandated rest and social recognition of new social status through rituals and gifts into more professional healthcare settings. Hence, it was assumed that women from non-Western countries were protected from depression by these culturally specific rituals and factors [2]. However, there is little evidence to support this. Not only do PMH conditions such as depression seem to occur universally, but recent research has investigated the cross-cultural validity of maternal depression and has found there to be a convergent validity for the construct of maternal depression in terms of the emotional and somatic distress in the setting of social adversity [3]. Therefore, PMH is a worldwide public health issue crossing cultural boundaries. However, conceptualisation of perinatal depression and anxiety as a psychiatric, or even a medical disorder is less applicable in South Asian cultures which have differing perceptions of PMH making it a challenging, yet important, construct to tackle in these cultural settings [3, 4].

Bangladesh, India and Pakistan are South Asian countries which have very different social and cultural norms to Western countries. Previous literature has found the prevalence of perinatal depression in these South Asian countries to range from 10% [5] to as high as 39% [6], compared to 7–15% in High Income Countries (HICs) [7]. These disparities stem from the poor understanding and high levels of stigma around mental health conditions in South Asia [8], which prevent pregnant women from seeking and accessing mental health care and reporting mental health issues due to shame and fear of judgment. It also impacts the way families and healthcare providers of pregnant women manage and view depression during pregnancy, consequently, affecting the care women receive. It is, therefore, imperative we investigate the perceptions and attitudes that these communities in Bangladesh, India and Pakistan have towards PMH to aid our understanding of why these countries have high rates of perinatal depression, what needs to be done to break the stigma and to make mental health care more available and accessible for pregnant women.

Perceptions and attitudes, which are heavily contingent and linked to socio-cultural context, influence how individual's understand mental illness and the way they respond to situations, activities and relationships [9]. South Asian cultures, like Bangladeshi, Indian and Pakistani, are very collectivist, in that social controls such as family, culture, religion and community, often dominate decision-making and strongly shape the attitudes that individuals hold [10]. Hence, the way such collectivist cultures perceive social roles and conceptualise identities will influence individuals' perceptions and attitudes towards certain phenomena such as PMH. A systematic review by Watson et al. [11] investigating ethnic minority women's, largely South Asian and Black African, experience of PMH conditions and services in the UK found that lack of awareness, ongoing stigma and cultural expectations all impacted women's receipt of adequate PMH support in the UK [11]. However, there is limited research investigating these experiences in Bangladesh, India and Pakistan where the social and structural settings are vastly different.

To our knowledge, there are no existing systematic reviews which have investigated perceptions and attitudes of perinatal (pregnant and postpartum) women, their families and health care providers towards PMH in Bangladesh, India and Pakistan. These groups are key actors in perinatal women receiving mental health care. Therefore, the aim of this systematic review is to identify, synthesise and appraise the available literature on perceptions and attitudes of perinatal women, their families and healthcare providers surrounding PMH in Bangladesh, India and Pakistan. This will inform the development of PMH services in these countries and interventions and programmes to improve mental health of perinatal women.

Methods

This systematic review followed the Enhancing Transparency in Reporting the synthesis of qualitive research (ENTREQ) guidelines (See Additional File 1).

Search strategy

Inclusion criteria were qualitative and quantitative studies with a qualitive component reporting perceptions and attitudes related to women in the perinatal period, their families and healthcare providers surrounding PMH (only including perinatal depression and anxiety as the most common PMH conditions). Studies must be conducted on populations in Bangladesh, India or Pakistan and be published in English between 1st January 2000 and 31st January 2021. This date restriction was imposed because scoping searches on Medline showed no appropriate studies before 2000, suggesting that investigation of maternal depression and anxiety in South Asia is a relatively recent area of research enquiry. Over time, clinical approaches and thinking towards maternal depression and depression as a whole, particularly in Low and Middle Income countries (LMICs), have developed and changed [3]. Therefore, this date restriction ensured that the literature in this review would still be clinically and socially relevant. Studies were excluded if they had

no qualitative element that could be extracted, not conducted in Bangladesh, India or Pakistan and published outside of the time frame investigated.

A systematic literature search was conducted via the electronic databases MEDLINE, Embase, PsycINFO, Scopus and Web of science using keywords and MeSH headings developed with an information specialist (See Additional File 2 for full OVID search strategy). This review relating to perceptions and attitudes forms part of a wider search strategy which also included a second review investigating the social determinants of antenatal depression in South Asia [12] (see PROSPERO Ref: CRD42020167903). Search results were exported into the reference manager Zotero and duplicates were removed. Titles and abstracts were independently screened by two reviewers (NI and AW) and compared. Those that met the criteria were obtained in full and assessed by the research team to determine inclusion. Grey literature was searched using the Open Grey database and also via charities and non-governmental organisations such as Building Resources Across Communities (BRAC), UNICEF and National Health Mission India. Supplementary searches included hand searching study reference lists included from databases and citation searches using Google Scholar. Database searches were completed on 31st January 2020 and updated on 4th January 2021.

Quality appraisal

Quality appraisal was carried out by two reviewers (NI and AW) and agreement was reached through discussion. Study quality was assessed using the 2018 Critical Appraisal Skills Programme (CASP) checklist for qualitative studies (see Additional File 3) [13]. A scoring of 1 mark for each question that was answered with "Yes" was allocated to provide an indicator of quality and enabled comparison between reviewers. Cut-off scores of ≥ 9 indicated 'strong' quality, 6–8 indicated 'adequate' quality and ≤ 5 indicated 'weak' quality for the CASP tool.

For the primarily quantitative studies included in this review, there are no CASP checklists for cross-sectional studies, therefore, the Newcastle–Ottawa Scale (NOS) adapted for cross-sectional studies which assesses information bias, selection bias and confounding bias was used to assess quality (see Additional File 4) [14]. The maximum quality score a paper could receive was 10; 0–3 reflected 'low' quality papers, 4–7 was 'medium' and 8–10 indicated 'high' quality.

Data extraction and synthesis

Data including bibliographic information, country, type of study, sampling size and method, participant characteristics, data collection method, data analysis method, quality assessment and findings were recorded on a standardised data extraction form. The integrated mixed methods synthesis approach developed by Sandelowski et al. [15] was adopted. This involves transforming the data which are then integrated and synthesised together. In this review, the data from the cross-sectional studies were transformed into qualitative data, as the studies used surveys or semi-structured interview questionnaires that consisted of phrases/sentences that summarised certain attitudes and perceptions. The transformation was done by extracting the statements that were mostly agreed with by the participants and that the authors highlighted in their findings and discussion and using those as qualitative statements to form themes. This method is appropriate when both quantitative and qualitative studies address the same research question.

Integration and synthesis of qualitative findings were carried out using thematic synthesis adapted from Thomas and Harden [16] by the lead reviewer (NI) in two iterative stages: 1) coding of extracted data: data from included studies were examined for meaning and content and codes were applied to capture these. As each new study was examined, similar codes were highlighted, and new codes were added to the bank; 2) Grouping of codes into themes: list of initial codes were analysed for meaning and organised into categories of similar or associated meanings. Each category was examined, and descriptive themes were generated and relationships between them were revealed. Themes were then interrogated by other reviewers (SF and JR).

Results

Included studies

Eight studies published between 2000 and 2021 met the inclusion criteria (Table 1). See Fig. 1 for the PRISMA flow diagram. Six studies were conducted in India [3, 17–21], two in Bangladesh [9, 22] and one in Pakistan [19]. Four studies were qualitative [3, 9, 19, 22] and three were cross-sectional studies [17, 18, 20, 21]. Sample sizes ranged from 21 to 332 using mostly purposive or convenience sampling. The included studies investigated perceptions and attitudes from four population groups: pregnant women [17–19, 21], postpartum women [3, 9, 17, 19, 21, 22], family members of pregnant women [3, 20] and healthcare providers [9, 21]. The CASP quality appraisal of qualitative studies found that all four qualitative studies [3, 9, 19, 22] were of 'adequate' quality. The NOS quality appraisal of cross-sectional studies found that three studies [17, 20, 21] were of 'medium' quality and one study [18] was of 'low' quality.

Findings

Five themes were synthesised concerning perceptions and attitudes surrounding PMH in Bangladesh, India

Table 1 Summary of included studies

Author, publication year	Country	Aims	Type of study, Sampling size & method	Participant characteristics	Data collection method	Data analysis method	Main themes and sub-themes identified	Quality
Edhborg et al., 2015	Bangladesh	"To explore and describe the experiences and concerns during the first 3–9 months following childbirth of those mothers who showed depressive symptoms 2–3 months postpartum, in a rural area in Bangladesh"	Qualitative 21 Purposive sampling	Mothers with depressive symptoms 2–3 months postpartum	Narrative interviews	Inductive content analysis	Perceived causes of PMH (Gender of baby, IPV, Relationship with husband/in-laws, economic difficulties, spiritual) Perceived symptoms of PMH (emotional, physical) Baby's health is most important Awareness of PMH is an enabler of accessing PMH care	Adequate[b]
Goyal et al., 2020	India	"To study psychiatric morbidity, its prevalence and cultural factors influencing illness understanding, help-seeking behaviour and barriers to care in perinatal women."	Mixed methods (cross-sectional with a qualitative component) 27 Purposive sampling	Perinatal women with depression or anxiety screened using EPDS or PASS	Cultural Formulation Interview	Thematic interpretations	Perceived causes of PMH (spiritual) Perceived symptoms of PMH (emotional, physical) Barriers to accessing PMH care (stigma, limited resources) Friendly and confidential counselling are enablers to accessing PMH care Religious and personal coping strategies	Medium[a]
Manjrekar et al., 2018	India	"To find out the awareness and perception of mental health problems in pregnant women residing in rural areas of India"	Cross-sectional 300 Convenience sampling	Antenatal women	Semi structured questionnaire interview	Descriptive statistics	Perceived causes of PMH (spiritual) Barriers to accessing PMH care (stigma, limited resources) Friendly and confidential counselling are enablers to accessing PMH care	Low[a]
McCauley et al., 2020	India & Pakistan	"To explore what women attending for routine antenatal care (ANC) or postnatal care (PNC) at healthcare facilities in India and Pakistan consider health and ill health to be in general, and, what they consider health and ill health to be, during and after pregnancy"	Qualitative 130 Purposive sampling	Antenatal and postnatal women	Focus group discussions	Thematic framework analysis	Perceived causes of PMH (Gender of baby, IPV, Relationship with husband/in-laws) Perceived symptoms of PMH (emotional) Baby's health is most important Barriers to accessing PMH care (stigma, limited resources) Enablers to accessing PMH care (awareness, friendly and confidential counselling)	Adequate[b]

Table 1 (continued)

Author, publication year	Country	Aims	Type of study, Sampling size & method	Participant characteristics	Data collection method	Data analysis method	Main themes and sub-themes identified	Quality
Poreddi et al., 2020	India	"To explore the knowledge and attitudes of family members towards postpartum depression."	Cross-sectional 202 Random sampling	Family members of postpartum women	Face to face interview using semi-structured questionnaire	Descriptive statistics	Perceived causes of PMH (Lack of practical support, Gender of baby, IPV, economic difficulties, spiritual) Perceptions of motherhood (baby's health and motherhood are sacred) Barriers to accessing PMH care (stigma) Enablers to accessing PMH care (friendly and confidential counselling)	Medium[a]
Ransing et al., 2020	India	"To assess the knowledge gap, perceptions, and misconceptions about perinatal depression at three different levels i.e. high-level service providers (e.g. specialists, general practitioners), mid-level health care providers (nurses, midwives) and service utilizers (perinatal women)."	Cross-sectional 332 Convenience sampling	270 Perinatal women 42 Nursing providers 20 Medical practitioners	Questionnaire and online survey forms	Descriptive statistics	Perceived causes of PMH (Lack of practical support, Gender of baby, relationship with husband/in-laws) Perceived symptoms of PMH (emotional) Motherhood is sacred Barriers to accessing PMH care (stigma) Enablers to accessing PMH care (awareness, friendly and confidential counselling)	Medium[a]
Rodrigues et al., 2003	India	"To describe attitudes and perceptions of mothers and husbands towards childbirth in Goa in order to explore the processes through which the relationship between social adversity and PND is mediated."	Qualitative 39 Purposive sampling	Postpartum mothers (19 with postpartum depression) and their husbands	In-depth interviews	Thematic analysis	Perceived causes of PMH (Lack of practical support, Gender of baby, IPV, relationship with husband/in-laws, economic difficulties) Perceived symptoms of PMH (physical, emotional) Baby's health is most important Barriers to accessing PMH care (stigma, limited resources)	Adequate[b]

Table 1 (continued)

Author, publication year	Country	Aims	Type of study, Sampling size & method	Participant characteristics	Data collection method	Data analysis method	Main themes and sub-themes identified	Quality
Williams et al., 2018	Bangladesh	"To understand the cultural attitudes, from both new mothers and maternal health professionals, towards mental health post-childbirth."	Qualitative 70 Purposive sampling	36 postpartum mothers 34 medical personnel	In-depth interviews	Thematic analysis	Perceived causes of PMH (Lack of practical support, IPV, economic difficulties) Perceived symptoms of PMH (physical) Perceptions of motherhood (baby's health and motherhood are sacred) Barriers to accessing PMH care (stigma, limited resources)	Adequate[b]

EPDS Edinburgh Postnatal Depression Scale

PASS Perinatal Anxiety Screening Scale

[a] Quality assessed using Newcastle–Ottawa Scale for cross-sectional studies (see Additional File 4)

[b] Quality assessed using the CASP tool for qualitative studies (see Additional File 3)

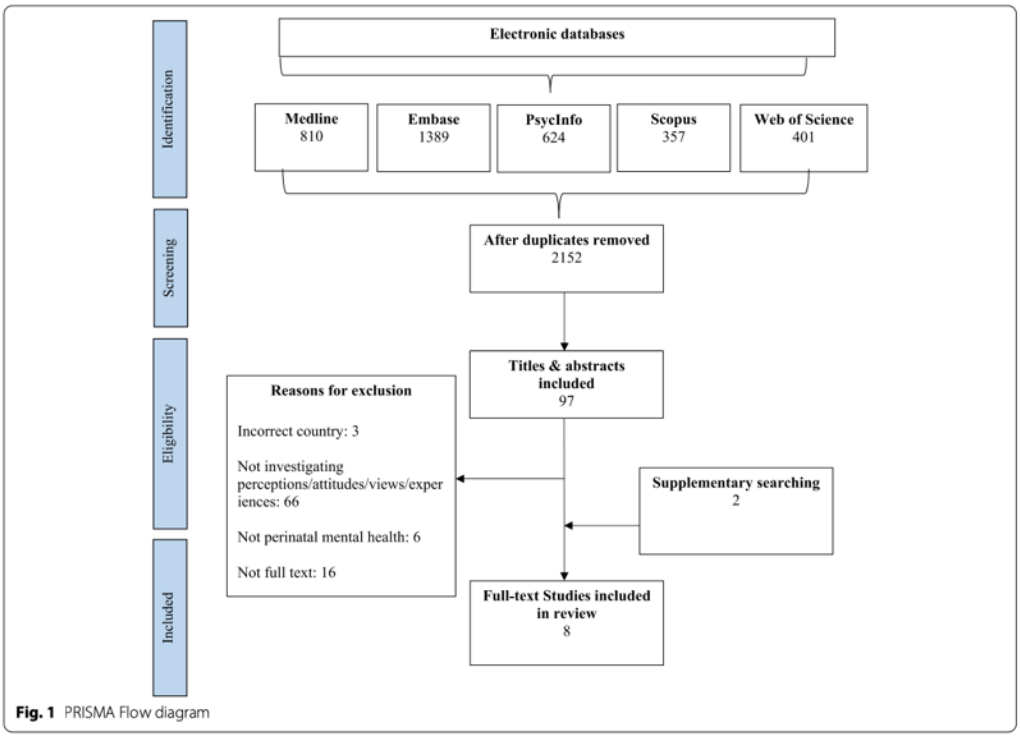


Fig. 1 PRISMA Flow diagram

and Pakistan: (1) Perceived causes of PMH, (2) Perceived symptoms of PMH, (3) Perceptions of motherhood, (4) Accessing PHM care and (5) Emotional sharing and coping strategies. Themes comprised sub-categories. Direct quotes cited from interviews or focus groups from the relevant included studies are used to illustrate the themes and their sub-categories.

Theme 1: Perceived causes of PMH

This theme comprises the perceived causes of PMH conditions such as depression and anxiety. The sub-categories include a range of different psychosocial, cultural and socioeconomic issues.

Lack of practical support Four studies reported findings suggesting lack of practical support from in-laws and husbands in helping out with household chores to be a cause of PMH conditions [3, 9, 20, 21]. This view was found among postpartum women [3, 9], nurse practitioners [21] and family members of postpartum women [20]. This is illustrated by this quote cited from an interview with a first-time mother in a study conducted in India:

> *"I have to do everything by myself. I need someone to help me. I haven't gone to stay with mother because then my husband will have trouble managing things; he will have to cook by himself. So I'll go later and stay a month with her." [3]*

Poor relationship with husband/in-laws Four studies found that a poor relationship with husband or in-laws was perceived to be a cause of PMH conditions [3, 19, 21, 22]. A quote cited from an interview with a postpartum mother in a study conducted in India highlights the importance women attach to the relationship with their husband and the care that he provides:

> *"Once you are married you are under the care of your husband. If your husband is not good then there is tension and you wonder why you ever got married." [3]*

Intimate partner violence (IPV) Five studies reported IPV to be perceived as a cause of PMH conditions by pregnant [19] and postpartum women [3, 9, 19, 22] and

family members of postpartum mothers [20]. IPV was found mainly to take the form of spousal emotional and physical violence both during and after pregnancy as highlighted in this quote from a postpartum mother in India.

> *"He used to hit me when I was pregnant. We had an argument and I said I was going to leave him. He said we should both die and save our lives. I said I would give back all the money he had spent on me. Then, he hit me badly; I fell on my stomach." [3]*

One study conducted in Bangladesh highlighted that along with spousal violence, women also feared violence from in-laws creating an abusive family environment for postpartum mothers [22].

> *"I am scared of them (the parents in-law), they look at me threateningly, and they want to pick issues. When they come home, I stay in my room and don't get out." [22]*

Gender of the baby Gender of the baby is a major concern for perinatal women in South Asian countries and having a female child has been highlighted as a perceived cause of PMH conditions in five studies [3, 19–22]. The stress caused by familial abuse when having a daughter is highlighted in this quote cited from a focus group discussion with postnatal women in a study conducted in India:

> *"This time also I have given birth to a girl child [and] because of this the family atmosphere is not good. My family members are abusing me and because of this I am going through a lot of stress." [19]*

Health care providers also stated that pressure from the husband and in-laws for a male child were causes of perinatal depression. This perception was more prominent in nursing providers than medical practitioners [21] as they tend to have more discussion time with perinatal women during antenatal and postnatal care visits.

Economic difficulties Poverty was reported to be an important cause of PMH conditions by perinatal women and their family members [3, 9, 20, 22]. In rural areas, husbands often work as farmers on others' lands which leads to unsteady income due to periods of unemployment [3]. For perinatal women, this results in worries around meeting medical costs during and after her pregnancy and being able to feed her baby properly. It also indicates their financial dependency on husbands.

> *"The main tension is because of the unemployment of my husband. Since it's summer now, he can go anywhere (and do odd jobs), but once it starts raining, where can he go…my elder son has jaundice. If one of them is better, then the other one falls ill. I'm worried because we have to buy the medicine prescribed by the doctor." [3]*

Spiritual Three studies reported perceptions of spiritual causes of PMH conditions. These views were found in perinatal women [17, 18] and family members [20]. Many thought that mental illness during pregnancy was caused by being possessed by God or some devil, committing sin, black magic or witchcraft.

Theme 2: Perceived symptoms of PMH

This theme comprises findings relating to the views of perinatal women, family members and health care providers on the symptoms and effects of PMH conditions.

Physical symptoms Four studies highlighted that many perinatal women attached importance to physical, somatic signs such as weakness, backache, headache and "brain stops" when describing the symptoms and effects of PMH conditions [3, 9, 17, 22]. This is illustrated by this quote cited from an interview with a postpartum mother when asked about symptoms of depression in a study conducted in India:

> *"The headaches, dizziness, cold, tiredness all are the effects of this tension." [3]*

It was also suggested by Goyal et al. [17] that women tend to focus more on these physical symptoms when trying to describe poor PMH to others such as healthcare providers or their families, as women perceive these to be more acceptable and treatable symptoms rather than symptoms such as worry, fear, loneliness and stress. Physical symptoms prevent women from completing household chores or looking after her baby which was also highlighted to be a concern within perinatal women [22].

Emotional symptoms However, some studies found that perinatal women and health care providers do consider emotional symptoms of PMH to be important [3, 17, 19, 21, 22]. Some women described symptoms of PMH as feeling sad, low mood, being depressed, crying, having tension or being mentally weak. These symptoms were acknowledged by women, but they would not always share these [17].

"Things are running in my head. I can't stop worrying about everything." [22].

Some pregnant women did not consider certain physical symptoms such as change in appetite, sleep problems and feeling tired as symptoms of depression [21]. Such symptoms are considered to be a normal part of pregnancy or of the postpartum period [19].

Theme 3: Perceptions of motherhood

Perceptions that individuals carry towards pregnancy and motherhood impacts the way communities view mental health issues within this population group as they have preconceived notions of what pregnancy should entail and how mothers should behave. In this theme, these perceptions are highlighted and how this impacts their attitudes towards PMH is explained.

Motherhood is sacred Motherhood is sacred and an integral part of being a woman in many South Asian communities as highlighted by three studies [9, 20, 21]. When a woman becomes a mother, her status is elevated in her family and society and it is acknowledged as an accomplishment as stated by a postpartum mother in a study conducted in Bangladesh:

"Now everything is changed, people say that now you are a mother, they give me more importance, and they are so happy with me." [9]

However, this places pressures on women to live up to this status. These societal expectations placed on mothers can result in criticism of women who may appear to struggle during pregnancy or postpartum, particularly to the point where she needs to access care [9]. In addition, these perceptions can be subject to certain conditions such as having female children which is often perceived as a status reducing characteristic for mothers [22].

Baby's health is most important Prominence is attached to a baby's health as an important aspect of maternal wellbeing as highlighted in five studies [3, 9, 19, 20, 22]. Women during and after pregnancy assess their mothering abilities, health and success on the basis of their baby's health.

"The most important thing is this that baby is healthy." [19].

"I am more worried now about the baby...if my baby stays healthy, I will be in peace." [22]

Theme 4: Accessing PMH care

This theme highlights some of the key barriers and enablers of access to PMH care as perceived by perinatal women, their family members and healthcare providers.

Mental health awareness A key enabler in improving access to PMH care is professional awareness of mental health and the understanding that perinatal women are at risk of mental health conditions [3, 9, 18, 19, 21]. Many healthcare providers had poor awareness and understanding of mental health and PMH and perceived that such conditions were not a serious health concern [3, 9, 21]. This is highlighted in this quote cited from an interview with a gynaecological intern in a study conducted in Bangladesh:

"There is no mental health in maternal health – we don't take mental health seriously, it's just not there." [9]

Awareness among perinatal women varies with many finding it difficult to understand the links between pregnancy or motherhood and depression.

"Why would she be sad? I don't understand." "I don't know, this does not happen." [9].

One study highlighted that the husbands of postpartum women have very limited awareness of PMH [3]. Husbands are less aware of the emotional needs of women during and after pregnancy and they often feel their only duty is to provide financial support [3]. Many husbands of women with postpartum depression were also ignorant that their wives were unwell in any way [3].

Stigma and shame Stigma and the related shame associated with PMH in Bangladesh, India and Pakistan is a key barrier for perinatal women accessing PMH care as highlighted by seven studies [3, 9, 17–21]. There is the attitude that emotional health problems should stay within the family and women would not tell their friends if they do attend mental health services because of the taboo attached to them [9].

Attitudes of family members of perinatal women also reflect these stigmatisations regarding women with postpartum depression characterising them as not being good mothers [20]. Nevertheless, many are ready to help reduce depression through family interventions. Nursing providers stated that postpartum depression is a sign of weakness and "women choose to get postpartum depression." [21].

Stigma results in women being ashamed or fearful to access care due to repercussions from family or judgment from healthcare providers and communities [19]. These concerns are highlighted in this quote cited from a focus group discussion with pregnant women in a study conducted in Pakistan:

> *'Some women will not talk about their problems as they are afraid. They think that their husband will know about it if they talk to doctor and will be angry with them." [19]*

Friendly and confidential counselling McCauley et al. [19] found that perinatal women would find counselling acceptable if healthcare providers providing the counselling were friendly, kept the information confidential and removed the stigma attached to PMH. Counselling is a more appropriate form of treatment compared to drugs and medications for healthcare providers to offer and perinatal women to receive [20, 21].

Counselling and social support have been acknowledged as helpful by many women but require much more cultural attention to tackle the shame and stigma associated with it [17]. Western protocols and counselling methods may not work as well in these South Asian settings as they need to consider religious, cultural and societal beliefs which shape a lot of the beliefs around PMH [3, 9].

Limited resources A key barrier in accessing adequate PMH care is the limited resources in these South Asian countries as reported by five studies [3, 9, 17–19]. One aspect of limited resources is the lack of time that healthcare providers have which prevents them from focusing on PMH [9, 19].

> *"A doctor has many patients...obviously, they don't have time for these details!" [19].*

Another element is poverty and lack of money in the health system to treat PMH conditions and within families to access such treatments [3, 17, 18]. Resources are very constrained within the healthcare system, with respect to mental health, and healthcare providers have to make choices about the care they deliver and physical health takes precedence [9].

> *"How can we focus on maternal depression when unattended birth and postpartum haemorrhage is such a problem, and we are a poor country?" [9]*

There is also limited education for healthcare providers on PMH with counselling not widely taught and no protocols on how to educate or identify women with postpartum depression [9]. This lack of education impacts the care available to perinatal women.

Theme 5: Emotional sharing and coping strategies

This theme highlights the ways in which perinatal women share their emotions and the "coping" strategies (term used by William et al. [9]) they adopt to manage their PMH conditions.

Sharing is pointless Many women reported that sharing their mental health issues is pointless or difficult because of resulting quarrels and judgment from others. [3, 9, 19] Family members tend to be disinterested in such matters and women feel that healthcare providers will not be able to do much as these are family related problems [3]. Hence, emotional sharing tends to be very limited in these South Asian countries.

> *"It's not necessary to share, I feel better to just keep things to myself. I am afraid of sharing."[9]*

Personal coping strategies With limited emotional sharing with others, perinatal women tend to resort to personal coping strategies [3, 9, 17]. A common form is through distraction from the issues that worry women by getting out of the house or playing with their baby [9]. This is highlighted by this quote cited from an interview with a Bangladeshi postpartum mother when asked how she copes with stress:

> *"Sometimes I go outside the gate and stand there, and then I come back." [9].*

Religious coping strategies Aside from coping through personal strategies, four studies, Rodrigues et al. [3], Williams et al. [9], Goyal et al. [17] and Manjrekar et al. [18] found that perinatal women would also turn to religious coping methods through prayers, faith-healing and crying for God's help. Religion is often a source of comfort and strength for many women in these South Asian countries.

Discussion

This review synthesised data from primary studies investigating perceptions and attitudes around PMH in Bangladesh, India and Pakistan, a topic largely overlooked in the literature to date. This review identified five themes: perceived causes of PMH, perceived symptoms of PMH, perceptions of motherhood, accessing PMH care and emotional sharing and coping strategies.

The perceptions and attitudes identified in this review are largely related to sociocultural factors, reflecting cultural collectivism and the local social construction of gender roles and power relations. This is evident from the perceived causes of PMH conditions as identified from the studies included in this review. These findings relate to evidence from cross-sectional studies which found significant associations between IPV [23–26], poor relationship with husband/in-laws [8, 26], low practical support [25, 26], female gender of the baby [5, 24, 27], and antenatal depression among women in South Asia. Social construction of gender roles within South Asian communities often places women at a disadvantage. Women are expected to carry out all the household chores, even during pregnancy, with little practical support and an inability to do this can result in repercussions from the family including being subjected to violence [24]. Even before a woman is born, society deems her to be of lower status and value than men as reflected by the preference for male children [28]. These gender inequalities are rooted within many South Asian communities and can strongly influence individuals' perceptions and attitudes.

Affective symptoms of PMH are acknowledged by perinatal women, however, discussing physical, more somatic symptoms is much more acceptable due to the perception that women with mental health conditions are 'weak' or 'mad'. This is common in Asian culture as highlighted in the review by Halbreich & Karkun [4] who investigated cross-cultural postpartum depression prevalence and symptoms. Since motherhood is sacred, is it not socially acceptable to be seen as weak. Thus, women are forced to hide their emotional symptoms from family members and healthcare providers with the fear that their child may be removed from their care. Even healthcare providers' attitudes are dominated by collectivist cultural controls where they infrequently ask about emotional or mental health symptoms as they perceive it to be inappropriate. These findings are consistent with the systematic review by Watson et al. [11] on South Asian and Black African women's experiences of PMH in the UK.

Barriers and enablers to accessing PMH care operate at a number of levels ranging across sociocultural to structural factors. Perinatal women who acknowledged mental health conditions were more open to discussing these with healthcare providers. Family members and healthcare providers who had adequate awareness around the importance of PMH also enabled perinatal women to access care. This is consistent with other studies which found that lack of awareness among women, their families and healthcare providers led to poor recognition of symptoms and delayed management [11, 13]. Therefore, increased training and education is required to increase awareness around PMH.

In addition, lack of awareness often stems from the high levels of stigma associated with mental health which is a significant barrier for accessing PMH care. This again highlights the theory of collectivist cultures [29] where societal stigma found within these South Asian countries dominate the attitudes people have towards PMH resulting in women who access PMH care being criticised. These findings are highlighted in previous literature where stigma is a key barrier to accessing PMH care [11, 30]. Women do not want to be labelled with mental health conditions as society will deem them to be weak and unsuitable mothers. This relates to the emotional sharing and coping strategies adopted by perinatal women in this review. Sharing is perceived as pointless as they won't be taken seriously by a society which belittles PMH, therefore, women resort to personal and religious coping strategies [3, 9, 17]. Distraction and praying are two methods; however, these are not sustainable. This calls for culturally appropriate coping interventions for long-term management.

Structural factors such as limited resources also create barriers for perinatal women receiving PMH care and healthcare providers delivering the care. Lack of time and limited health workforce means healthcare providers are forced to focus only on physical health. Postpartum haemorrhage is a leading cause of maternal mortality in South Asian countries [31]. Therefore, immediate attention must be given to such physical health concerns. Poverty plays a role in the limited resources in countries such as Bangladesh who only spend 0.5% of the health budget on mental health [32]. There is limited money to spend on mental health services and facilities which are accessible for perinatal women from all regions. These findings are consistent with other LMICs [30].

These findings have implications for informing cross-sectoral prevention and management interventions for PMH in these South Asian countries. The education sector will be a key actor in the development of these interventions addressing many of the perceptions and attitudes identified in this review. Improved education for healthcare providers at all levels is vital for increasing awareness of PMH and better knowledge of the symptoms and management options [33]. Healthcare providers working directly with perinatal women should be given adequate training around friendly and confidential counselling. Psychosocial education aimed to tackle IPV, gender inequalities and stigma within communities could be provided through Government and non-governmental social programmes. Family planning services are also key to address the issue of unplanned pregnancies. This can be achieved through Maternal, Neonatal and Child Health platforms who can provide these services at a community and household level through community health workers who are

more aware of the rural settings and culture. The Social welfare sector should be integrated to tackle poverty, housing and unemployment, for example, by encouraging microcredits for women to start their own small businesses and promoting health knowledge [34].

To our knowledge this is the first systematic review of literature focusing on perceptions and attitudes around PMH in Bangladesh, India and Pakistan. A wide range of databases were searched using a comprehensive search strategy and supplementary searches were carried out. However, we acknowledge some possible limitations. Due to resource constraints, articles were restricted to those in the English language which may produce language bias. Nevertheless, journals from Bangladesh, India and Pakistan tend to increasingly publish in the English language or contain translations making them accessible for a wider audience. Second, although most studies were deemed to be of 'adequate' or 'medium' quality, there were gaps in reporting, particularly with regards to possible researcher bias during data collection and analysis and what implications this may have on the findings and conclusions in the included studies. Finally, only eight studies were identified for inclusion in this review, with only one from Pakistan which may question the relevance of the themes in a Pakistani setting. Nevertheless, qualitative reviews adopt more interpretative approaches using depth and richness of data rather than breadth which privilege smaller number of studies [35]. Given this, Booth et al. [35] suggests a preferred number of between 6 and 14 studies for a qualitative review.

Conclusion

There is a complex range of perceptions and attitudes around PMH which influence women's experiences and access to PMH care within Bangladesh, India and Pakistan. Sociocultural expectations underpin many of the perceptions and attitudes identified in this review including the importance of familial and societal causes of PMH, emphasis on physical symptoms, sacredness of motherhood, lack of awareness, stigma, shame, limited resources allocated for mental health and lack of emotional sharing. These interlinking perceptions reflect the cultural collectivist nature of South Asian cultures preventing proper identification and adequate care for perinatal women with PMH conditions. These findings can be used to inform policy and practice through targeted interventions to tackle stigmatising attitudes and increasing education and training for healthcare providers. Further research should explore perceptions and attitudes of perinatal women along with key family members, such as husband and mother-in-law, and community-based healthcare providers in Bangladesh, where there is limited qualitative research, particularly during the antenatal period.

Abbreviations

CASP: Critical Appraisal Skills Programme; IPV: Intimate Partner Violence; HICs: High Income Countries; LMICs: Lower- and Middle-Income Countries; NOS: Newcastle-Ottawa Scale; PMH: Perinatal Mental Health.

Supplementary Information

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Additional file 1. The ENTREQ checklist.

Additional file 2. Search terms for OVID database.

Additional file 3. Summary of CASP tool used for quality appraisal of qualitative studies.

Additional file 4. Summary of NOS tool used for quality appraisal of cross-sectional studies.

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Authors' contributions

JR, SF and NI conceived the study. NI conducted the searches, screening, quality appraisal and data extraction, analysis and synthesis, and was the major contributor to the manuscript. AW conducted the screening and quality appraisal. JR and SF supported the analysis and synthesis of data and contributed to the drafting of the manuscript. All authors read and approved the final manuscript.

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Availability of data and materials

All data from the included studies analysed in this review are publicly available and included in this published article.

Declarations

Ethics approval and consent to participate

Not Applicable.

Consent for publication

Not Applicable.

Competing interests

All authors declare no competing interests.

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Appendix 3 - Social Determinants and Prevalence of Antenatal Depression among Women in Rural Bangladesh: A Cross-Sectional Study

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Article

Social Determinants and Prevalence of Antenatal Depression among Women in Rural Bangladesh: A Cross-Sectional Study

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Abstract: The prevalence of antenatal depression in Bangladesh ranges from 18 to 33%. Antenatal depression has negative impacts on the mother and child such as suicidal ideations, low birth weight, and impaired fetal development. This cross-sectional study aims to determine the prevalence and social determinants of antenatal depression in rural Sylhet, Bangladesh. Data were collected from 235 pregnant women between March and November 2021. The validated Bangla Edinburgh Postnatal Depression Scale was used to measure antenatal depressive symptoms (ADS). Background information was collected using a structured questionnaire including the Duke Social Support and Stress Scale, pregnancy choices, and WHO Intimate Partner Violence questions. Point-prevalence of antenatal depression was 56%. Intimate partner violence (IPV) before pregnancy (adjusted odds ratio (AOR) 10.4 [95% confidence interval (CI) 2.7–39.7]) and perceived husband's male gender preference (AOR 9.9 [95% CI 1.6–59.6]) were significantly associated with increased odds of ADS among pregnant women. Increased family support was a significant protective factor for ADS (AOR 0.94 [95% CI 0.91–0.97]). Antenatal depression commonly occurs in rural Sylhet, Bangladesh, highlighting the need for improved screening and management within these settings. The findings suggest the need for community-based interventions for women with low family support and experiencing intimate partner violence, and educational programs and gender policies to tackle gender inequalities.

Keywords: perinatal mental health; antenatal depression; Bangladesh; prevalence; social determinants

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1. Introduction

The Global Burden of Disease Study in 2017 found depression to have prevailed as a leading cause of non-fatal health loss for almost three decades [1], with women being about twice as likely than men to develop depression in their lifetime [2]. Although pregnancy is mostly viewed as a positive experience, women have increased vulnerability to depression during and immediately after pregnancy [3]. The literature mostly focuses on the postnatal period, whereas evidence suggests that two-thirds of postnatal depression actually presents antenatally [4,5], making the antenatal period a key phase to target with screening and interventions. Antenatal depression, symptomized by persistent low mood, crying, loss of appetite, loss of sleep, and sadness, among other symptoms, has negative consequences on the mother and child such as suicidal ideations, low birth weight, and impaired fetal development [6].

Bangladesh is one of the most densely populated countries and is listed as a low–middle income country (LMIC) by the World Bank [7]. The prevalence of antenatal depression in Bangladesh has been found to range between 18 and 33% [3,8], which is higher than worldwide estimates of 10–20% [9]. This may be due to the lack of mental healthcare integration into antenatal healthcare, poor understanding and stigma surrounding mental health conditions, and higher socioeconomic deprivation [10]. Despite this burden, mental

health remains a neglected topic within the Bangladeshi population, government, and health system. Only 0.5% of the total health budget is allocated to mental health [11]. This is a small proportion considering the burden attributable to psychiatric disorders in Bangladesh is approximately 11.2% [12]. Furthermore, there is only one mental health institute in Bangladesh, the National Institute of Mental Health (NIMH), and limited mental health hospitals, the majority of which are located in urban areas with the density of psychiatric beds in and around the capital Dhaka being five times higher than anywhere else [12]. This raises a problem as 70% of the population live in rural areas. These areas are often served by informal providers and non-governmental organisations, such as the Maternal Aid Association (MAA) [12]. This creates limited perinatal mental healthcare services for pregnant women in rural regions of Bangladesh.

Previous studies in Bangladesh have found certain social determinants associated with antenatal depression. Social determinants are considered to be mostly responsible for health inequalities; hence, this study focuses on such determinants rather than obstetric risk factors, especially given the social and cultural dynamics within antenatal depression. Both Nasreen et al. [3] and Gausia et al. [8] found intimate partner violence (IPV) and low family support to be significantly associated with antenatal depression in pregnant women in rural Bangladesh. Gausia et al. [8] also found male gender preference to play an important role in antenatal depression. It is important to investigate these social determinants of antenatal depression, especially in a country such as Bangladesh which often neglects these issues, to understand and tackle the underlying mechanisms that play a role in antenatal depression. These previous studies [3,8] were both conducted in the capital of Bangladesh, Dhaka. There are no known studies conducted in the Sylhet division, which has the highest fertility rate and lowest contraceptive use [13,14]. Contraceptive prevalence rate is only 4.8% in Sylhet compared to the national average of 62.4% [15]. This is a concern as high fertility rates and low contraceptive rates highlight lack of education, family planning services, and gender equity within this region. It is essential to carry out studies in Sylhet to understand these social determinants and how they contribute to antenatal depression. Therefore, the aim of this study was to determine the prevalence and social determinants of antenatal depression among women in rural Sylhet, Bangladesh.

2. Materials and Methods

2.1. Study Design, Setting, and Sampling

This study followed a cross-sectional survey design to estimate a point-prevalence of antenatal depression and analyse the associated social determinants. The study setting was MAA health camps in Moulvibazar. Moulvibazar is a town in the Moulvibazar district within the Sylhet division located in north-eastern Bangladesh (Figure 1) with a population of approximately 41,358. Pregnant women attending and registering for the health camps were the study population and sampling frame used for this study. Convenience sampling was used to recruit participants, whereby pregnant women aged 18 or older in any trimester of their pregnancy attending the MAA health camps were eligible to take part in the survey with informed consent. Women with a known pre-existing diagnosis of any mental health condition were excluded as this would skew the measurement of antenatal depression, creating measurement bias. Women below the age of 18 were excluded due to ethical concerns. Using the sample size for prevalence studies calculation [16], a sample size of 227 was calculated to be adequate to estimate the prevalence of antenatal depression.

2.2. Survey Instrument

The survey consisted of five sections: (a) general background information, (b) Duke Social Support and Stress Scale (DUSOCS), (c) pregnancy choices, (d) Edinburgh Postnatal Depression Scale—Bangla (EPDS-B), and (e) WHO IPV. The survey was first written in English to develop the constructs and items. The English version was sent to experts in global health, maternal health, and sociology who have experience and knowledge on data

collection, particularly in LMICs within South Asia. The experts provided feedback on the design, clarity, and content of the survey, providing initial face and content validity.



Figure 1. Map of Bangladesh [17] showing divisions and districts (red arrow pointing to study location).

A systematic cultural adaptation process based on Khan and Avan [18] was used to develop, translate, and validate the survey. This involved six steps and aimed to ensure that each item was assessing the construct it was attempting to understand. Each item was first written in English, and the process of adaptation was (1) translation into Bangla independently by NI and RI; (2) comparing these translations, assessing technical equivalence of these, and then producing final translations by consensus for testing; (3) field research with project staff and mothers of young children to test understanding of the translations and to improve them; (4) finalisation of the tool for pre-testing; (5) pre-testing in the community to assess usability; and (6) assessor training and pilot-testing.

2.3. Dependent/Outcome Variable—Antenatal Depressive Symptoms (ADS)

ADS was measured using the EPDS-B. The EPDS is a self-report questionnaire consisting of ten items, each with four possible responses related to the subject's mood or feelings

in the last seven days [19]. Although originally meant for screening of postnatal depression, studies have shown it to be just as appropriate for use in the antenatal period [20]. Each item is scored between 0 and 3, with a total EPDS score ranging between 0 and 30. The Bangla translated version of the original EPDS has been validated in a Bangladeshi setting [19]. The validation study on EPDS-B against standard diagnosis of depression using the Structured Clinical Interview (SCID) for DSM-IV among Bangladeshi women showed a sensitivity of 89% and a specificity of 87% at cut-off scores of 10 [19]. Therefore, pregnant women scoring 10 or above in the EPDS-B were categorised as having ADS and pregnant women scoring less than 10 were categorised as non-ADS. The EPDS-B has a Cronbach's alpha of 0.84, suggesting good internal consistency [19].

2.4. Independent Variables—Social Determinants

General background information such as trimester of pregnancy, age, marital status, highest education level, incoming-earning activity, and husband's employment were collected in section A of the survey.

Section B consisted of the social support section of the DUSOCS, which captures an individual's perceptions of how supportive their relationships are [21]. It also allows for the identification of one's most supportive relationship. The DUSOCS creates six scores: family support, family stress, non-family support, non-family stress, total social support, and total social stress. For this study, we were interested in family support. A 3-point scale was used ("none = 0", "some = 1", and "a lot = 2") in which the pregnant women rate their family members as people who give personal support (6 items). The DUSOCS was slightly adapted, as recommended by the literature, for this study in order to capture all the important relationships for pregnant women in rural areas of Bangladesh [22]. All scores range from 0 to 100, with higher scores indicating higher levels of support. The family support items has a test-retest reliability of 0.76 and Cronbach's alpha of 0.7. This suggests excellent reliability [21].

Section C entailed questions around certain pregnancy choices: gender preference of both mother and father, unwanted pregnancies, and fear of childbirth.

The final section consisted of questions relating to IPV. These questions covered the three main areas of violence as defined by the WHO: verbal, physical, and sexual violence [23]. A 4-point scale was used ("no = 0", "yes, once = 1", "yes, a few times = 2", and "yes, many times = 3") in which pregnant women rate whether (and how often) they experienced each type of IPV both before their pregnancy and during the current pregnancy. The responses were coded to "yes" or "no" for all forms of IPV before pregnancy and during pregnancy to create 2 separate variables (IPV before and IPV during pregnancy) due to the small sample sizes for some of the 4-point responses for individual types of violence.

2.5. Data Collection Procedure

Data were collected by MAA researchers in the health camps and the new clinic in Moulvibazar, Sylhet, from 10 March 2021 to 2 November 2021. Pilot data collection using the survey from 36 pregnant women was conducted on 10 March 2021 in the Moulvibazaar health camp, followed by a meeting with the MAA researchers to gain feedback on the clarity, design, and content of the survey within the target population. No changes to the survey were deemed necessary by the MAA team; therefore, data from the pilot phase were included in the final sample and data collection was resumed.

Information regarding the study were verbally administered to the participants in Bengali and both written and verbal informed consent was acquired. Survey interviews were conducted in a private area of the health camps/clinic and care was taken to ensure participants were comfortable whilst answering the questions. The interviews lasted on average 15 minutes.

2.6. Statistical Analysis

Data from Excel were exported into the Statistical Package for the Social Sciences (SPSS) for coding and analysis. Categories were created from the total EPDS-B scores to indicate those with EPDS-B scores of 10 or more to have ADS and those with lower than 10 to be non-ADS. Means and standard deviations were used to describe normally distributed continuous data, and median and interquartile ranges (IQR) were used for non-normally distributed continuous data. Frequencies and percentages were used to describe categorical variables.

An independent samples t-test was used to compare means between groups. Bivariate analyses (chi-squared and Fisher's exact) were conducted between each independent variable and the outcome variable of ADS and non-ADS independently to identify possible determinants that were associated with ADS. The minimum sample size for binary logistic regression was calculated to be 227 using the pmsampsize package on STATA as recommended by Riley et al. 2020 [24]. Therefore, the use of a logistic regression model was justified for this study. The independent variables that were significantly associated with ADS in the bivariate analyses were considered as possible social determinants and were entered into a multivariate logistic regression model, and adjusted odds ratios (AOR) and 95% confidence intervals (CI) were produced. The minimum variance inflation factors (VIFs) from the regression suggested no severe multicollinearity between the independent variables.

2.7. Ethical Considerations

This study was approved by the Faculty of Medical Sciences Research Ethics Committee, part of Newcastle University's Research Ethics Committee, ref 2049/4634/2020. This committee contains members who are internal to the faculty but independent to the study. This study was reviewed by members of the committee, who must provide impartial advice and avoid significant conflicts of interest. The study was also approved by the Sylhet Women's Medical College Research Ethics Board in Bangladesh.

3. Results

Overall, there were 235 respondents with ages ranging from 18 to 38 years and a median age of 25. The pregnant women were in varying stages of their pregnancy, with 100 women (42.6%) in their second trimester, 71 women (30.3%) in their third trimester, and 57 women (24.2%) in their first trimester. Seven women did not know which trimester they were in. All the respondents were married, with the majority (75.8%) having no education or having ended their education before completing class 10, i.e., having not gained their Secondary School Certification (SSC) qualifications (equivalent to GCSE in the UK). The remaining had some level of education SSC and above. The majority of respondents, 97.9%, do not undertake any paid work, with the husband being the sole breadwinner for the family through work such as fishing, farming, construction, or business (96.6% of respondents' husbands undertake paid work). See Table 1 for a full summary of respondents' characteristics.

Table 1. Summary of respondent (pregnant women) characteristics.

Variable	Study Sample (n = 235)
Age	
Median age in years, (IQR)	25 (6)
Trimester, n (%)	
First	57 (24.2)
Second	100 (42.6)
Third	71 (30.3)
Unknown	7 (3.0)
Marital status, n (%)	
Married	235 (100)

Table 1. *Cont.*

Variable	Study Sample (*n* = 235)
Highest level of education, *n* (%)	
No schooling/Pre-SSC	178 (75.8)
SSC or above	56 (23.8)
Do not know	1 (0.4)
Paid work, *n* (%)	
Yes	5 (2.1)
No	230 (97.9)
Gender of baby, *n* (%)	
Girl	7 (3.0)
Boy	9 (3.8)
Do not know	207 (88.1)
Prefer not to say	12 (5.1)
Gender preference, *n* (%)	
Girl	31 (13.2)
Boy	51 (21.7)
No preference	133 (56.6)
Prefer not to say	20 (8.5)
Perceived husband's gender preference, *n* (%)	
Girl	30 (12.8)
Boy	65 (27.7)
No preference	119 (50.6)
Prefer not to say	21 (8.9)
Want pregnancy, *n* (%)	
Yes	142 (60.4)
No	20 (8.5)
Prefer not to say	73 (31.1)
Perceive husband wants pregnancy, *n* (%)	
Yes	152 (64.7)
No	17 (7.2)
Prefer not to say	66 (28.1)
Worried about childbirth/labour, *n* (%)	
Yes	200 (85.1)
No	17 (7.2)
Prefer not to say	18 (7.7)
Family support	
Family support score, mean (SD)	50.0 (17.3)
Intimate Partner Violence (IPV), *n* (%)	
Before pregnancy	
Yes	64 (27.2)
No	86 (36.6)
Prefer not to say	85 (36.2)
During pregnancy	
Yes	32 (13.6)
No	110 (46.8)
Prefer not to say	93 (39.6)
Antenatal Depressive Symptoms (ADS), *n* (%)	
Yes	133 (56.6)
No	102 (43.4)

IQR = Interquartile Range; SD = standard deviation.

The majority of the respondents, 207 (88.1%), did not know the gender of the baby they were having. Nine women (3.8%) stated they were having a boy and seven (3.0%) stated they were having a girl. The rest of the women, 12 (5.1%), preferred not to say what gender of the baby they were having. The majority of the women, 133 (56.6%), stated they had no preference for the gender of the baby and the majority, 119 (50.6%), perceived their husband to also have no preference. Almost a quarter of the women, 51 (21.7%), stated they desired to have a boy compared to 31 (13.2%) who wanted a girl, and 65 (27.7%) perceived

their husband's preference to be a son compared to 30 (12.8%) who perceived their husband to want a girl.

Most pregnant women, 142 (60.4%), stated that their pregnancy was wanted at this time compared to 20 (8.5%) who stated they did not want the pregnancy. Similarly, 152 (64.7%) pregnant women perceived their husband to also want the pregnancy compared to 17 (7.2%) who felt their husband did not want the baby at this time. The remaining 73 (31.1%) pregnant women preferred not to say whether the pregnancy was wanted or not and 66 (28.1%) preferred not to say whether they felt their husband wanted the baby or not. A total of 200 (85.1%) pregnant women were worried about childbirth/labour compared to 17 (7.2%) who stated they were not worried. The remaining 18 (7.7%) preferred not to say.

From the DUSOCS results, the family support score was used for the analysis, which ranged from 7.1 to 100. The mean family support score was 50.0 (SD 17.3). In this study, only 12.3% and 12.8% of pregnant women stated they receive no support from their husband and in-laws, respectively. The remaining receive some (45.5% and 66.8%) or a lot (42.1% and 20.4%) of support from their husband and in-laws, respectively.

From the IPV questions, 64 (27.2%) pregnant women stated they have experienced IPV before their current pregnancy compared to 86 (36.6%) who stated they have not experienced IPV before their current pregnancy. Furthermore, 110 (46.8%) stated they have experienced IPV during their current pregnancy compared to 32 (13.6%) who stated they have not experienced IPV during their current pregnancy.

Of the 235 respondents, 133 (56.6%) had total EPDS-B scores of 10 or above, indicating a point-prevalence of antenatal depression of 56.6% (95.5% CI 50.0–63.0%). Figure 2 shows the distribution of total EPDS-B scores among respondents, which ranged from 0 to 23. The mean EPDS-B score among all respondents was 10.5 (SD 5.2). The mean EPDS-B score for respondents without ADS and those with were 5.6 (SD 2.2) and 14.3 (SD 3.2), respectively. The difference of −8.8 (95% CI −9.5–−8.0) in the mean scores was statistically significant ($t = -23.7$, $df = 233$, $p < 0.001$), indicating those with depressive symptoms had significantly higher EPDS-B scores than those without depressive symptoms.

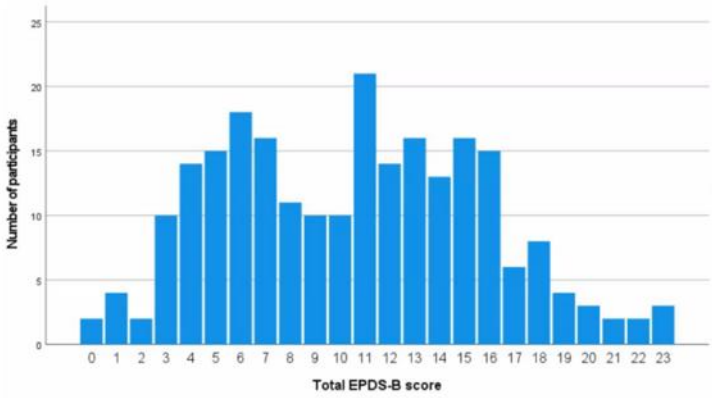


Figure 2. Bar graph showing the distribution of total EDPS-B scores.

A number of independent potential social determinants for antenatal depression were examined in this study. From the bivariate analyses, IPV before pregnancy, IPV during pregnancy, pregnant woman's gender preference, perceived husband's gender preference for the baby, and family support were significantly associated with ADS (Table 2).

Table 2. Bivariate analysis of social determinants with antenatal depressive symptoms among pregnant women in MAA health camps in rural Sylhet.

Variables	ADS n = 133 (%)	Non-ADS n = 102 (%)	p-Value [c]
Trimester			
First	33 (24.8)	24 (23.5)	
Second	56 (42.1)	44 (43.1)	0.79
Third	37 (27.8)	34 (33.3)	
Education			
No schooling/Pre-SSC	106 (79.7)	72 (70.6)	0.084
SSC or above	26 (19.5)	30 (29.4)	
IPV before pregnancy			
Yes	53 (39.8)	11 (10.8)	<0.001 *
No	37 (27.8)	49 (48.0)	
IPV during pregnancy			
Yes	26 (19.5)	6 (5.88)	0.002 *
No	56 (42.1)	54 (52.9)	
Gender preference			
Boy	38 (28.6)	13 (12.7)	
Girl	15 (11.3)	16 (15.7)	0.013 *
No preference	69 (51.9)	64 (62.7)	
Perceived husband's gender preference			
Boy	51 (38.3)	14 (13.7)	
Girl	15 (11.3)	15 (14.7)	<0.001 *
No preference	57 (42.9)	62 (60.8)	
Want baby			
Yes	89 (66.9)	53 (52.0)	0.524
No	14 (10.5)	6 (5.9)	
Perceive husband wants pregnancy			
Yes	99 (74.4)	53 (52.0)	0.606
No	10 (7.5)	7 (6.9)	
Variables	**ADS mean (SD)**	**Non-ADS mean (SD)**	**p-value [t]**
Age	24.33 (5.43)	24.23 (4.20)	0.871
Family support	46.4 (18.7)	52.3 (14.7)	0.009 *

[c] = chi-square test for association or Fisher's exact test. [t] = independent samples t-test. * = statistical significance at 0.05 level. SD = standard deviation. Percentages do not add to 100 as an answer of 'prefer not to say' was not included in the analyses. SSC = Secondary School Certification; ADS = antenatal depressive symptoms.

After performing the multivariate logistic regression (Table 3), IPV before pregnancy, family support, and perceived husband's gender preference were found to be significant determinants. Pregnant women who have experienced IPV before their pregnancy had 10 times higher odds of ADS compared to pregnant women who have not experienced IPV before pregnancy (AOR 10.4 95% CI 2.7–39.7). ADS was significantly associated with low family support, whereby as family support increases, the odds of ADS decrease (AOR 0.94 [95% CI 0.91–0.97]). Pregnant women who perceived their husbands to want a son had almost 10 times higher odds of antenatal depression compared to pregnant women who perceived their husband to have no gender preference (AOR 9.9 [95% CI 1.6–59.6]).

Table 3. Multivariate logistic regression model showing associated factors of antenatal depressive symptoms among pregnant women in MAA health camps in rural Sylhet.

Variables	AOR (95% CI)	*p*-Value [c]
IPV before pregnancy		
Yes	10.4 (2.7–39.7)	<0.001 *
No	1	
IPV during pregnancy		
Yes	1.4 (0.3–7.1)	0.656
No	1	
Gender preference		
Boy	0.21 (0.04–1.2)	0.085
Girl	0.34 (0.04–3.2)	0.348
No preference	1	
Perceived husband's gender preference		
Boy	9.9 (1.6–59.6)	0.012 *
Girl	2.4 (0.29–19.1)	0.418
No preference	1	
Variables		***p*-value [t]**
Family support	0.94 (0.91–0.97)	<0.001 *

[c] = chi-square test for association or Fisher's exact test. [t] = independent samples *t*-test. AOR = adjusted odds ratio. * = statistical significance at 0.05 level. SD = standard deviation. CI = confidence interval.

4. Discussion

4.1. Key Findings

This study aimed to determine the prevalence and identify social determinants of antenatal depression in rural Sylhet, Bangladesh. This study found a point-prevalence of antenatal depression of 56.6%. Low family support was found to be a significant determinant of ADS, while perceived husband's male gender preference and IPV were found to significantly increase the odds of ADS.

The prevalence of antenatal depression determined in this study is higher than previous studies conducted by Gausia et al. [8] and Nasreen et al. [3], which found a prevalence of antenatal depression in rural Bangladesh of 33% and 18.3%, respectively. A higher prevalence found in this current study may be due to methodological differences such as study location and population. Previous studies were conducted in Dhaka, the capital of Bangladesh, whereas our study was conducted in the division and district of Sylhet, which is a rural area. The Sylhet division has the highest fertility rate and lowest contraceptive use [13,14], with a contraceptive prevalence rate of 4.8% in Sylhet compared to the national average of 62.4% [15]. This is a concern as high fertility rates and low contraceptive rates highlight lack of education, family planning services, and gender equity within this region. Furthermore, the Sylhet region has geographical characteristics such as hilly areas, which make it difficult for universal health coverage in Bangladesh [15]. This is reflected by the 25.7% difference in antenatal care coverage between Khulna (second largest of the eight divisions in Bangladesh) and Sylhet, with Sylhet having low levels of antenatal care [15]. Therefore, these differences may play a role in the higher prevalence of antenatal depression in our study compared to previous studies. Nevertheless, differences may also be due to smaller sample size in our study, which could affect study power.

The association between antenatal depression and low family support is reflected in many previous studies [3,5,8,25,26]. Family support is of special importance in Bangladeshi communities, where women often live with their in-laws after marriage [3]. Therefore, family dynamics play a key role in a pregnant woman's emotional well-being and support from her newly formed family, especially her husband and mother-in-law, is essential in making her feel accepted and protected during her pregnancy. Furthermore, women are expected to carry out the household chores, which can become more difficult during pregnancy. Therefore, practical support, as well as emotional support, from family members

during this time can be a source of ease, preventing depressive symptoms. Practical support can be displayed through helping with household chores such as cooking, washing, and cleaning. This type of support may also be seen as a sign of approval and acceptance from the in-laws, which can have positive impacts on the mental health of pregnant women [3].

IPV has been well established in the literature as an important social determinant of antenatal depression [27]. IPV experienced before pregnancy was significantly associated with ADS and was identified to be the most important social determinant of ADS in this study. However, IPV experienced during pregnancy, although showing increased odds of ADS, was not statistically significant. Nasreen et al. [3] and Gausia et al. [8] both found that IPV during pregnancy was significantly associated with ADS in Bangladesh. Therefore, it would not be appropriate to conclude that IPV during pregnancy is not associated with ADS in this study as the non-significance may be due to the small sample size of the IPV during pregnancy sub-groups, which may be underpowered [28]. Pregnant women undergo social, emotional, and physical isolation and pressure when experiencing IPV, which is heightened during pregnancy due to worries about the unborn baby's health [29]. Further studies with larger sample sizes are required to improve the power related to sub-group analyses.

The husband's gender preference for the baby as perceived by the pregnant woman has not been widely investigated in Bangladesh. A study of pregnant women in Iran [30] found that the husband's preference for a male child is significantly associated with antenatal depression in women. Similarly, a study of pregnant women in China [31] found that no gender preference in the spouse was associated with a significant reduction in persistent depression. These findings are also reflected in our study, whereby pregnant women who perceived their husband to want a son compared to having no gender preference have higher odds of antenatal depression. However, there was no significant association between the woman's own gender preference and antenatal depression, which is contradictory to many previous studies [8,28,32–34]. Male gender preference is a common phenomenon in Bangladeshi communities and has been associated with antenatal depression [26]. This again reflects the importance of the husband in decision making and power dynamics in Bangladeshi families, whereby the husband's preference is taken to be more important. A husband's desire for a son may create stress for many pregnant women if they are unable to give him a son, which can result in serious repercussions. This finding demonstrates the need for more whole-family interventions as a pregnant woman's family, in particular her husband, plays a key role in her mental well-being.

These social determinants identified within this study are complex and interlinked factors. Socially constructed gender roles favour and elevate the status of men, creating gender inequalities in Bangladesh [35]. These gender inequalities can be displayed through IPV where men feel they have power over women, pressure on pregnant women from her husband or in-laws to give birth to a son who can provide for the family in the future, and lack of family support during pregnancy. Stigma around mental health and women's health also prevents women from speaking up about violence, family issues, and their emotions, resulting in many pregnant women feeling a lack of support and being vulnerable to depressive symptoms during their pregnancy [36].

4.2. Strengths and Limitations

This study has included a widely used and validated scale to measure antenatal depression in pregnant women in Bangladesh [19]. This increases the accuracy of measuring antenatal depression and allows for effective comparisons to be made with previous studies. Another strength of this study is that the survey was reviewed by experts and piloted within the population of interest, which allowed us to test the suitability and appropriateness of the survey in that setting and allowed the MAA volunteers to gain training in delivering the survey.

Furthermore, considering the likely low levels of literacy in rural Bangladesh, the survey was delivered by trained MAA volunteers in Bengali and not self-administered.

This ensured that pregnant women clearly understood the survey as there was someone there to explain the instructions and questions to them. Nevertheless, this may introduce some response bias as pregnant women may have provided responses that they felt were socially acceptable or not provided honest answers to sensitive questions due to the limited anonymity. Women may feel scared or judged whilst answering questions regarding IPV, pregnancy choices, and EPDS-B items, especially if their family members were present around them. This was highlighted by the MAA researchers after the pilot surveys who found that women were less likely or hesitant to answer questions related to IPV if they came to the health camps with their family. This is also evident from the high proportion of 'prefer not to say' responses. Response bias may have led to an under-reporting of certain social determinants and ADS. To minimise this bias, MAA volunteers reassured pregnant women that their answers would remain anonymous and confidential, and they built a good rapport with the women beforehand.

As the study was cross-sectional, temporality and causality cannot be established as both the outcome and social determinants were measured at the same time. However, this study design was still deemed appropriate and addresses the objective of identifying the social determinants which have an association with antenatal depression in Bangladesh, which can be used to recommend the development of tailored interventions. Nevertheless, given the small sample size, the results cannot be generalizable to the whole population. However, the results are effective in providing an initial insight into the social determinants of antenatal depression in rural Sylhet, Bangladesh, upon which further research can be conducted.

5. Conclusions

The findings from this study suggest a high prevalence of antenatal depression in rural Sylhet, Bangladesh, highlighting the need for interventions and policies to tackle such high rates. IPV, low family support, and perceived husband's male gender preference were found to be associated with antenatal depression. This highlights the need for more tailored interventions for pregnant women with low family support or those facing IPV, ensuring they are receiving the support they need and educating families on pregnancy support. There is also a need for educational programs and gender policies to tackle gender inequalities within these communities. Further research should include qualitative in-depth interviews with pregnant women and their families to understand the mechanisms within these social determinants.

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Institutional Review Board Statement: This study was approved by the Faculty of Medical Sciences Research Ethics Committee, part of Newcastle University's Research Ethics Committee, ref 2049/4634/2020. This committee contains members who are internal to the faculty. This study was reviewed by members of the committee, who must provide impartial advice and avoid significant conflicts of interest. The study was also approved by Sylhet Women's Medical College Research Ethics Board in Bangladesh.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The datasets used and analysed during the current study are available from the corresponding author upon reasonable request.

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Conflicts of Interest: The authors declare no conflict of interest.

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Appendix 4 – MOOSE checklist for Meta-analysis of Observational Studies

Item No	Recommendation	Reported yes/no	Reported in section, page number
	Reporting of background should include		
1	Problem definition	Yes	2.2.2, 5-6
2	Hypothesis statement	No	N/A
3	Description of study outcome(s)	Yes	2.3.1, 7-8
4	Type of exposure or intervention used	Yes	2.3.1, 8-9
5	Type of study designs used	Yes	2.3.1, 8-9
6	Study population	Yes	2.3.1, 8-9
	Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	Yes	2.3.1, 11-12
8	Search strategy, including time period included in the synthesis and key words	Yes	2.3.1, 7-9 & Appendix 5
9	Effort to include all available studies, including contact with authors	Yes	2.3.1, 11
10	Databases and registries searched	Yes	2.3.1, 11
11	Search software used, name and version, including special features used (eg, explosion)	Yes	2.3.1, 11
12	Use of hand searching (eg, reference lists of obtained articles)	Yes	2.3.1, 11
13	List of citations located and those excluded, including justification	Yes	Fig 2
14	Method of addressing articles published in languages other than English	Yes	2.3.1, 7
15	Method of handling abstracts and unpublished studies	Yes	2.3.1, 7
16	Description of any contact with authors	No	N/A
	Reporting of methods should include		

17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	No	N/A
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	Yes	2.3.4, 14-15
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	Yes	2.3.4, 14-15
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	Yes	2.3.4, 14-15
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	Yes	2.3.3 13-14 Appendix 7 & 8
22	Assessment of heterogeneity	Yes	2.3.4, 14 2.4
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	Yes	2.3.4, 14-15
24	Provision of appropriate tables and graphics	Yes	2.4 Fig 2
	Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Yes	Fig 3-18
26	Table giving descriptive information for each study included	Yes	Table 5
27	Results of sensitivity testing (eg, subgroup analysis)	Yes	2.4.14, 61-62

28	Indication of statistical uncertainty of findings	Yes	Confidence intervals provided throughout narrative results, tables and figures in the main manuscript
	Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	Yes	2.4
30	Justification for exclusion (eg, exclusion of non-English language citations)	Yes	2.5.3, 70
31	Assessment of quality of included studies	Yes	2.4.1, 28 Appendix 8
	Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	Yes	2.5, 62-67
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	Yes	2.5, 62-67
34	Guidelines for future research	Yes	2.5, 62-67
35	Disclosure of funding source	Yes	Funding declaration

Appendix 5 - MEDLINE, Embase & PsychInfo search strategy via OVID

1. perspective*.mp.
2. views.mp.
3. opinions.mp.
4. attitude*.mp.
5. perception*.mp.
6. factors*.mp.
7. "risk factors".mp.
8. determinants.mp.
9. dimensions.mp.
10. correlates.mp.
11. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10
12. Maternal Health/ or maternal.mp.
13. perinatal.mp.
14. antenatal.mp. or Pregnancy/
15. pregnan*.mp.
16. postpartum.mp.
17. women.mp.
18. 12 or 13 or 14 or 15 or 16 or 17
19. depress*.mp.
20. Depression, Postpartum/
21. Anxiety/ or anxiety.mp.
22. anxi*.mp.
23. stress.mp.
24. mood.mp.
25. 19 or 20 or 21 or 22 or 23 or 24
26. Bangladesh/ or bangladesh*.mp.
27. Pakistan/ or pakistan*.mp.
28. India/ or india*.mp.
29. "south asia*".mp.
30. 26 or 27 or 28 or 29
31. 11 and 18 and 25 and 30

Appendix 6 - Data extraction form template

Adapted from Cochrane data extraction template

Reviewer	
Title of paper	
Author and year	
Country	Bangladesh ☐ India ☐ Pakistan ☐
Methodology	Cohort ☐ Case-control ☐ Cross-sectional ☐
Setting	Hospital ☐ Community-based ☐
Inclusion criteria for women	
Exclusion criteria for women	
Recruitment and sampling method	
Sample size of pregnant women	

Outcome	**Ascertainment of outcome (what tool was used)**	**Was outcome tool validated?**	**What cut-off score was used to categorise the outcome**
Depression ☐ Anxiety ☐ Depression & Anxiety ☐			

Social determinants collected from participants	Ascertainment of social determinants (e.g questionnaires etc)	Which social determinants included in analysis?

Results

Social determinant	Unadjusted (*=significant)	Adjusted (*=significant)	Factors adjusted for

Appendix 7 - Newcastle-Ottawa Quality Assessment Scale adapted for cross-sectional studies[1]

Selection (maximum 5 stars)

1) Representativeness of the sample
    a. Truly representative of the average in the target population/sampling frame (all subjects or random sampling) *
    b. Somewhat representative of the average in the target population/sampling frame (non-random sampling) *
    c. Selected group of pregnant women
    d. No description of sampling strategy
2) Sample size
    a. Justified and satisfactory *
    b. Not justified
3) Non-respondents
    a. Proportion of target sample recruited attains pre-specified target or basic summary of non-respondent characteristics in sampling frame recorded. *
    b. Unsatisfactory recruitment rate, no summary data on non-respondents.
    c. No description of the response rate or the characteristics of the respondents and the non-respondents
4) Ascertainment of the exposure (risk-factors)
    a. Validated questionnaires used/record linkage **
    b. Non-validated questionnaires, but the questions are available or described. *
    c. No description of the measurement tool

Comparability[a] (maximum 2 stars)

5) The subjects in different outcome groups are comparable, based on the study design or analysis. Confounding factors are controlled.
    a. The study controls for the important factors (as identified by univariate analysis in the study or justified). **
    b. The study controls for factors but no justification as to why they were chosen. *

[1] Herzog R, Alvarez-Pasquin MJ, Diaz C, Del Barrio JL, Estrada JM, Gil A. Are healthcare workers' intentions to vaccinate related to their knowledge, beliefs and attitudes? A systematic review. BMC Public Health. 2013. 154. https://doi.org/10.1186/1471-2458-13-154

c. No factors controlled for.

Outcome (Depression/anxiety) (maximum 3 stars)

6) Assessment of the outcome[b]
    a. Validated measurement tool with depression/anxiety categories justified and appropriate **
    b. Record linkage **
    c. Validated measurement tool but depression/anxiety categories not justified. *
    d. No description of ascertainment method.
7) Statistical analysis
    a. The statistical test used to analyze the data is clearly described and appropriate, and the measurement of the association is presented, including confidence intervals and the probability level (p value). *
    b. The statistical test is not appropriate, not described, or incomplete

This scale has been adapted from the Newcastle-Ottawa Quality Assessment scale for cohort studies to provide quality assessment for cross-sectional studies.[1]
Red font is where the form has been adapted to make questions relevant to this systematic review.

[a]We have not selected one factor that is the most important for comparability, because the variables are not the same in each study. Thus, the study should justify why they have controlled for certain factors e.g. if they were significant in univariate analysis.

[b]Original question six options: a) independent blind assessment b) record linkage c) self-report d) no description, are not applicable to this study as depression/anxiety were the outcomes of interest which are usually measured using a depression/anxiety scale which in nature contain self-reported questions. Therefore, the options were adapted to suit this systematic review.

Appendix 8 - Quality assessment of all included studies

Paper	Newcastle Ottawa Scale question number and score allocated[a]									Independent reviewer initials[d]
	1	2	3	4	5	6	7	Total stars[b]	Category[c]	
Ajinkya 2013	a*	a*	c	c	c	a**	b	4	Medium	NI & AW
Ali 2012	d	b	c	b*	a**	a**	a*	6	Medium	NI & AW
Ayaz 2019	d	b	c	b*	c	c*	a*	3	Low	NI &AW
Ayyub 2018	b*	a*	a*	a**	a**	a**	a*	10	High	NI & AW
Babu 2018	a*	a*	c	a**	a**	c*	a*	8	High	NI & AW
Bavle 2016	d	b	c	a**	c	a**	b	4	Medium	NI & AW
Dahiya 2020	a*	b	c	b*	b*	a**	a*	6	Medium	NI & AW
Din 2016	d	b	c	b*	a**	c*	a*	5	Medium	NI & AW
Gausia 2009	a*	a*	c	b*	a**	a**	a*	8	High	NI & AW
George 2016	a*	a*	a*	a**	a**	a**	a*	10	High	NI & AW
Ghaffar 2017	a*	a*	b	b*	a**	a**	a*	8	High	NI & AW
Goyal 2020	b*	a*	a*	b*	c	a**	b	6	Medium	NI & SF
Gul 2017	b*	a*	a*	a**	c	a**	b	7	Medium	NI & SF
Hedge 2013	d	b	c	b*	c	c*	b	2	Low	NI & SF
Humayun 2013	a*	a*	c	b*	c	c*	a*	5	Medium	NI & AW
Imran 2010	b*	a*	a*	b*	c	a**	b	6	Medium	NI & SF
Jafri 2017	a*	b	c	c	c	c*	b	2	Low	NI & SF

Jamal 2018	a*	a*	c	b*	b*	c*	b	5	Medium	NI & SF
Karmaliani 2009	d	b	c	b*	a**	a**	a*	6	Medium	NI & AW
Maselko 2018	a*	b	c	a**	b*	a**	a*	7	Medium	NI & SF
Mir 2012	a*	a*	a*	a**	a**	c*	a*	9	High	NI & SF
Nasreen 2011	a*	a*	a*	a**	a**	a**	a*	10	High	NI & AW
Nath 2019	a*	b	c	a**	a**	a**	a*	8	High	NI & AW
Niaz 2004	a*	b	c	c	c	c*	b	2	Low	NI & AW
Rabia 2017	b*	a*	a*	b*	c	c*	b	5	Medium	NI & AW
Rahman 2003	a*	b	c	a**	a**	c*	a*	7	Medium	NI & AW
Sabir 2019	b*	a*	a*	c	c	c*	b	4	Medium	NI & AW
Safi 2013	d	b	c	b*	c	a**	b	3	Low	NI & AW
Sheeba 2019	a*	b	c	a**	a**	a**	a*	8	High	NI & AW
Shehroz 2019	d	b	c	c	c	c*	b	1	Low	NI & AW
Shidhaye 2017	a*	a*	a*	a**	a**	a**	a*	10	High	NI & AW
Srinivasan 2015	b*	b	c	a**	c	a**	a*	6	Medium	NI & AW
Surkan 2018	a*	a*	a*	b*	a**	a**	a*	9	High	NI & AW
Zia 2018	d	b	c	b*	c	c*	b	2	Low	NI & AW

[a]Newcastle-Ottawa question numbers 1-7, answers a-d, and associated number of stars (*) are detailed in Appendix
[b]Minimum number of possible stars to be awarded = 0, maximum number of possible stars to be awarded = 10.
[c]Categories were allocated as: Low = 0-3 stars, Medium = 4-7 stars, High = 8-10 stars.
[d]Reviewers' initials relate to manuscript authors: NI, Nafisa Insan; AW, Anthony Weke, SF, Simon Forrest

Appendix 9 – The ENTREQ checklist

Item	Guide and Description	Reported on page #
Aim	State the research question the synthesis addresses	73
Synthesis methodology	Identify the synthesis methodology or theoretical framework which underpins the synthesis, and describe the rationale for choice of methodology (e.g. meta-ethnography, thematic synthesis, critical interpretive synthesis, grounded theory synthesis, realist synthesis, meta-aggregation, meta-study, framework synthesis).	76-77
Approach to searching	Indicate whether the search was pre-planned (comprehensive search strategies to seek all available studies) or iterative (to seek all available concepts until theoretical saturation is achieved).	73
Inclusion criteria	Specify the inclusion/exclusion criteria (e.g. in terms of population, language, year limits, type of publication, study type).	75
Data Sources	Describe the information sources used (e.g. electronic databases (MEDLINE, EMBASE, CINAHL, psychINFO, Econlit), grey literature databases (digital thesis, policy reports), relevant organisational websites, experts, information specialists, generic web searches (Google Scholar), hand searching, reference lists) and when the searches were conducted; provide the rationale for using the data sources.	73
Electronic Search strategy	Describe the literature search (e.g. provide electronic search strategies with population terms, clinical or health topic terms, experiential or social phenomena related terms, filters for qualitative research and search limits).	Appendix 5

Study screening methods	Describe the process of study screening and sifting (e.g. title, abstract and full text review, number of independent reviewers who screened studies)	73
Study characteristics	Present the characteristics of the included studies (e.g. year of publication, country, population, number of participants, data collection, methodology, analysis, research questions).	77-82
Study selection results	Identify the number of studies screened and provide reasons for study exclusion (e.g. for comprehensive searching, provide numbers of studies screened and reasons for exclusion indicated in a figure/flowchart; for iterative searching describe reasons for study exclusion and inclusion based on modifications t the research question and/or contribution to theory development).	74
Rationale for appraisal	Describe the rationale and approach used to appraise the included studies or selected findings (e.g. assessment of conduct (validity and robustness), assessment of reporting (transparency), assessment of content and utility of the findings)	75-76 Appendix 10 & 11
Appraisal items	State the tools, frameworks and criteria used to appraise the studies or selected findings (e.g. Existing tools: CASP, QARI, COREQ, Mays and Pope [25]; reviewer developed tools; describe the domains assessed: research team, study design, data analysis and interpretations, reporting).	75-76
Appraisal process	Indicate whether the appraisal was conducted independently by more than one reviewer and if consensus was required.	75
Appraisal results	Present results of the quality assessment and indicate which articles, if any, were weighted/excluded based on the assessment and give the rationale.	Table 11

Data extraction	Indicate which sections of the primary studies were analysed and how were the data extracted from the primary studies? (e.g. all text under the headings “results /conclusions” were extracted electronically and entered into a computer software).	76
Software	State the computer software used, if any.	76
Number of reviewers	Identify who was involved in coding and analysis.	76
Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts).	76
Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary).	76
Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive.	76
Quotations	Provide quotations from the primary studies to illustrate themes/constructs, and identify whether the quotations were participant quotations or the author’s interpretation	85-93
Synthesis output	Present rich, compelling and useful results that go beyond a summary of the primary studies (e.g. new interpretation, models of evidence, conceptual models, analytical framework, development of a new theory or construct).	85-93

Appendix 10 – Summary of CASP tool used for quality appraisal of qualitative studies

Author/year	Clear Aims	Appropriate methodology	Appropriate research design	Appropriate recruitment strategy	Appropriate data collection methods	Researcher-participant relationship considered	Ethical issues considered	Rigorous data analysis	Clear findings	Value of research	Overall assessment
Edhborg et al, 2015	Y	Y	Y	Y	Y	?	Y	?	Y	Y	Adequate
McCauley et al, 2020	Y	Y	Y	Y	Y	?	?	Y	Y	Y	Adequate
Rodrigues et al, 2003	Y	Y	Y	Y	Y	?	?	?	Y	Y	Adequate
Williams et al, 2018	Y	Y	Y	?	Y	?	Y	Y	N	Y	Adequate

Y= Yes
? = Insufficient information to make an assessment
N = No

Scoring

A scoring of 1 mark for each question that was answered with "Y" was allocated to provide an indicator of quality and enable comparison between reviewers. Cut-off scores of ≥9 indicated 'strong' quality, 6-8 indicated 'adequate' quality and ≤5 indicated 'weak' quality for the CASP tool.

Appendix 11 – Summary of NOS tool used for quality appraisal of cross-sectional studies

Paper	Newcastle Ottawa Scale question number and score allocated[a]								
	1	2	3	4	5	6	7	Total stars[b]	Category[c]
Goyal et al, 2020	b*	a*	a*	b*	c	a**	b	6	Medium
Manjrekar et al, 2018	b*	b	c	b*	c	c*	b	3	Low
Poreddi et al, 2020	a*	b	a*	b*	c	a**	a*	6	Medium
Ransing et al, 2020	b*	b	c	a**	c	c*	a*	5	Medium

[a]Newcastle-Ottawa question numbers 1-7, answers a-d, and associated number of stars (*) are detailed below
[b]Minimum number of possible stars to be awarded = 0, maximum number of possible stars to be awarded = 10.
[c]Categories were allocated as: Low = 0-3 stars, Medium = 4-7 stars, High = 8-10 stars.

Appendix 12 – STROBE checklist for pregnant women's cross-sectional survey study

	Item No	Recommendation	Page No
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract	119
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found	N/A as this is part of the wider PhD thesis
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	118-119
Objectives	3	State specific objectives, including any prespecified hypotheses	118
Methods			
Study design	4	Present key elements of study design early in the paper	119
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	119-123
Participants	6	(*a*) Give the eligibility criteria, and the sources and methods of selection of participants	120
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	121-123
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	121-123
Bias	9	Describe any efforts to address potential sources of bias	123-124 139
Study size	10	Explain how the study size was arrived at	120
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	124-125
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding	124-125
		(*b*) Describe any methods used to examine subgroups and interactions	N/A
		(*c*) Explain how missing data were addressed	N/A

		(*d*) If applicable, describe analytical methods taking account of sampling strategy	N/A
		(*e*) Describe any sensitivity analyses	N/A
Results			
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	N/A due to the intermediary data collection of women attending the MAA health camps/clinics
		(b) Give reasons for non-participation at each stage	N/A
		(c) Consider use of a flow diagram	N/A
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	126
		(b) Indicate number of participants with missing data for each variable of interest	N/A
Outcome data	15*	Report numbers of outcome events or summary measures	126-129
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	133-136
		(*b*) Report category boundaries when continuous variables were categorized	131
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	N/A
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	N/A
Discussion			
Key results	18	Summarise key results with reference to study objectives	137-138
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	139
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	137-138

Generalisability	21	Discuss the generalisability (external validity) of the study results	252
Other information			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	6

Appendix 13 – Pregnant women's survey

Survey for pregnant women

Antenatal depression in Bangladesh: A mixed methods study of the social determinants & perspectives and their role in maternal mental healthcare

Research team: Lead investigator: Nafisa Insan
Co-investigators: Professor Judith Rankin & Professor Simon Forrest
Organisation: Newcastle University
Collaborators: Maternal Aid Association (MAA)
Email: n.l.insan2@newcastle.ac.uk

INFORMED CONSENT (read out to the mothers)

I would like to ask you a few questions to help me understand your mental health during your pregnancy. Understanding your experiences can help in developing programmes for mental health care during pregnancy in the future.

Your participation in this research is completely voluntary. If you choose not to participate, this will not affect the care you receive. If you want to stop the survey at any point for any reason, I will stop it and provide any support you may need. We can resume the survey only if and when you are comfortable to do so.

Everything you say will be kept confidential and only the research team will have access to this information. If you do not want to answer a particular question, then we will skip it. You are free to withdraw at any point and this will not affect the care you receive. Would you like to ask any questions?

If you agree, please put your/participant initials in the box

1. I confirm that the information sheet for this study has been read out to me. I have had the opportunity to consider the information, ask questions and had these answered satisfactorily.	
2. I understand that my participation is voluntary and that I am free to withdraw at any time without giving any reason.	
3. I understand that relevant sections of the data collected during the study may be looked at by individuals who are part of the research team, other than the lead researcher. I give permission for these individuals to have access to the data collected.	
4. I agree to the interview being audio recorded and transcribed by a company which is external to Newcastle University, and the use of anonymised quotes in research reports and publications.	
5. I agree to take part in this study.	

Name of participant	Date	Signature/thumb print of participant

For researcher:

I have accurately provided the information sheet to the potential participant, and to the best of my ability made sure that the participant understands the details of the study. I confirm that the participant was given an opportunity to ask questions about the study, and all the questions asked by the

participant have been answered correctly and to the best of my ability. I confirm that the individual has not been coerced into giving consent, and the consent has been given freely and voluntarily.

_____	_____	_____
Name of interviewer	Date	Signature of interviewer

Please record the following information about the participant

Participant ID given when they register for health camp: ……

District: …….

Upazilla/Thana: …….

Pariticpants's pregnancy status:
☐First trimester (1-12 weeks pregnant)
☐Second trimester (13-26 weeks pregnant)
☐Third trimester (27+ weeks pregnant)

Section A: Sociodemographic information

I would like to start by asking you a few general background questions

1. Firstly, how old are you in years? ……
   ☐Don't know

2. May I know your current marital status? *(Select one)*
   ☐Married
   ☐Unmarried
   ☐Divorced
   ☐Separated from husband
   ☐Widow
   ☐Prefer not to say

3. What is the highest level of education you have received? *(Select one)*
   ☐No class passed/no schooling
   ☐Which class passed ……
   ☐Secondary school certificate (SSC)
   ☐High secondary school certificate (HSC)
   ☐Graduate
   ☐Masters
   ☐Higher than masters
   ☐Don't know

4. Do you do any paid work? This is work that gives you any sort of income
   ☐Yes
   ☐No

5. Does your husband do any paid work? If yes, what does he do? (*Skip if participant is unmarried*
   ☐Yes …….
   ☐No paid work

Participant ID:

Section B: Social support network

This next section is about your family and social support

6. We are interested to know about people who give you personal support **during your pregnancy.** These are people who are helpful, who listen to you, or who back you up when you are in trouble. Please tell me if the people I mention provide you with no support, some support or a lot of support during your pregnancy. If there is no such person in your life, please tell me

		None	Some	A Lot	N/A
a.	Your husband (or father of child)	☐	☐	☐	☐
b.	Your own children	☐	☐	☐	☐
c.	Your own parents/grandparents	☐	☐	☐	☐
d.	Your own brothers/sisters	☐	☐	☐	☐
e.	Other blood relatives (uncle/aunt)	☐	☐	☐	☐
f.	Your in-laws	☐	☐	☐	☐
g.	Your neighbours	☐	☐	☐	☐
h.	Your co-workers	☐	☐	☐	☐
i.	Members in your masjid or community groups	☐	☐	☐	☐
j.	Your other friends	☐	☐	☐	☐

7. Is there **one** particular person whom you trust and to whom you can go with personal difficulties?
   ☐Yes
   ☐No (Skip to question 9)

8. Please can you tell me who this person is? (from the list above)
   ……..

9. Now I would like to ask you about people who cause personal stress. A person who stresses you is one who causes problems for you or makes your life more difficult. For each person I mention, please tell me how much stress they cause during your pregnancy: no stress, some stress or a lot of stress. If there is no such person in your life, please tell me.

		None	Some	A Lot	N/A
a.	Your husband (or father of child)	☐	☐	☐	☐
b.	Your own children	☐	☐	☐	☐
c.	Your own parents/grandparents	☐	☐	☐	☐
d.	Your own brothers/sisters	☐	☐	☐	☐
e.	Other blood relatives (uncle/aunt)	☐	☐	☐	☐
f.	Your in-laws	☐	☐	☐	☐

Participant ID:

g. Your neighbours	☐	☐	☐	☐
h. Your co-workers	☐	☐	☐	☐
i. Members in your masjid/ community groups	☐	☐	☐	☐
j. Your other friends	☐	☐	☐	☐

10. Is there **one** particular person around you who causes you the most stress during your pregnancy?
    ☐Yes
    ☐No (Skip to section C)

11. Please can you tell me who this person is that causes you the most stress? (From list above)
    ……..

Section C: Pregnancy choices

This next section is about your pregnancy choices, please remember everything you say will stay between you and me

12. Do you know if you are having a girl or a boy, (if yes, may I know what the gender is?)
    ☐Boy
    ☐Girl
    ☐Don't know
    ☐Prefer not to say

13. Which would you prefer?
    ☐Boy
    ☐Girl
    ☐No preference
    ☐Prefer not to say

14. What do you think your husband (or father of your child) would prefer?
    ☐Boy
    ☐Girl
    ☐No preference
    ☐Prefer not to say

15. Did you want to have a baby at this time?
    ☐Yes
    ☐No
    ☐Prefer not to say

16. Did your husband (or father of your child) want to have a baby at this time?
    ☐Yes
    ☐No
    ☐Don't know
    ☐Prefer not to say

17. Are you worried about childbirth/labour?

- ☐ Yes
- ☐ No
- ☐ Prefer not to say

Section D: Edinburgh Postnatal Depression Scale (EPDS)

Moving onto the next section which is about how you have been feeling emotionally **in the past 7 days**.

18. Have you been able to laugh and see the funny side of things?
    - ☐ As much as I always could (0)
    - ☐ Not quite as much now (1)
    - ☐ Definitely not so much now (2)
    - ☐ Not at all (3)
19. Have you looked forward with enjoyment to things?
    - ☐ As much as I ever did (0)
    - ☐ Rather less than I used to (1)
    - ☐ Definitely less than I used to (2)
    - ☐ Hardly at all (3)
20. Have you blamed yourself unnecessarily when things went wrong?
    - ☐ Yes, most of the time (3)
    - ☐ Yes, some of the time (2)
    - ☐ Not very often (1)
    - ☐ No, not ever (0)
21. Have you been anxious or worried for no good reason?
    - ☐ No, not at all (0)
    - ☐ Hardly ever (1)
    - ☐ Yes, sometimes (2)
    - ☐ Yes, very often (3)
22. Have you felt scared or panicky for no good reason?
    - ☐ Yes, quite a lot (3)
    - ☐ Yes, sometimes (2)
    - ☐ No, not much (1)
    - ☐ No, not at all (0)
23. Have you felt that things have been getting on top of you?
    - ☐ Yes, most of the time I haven't been able to cope at all (3)
    - ☐ Yes, sometimes I haven't been coping as well as usual (2)
    - ☐ No, most of the time I have coped quite well (1)
    - ☐ No, I have been coping as well as ever (0)
24. Have you felt so unhappy that you had difficulty sleeping?
    - ☐ Yes, most of the time (3)
    - ☐ Yes, some of the time (2)
    - ☐ Not very often (1)
    - ☐ No, not at all (0)
25. Have you felt sad or miserable?
    - ☐ Yes, most of the time (3)
    - ☐ Yes, some of the time (2)
    - ☐ Not very often (1)
    - ☐ No, not at all (0)

Participant ID:

26. Have you been so unhappy that you have been crying?
    - ☐ Yes, most of the time (3)
    - ☐ Yes, quite often (2)
    - ☐ only occasionally (1)
    - ☐ No, never (0)
27. Have you had thoughts of harming yourself?
    - ☐ Yes, quite often (3)
    - ☐ Sometimes (2)
    - ☐ Hardly ever (1)
    - ☐ Never (0)

Section E: Intimate partner violence

(If husband has died or the participant is not married then please skip this section)

I want to now ask you about some difficult things that might have happened to you with your husband (or father of the child). These questions are about things that happen to many women. Anything you tell me will stay between me and you.

Has your husband (or father of your child):

28. Insulted you or made you feel bad about yourself?

Before pregnancy	**During pregnancy**
☐ Yes, once	☐ Yes, once
☐ Yes, a few times	☐ Yes, a few times
☐ Yes, many times	☐ Yes, many times
☐ No	☐ No
☐ Prefer not to say	☐ Prefer not to say

29. Physically hurt you in any way e.g. slapped, kicked, punched, burned etc.?

Before pregnancy	**During pregnancy**
☐ Yes, once	☐ Yes, once
☐ Yes, a few times	☐ Yes, a few times
☐ Yes, many times	☐ Yes, many times
☐ No	☐ No
☐ Prefer not to say	☐ Prefer not to say

30. Forced you to have sexual intercourse when you did not want to?

Before pregnancy	**During pregnancy**
☐ Yes, once	☐ Yes, once
☐ Yes, a few times	☐ Yes, a few times
☐ Yes, many times	☐ Yes, many times
☐ No	☐ No
☐ Prefer not to say	☐ Prefer not to say

Do you have any more comments you would like to add about anything we have discussed today?

……

……

Thank you for taking part and answering those questions!

Instruction to interviewer: Please take time to debrief the participants by asking how they are feeling, remind them that everything they have said will remain confidential and make sure they are okay before they leave.

Appendix 14 – Duke Social Support and Stress Scale scoring system

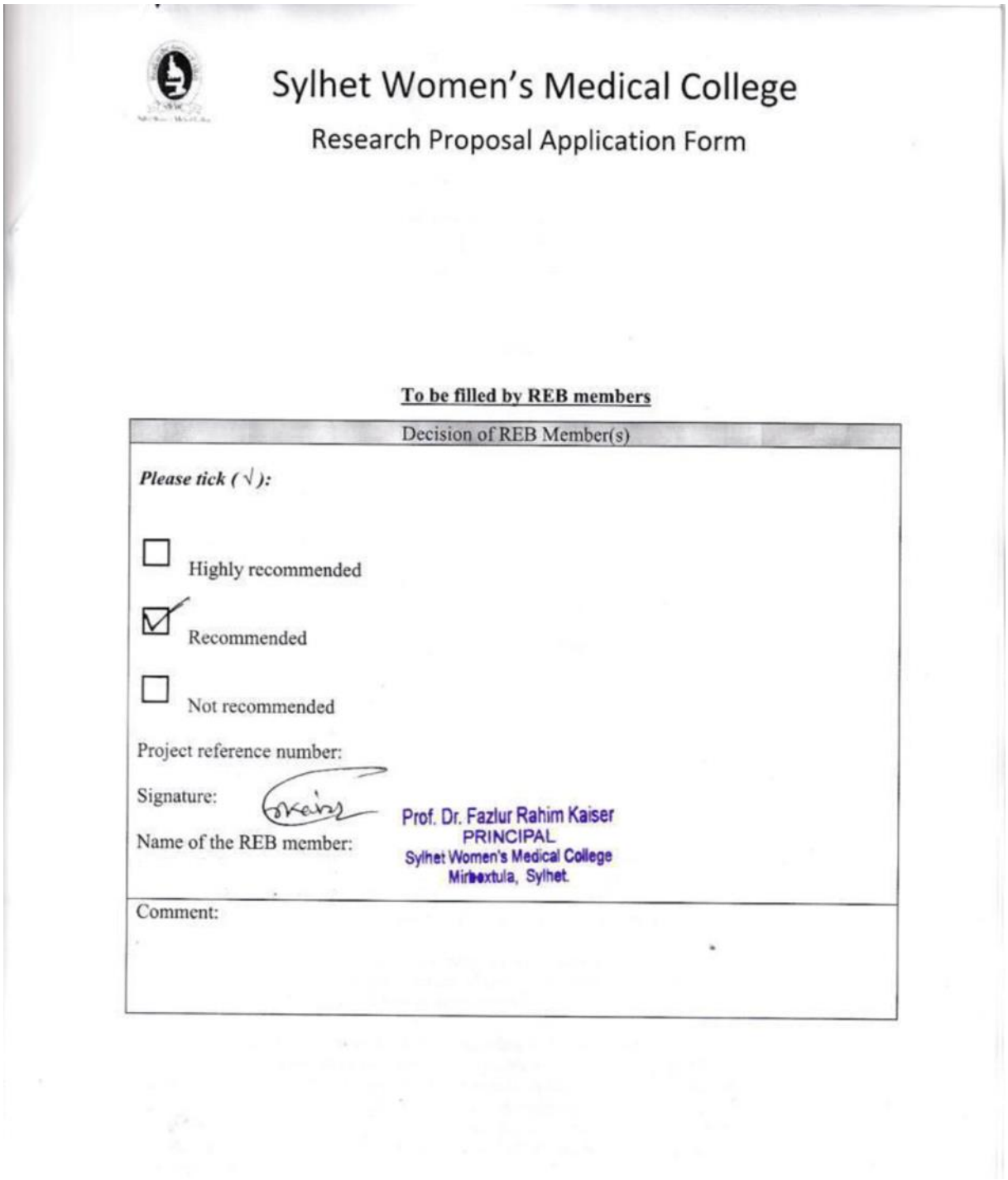
Sum the raw scores (range between 0-2) for the six family members:

1. Husband
2. Existing children
3. Parents
4. Siblings
5. Blood relatives
6. In-laws

If special supportive person is a family member add 2 to the sum above

Divide this final score by 14 and multiply by 100to obtain family support score on a scale of 0-100

Appendix 15 – Ethics approval from Sylhet Women's Medical College Research Ethics Board



Sylhet Women's Medical College

Research Proposal Application Form

To be filled by REB members

Decision of REB Member(s)
Please tick (√):   ☐ Highly recommended   ☑ Recommended   ☐ Not recommended   Project reference number: Signature: Name of the REB member: Prof. Dr. Fazlur Rahim Kaiser, PRINCIPAL, Sylhet Women's Medical College, Mirboxtula, Sylhet.
Comment:

Appendix 16 – STROBE checklist for HCP's cross-sectional survey study

	Item No	Recommendation	Page No
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract	143
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found	N/A as this is part of the wider PhD thesis
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	143
Objectives	3	State specific objectives, including any prespecified hypotheses	143-144
Methods			
Study design	4	Present key elements of study design early in the paper	144
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	144-147
Participants	6	(*a*) Give the eligibility criteria, and the sources and methods of selection of participants	144
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	145-147
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	145-147
Bias	9	Describe any efforts to address potential sources of bias	147 173
Study size	10	Explain how the study size was arrived at	N/A
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	147-148
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding	147-148
		(*b*) Describe any methods used to examine subgroups and interactions	N/A
		(*c*) Explain how missing data were addressed	147
		(*d*) If applicable, describe analytical methods taking account of sampling strategy	N/A
		(*e*) Describe any sensitivity analyses	N/A
Results			

Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	N/A as this was an online survey
		(b) Give reasons for non-participation at each stage	N/A
		(c) Consider use of a flow diagram	N/A
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	149-150
		(b) Indicate number of participants with missing data for each variable of interest	147
Outcome data	15*	Report numbers of outcome events or summary measures	151-166
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	151-166
		(*b*) Report category boundaries when continuous variables were categorized	151-166
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	N/A
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	N/A
Discussion			
Key results	18	Summarise key results with reference to study objectives	167-172
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	172-173
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	173-174
Generalisability	21	Discuss the generalisability (external validity) of the study results	173
Other information			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	6

Appendix 17 – Healthcare providers' online survey

Questionnaire for healthcare providers

Antenatal depression in Bangladesh: A mixed methods study of the social determinants and perspectives and their role in maternal mental health care

Screening

Before you take part in this survey, please can you confirm:

☐ You are aged 18 or over

☐ You provide healthcare for pregnant women in Bangladesh e.g. you are a doctor, nurse, midwife, community health worker, health attendant, member of charity organisation caring for pregnant women or medical student

What is the study about?

My name is Nafisa Insan and I would like to invite you to take part in this research study. This study is being conducted as part of a PhD project at Newcastle University in the UK, in collaboration with the Maternal Aid Association (MAA). The purpose of this study is to get a better understanding of mental health during pregnancy and the healthcare available for pregnant women with mental health conditions in Bangladesh.

Healthcare providers, like yourself, play a key role in access and delivery of mental health care for pregnant women; therefore, we are interested to understand your knowledge, attitudes and learning needs around mental health. We hope the results from this survey will create a better understanding of the health care available for pregnant women living with mental health conditions and how we can improve this through programmes and training.

What will it involve for me to take part?

The survey should take you approximately 15 minutes to complete. You have the right to withdraw from the study at any point during the survey. All responses are anonymous unless you decide to provide your email address at the end of the survey which is completely voluntary. Please answer all the questions and complete the survey only once.

Will my responses be kept safe and secure?

All the data from this survey will be kept on a password protected database on a password protected computer and will be kept strictly confidential and anonymous in the PhD thesis and any reports/papers we publish.

Who is this study funded and approved by?

This study is funded by the UK Economic and Social Research Council (ESRC).

This study was approved by the Faculty of Medical Sciences Research Ethics Committee, part of Newcastle University's Research Ethics Committee. This committee contains members who are internal to the Faculty. This study was reviewed by members of the committee, who must provide impartial advice and avoid significant conflicts of interests.

This study was also approved by the Sylhet Women's Medical College Institutional Review Board.

Who can I contact if I have any questions or queries?

If you have any questions please contact me, Nafisa Insan, by emailing n.l.insan2@newcastle.ac.uk.

We really appreciate your participation!

If you have understood the information provided and wish to complete the survey, please indicate your consent below and click 'Next' to begin the survey

I confirm and understand that:

- I have read and understood the information provided for this study
- I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily
- I have met the eligibility criteria: aged 18 or over and are a healthcare provider for pregnant women in Bangladesh
- My participation is voluntary and that I am free to withdraw at any time without giving any reason
- The data collected will be looked at by the lead researcher and co-investigators

I agree to have read and understood the information provided above and consent to take part in this survey:

☐Yes

Section A: About you

This section asks about you

1. How old are you? (Age in years) ……..

2. How do you describe your gender?
   ☐Female
   ☐Male
   ☐Other
   ☐Prefer not to say

3. What is your healthcare provider role? *(Please tick one)*
   ☐Doctor/Physician
   ☐Nurse
   ☐Midwife
   ☐Community health worker/Health attendant
   ☐Medical student
   ☐Other (Please specify) ……..

4. Where do you conduct your main maternal healthcare work in Bangladesh? *(Please tick one)*
   ☐Medical University
   ☐Government Medical College hospital
   ☐Speciality hospital
   ☐District hospital

☐Maternal & Child Welfare centre
☐Upazilla health complexes
☐Union health and family welfare centre
☐Community clinics
☐NGO/Charity work in rural villages
☐Other (Please specify) …….

5. Approximately how many years have you worked in maternal healthcare? …….

6. Do you have any experience providing healthcare for people with **mental health** conditions? (This can be paid or unpaid work)
☐Yes
☐No

7. Have you had any education or training that has increased your knowledge about **mental health** within the past two years?
☐Yes
☐No

Section B: Attitudes & views around mental health

This section is about your views on mental health and mental health care for pregnant women in Bangladesh. There are no right or wrong answers, please give your opinion.

8. One of the main causes of **mental illness** is a lack of self-discipline and will power

Strongly Disagree	disagree	neutral	agree	strongly agree
☐	☐	☐	☐	☐

9. I would feel comfortable talking to someone who has a mental health condition

Strongly Disagree	disagree	neutral	agree	strongly agree
☐	☐	☐	☐	☐

10. The physical health of pregnant women is **more important** than their mental health

Strongly Disagree	disagree	neutral	agree	strongly agree
☐	☐	☐	☐	☐

11. Mental illness is an illness which requires medical attention

Strongly Disagree	disagree	neutral	agree	strongly agree
☐	☐	☐	☐	☐

12. Pregnant women with mental health conditions are a burden to their families

Strongly Disagree	disagree	neutral	agree	strongly agree
☐	☐	☐	☐	☐

13. A pregnant woman who shows signs of mental health issues should be immediately hospitalised

Strongly Disagree	disagree	neutral	agree	strongly agree
☐	☐	☐	☐	☐

14. Mental health conditions in pregnant women is not a serious health concern that healthcare providers need to address

Strongly Disagree	disagree	neutral	agree	strongly agree
☐	☐	☐	☐	☐

15. Pregnant women with depression or anxiety will never recover enough to have a good quality of life

Strongly Disagree	disagree	neutral	agree	strongly agree
☐	☐	☐	☐	☐

16. Mental health issues should be discussed more openly within Bangladeshi families and communities

Strongly Disagree	disagree	neutral	agree	strongly agree
☐	☐	☐	☐	☐

Section C: Questions about maternal depression

These questions are **not** a test but simply to get an idea of your understanding around antenatal depression and anxiety for women living in Bangladesh. Please select an option even if you are unsure about the answer

17. The prevalence (percentage of the population with the condition at a given time) of antenatal depression in Bangladesh is approximately *(Please tick one):*
    ☐5-15%
    ☐20-33%

☐35-40%
☐45-50%

18. Which of the following can happen as a result of antenatal depression and anxiety? *(Please tick one)*
    ☐Postpartum depression
    ☐Delayed child development
    ☐Low birth weight for the baby
    ☐All of the above

19. Which of the following is **not** considered a common risk factor for antenatal depression and anxiety? *(Please tick one)*
    ☐Intimate partner violence
    ☐High number of previous pregnancies
    ☐Unplanned pregnancy
    ☐Low social support

20. Which of the following is the **most common** symptom of depression during pregnancy? *(Please tick one)*
    ☐Irritability around others
    ☐Attention seeking from family and friends
    ☐Reliving past experiences and events
    ☐Feelings of isolation, loneliness & sadness

21. Which one of the following statements is **true**? *(Please tick one)*
    ☐Antenatal depression always continues into the postpartum period
    ☐Women with antenatal depression have a higher chance of developing postpartum depression
    ☐Women who are depressed during pregnancy do not require specific treatment
    ☐Antenatal depression will resolve with the birth of the baby

22. What is the Edinburgh Postnatal Depression Scale (EPDS)?*(Please tick one)*
    ☐A diagnostic tool for antenatal and postnatal depression and anxiety
    ☐A screening tool for antenatal and postnatal depression and anxiety
    ☐A management tool for antenatal and postnatal depression and anxiety
    ☐None of the above

23. What is the recommended treatment/management method for mild to moderate antenatal depression in Bangladesh? *(Please tick one)*
    ☐Understanding and empathy
    ☐Psychosocial interventions & counselling
    ☐Hospitalisation and medication
    ☐Naturotherapy and relaxation

Section D: Perinatal mental health learning needs

This section is about your views on what you think would enable you to provide better care for pregnant women in Bangladesh. There are no right or wrong answers, it is completely your opinion.

24. Do you feel confident in supporting pregnant women with depression or anxiety?
    - ☐ Yes, fully confident
    - ☐ Yes, mostly confident
    - ☐ No, not that confident
    - ☐ No, not confident at all

25. Do you have sufficient access to information on antenatal depression and anxiety in women in Bangladesh?
    - ☐ Yes
    - ☐ No

26. Which **mental health disorders** do you think you require additional education on? *(please select as many of the following mental health disorders you may want more education on)*
    - ☐ Personality disorders
    - ☐ Bipolar
    - ☐ Substance related disorders
    - ☐ Tokophobia (fear of childbirth)
    - ☐ Schizophrenia
    - ☐ Anxiety disorders
    - ☐ Depression
    - ☐ Other
    - ☐ No further education required on any of the above mental health disorders

27. Which **topics on mental health** do you think you require additional education on? *(please select as many of the following topics which you need more education on)*
    - ☐ Impact of childbearing on mental illness
    - ☐ Service options
    - ☐ Signs and symptoms
    - ☐ Roles of other healthcare providers
    - ☐ Child safety
    - ☐ Other
    - ☐ No further education required on any of the above topics

28. Which skills do you think you require additional training on? *(please select as many of the following skills which you feel you need more training on)*
    - ☐ Handing stress and aggression
    - ☐ Assessment of mental health
    - ☐ Assessment of risk of mental illness
    - ☐ Clinical management of mental health disorders
    - ☐ Discharge planning
    - ☐ Communication skills
    - ☐ Working with families and carers
    - ☐ Support with breastfeeding
    - ☐ Other
    - ☐ No further education required on any of the above skills

29. What is your preferred method of learning?
    - ☐ Online packages
    - ☐ Lectures with guest speaker
    - ☐ Series of seminars/workshop

How to contribute further to this research project

We would like to invite you take part in an online interview with the lead researcher, Nafisa Insan, so that we can learn more about healthcare providers' role in providing care for pregnant women with mental health conditions. It is completely voluntary, if you would be willing to participate in an interview (in English or Bangla), we would really appreciate if you would provide your email address so we can contact you with all the information. Your email will only be used to contact you about the interview and not for any other reasons. It will be kept confidential.

1. Email …….
2. Anything else you would like to add about the topics discussed in this survey?
   ……..

Thank you very much for completing the survey!

Appendix 18 – COREQ checklist for qualitative study

No. Item	Guide questions/description	Reported on Page #
Domain 1: Research team and reflexivity		
Personal Characteristics		
1. Interviewer/facilitator	Which author/s conducted the interview or focus group?	177
2. Credentials	What were the researcher's credentials? E.g. PhD, MD	N/A
3. Occupation	What was their occupation at the time of the study?	N/A
4. Gender	Was the researcher male or female?	177
5. Experience and training	What experience or training did the researcher have?	177
Relationship with participants		
6. Relationship established	Was a relationship established prior to study commencement?	176-178
7. Participant knowledge of the interviewer	What did the participants know about the researcher? e.g. personal goals, reasons for doing the research	176-178
8. Interviewer characteristics	What characteristics were reported about the interviewer/facilitator? e.g. Bias, assumptions, reasons and interests in the research topic	177

Domain 2: study design		
Theoretical framework		
9. Methodological orientation and Theory	What methodological orientation was stated to underpin the study? e.g. grounded theory, discourse analysis, ethnography, phenomenology, content analysis	176
Participant selection		
10. Sampling	How were participants selected? e.g. purposive, convenience, consecutive, snowball	177-178
11. Method of approach	How were participants approached? e.g. face-to-face, telephone, mail, email	177-178
12. Sample size	How many participants were in the study?	182
13. Non-participation	How many people refused to participate or dropped out? Reasons?	177-178
Setting		

14. Setting of data collection	Where was the data collected? e.g. home, clinic, workplace	180-181 .
15. Presence of non-participants	Was anyone else present besides the participants and researchers?	181
16. Description of sample	What are the important characteristics of the sample? e.g. demographic data, date	184
Data collection		
17. Interview guide	Were questions, prompts, guides provided by the authors? Was it pilot tested?	179-180
18. Repeat interviews	Were repeat interviews carried out? If yes, how many?	N/A
19. Audio/visual recording	Did the research use audio or visual recording to collect the data?	181
20. Field notes	Were field notes made during and/or after the interview or focus group?	N/A
21. Duration	What was the duration of the inter views or focus group?	181
22. Data saturation	Was data saturation discussed?	179
23. Transcripts returned	Were transcripts returned to participants for comment and/or correction?	N/A
Domain 3: analysis and findings		
Data analysis		
24. Number of data coders	How many data coders coded the data?	182-183
25. Description of the coding tree	Did authors provide a description of the coding tree?	182-183
26. Derivation of themes	Were themes identified in advance or derived from the data?	182-183
27. Software	What software, if applicable, was used to manage the data?	N/A
28. Participant checking	Did participants provide feedback on the findings?	N/A
Reporting		
29. Quotations presented	Were participant quotations presented to illustrate the themes/findings? Was each quotation identified? e.g. participant number	187-219
30. Data and findings consistent	Was there consistency between the data presented and the findings?	187-219
31. Clarity of major themes	Were major themes clearly presented in the findings?	187-219
32. Clarity of minor themes	Is there a description of diverse cases or discussion of minor themes?	187-219

Appendix 19 – Topic guide for semi-structured interviews with healthcare providers

Interview topic guide for Healthcare providers

Antenatal depression in Bangladesh: A mixed methods study of the social determinants and perspectives and their role in maternal mental health care

ID no:
Job/position/role:
Country of residence:

Introduction

Can I confirm that you have had the opportunity to read the information sheet about this study and were given the chance to ask any questions and had them answered? And, can I confirm that you give consent to participate in this study?

Thank you for agreeing to participate in this research and to undergo this interview with me. I want to understand your knowledge and views around maternal mental health, management options for pregnant women who experience mental health disorders in Bangladesh and hear any suggestions that you may have for change in how we respond or policies and practice. Please expand as much as you want on any of the questions, I really want to understand your thoughts and experiences. Please be assured that anything you say will stay anonymous and confidential.

1. To start with would you tell me a bit about the maternal healthcare work that you do in Bangladesh. Why maternal health?
   - *Role, locations, services, programmes etc*
   - *Motivations*

Knowledge base, attitudes and perceptions of mental health overall

2. When did you first get taught about mental health conditions?
   - *What was the teaching like? What types of conditions were covered; how long, how much was covered?*

3. What were your views on mental health conditions such as depression and anxiety when you first learned about them?
   - *Did it interest you? How important do you think mental health is?*
   - *Have these views changed now and if so, how and why/what has changed your views?*

Experience and views on maternal mental health

4. During your experience of providing antenatal care for women in Bangladesh, have you ever come across pregnant women with mental health conditions?

- *What were these mental health conditions?*
- *How did you screen/diagnose and treat?*

5. Do you think women in Bangladesh are open to discussing their mental health issues during pregnancy?
   - *Do women ever speak to you about their mental health? How do pregnant women tend to talk about their mental health?*
   - *Why? Why not? Can you give an example of a time when a woman chose to speak/not to speak about her mental health?*

6. Do you think mental health should be assessed and managed in antenatal care for pregnant women in Bangladesh?
   - *What role would you play? How should it be managed?*

Social determinants of maternal mental health

7. Do you see social factors playing a role in antenatal depression?
   - Can you give me some examples of some social factors that might influence antenatal depression?
   - *Probe some answers like intimate partner violence, gender preference, unplanned pregnancies if they do not have any answers*

8. Do you think knowledge of social determinants amongst HCP could impact mental health and healthcare for pregnant women in Bangladesh? If so, in what ways?
   - *Heath inequalities*
   - *Screening and identifying women at risk*
   - *Management through psychosocial and community-based interventions*
   - *Training and educating women on factors such as family planning, gender-based violence etc*

Management for pregnant women who experience mental health disorders

9. What methods or tools are you aware of to screen and manage depression and anxiety within pregnant women in Bangladesh?
   - *Which tools do you use? Why?*
   - *Screening tools – EPDS Bangla*
   - *Psychosocial and community-based interventions*
   - *Medications*

Suggestions for change to provide better maternal mental health care

10. How do you think maternal mental healthcare can be improved in Bangladesh? What changes would you suggest?
    - *What three things do you think would make the greatest difference to improving pregnant women's mental health? In what ways? What needs to change to enable that difference to occur?*
    - *Training & education for healthcare providers and pregnant women*
    - *Increase in staff*
    - *Routine mental health guidelines*
    - *Multidisciplinary teams' approach*

Debrief

Thank you for participating in this study. From this research, I aim to learn more about the knowledge base and attitudes towards antenatal depression of healthcare providers in Bangladesh. I hope to understand the barriers and facilitators for pregnant women in accessing mental healthcare and the barriers and facilitators for healthcare in delivering mental healthcare for pregnant women. This will enable us to develop appropriate training programmes to ensure that healthcare providers are equipped to screen and manage depression during pregnancy.

The data you have provided will remain anonymous and confidential. Any writing up of the data cannot be traced back to your personal identity. Your data will not be available to anyone outside of the research team. If you want to withdraw at any point, you can email me at n.l.insan2@newcastle.ac.uk and all your data will be permanently and securely deleted.

If you would like further information about the study or would like to know about what my findings are when all the data have been collected and analysed then please contact me on the email I have provided. Thank you again for your participation.

Appendix 20 – Information sheet and consent form for healthcare providers' participating in semi-structured interviews

Information for healthcare providers about the study and a form to consent to participate

Antenatal depression in Bangladesh: A mixed methods study of the social determinants and perspectives and their role in maternal mental health care

Lead investigator: Nafisa Insan
Email: n.l.insan2@newcastle.ac.uk
Co-investigators: Professor Judith Rankin & Professor Simon Forrest
Organisation: Newcastle University
Collaborators: Maternal Aid association (MAA)

Part I: Information about the study

I would like to invite you to take part in our research study which aims to understand healthcare provider's knowledge and attitudes around antenatal depression and the health care available for pregnant women with mental health conditions in Bangladesh. It is up to you if you would like to take part in this study or not. Before you decide, it is important you understand why the research is being done and what being involved in the study would mean for you. Please take time to read the following information and if there is anything that is not clear, or if you would like more information, please do not hesitate to contact me (Nafisa Insan); my details are given below.

What is the purpose of the study?

The purpose of this study is to get a better understanding of mental health during pregnancy and the healthcare available for pregnant women who experience mental health conditions in Bangladesh. Healthcare providers play a key role in care for the mental health for pregnant women; therefore, it is important to know about your knowledge, attitudes, experience and learning needs around mental health during pregnancy. We would also like to know whether you have any suggestions that you may have for changing how healthcare is currently given.

Why have I been invited?

We are inviting you to take part because you are a healthcare provider for pregnant women in Bangladesh and have experience of caring or organising care for pregnant women. We think that you can contribute to the understanding of delivery and uptake of mental health care by pregnant women.

Do I have to take part?

It is up to you whether you take part in this research or not.. If you agree to take part, you still have the option to withdraw from it up until the interview has been completed without giving a reason. If you choose not to take part, this will not affect you in any way.

What will happen if I decide to take part?



I will set up a meeting at a time that is good for you to take part in an informal interview. The interview will take place online (zoom/skype/teams/telephone) with myself, Nafisa, as the lead researcher. I want you to feel comfortable in giving your views and opinions on this topic knowing that there is no right or wrong answer, I want you to be as honest as you can. If you do not want to answer a particular question(s), then I can move on. The interview will be audio recorded with your

consent; however, you will not be identifiable on the recording. Recording the interview allows us to analyse the data from the interviews.
If you decide at the end of the interview that you no longer want to be included in the study, I will delete all your information and the interview recording.

How much of my time will this take?

The interview will last about 45 minutes.

Is there anything else that I should be aware of?

I will be asking about your own views, which you may find to be quite personal. I will only take your name and professional role, which will be kept secure and will not be shared with anyone outside the research team. Your name will not be given on any data from this interview. If you do not want to answer a question for any reason that is completely fine, no questions will be asked about the reason, I will simply move on.

What are the possible benefits of taking part?

We hope the results from this research will lead to a better understanding of the health care available in Bangladesh to pregnant women with mental health conditions and how we can improve this.

Will my taking part in the study be kept confidential?

We will talk to you online. Recording of the interview will only start when you agree. I will note down your name and job role, however, I will not be sharing any of this information or the answers you give to anyone outside of the research team. You will not be identifiable on any of the data .

What will happen with the interview recordings?

Nothing you tell me will be shared with anybody outside of the research team. The interview recordings and your personal details will be stored securely, and password protected and can only be accessed by me. The recordings will be written out as words on paper and then I will analyse these. This is the research data and will only be accessible by me and the co-investigators. The results of what I find will be published so that other researchers and people interested in the research area may learn from this study. The recordings will be destroyed after they have been transcribed and transcribed interviews will be archived after the study.

What will happen if I don't want to carry on with the study?

You do not have to take part in this research if you do not want to. You may stop participating at any point during the interview and if you do not want the information you have provided to be shared, I will permanently delete all the recordings and data you have provided. You may withdraw from the study up until competition of the interview when you will be asked again to confirm if you are happy for us to use the recording.

What will happen to the results of this study?

This study will contribute to the fulfilment of a PhD/doctoral thesis. We intend on publishing our results in scientific journals, presenting our findings at research conferences and writing up reports for MAA. You will not be identifiable from any of the reports or publications.

Who is funding this study?

This study is funded by the UK's Economic and Social Research Council (ESRC).

Who to contact?

If you have any questions, please contact me on the following: n.l.insan2@newcastle.ac.uk

Who has reviewed the study?

This study was approved by the Faculty of Medical Sciences Research Ethics Committee, part of Newcastle University's Research Ethics Committee. This committee contains members who are internal to the Faculty. This study was reviewed by members of the committee, who must provide impartial advice and avoid significant conflicts of interests.

This study is also approved by the Sylhet Women's Medical College Institutional Review Board

Thank you for reading this information. Please see below for the consent form. I look forward to receiving your reply.

Part II: Participant Consent Form

Antenatal depression in Bangladesh: A mixed methods study of the social determinants and perspectives and their role in maternal mental health care

Lead investigator: Nafisa Insan
Email: n.l.insan2@newcastle.ac.uk
Co-investigators: Professor Judith Rankin & Professor Simon Forrest
Organisation: Newcastle University
Collaborators: Maternal Aid Association (MAA)

If you agree, please put your initial in the box

6. I confirm that I have read the information sheet for this study. I have had the opportunity to consider the information, ask questions and had these answered satisfactorily.	
7. I understand that my participation is voluntary and that I can withdraw at any time up until the interview is completed without giving any reason	
8. I understand that relevant parts of the data collected during the study will only be looked at by the lead researcher (Nafisa Insan) and co-investigators. I give permission for these individuals to have access to the data collected .	
9. I agree to the interview being audio recorded and transcribed with assistance from external services and the use of anonymised quotes in research reports and publications.	
10. I agree to take part in this study.	

Name of participant	Date	Signature of participant

I have accurately provided the information sheet to the potential participant, and to the best of my ability made sure that the participant understands the details of the study. I confirm that the participant was given an opportunity to ask questions about the study, and all the questions asked by the participant have been answered correctly and to the best of my ability. I confirm that the individual has not been coerced into giving consent, and the consent has been given freely and voluntarily. A copy of this Information and Consent Form has been provided to the participant.

Name of researcher	Date	Signature of researcher